

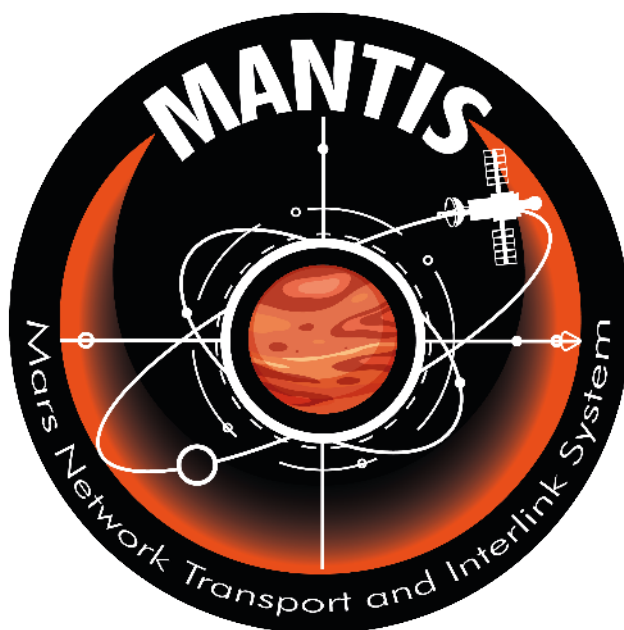


Politecnico di Milano

Applied Space Mission Analysis and Design

Course (A.Y. 2025/2026)

Final Report **MANTIS Mission**



MArs Network Transport and Interlink System

Authors:

Addeo Maria Teresa	10820497	mariateresa.addeo@mail.polimi.it
Atoche Angela	11061016	angelasofia.atocche@mail.polimi.it
Bove Giovanni	11078130	giovanni.bove@mail.polimi.it
Cammarota Caterina	10812113	caterina.cammarota@mail.polimi.it
Coppola Riccardo	10773170	riccardo.coppola@mail.polimi.it
Perniciaro Angelica	10823539	angelica.perniciaro@mail.polimi.it
Pezzi Francesco	11093867	francesco2.pezzi@mail.polimi.it
Pirozzi Federica	10737390	federica.pirozzi@mail.polimi.it
Scoleri Erika	10711477	erika.scoleri@mail.polimi.it
Severo Cristian	10842151	cristian.severo@mail.polimi.it
Torrano Francesco	11076885	francesco.torrano@mail.polimi.it

Professor: Michèle Roberta Lavagna

Lecturers: Andrea Brandonisio, Matteo d'Ambrosio, Giovanni Pio Parracino

Team Roles and Authorship Declaration



Team Member	Primary Role	Contributions
Addeo Maria Teresa	OBDH	TTMTC
Atoche Angela	COST&RISK	PS
Bove Giovanni	MECHS	CONF, EPS
Cammarota Caterina	CONF, STRUCT	MA, ADCS
Coppola Riccardo	GNC	MA, TCS, EPS
Perniciaro Angelica	PS	LAUNCHERS
Pezzi Francesco	MA	GNC, TCS
Pirozzi Federica	TTMTC	OBDH
Scoleri Erika	TCS	PS
Severo Cristian	EPS	PS
Torrano Francesco	ADCS	TTMTC, CONF

Table 1: Authorship declaration and team roles for the MANTIS mission.

Contents

1	Introduction	1
1.1	Mission Objectives	1
1.2	High-Level Mission Requirements	1
2	Mission Environment Analysis	2
2.1	Earth Vicinity	2
2.2	Interplanetary Cruise	2
2.3	Mars System Environment	4
2.4	Quantified Environmental Effects on Architectures	6
3	State of the Art	7
3.1	Criticalities and Lessons Learnt	8
3.1.1	Past Martian Missions	8
3.1.2	Space Tugs	9
3.2	Level of Autonomy	9
3.3	Rough Order of Magnitude of System Indicators	10
4	Mission Architecture	11
4.1	Mission Architecture Alternatives	11
4.1.1	Mission Alternatives Criteria	12
4.1.2	Qualitative Analysis	12
4.1.3	Quantitative Analysis	13
5	Preliminary Mission Definition	15
5.1	Functional Decomposition	15
5.2	Mission Drivers	16
5.3	Mission Phases	16
5.4	Concept of Operations (ConOps)	17
6	Launcher selection	18
6.1	European Launch Vehicle Option: Ariane 64	18
6.2	Non-European Launch Vehicle Options	18
7	Mission Analysis	20
7.1	Orbits of interest	20
7.2	Relevant perturbations	20
7.3	Trade-off	21
7.3.1	Interplanetary transfer	21
7.3.2	Mars Orbit Injection (MOI)	22
7.3.3	Orbit Lowering	22
7.4	Interplanetary trajectory design	22
7.4.1	Earth Departure	23
7.4.2	B-Plane Target	24
7.4.3	Mars Orbit Injection	24
7.4.4	Monte Carlo simulation	25
7.5	Client Alternatives	26
7.6	Aerobraking	27
7.7	Relay Orbit	28

7.7.1	Trade-off	28
7.7.2	Station Keeping Strategy	29
7.8	ΔV Budget	30
8	Attitude Determination and Control	31
8.1	Modes	31
8.2	Pointing Budget	32
8.3	Sensors Selection	32
8.4	Relay Architecture	33
8.4.1	Pointing Strategy	33
8.4.2	Actuators Sizing	33
8.4.3	Configuration	35
8.5	Carrier Architecture	36
8.5.1	Actuators Sizing	36
8.5.2	Propellant Mass Estimation	39
8.5.3	Aerobraking Strategy Selection	39
8.5.4	Configuration	40
8.6	Performance	40
8.6.1	Aerobraking Performance	41
8.6.2	Relay Pointing Performance	41
9	Propulsion Subsystem	42
9.1	Carrier Propulsion Subsystem	42
9.1.1	Criteria & Trade-off	42
9.1.2	Modeling & Assumptions	42
9.1.3	Dual-mode Architecture	44
9.1.4	Thrusters configuration	44
9.1.5	Number Selection and Redundancy Analysis	46
9.1.6	Feeding Strategy	47
9.1.7	Effets of Carrier PS on s/s	51
9.2	Relay Propulsion Subsystem	51
10	Tracking, Telemetry and Telecommand	53
10.1	Preliminary Identification of Data Budgets to Download	53
10.2	Communication Strategy	53
10.3	Relay	56
10.3.1	Frequency Trade-Off: X-Band vs Ka-Band	56
10.3.2	Parabolic Antenna Sizing for Scientific Data Downlink	57
10.3.3	Ground Segment Verification with ESA ESTRACK	58
10.3.4	Communication Architecture	59
10.4	Carrier	59
10.4.1	Relay–Carrier Visibility During Payloads Release	60
10.4.2	Frequency Trade Off: UHF vs S-Band	61
10.5	Architecture	63
10.5.1	X-Band RF Chain	63
10.5.2	UHF RF chain	64
10.6	Tracking Strategy	65
10.6.1	Relay-Earth	65
10.6.2	Relay-Payload	65

11 Guidance, Navigation, and Control	67
11.1 MOI Uncertainty Analysis	67
11.2 Navigation Strategy	68
11.2.1 Pre-Relay Release Navigation	69
11.2.2 Relay Navigation	70
11.2.3 Post-Relay Release	71
12 On Board Data Handling	73
12.1 Design Drivers and Component Selection Rationale	73
12.2 Relay OBDH Architecture	73
12.3 Component Selection	75
12.3.1 FPGA adoption rationale	75
12.3.2 MRAM adoption rationale	76
12.4 Relay State Diagram	76
12.5 Relay Operation	76
12.6 Carrier Operation	77
12.7 Relay Solid State Recorder (SSR) sizing	77
13 Electric Power Subsystem	78
13.1 Introduction	78
13.2 Power Analysis Methodology	78
13.2.1 Margin Philosophy	78
13.3 Power Budget Breakdown	79
13.4 Primary Power Source	80
13.5 Secondary Power Source	82
13.6 EPS Architecture	83
13.7 Effects and Constraints	85
14 Thermal Control	86
14.1 Requirement Temperatures	86
14.2 Internal heat dissipation	87
14.3 Analysis cases	88
14.3.1 Interplanetary Cruise	88
14.3.2 Release Operations	88
14.3.3 Relay nominal orbit	89
14.3.4 Identification of Design Cases	89
14.4 Multinodal thermal analysis	90
14.4.1 Stacked configuration Nodalization Scheme	90
14.4.2 Relay Nodalization Scheme	91
14.4.3 Radiative Environment and Heat Exchange Modeling	91
14.5 Design Description	91
14.5.1 Carrier	91
14.5.2 Relay	92
14.5.3 Thermal Properties	94
14.6 Temperature Prediction	94
14.6.1 Interplanetary Cruise	94
14.6.2 Release Phase	95
14.6.3 Relay Nominal Orbit	95
15 Mechanisms	96

15.1	Relay Release and Support Frame	96
15.1.1	System Role and Architectural Philosophy	96
15.1.2	Retention, Actuation and Controlled Separation	97
15.1.3	Operational Logic, Monitoring and Redundancy	97
15.1.4	Maturity Assessment	99
15.2	Dual-Leaf Payload Bay Door	99
15.2.1	System Role and Architectural Rationale	99
15.2.2	Operational Logic and State Architecture	100
15.2.3	Deployment Kinematics and Dynamic Verification	101
15.2.4	Actuation and Mechanical Implementation	102
15.2.5	Architectural Trade-Off	102
15.2.6	Functional Responsibility of Main Components	103
15.2.7	Maturity Assessment	103
15.3	CPDI + PAI (Motorized Lightband) – Mission Service Interface and Payload Release .	103
15.3.1	Architectural role of CPDI and standardization strategy	103
15.3.2	PAI mechanism: Non-Explosive Motorized Lightband	103
15.3.3	Release command and safety logic: finite-state model	105
15.3.4	Clearance analysis on separation impulse and spring sizing traceability	105
15.3.5	Separation Dynamics Maturity Roadmap	107
16	Configuration	108
16.1	MANTIS general configuration and constraints	108
16.2	Carrier external configuration	108
16.3	Carrier internal configuration	110
16.4	Relay internal and external configuration	111
17	Structures	112
17.1	Material selection	112
17.2	Carrier and Relay internal structure	113
17.3	Structural analysis	114
17.3.1	Modal Analysis	114
17.3.2	Quasi Static analysis	115
17.3.3	Sine Analysis	115
17.3.4	Random Analysis	115
17.3.5	Analysis Output	116
17.4	Possible Further Analysis	116
18	PHAROS-M Interface	117
19	Mission Budgets	118
19.1	Power Budget	118
19.2	Mass Budget	119
20	Conclusions	120
A	Mission Analysis	121
A.1	Baseline 1: ΔV Budget	121
A.2	Baseline 2: ΔV Budget	121
A.3	Baseline 3: ΔV Budget	122
B	ADCS	123

B.1	Carrier Modes State Machine	123
B.2	Relay Modes State Machine	123
B.3	Aerobraking Dynamical Simulator	124
B.4	Relay Dynamical Simulator	125
C	PS results	127
C.1	Baseline 1	127
C.2	Baseline 2	128
D	TTMTC	129
D.1	Redundancy, Robustness and Autonomy	129
D.2	Trade off X and Ka band: comparison link budget	130
D.3	Link Budget Analysis	130
D.3.1	X-band horn antenna	131
D.3.2	X-band parabolic antenna	133
D.3.3	UHF Relay Link: Carrier	134
D.3.4	UHF Relay Link: Relay	136
D.4	Components of the RF chain	137
D.4.1	X band	137
D.4.2	Relay: UHF	138
D.4.3	Carrier: UHF	138
E	EPS	140
E.1	Power Budget Tables	140
E.2	Batteries Dynamical Simulator	142
F	Mass Budget	143

List of Figures

1	Solar proton fluence from the ESP–PSYCHIC model (1-year, 95% CL).	3
2	Predicted SEU rate for COTS and radiation-hardened memories.	3
3	Interplanetary meteoroid cumulative flux derived from the Grün model (1-year exposure).	4
4	Crustal magnetic field model at 500 km.	5
5	Architecture Tree.	11
6	AHP Monte Carlo analysis results.	13
7	Mission functional decomposition.	15
8	Mission ConOps.	18
9	Ariane 64, Payload Mass vs v_∞	19
10	American launch vehicles, Payload Mass vs $C3$	19
11	Orbital parameters evolution.	21
12	Earth Mars porkchop plot over the time window 2030-2036.	22
13	Earth–Mars Lambert transfer in the Sun-centered ecliptic J2000 frame.	23
14	Earth-centered departure hyperbolas in the J2000 reference frame.	23
15	Sun-centered transfer trajectory in the Ecliptic J2000 reference frame.	24
16	Mars-centered orbit in the Mars inertial reference frame.	25
17	Monte Carlo simulation results under TCM uncertainties.	25
18	Monte Carlo simulation results under two TCM uncertainties.	26
19	Monte Carlo histograms for TCM magnitude.	26
20	Client Alternatives.	27
21	Orbit and pericenter altitude profile around Mars.	27
22	Ground visibility comparison for the three candidate relay orbits.	29
23	Eclipse Duration Map over 15 years lifetime.	30
24	Earth Occultation Duration Map over 15 years lifetime.	30
25	Relay sensors and actuators configuration: CSS in yellow, Star Trackers in blue, thrusters in red.	35
26	Aerobraking strategies. On the left, maximum area case; on the right, minimum area case.	37
27	Carrier sensors and actuators: CSS in yellow, star trackers in blue, thrusters in red.	40
28	Monte Carlo analysis of the aerobraking phase. Left: full transient response. Right: zoom at peak aerodynamic loading (100 km altitude).	41
29	Monte Carlo analysis of the telecommunication phase. Left: acquisition from large initial errors. Right: steady-state pointing accuracy after convergence.	41
30	Thrusters configuration geometry.	45
31	3D Plume impingement visualization.	45
32	TCMs single failure managment.	46
33	Post-MOI single failure managment.	46
34	MOI single failure managment.	46
35	Tanks' layout.	47
36	Three-stage pyrotechnic ladders activation scheme.	49
37	Carrier Propulsion Schematics.	50
38	Relay Propulsion Schematics.	52
39	Communication strategy: Near Earth Commissioning.	55
40	Communication strategy: Interplanetary and MOI.	55
41	Communication strategy: Relay Operations.	56
42	Minimum input and transmitting power required to overcome system margins, shown as a functions of antenna diameter and pointing error for X-band and Ka-band.	57

43	Atmospheric losses X band: worst case.	59
44	Data rate dependency on the power transmitted and diameter of the antenna.	59
45	Relay – Carrier Visibility Analysis	61
46	Parametric link-closure analysis as a function of payload TX power P_{TX} and relay helical antenna diameter D . Green regions indicate closed links ($R_{max} \geq R_{HK}$), while red points represent configurations that fail to meet the required (E_b/N_0) margin. . .	62
48	RF chain: X band.	64
49	Radiation pattern: Relay UHF helix antenna.	64
50	RF chain: UHF band.	65
51	Modulation index behaviour.	65
52	Monte Carlo Apoapsis Dispersion.	67
53	Inclination, AOP and RAAN Monte Carlo Dispersions.	68
54	Carrier OD Pre-Relay Release.	70
55	Relay OD.	71
56	Carrier OD Post-Relay Release.	72
57	Relay OBDH architecture.	73
58	Carrier OBDH architecture.	74
59	OBDH State Diagram.	76
60	Subsystem Power Consumption per Phase (Carrier).	79
61	Subsystem Power Consumption per Phase (Relay).	80
62	Carrier Solar Arrays Configuration.	81
63	Relay Solar Arrays Configuration.	81
64	Carrier Batteries Dynamical Simulation.	83
65	Relay Batteries Dynamical Simulation.	83
66	EPS Architecture.	84
67	Aerodynamic heat flux evolution during the Release Operation (left) and environmental heat flux during the longest eclipse period (right).	89
68	Evolution of the environmental fluxes during worst cases in Nominal Orbit.	89
69	Stacked configuration internal thermal nodes.	90
70	Relay internal thermal nodes.	91
71	Carrier external thermal design.	92
72	Relay external thermal design.	93
73	Predicted Temperature for the most critical nodes during Interplanetary.	94
74	Predicted Temperature for the most critical nodes during Release Phase.	95
75	Predicted Temperature for the most critical nodes during Relay Nominal Orbit.	95
76	C-shaped Relay Release and Support Frame mounted on the dual-leaf bay door. The open-side geometry avoids the formation of a closed structural loop across the hinge line, reducing sensitivity to differential leaf deformation while preserving primary load-path stiffness.	97
77	Nominal relay separation along the spacecraft +Z axis. The guided initial motion phase ensures controlled extraction prior to transition to torque-free rigid-body dynamics. .	98
78	High-level operational state machine of the Relay Release and Support Frame. The architecture separates nominal non-explosive actuation from the cold-redundant pyrotechnic backup path, with time-bounded verification and FDIR-gated escalation logic. .	98
79	Dual-leaf payload bay door in deployed configuration. Symmetric $\pm X$ rotation ensures balanced disturbance and deterministic clearance for relay extraction.	99
80	Supervisory state machine coordinating dual-leaf deployment and gating relay release. .	100
81	Leaf A local state machine.	101
82	Leaf B local state machine.	101

83	CPDI with single PAI (baseline configuration). The CPDI provides the carrier-side accommodation plane; the PAI defines the standardized separation interface towards the payload.	104
84	CPDI with multiple PAIs. The CPDI pattern is mission-tailored, while each PAI remains a standardized module, enabling modular payload integration without modifying the carrier platform.	104
85	Motorized lightband PAI: functional layout of the upper ring (payload-side), bottom frame (carrier-side), engagement leaves and retaining ring. Release is achieved by retracting the retaining ring, enabling leaf collapse via spring plungers and freeing the upper ring for separation.	105
86	Motorized lightband PAI – high-level release FSM. The model separates SAFE/DISARMED, ARMED and RELEASE IN PROGRESS states, and makes explicit the time-bounded monitoring, fault latching, and the pre-unlatch point of no return (abort permitted only before irreversibility).	106
87	Separation dynamics refinement roadmap. Level 2 introduces explicit spring force modeling during the push phase, while Level 3 addresses statistical robustness and worst-case identification.	107
88	(a) Stowed configuration.	109
89	(b) Interplanetary configuration.	109
90	(c) Post-relay release.	109
91	(d) Final release configuration.	109
92	MANTIS Spacecraft configurations during different mission phases.	109
93	Carrier external configuration and characteristic dimensions in meters.	110
94	Carrier internal configuration, with a focus on the Carrier Bus and Payload Bay. . . .	110
95	External and bottom configurations of the Relay spacecraft, showing antenna placement and interface details.	111
96	Relay internal configuration.	111
97	Internal structural skeleton of the Carrier Bus.	113
98	Internal structural skeleton of the Relay spacecraft.	113
99	Finite Element Model (FEM) mesh of the primary MANTIS structure.	114
100	First lateral mode (X-axis).	114
101	First lateral mode (Y-axis).	114
102	First longitudinal mode (Z-axis).	114
103	APSD for random analysis.	115
104	Subsystem Power Consumption per Phase (Carrier).	118
105	Subsystem Power Consumption per Phase (Relay).	118
106	State Machine of Carrier ADCS modes.	123
107	State Machine of Relay ADCS modes.	123
108	Carrier Dynamical Simulator.	124
109	Carrier Dynamical Simulator controller logic.	125
110	Relay Dynamical Simulator.	126
111	Relay Dynamical Simulator.	126
112	Carrier Power Budget.	140
113	Relay Power Budget.	141
114	Batteries Dynamical Power Simulator.	142
115	Relay Mass Budget.	143
116	Carrier Mass Budget.	144
117	Carrier Mass Budget.	145

List of Tables

1	Authorship declaration and team roles for the MANTIS mission.	i
2	Quantified environmental effects and design reference margins for the MANTIS carrier.	6
3	Missions and Space tugs overview.	7
4	Quantitative Analysis Results.	14
5	Hyperbolic excess velocity provided by Falcon Heavy configurations	19
6	Lambert solution for Earth–Mars transfer.	23
7	Earth departure hyperbola relevant parameters.	23
8	Mission relevant dates and phases duration.	24
9	MOI and achieved orbit relevant parameters.	25
10	Allocated ΔV budget for Carrier and Relay.	30
11	Carrier pointing budget.	32
12	Relay pointing budget.	32
13	Sensors selection.	33
14	Worst-case disturbance torques during aerobraking for the three bounding spacecraft configurations. For each case, the table reports the pericenter velocity and the peak gravity-gradient and aerodynamic drag torques corresponding to both maximum-area and minimum-area strategies.	37
15	Estimated number of aerobraking passages and corresponding ADCS propellant consumption for the two investigated aerobraking strategies.	39
16	Propellant mass (without 5.5% margin), Maneuver time and acceleration for Baseline 3.	43
17	PS Carrier’s thruster selection.	44
18	Tanks geometric specifications.	47
19	Propulsion System Specifications and Tank Data	51
20	Mars Relay Network telemetry data rate estimation.	53
21	X band link budget considering ESA ESTRACK ground station and data science in downlink.	58
22	Link budget design of UHF band downlink.	61
23	Link budget design of UHF band uplink.	61
24	Link budget of S band downlink.	63
25	Link budget of S band uplink.	63
26	Carrier navigation approach pre-Relay release.	69
27	Carrier OD position errors pre-Relay release.	70
28	Relay Navigation approach.	70
29	Relay OD position errors.	71
30	Carrier navigation approach post-Relay release.	72
31	Carrier OD position errors post-Relay release.	72
32	Selected OBDH components for the Relay spacecraft.	75
33	Subsystem Power Distribution during MOI.	79
34	Subsystem Power Distribution during High-Rate Operations.	80
35	Solar arrays design parameters.	82
36	Batteries design parameters.	82
37	Carrier Temperature Range.	86
38	Relay Temperature Range.	87
39	Carrier internal heat loads.	88
40	Relay internal heat loads.	88
41	Optical properties of materials/surfaces.	94
42	Geometric and Mechanical Properties of the Sandwich Panel.	113

43	Output of modal analysis and compliance with launcher requirements [1].	114
44	Quasi-Static test results: comprehensive stress and displacement data for all axes. . .	115
45	Sine vibration test results: comprehensive stress and displacement data for all axes. .	115
46	Random vibration analysis results: 3-Sigma peak stresses and displacements.	115
47	Summary of Maximum Stresses and Margins of Safety for specific material.	116
48	Relay Mass Budget.	119
49	Carrier Mass Budget.	119
50	Comparison between system launch mass and Ariane 64 launchable mass.	119
51	Propellant mass, maneuver time and acceleration for Baseline 1.	127
52	Propellant mass, maneuver time and acceleration for Baseline 2.	128
53	Mission execution autonomy levels per mission phase (ECSS-E-ST-70-11C compliant). .	129
54	Assumptions for link budget.	130
55	Link budget preliminary design.	130
56	Downlink: Interplanetary.	131
57	Uplink: Interplanetary.	131
58	Downlink: Nominal mode around Mars.	132
59	Uplink: Nominal mode around Mars.	132
60	Downlink: High data rate.	133
61	Downlink: Near Earth Commissioning.	134
62	Uplink: Near Earth Commissioning.	134
63	Downlink: Carrier-relay separation and release operations.	135
64	Uplink: Carrier-relay separation and release operations.	135
65	Downlink: telecommand for payloads.	136
66	Uplink: High data rate with PHAROS-M.	136
67	Uplink: High data rate maximum.	137
68	Relay X-band Direct-To-Earth communication components.	137
69	Relay UHF proximity communication components.	138
70	Carrier UHF communication components.	138
71	Carrier TMTC data budget with contact assumptions.	139

Nomenclature

List of acronyms

ADCS Attitude Determination and Control Subsystem

AHP Analytic Hierarchy Process

AKE Absolute Knowledge Error

AOCS Attitude and Orbit Control Subsystem

APE Absolute Performance Error

APSD Acceleration Power spectral density

ARKE Absolute Rate Knowledge Error

ARPE Absolute Rate Pointing Error

ASO Areostationary Orbit

AU Astronomical Unit

BCDU Battery Charge/Discharge Unit

BLDC Brushless DC Motor

BOL Beginning of Life

BOL Beginning of life

CFRP Carbon Fiber Reinforced Polymer

CG Center of Gravity

CLA Coupled Load Analysis

CL Confidence Level

CoM Center of Mass

CoP Center of Pressure

CPDI Carrier-Payload Deck Interface

CSS Coarse Sun Sensor

DC Direct Current

DET Direct Energy Transfer

DoD Depth of Discharge

DSN Deep Space Network

ECSS European Cooperation for Space Standardization

EDAC Error Detection And Correction

EOL End of life

EOL End-of-Life

EPS Electric Power Subsystem

ESA European Space Agency

ESD Electrostatic Discharge

FDIR Fault Detection, Isolation and Recovery

FEM Finite Element Method

FOV Field Of View

FS Factor of Safety

GCR Galactic Cosmic Rays

GFE Government Furnished Equipment

GG Gravity Gradient

GNC Guidance, Navigation and Control

GS Ground Station

GTO Geostationary Transfer Orbit

HDRM Hold Down and Release Mechanism

HEPO High Elliptic polar Orbit

He Helium

HGA High-Gain Antenna

HK Housekeeping

IMFM Interplanetary Meteoroid Flux Model

IMU Inertial Measurement Unit

IPIRM Interplanetary Ionizing Radiation Model

IR Infrared Radiation

ITU International Telecommunication Union

JPL Jet Propulsion Laboratory

KA Keep-Alive (payload thermal/power survival mode)

KSC Kennedy Space Center

LCL Latching Current Limiter

LEOP Launch and Early orbit phase

LEO Low Earth Orbit

LMO Low Mars Orbit

LM Launch mass

LSS	Lightband Separation System	RAAN	Right Ascension of Ascending Node
LUM	Launcher User Manual	RTG	Radioisotope Thermoelectric Generator
LVA	Launch vehicle adapter	RW	Reaction Wheel
MANTIS	Mars Network Transport and Inter-link System	SAA	South Atlantic Anomaly
MARCONI	Communication and Navigation Infrastructure	SADM	Solar Array Drive Mechanism
MAVEN	Mars Atmosphere and Volatile Evolution Mission	SA	Solar Array
MA	Mission Analysis	SEE	Single-Event Effects
MECH	Mechanism Subsystem	SEP	Solar Electric Propulsion
MEX	Mars Express	SEU	Single Event Upset
MGS	Mars Global Surveyor	SK	Station Keeping
MLI	Multi-Layer Insulation	SNR	Signal to Noise Ratio
MMH	Monomethylhydrazine	SoC	State of Charge
MMO	Medium Mars Orbit	SOI	Sphere of Influence
MOI	Mars Orbit Insertion	SPENVIS	Space Environment Information System
MON	Mixed Oxides of Nitrogen	SPE	Solar Proton Events
MoS	Margin of Safety	SRP	Solar Radiation Pressure
MPO	Medium Polar Orbit	SSM	Second Surface Mirror
MRO	Mars Reconnaissance Orbiter	STR	Structures
NC	Normally Closed valve	TCM	Trajectory Correction Maneuver
NO	Normally Open valve	TCS	Thermal Control Subsystem
NTO	Nitrogen Tetroxide	TID	Total Ionizing Dose
OBDH	On Board Data Handling	TRL	Technology Readiness Level
ODY	Mars Odyssey	TTMTC	Tracking, Telemetry and Telecommand
PAI	Payload Accomodation Interface	UHF	Ultra High Frequency
PCDU	Power Control and Distribution Unit	PLL	Phase Locked Loop
PID	Proportional-Integral-Derivative	List of symbols	
PL	Payload	α_P	Payload visibility half-angle [deg]
PPT	Peak Power Tracking	α_S	solar absorptivity
PPU	Power Processing Unit	ΔV	Delta-V required [m/s]
PS	Propulsion Subsystem	ΔV	Separation velocity imparted by springs [m/s]
PWPF	Pulse-Width Pulse-Frequency		

$\dot{\Omega}$	Right ascension of ascending node rate [deg/day]	E_z	Equivalent Young's modulus out-of-plane [Pa]
$\dot{\omega}$	Argument of pericenter rate [deg/day]	E_{bat}	Battery energy capacity [Wh]
$\dot{m}$	Mass flow rate [kg/s]	F	Thrust [N]
ϵ_n	Nozzle expansion ratio [-]	g_0	Standard gravity (9.80665 m/s ²)
ν	Poisson's ratio [-]	G_c	Core Shear Modulus [Pa]
ν	True anomaly [deg]	G_{RX}	Receiver antenna gain [dBi]
Ω	Right ascension of ascending node [deg]	G_{TX}	Transmitter antenna gain [dBi]
ω	Argument of pericenter [deg]	G_{xy}	In-Plane Equivalent Shear Modulus [Pa]
ρ	density	G_{xz}	Out-of-Plane Equivalent Shear Modulus [Pa]
ρ_c	Core Density [kg/m ³]	G_{yz}	Out-of-Plane Equivalent Shear Modulus [Pa]
ρ_f	CFRP Density [kg/m ³]	h	Orbital altitude above Mars [km]
ρ_{eq}	Equivalent Density [kg/m ³]	h_c	Core thickness [m]
σ_{actual}	Derived Stress [Pa]	h_{tot}	Total Panel Thickness [m]
$\sigma_{ultimate}$	Ultimate stress [Pa]	i	Orbital inclination [deg]
θ_g	Gimbal angle [deg]	I_{sp}	Specific impulse [s]
a	Semimajor axis [km]	J	Relay orbit performance index [-]
A_e	Nozzle exit area [m ²]	J_2	Second-order zonal harmonic [-]
A_{SA}	Solar-array area [m ²]	J_4	Fourth-order zonal harmonic [-]
$a_{3\text{Body}}$	Third-body perturbation acceleration [km/s ²]	L_q	Link-quality proxy [-]
a_{Drag}	Atmospheric drag acceleration [km/s ²]	L_{atm}	Atmospheric loss [dB]
a_{SRP}	Solar radiation pressure acceleration [km/s ²]	L_{FS}	Free-space loss [dB]
C^*	Characteristic velocity [m/s]	m_0	Initial mass [kg]
C_3	Characteristic launch energy [km ² /s ²]	M_d	Dry mass [kg]
C_F	Thrust coefficient [-]	m_f	Final mass [kg]
C_H	Heat transfer coefficient	M_p	Payload mass [kg]
e	Orbital eccentricity [-]	m_p	Propellant mass [kg]
E_b/N_0	Energy-per-bit to noise density ratio [dB]	m_{bat}	Battery mass [kg]
E_f	Facesheet Young's modulus [Pa]	m_{dry}	Dry mass after release [kg]
E_x	Equivalent Young's modulus in-plane [Pa]	m_{wet}	Wet mass [kg]
E_y	Equivalent Young's modulus in-plane [Pa]	n	Mean motion of the spacecraft [s ⁻¹]
		P	Power [W]

p	Semi-latus rectum [km]	T	Orbital period [h]
P_c	Chamber pressure [Pa]	t_f	Facesheet thickness [m]
P_{ecl}	Power demand during eclipse [W]	t_{ecl}	Eclipse duration [h]
P_{loads}	Total power consumed by active subsystems [W]	v	velocity
P_{SA}	Solar-array delivered power [W]	V_p	Propellant volume [m ³]
P_{He}	Helium tank pressure [Pa]	v_{∞}	Hyperbolic excess velocity [km/s]
q_{conv}	convective heat flux	V_{He}	Helium tank volume [m ³]
R	Reference radius of the central body [km]	X_d	Line efficiency in sunlight [-]
r	Orbital radius [km]	X_e	Line efficiency in eclipse [-]
r_{tank}	Tank radius [m]	ε_{IR}	infrared emissivity
$S(r)$	Solar irradiance at distance r from the Sun [W/m ²]	DoD	Depth of Discharge [-]
		SNR	Signal-to-noise ratio [dB]

1 Introduction

The **MANTIS** (Mars Network Transport and Interlink System) mission originates from the need to move beyond single-use interplanetary transfer vehicles toward a reusable, service-oriented orbital infrastructure for Mars. Instead of conceiving the transfer stage as a disposable element, MANTIS investigates the feasibility of a modular "space tug" capable of performing three sequential roles: high-capacity interplanetary transport, coordinated deployment of multiple assets in heterogeneous Martian orbits, and long-term reconfiguration into a telecommunications relay platform. This report documents the feasibility study conducted.

A central design driver has been modularity. The system is conceived as a recurring transportation service synchronized with Earth–Mars synodic opportunities, enabling periodic deployment of scientific or commercial assets while preserving architectural stability across campaigns. This approach required resolving several critical engineering challenges, including dry-mass optimization under stringent launcher constraints, and link-budget closure for client spacecraft operating in different orbital regimes.

1.1 Mission Objectives

The study was structured around the following primary technical objectives:

- **Interplanetary Feasibility:** To validate the dynamical feasibility of transporting a high aggregate payload mass from Earth to Mars, including transfer trajectory definition, Mars Orbit Insertion (MOI), and subsequent orbit shaping.
- **Transport Efficiency:** To maximize the payload mass fraction deliverable to the Martian environment while maintaining compliance with launcher constraints and subsystem mass margins.
- **Service-Oriented Architecture:** To define a scalable and interface-compatible bus capable of supporting heterogeneous payloads across successive mission campaigns, minimizing structural redesign and enabling infrastructure-level reuse.

1.2 High-Level Mission Requirements

The MANTIS architecture was constrained by a set of high-level requirements progressively refined during the design iterations:

- The system shall transfer a minimum payload mass of 900 kg in Martian environment.
- The system shall deploy at least three independent payloads across a minimum of orbital bands, in altitude or orientation.
- The system shall provide relay communication services to Earth for all deployed assets over their entire operational lifetimes.
- The system design shall be frozen and flexible to accommodate different payload sets and transfer scenarios.

2 Mission Environment Analysis

The mission is divided into three environmental regimes: near-Earth phase, interplanetary cruise, and Mars operational phase. Only dominant physical drivers affecting sizing, reliability, and operations are retained.

2.1 Earth Vicinity

Before injection into interplanetary space the spacecraft operates inside the near-Earth space environment, dominated by trapped radiation, geomagnetic effects and microgravity conditions.

Radiation and Magnetosphere Near Earth, the dominant hazard is trapped radiation within the Van Allen belts. The inner belt is proton-dominated (hundreds of MeV), while the outer belt contains MeV electrons. The South Atlantic Anomaly locally increases fluxes by about one order of magnitude. The geomagnetic field shields roughly 90% of galactic cosmic rays [2] but introduces disturbances: geomagnetic storms can induce currents in large conductive elements and generate transient single-event upsets. The OBDH and power subsystems therefore require radiation-tolerant electronics, EDAC techniques, and TID verification.

Microgravity In orbital conditions, microgravity levels are below 10^{-6} g. Although structural loads are minimal, microgravity influences fluid behavior, particularly for propellant management and eliminates natural convection in thermal processes. This requires specialized thermal and fluidic design, such as capillary-based propellant systems and conduction-dominated heat transfer solutions.

2.2 Interplanetary Cruise

The transfer to Mars is the longest continuous exposure to deep-space conditions and sets most sizing requirements.

Radiations GCR and episodic Solar Protons Events (SPE) together define the cumulative and transient exposure levels along the transfer path. According to the *NASA Mars Transportation Environment Definition Document* [3], radiation represents the primary environmental driver for deep-space transit, while plasma effects are secondary. The same report provides quantitative estimates for TID accumulation and Single-Event Effects (SEE) rates along the Earth–Mars leg. The Interplanetary Ionizing Radiation Model (IPIRM) offers a practical computational basis for estimating shielding requirements and SEE rates [4].

The radiation environment along the Earth–Mars cruise was modeled in **SPENVIS** using ESA-standard tools (ESP–PSYCHIC, ISO-15390, CREME-96, SHIELDOSE-2/IRENE), configured for a one-year deep-space exposure outside the geomagnetosphere with a reference shielding of 1 g/cm² Al. This setup reproduces free-space conditions at 1 AU and was mapped to the 1–1.6 AU transfer using SPENVIS exposure factors, consistently with the actual interplanetary trajectory.

The Solar Particle Events (SPEs) constitute the dominant source of ionizing radiation. The **ESP–PSYCHIC** cumulative fluence (95% CL) reaches $\sim 10^{11}$ cm⁻² for protons above 10 MeV, with the 10–100 MeV band driving most of the ionizing dose and the >100 MeV tail governing single-event susceptibility. The resulting spectrum is shown in Figure 1.

The Single-event effects (SEEs) were assessed by combining ESP–PSYCHIC protons with **ISO-15390** Galactic Cosmic Rays (GCR). GCR fluxes, dominated by protons and helium, with carbon representing heavier species, follow the expected power-law decay and set the steady background for SEE and

Total Ionizing Dose (TID). Proton-induced upsets dominate the cruise phase: ~ 30 events/bit/yr for **COTS-SRAM** and < 0.3 events/bit/yr for hardened devices, consistent with heritage interplanetary missions. The corresponding SPENVIS-estimated rates are reported in Figure 2.

The TID, computed with **SHIELDOSE-2/IRENE**, is $\sim 3 \times 10^3$ rad(Si) behind 1 g/cm^2 Al. This reduces below 500 rad(Si)/yr for $3\text{--}5 \text{ g/cm}^2$, confirming that a few millimeters of aluminum provide adequate protection for avionics during cruise.

Peaks were additionally bounded using **CREME-96** “peak 5-min” and “worst-week” envelopes at solar maximum. When scaled with a conservative radial sensitivity $r^{-\alpha}$ ($\alpha \in [1, 2]$), these peaks define the upper bound for instantaneous SEE rates and drive FDIR response-time allocation, while the worst-week case constrains extended-storm resilience and cumulative-dose margins.

The results define the radiation envelope for the mission, setting the minimum shielding and hardness requirements to be propagated into subsystem-level trade-offs during the next design phase.

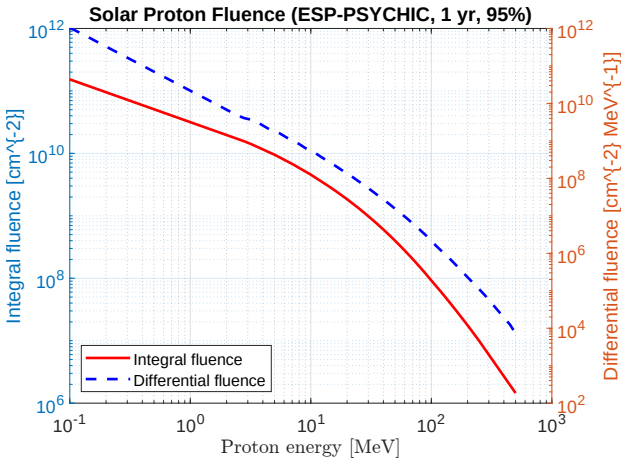


Figure 1: Solar proton fluence from the ESP-PSYCHIC model (1-year, 95% CL).

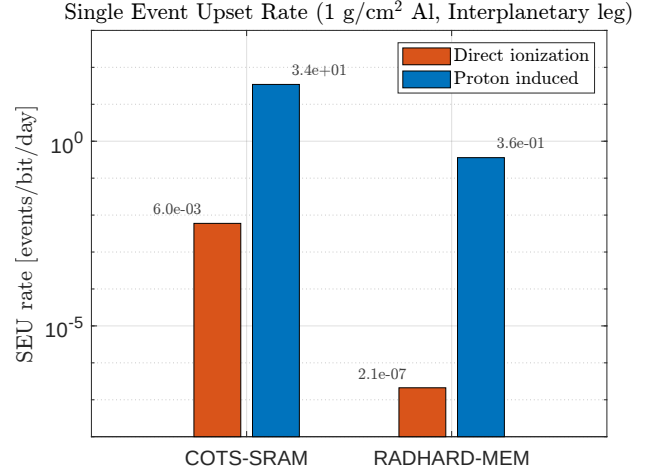


Figure 2: Predicted SEU rate for COTS and radiation-hardened memories.

Thermal Environment The carrier will likely experience alternating periods of sunlight and shadow depending on its attitude, which will lead to hot-cold cycling of internal components. Such cycles are known to induce thermal gradients, calibration drifts in sensors and accelerated battery calendar aging. Moreover, SRP acts continuously over the months-long journey. Although small in magnitude, SRP exerts persistent torques that can saturate reaction wheels over time, requiring periodic desaturation maneuvers. This cumulative effect competes with other attitude-control demands, such as maintaining thermal balance and Earth-pointing for communication.

Meteoroids and Interplanetary Dust Over the cruise, the carrier’s large cross sectional area increases its probability of encountering high velocity impacts. IMFM provides statistically validated fluence distributions by mass and velocity^[5]. Although widely used, the NASA environment report^[3] notes that these models may underestimate the flux of high mass, high speed particles by factors of up to 100, particularly for oblique trajectories and cometary stream crossings. The Mariner IV mission, which recorded a sudden spike in micrometeoroid impacts during a known stream encounter, remains a notable reminder that periods of elevated risk can occur along Earth Mars transfer trajectories^[6]. Hence, while model-based averages are invaluable for statistical design, mission-specific risk analysis must still account for dynamic, time-correlated exposure.

The micrometeoroid environment along the Earth-Mars trajectory was characterized using the **Grün**

interplanetary model [7], which provides cumulative flux distributions as a function of particle mass and heliocentric distance. The model was run for a one-year exposure period at a mean heliocentric distance of 1–1.5 AU, assuming a particle density of 2.5 g/cm³ and neglecting planetary gravitational focusing, thus representing a conservative case for the estimation of the flux. The cumulative flux follows the characteristic power-law trend $F(> m) \propto m^{-\alpha}$, typical of interplanetary dust populations, with a pronounced decrease in particle number for increasing mass. For particles with masses above 10⁻⁶ g, the flux is on the order of 10⁻⁴ m⁻²yr⁻¹, consistent with long-term in-situ measurements at 1 AU. Large impacts are statistically negligible, while sub-millimetre grains dominate the total population and represent the main driver of surface degradation and minor erosion. Although individual impacts are non-catastrophic, the cumulative effect can cause gradual optical and thermal degradation on coatings, radiators, and exposed sensors over multi-year operations. The resulting environment aligns with micrometeoroid levels observed in previous Mars transfer missions, confirming that standard structural and material protection strategies provide sufficient robustness. A mean areal density of ~1 g/cm² for external panels ensures adequate penetration resistance, while optical surfaces and small exposed components can be protected through local coatings or sacrificial layers. Overall, the Grün-model assessment indicates that meteoroid risk is non-critical for the MANTIS spacecraft but must be included in the material durability and contamination-control evaluations for the cruise phase.

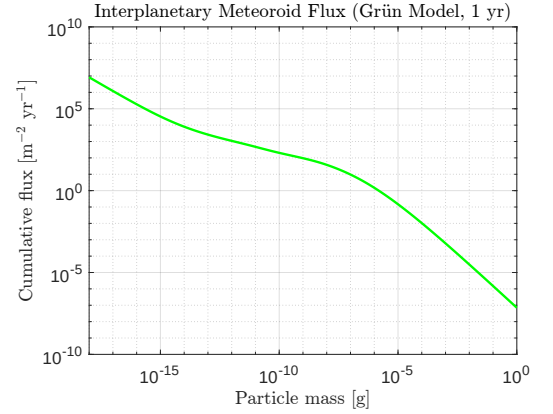


Figure 3: Interplanetary meteoroid cumulative flux derived from the Grün model (1-year exposure).

Surface Charging and Contamination While solar wind plasma density is low along the interplanetary leg, it remains capable of inducing differential charging across large, irregular surfaces. This becomes particularly relevant if high voltage propulsion is active, as it increases the likelihood of ESD events. Similarly, the combination of long duration outgassing and confined geometry inside the payload bay can lead to molecular contamination, with thin films accumulating on optical or thermal surfaces. For a mission carrying multiple satellites in proximity, these contamination and ESD phenomena can propagate across payloads [3].

2.3 Mars System Environment

The Martian environment governs long-term operations after orbit insertion.

Orbit Mars follows an elliptical orbit around the Sun at a distance varying between 1.38 and 1.67 AU and with a noticeably higher eccentricity (0.0935) with respect to the Earth's (0.0167) [8]. The different orbit periods between Earth (365.256 days) and Mars (686.972 days [8]) determine the synodic period (time required by two planets to realign with the Sun), whose value is 779.94 days. This has a strong effect on the design of an interplanetary transfer: to reduce the most the time of flight, favorable launch windows occur once every synodic period, when Earth and Mars are in opposition with respect to the Sun.

Moreover, the variable distance between the two planets affects telecommunications between the space segment and the ground station. In fact, the distance varies from a minimum of $54.6 \cdot 10^6$ km to a maximum of $401.4 \cdot 10^6$ km [8], causing the time taken to send a signal to range from 3 to 22 light-minutes. Furthermore, high free space path loss requires high gain antenna to be compensated.

Finally, the synodic period also determines times during which Earth and Mars are located on opposite sides of the Sun, called solar conjunctions. When the Sun-Earth-Mars angle falls below 5 deg, communications experience a huge amount of noise, while below 0.25 deg the Sun completely blocks telecoms. To overcome the noise, a common practice is to reduce the amount of data exchanged and set the ground station to full transmission power. In addition, space agencies typically upload all commands required to cover the entire conjunction interval before it begins [9].

These periods, which occur once every synodic period, can last more than a month: for example, the second solar conjunction experienced by MRO lasted 1 month and 6 days [10].

It is evident that solar conjunctions affect the design of the system, in particular TTMTTC and OBDH subsystems: the first should guaranty the most data exchange despite the noise caused by solar corona, while the latter should be able to store enough data until the downlink can be completely restored.

Atmosphere The Martian atmosphere is extremely rarefied, with a surface density of 0.02 kg m^{-3} [8] and a rapid exponential decay with altitude. At orbital heights, atmospheric drag becomes negligible, avoiding continuous propulsive compensation, which reduces propellant consumption and simplifies long term orbital maintenance.

However, the same atmospheric scarcity makes aerobraking less effective. For instance, MRO performed the aerobraking at approximately 110 km of altitude, requiring nearly 450 passages over a period of five months^{[10][11]}; Mars Global Surveyor (MGS) underwent a six month aerobraking phase^[12]; Mars 2001 Odyssey completed the aerobraking in 75 days and 336 orbits^[13].

Radiations and Magnetism Mars does not have a magnetic field, meaning that a probe will experience the same effects encountered during the interplanetary transfer in terms of radiations (GCR, SPE and SRP) and solar wind plasma. Furthermore, past missions observed the presence of an ionosphere as a result of the interaction between solar wind and Martian upper atmosphere [14][15]. This region, composed of weakly ionized plasma, is strongly influenced by solar activity [16][17] and Mars crustal magnetism [18]. In addition to the effects on the spacecraft due to plasma, it was proved that the ionosphere strongly interferes with Very-Low-Frequency (VLF) and Extremely-Low-Frequency (ELF) communications, which must be taken into account when planning links with satellites in low orbits using those frequencies [19].

Regarding the crustal magnetic field, a model based on [20] was developed to investigate potential perturbations arising from this phenomenon. Figure 4 presents the intensity of the Martian magnetic field at an altitude of 500 km above the surface, mapped as a function of latitude and longitude. The maximum value of the crustal magnetic field recorded is 114.86 nT. Assuming a magnetic dipole moment equal to $[0.1, 0.1, 0.1]^T \text{ Am}^2$ as worst-case scenario, the corresponding maximum magnetic disturbance torque is approximately $1.14 \cdot 10^{-8} \text{ Nm}$. Although such a torque could become significant over extended time periods, this intensity occurs only at specific locations (namely, near the center of the yellow region in Figure 4). Elsewhere, the torque is predicted to be sufficiently small to be entirely negligible. Therefore, the crustal magnetic field can be considered an insignificant source of disturbance throughout the mission.

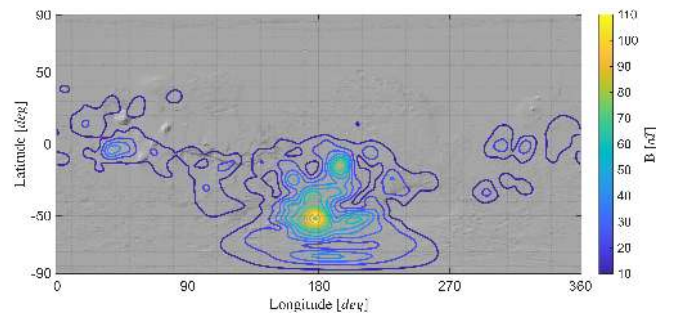


Figure 4: Crustal magnetic field model at 500 km.

Thermal Environment The eccentricity of the Martian orbit determines a variation in the incoming solar radiation flux, ranging from a minimum of 490.2 W m^{-2} at aphelion to a maximum of 713.2 W m^{-2} at perihelion. As a result, a spacecraft will receive different amounts of energy throughout the Martian year, affecting both power generation and thermal equilibrium. Other thermal sources are planetary albedo and IR emission from the planet.

Phobos and Deimos Mars has two moons: Phobos and Deimos. Thermal fluxes originating from their albedo are negligible due to their extremely low reflectivity. Nonetheless, their presence should be taken into account when designing the approach to Mars and the orbit injection to prevent potential close encounters. In addition, orbits that pass in their vicinity are affected by their gravitational perturbation.

2.4 Quantified Environmental Effects on Architectures

Environmental Driver	Key Metric (1 yr)	Reference Margin / Model	Design Implication
TID	$\sim 3 \times 10^3 \text{ rad(Si) @ } 1 \text{ g/cm}^2$	95% CL, ESP-PSYCHIC + ISO-15390	3–5 g/cm ² Al to reduce dose <500 rad(Si)/yr
SEE	$R_{\text{SEU}} \approx 30 \text{ events/bit/yr (COTS)}$	ESP-PSYCHIC + ISO-15390, $\sigma_{\text{SEU}} = 10^{-6} \text{ cm}^2/\text{bit}$	EDAC, latch-up protection, rad-hard logic
GCR	Flux $\sim 10^{-2} \text{ cm}^{-2}\text{s}^{-1}$ (E>100 MeV/n)	ISO-15390, 1 AU equivalent	Defines baseline SEE/TID background
Micrometeoroids	$F(> 10^{-6} \text{ g}) \approx 10^{-4} \text{ m}^{-2}\text{yr}^{-1}$	Grün interplanetary model	Non-critical; 1 g/cm ² outer shell sufficient
Thermo-optical ageing	$\Delta\alpha/\alpha_0 \approx +3\%/yr$	Empirical data from [7]	Surface coatings, MLI or protective films
Solar Proton Peak (5-min)	$\Phi_p(> 10 \text{ MeV}) \sim 10^4 \text{ cm}^{-2}\text{s}^{-1}$	CREME-96, solar maximum envelope	Transient SEE peaks, drives FDIR timing
Solar Proton worst-week	$\Phi_p(> 10 \text{ MeV}) \sim 10^3 \text{ cm}^{-2}\text{s}^{-1}$	CREME-96, statistical average	Cumulative TID check for extended storms

Note: Values are referenced to 1 AU free-space conditions and scaled for the Earth–Mars transfer (1–1.6 AU) according to SPENVIS exposure factors.

Table 2: Quantified environmental effects and design reference margins for the MANTIS carrier.

3 State of the Art

In the State of the Art an analysis of comparable missions in terms of environment, goals and technologies is conducted. Several missions to Mars have been conducted in the past, providing many references to draw upon. By studying these past missions, the research aims to retrieve useful information about the technological solutions adopted and to avoid the occurrence of previous failures. Moreover, an investigation is conducted on existing space tugs in order to identify their characteristics and main aspects. However, all these space tugs have operated only in missions around Earth. An overview analysis is reported in chronological order in Table 3.

Past Martian Missions		
Name	Lifetime	Goal
Mariner 9 ^[21]	1971 - 2020	First spacecraft to successfully orbit Mars.
Mars 2001 ^[22]	2004 - ongoing	Global mapping of Martian surface composition. Serve as a communication relay.
Mars Express ^[23]	2003 - ongoing	Provide a complete map of the chemical composition of the atmosphere, study Mars' innermost moon Phobos and the history of water across the globe.
Mars Reconnaissance Orbiter ^[24]	2005 - ongoing	Provide a high data rate communication relay. Examine potential future landing sites for future surface missions.
Curiosity ^[25]	2011 - ongoing	Rover collects soil samples to determine if Mars was ever able to support life.
Mars Orbiter Mission ^[26]	2013 - 2022	Examine Mars' surface features, morphology, mineralogy and atmospheric composition.
MAVEN ^[27]	2013 - ongoing	Investigate how atmospheric evolution affected Mars' climate and the previous presence of liquid water on its surface, potentially determining its prior habitability.
Exomars ^[28]	2016 - 2022	Analyze atmospheric trace gases.
Perseverance ^[29]	2020 - ongoing	Rover collects rocks and sought signs of ancient life that will return to Earth on a future mission.
Tianwen-1 ^[30]	2021 - 2023 (orbiter)	Study Martian topography, soil composition and magnetic field.
Space tugs		
Name	Lifetime	Configuration
SHERPA ^[31]	2015 - 2022	Modular spacecraft on which can be mounted one or more rings of different types of payload, maximising modularity.
ION satellite carrier ^[32] ^[33]	2020 - 2024	Satellite dispenser capable of hosting a combination of CubeSats that fits the volume. It can release the hosted satellites individually.
Vigoride ^[34]	2022 - ongoing	Platform containing different configurations of satellites and housing a large volume thanks to its compactness, without extending outside of the main body.
ESPASat ^[35]	2018 - ongoing	Platform with modular highly capable infrastructure for hosting operational payloads. The platform's ports are capable of accommodating any combination of up to six hosted and 12 separable payloads.

Table 3: Missions and Space tugs overview.

Future Mission Focusing on the future, the **LightShip-1 Mission**, currently in Phase A development, is also investigated. Special attention is devoted to analyzing its features, given their strong resemblance to MANTIS. It is conceived as an interplanetary transfer service by the European Space Agency (ESA) and will serve as a space tug for transporting payloads to Mars, providing communication, navigation and weather monitoring services from a high Mars orbit. The launch date is planned for 2032. It takes away two of the constraints that would make low-cost missions to Mars extremely difficult: providing the propulsion needed for its passenger spacecraft to transfer and enter Mars' orbit, and hosting the Mars Communication and Navigation Infrastructure (MARCONI) to offer a dedicated data relay service. ESA's LightShip concept aims to open up access to Mars for a wider community, since building spacecraft for Mars missions is considered a task for large system integrators. This tug would also bring a wider participation in the exploration domain due to its modular design adaptable to a range of payloads and exploration goals [36].

3.1 Criticalities and Lessons Learnt

From the analysis of the criticalities of past Martian missions and Space tug missions, it is possible to gather design strategies useful for the development of MANTIS.

3.1.1 Past Martian Missions

Although most of the missions analyzed pursue different objectives than MANTIS, several anomalies encountered during their operational lifetimes provide valuable lessons that can help prevent similar issues in future missions.

- **Mariner 9:** experienced a frequency shift in the S-band, which reduced the receiver's tracking rate capability and degraded the performance of the ranging channel [37]. This anomaly highlights the potential limitations of a single communication band under dynamic conditions. It suggests the need to consider the use of alternative or multiple frequency bands to enhance link robustness and ensure continuous telemetry and command operations.
- **Mars Express:** lander Beagle 2 experienced a failure because, although it landed on Mars, it was unable to communicate with Earth. This failure was caused by two of its four solar panels that did not deploy, trapping the communications antenna and silencing the probe [38]. This emphasizing the importance of implementing redundancy or closed-loop verification mechanisms for deployment sequences.
- **Mars Reconnaissance Orbiter:** faced anomalies related to its communication system and degradation of electromechanical components. These issues underline the necessity of rigorous Assembly, Integration, and Verification procedures to ensure component integrity and interface compatibility.
- **MAVEN:** unforeseen events required the implementation of additional maneuvers, particularly collision avoidance maneuvers. Such occurrences demonstrate the importance of maintaining adequate safety margins not only in hardware design but also in software and control logic ensuring robustness to unexpected operational demands.
- **ExoMars:** encountered unexpected sensor dynamics which caused the spacecraft to enter an incorrect flight phase, leading to premature parachute deployment, early engine cutoff, and a hard landing; additionally, propulsion anomalies due to wiring inversion were reported [39]. This highlights the need to integrate both time-driven and event-driven control architectures in critical phases using hybrid redundancy, thereby preventing cascading failures when timing or sensor data become inconsistent.
- **Perseverance:** communication delays caused a discrepancy between real-time events and ground reception; by the time the mission team received atmospheric entry confirmation, the rover had

already landed ^[40]. This scenario underscores the critical importance of either ensuring near-real-time communication or, alternatively, implementing high-level onboard autonomy for time critical maneuvers to counteract latency constraints and mitigate mission risks.

3.1.2 Space Tugs

To establish a baseline for the MANTIS carrier design, it is useful to analyze the critical aspects of existing space tug systems.

- **SHERPA Rideshare Mission:** its modularity, has proven to be expensive due to the need for a dedicated avionics system, which significantly increased both the mission budget and development timeline. SHERPA's modular design enhances mission flexibility and performance, but it also introduces greater complexity in configuration; therefore, making launcher interface requirements more challenging. Additionally, due to externally mounted payloads the probability of collision with space debris is increased. Finally, the SHERPA mission lacks onboard propulsion and attitude control system, which, in the MANTIS case, could be problematic for accurately reaching the payload's target orbit ^[31].
- **ION satellite carrier:** this tug has the most extensive heritage among the tugs analyzed. The absence of external parts gives reliability to the design, however, reducing versatility in terms of payload volume and limiting the number of satellites hosted ^[32]. The absence of externally mounted payloads provides higher protection and shielding against the environmental hazards. Consequently, choosing this configuration for MANTIS would enhance the survivability of the space segment, and payloads, which is identified as a key driver of the mission.
- **Vigoride:** compact design and affordable costs is what characterizes this tug. Nevertheless, the compactness of the configuration results in a restriction on the number of payloads that can be accommodated, making it unsuitable for the requirements of the MANTIS mission ^[41].
- **ESPASstar:** a configuration similar to the SHERPA's one. The platform has a large mass capacity, indeed each port of the platform can sustain about 320 kg. Having six different ports, it can host an overall mass of 1920 kg, however, increasing the complexity of the configuration ^[35]. Therefore, this last platform configuration can be suitable for MANTIS, which shall transfer a payload of at least 900 kg as stated in the high level mission requirements. ESPASstar has the capability to host a high number of payloads, having a larger payload capacity with respect to other configurations.

3.2 Level of Autonomy

As a result of the study, high levels of autonomy are required to compensate for the unavoidable communication delay caused by the signal travel time between Earth and Mars. This latency varies significantly according to the relative orbital positions, ranging from approximately 3 minutes at minimum distance (inferior conjunction) to 22 minutes at maximum distance (superior conjunction). Historical precedents like MRO demonstrated the capability to autonomously maintain orbital position and attitude control during maneuvers where real-time ground loop control is impossible. Similarly, MAVEN utilized onboard sequences for scientific data acquisition. However, it is crucial to distinguish between time-tagged command execution and autonomous decision making. Following ECSS standards, the mission requires different autonomy levels depending on the operational phase. Routine operations, such as orbital maneuvers commanded weeks in advance, will rely on Level E2 (execution of pre-planned command schedules). Conversely, critical phases and emergency scenarios necessitate a higher level of onboard intelligence. Therefore, it is crucial to incorporate Fault Detection, Isolation and Recovery (FDIR) logic capable of operating at Level E3 (event-driven autonomy) or even Level

E4 (goal-oriented autonomy) to manage emergencies and communication outages without ground intervention.

3.3 Rough Order of Magnitude of System Indicators

The Rough Order of Magnitude for several indicators relevant to the MANTIS mission is discussed with reference to past and current missions.

Mass and Configuration Regarding system mass, EXOMARS 2022 represents the only Mars mission with a comparable payload, featuring a total system mass of 3641 kg ^[28]. In contrast, Earth-based space tugs like SHERPA 400 provide a lower bound reference, with a mass of approximately 1000 kg while carrying a payload of about 1500 kg ^[42].

Power Generation Power requirements represent a critical driver due to the increased distance from the Sun. Since solar flux intensity diminishes with the square of the distance from the Sun, the available irradiance at Mars is substantially lower than that available at Earth. The MRO mission generates approximately up to 2000 W in Mars orbit to support high power communications and instrumentation ^[43], setting a realistic target for a high capacity relay platform.

Communication Missions foreseen for the Moon, such as the Lunar Pathfinder data relay orbiter, provide a valuable functional baseline for the service architecture; however, their hardware specifications must be dimensioned with appropriate margins to withstand the Martian environment. The transition from cislunar to interplanetary space imposes a dramatic increase in free space path loss. As a consequence, while a Lunar orbiter can close a link to Earth with a low gain antenna, MANTIS will probably require a high gain antenna to transmit scientific data and to compensate for the path loss over the Mars-Earth distance.

Lifetime and Cost Typical lifetimes of Martian missions, as reported in Table 3, generally exceed 10 years for orbiters, which is consistent with the goal of providing a long term infrastructure service. Finally, concerning cost, traditional European missions such as Mars Express required approximately 300 million euros ^[23]. In comparison, the NASA MRO mission, more similar to MANTIS due to its dual payload delivery and relay role, totaled roughly 716.6 million US dollars ^[44]. However, MANTIS aims to leverage the commercial approach of new actors (e.g., LightShip), where the objective is to minimize recurring costs to provide a repeatable and economically viable service.

These values can serve as useful references for preliminary estimations of system parameters.

4 Mission Architecture

4.1 Mission Architecture Alternatives

An assessment of the mission from a high-level architecture standpoint presents various possibilities. The specific architecture alternatives selected for analysis are:

- Spacecraft Elements:** In the **Single-Element** configuration, the architecture is monolithic, integrating both the carrier and relay functionalities into a single spacecraft. This consolidated element, referred to as the Carrier, is responsible for executing the entire mission profile, including the interplanetary transfer, payload release, and subsequent relay operations. Conversely, the **Two-Element** configuration separates these functionalities into two distinct satellites: the Carrier and the Relay. The Carrier is dedicated to the transfer and release phases, while the Relay operates independently, initiating its primary communications functions after the deployment within the Martian Sphere of Influence.
- Propulsive Architecture:** In the **Mono-Stage** configuration, the propulsion system is implemented as a single module, responsible for executing the entire sequence of maneuvers from the interplanetary transfer to the Martian environment. In contrast, the **Multi-Stage** configuration utilizes a segmented design where the propulsion system is divided into distinct stages. These stages are released upon completion of their specific maneuver execution to optimize mass efficiency.
- End-of-Life:** In the **Come-back** configuration, the Carrier performs a return transfer to an Earth parking orbit following the completion of its primary mission objectives. Conversely, in the **Don't Come-back** configuration, the Carrier is a disposable asset and its operational life ends after the completion of its functionalities in the Martian environment.

Figure 5 represents the MANTIS **Architecture Tree**, showing the eight mission architectures derived from the combination of these high-level design alternatives.

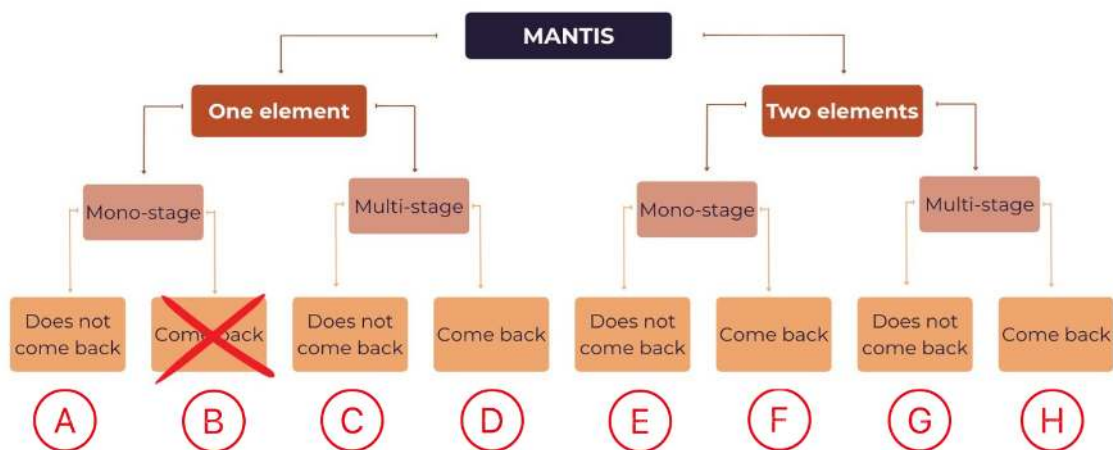


Figure 5: Architecture Tree.

Alternative B (Single-Element, Mono-Stage, Come-back) was immediately eliminated from the reasonable alternatives due to its functional incompatibility. The Carrier shall remain in Martian orbit to perform relay operations; however, the monolithic, mono-stage design lacks any separable component capable of executing an Earth return without abandoning the primary mission.

4.1.1 Mission Alternatives Criteria

The criteria defined to lead the trade-off analysis between the mission alternatives options are chosen with respect to the high-level goals of the mission. In this context, the selected criteria are presented:

- **Payload Mass:** this criterion underlines the maximum payload mass that can be accommodated within the carrier, ensuring that the system can successfully complete the mission. It refers to the total launchable mass given in the launcher user guide, minus the system mass.
- **Achievable Orbital Energy:** this criterion defines the capability and flexibility of the system in reaching different energetic orbital bands starting from a prescribed parking orbit around Mars, and under equal payload mass transported.
- **Complexity:** the complexity, in this scenario, is defined as the number of critical operations that characterize the different architectures. These are considered to be the detachment of a stage or element, the docking and refueling.
- **Cost:** the cost criterion represents costs of both the production, assembly, launch, and in-orbit operations. Enhancing the incurred costs by the clients with respect to the ones of the agency.
- **Building time:** the time criterion chosen in the trade-off expresses the time needed in the production of one entire unit, taking into account the possibility of the parallelization of the work for the Two-elements and multi-stage cases.

The architectural trade-off analysis is conducted in two phases. Initially, a qualitative selection is performed using the Analytic Hierarchy Process (AHP)^[45] to identify the three most promising candidates. Then, a quantitative assessment is executed, determining numerically each evaluation criteria to rigorously determine the optimal architecture out of the three options.

4.1.2 Qualitative Analysis

First, the qualitative trade-off analysis is performed by applying the AHP method to determine the best three architectures out of the seven presented in the architecture tree. The Criteria Matrices have been defined to compute both the weights of the criteria and the relative preferences of one alternative over another for each criteria. Thanks to this analysis a first qualitative ranking of the architectures is achieved. However, since this is a qualitative approach, its robustness must be verified before relying on it. To this end, a Monte Carlo-style analysis has been implemented over the AHP method by varying both the pairwise components of the Criteria Matrices and their weights. A proper Monte Carlo simulation is not feasible due to the large number of variables to be considered, so a simpler method is used. In this approach, random perturbation terms are applied for every simulation; in particular, variations of up to $\pm 30\%$ for the pairwise components and $\pm 20\%$ on the criteria weights were considered. The results of this analysis after 1000 simulations are presented in Figure 6. The simulation shows that, in the vast majority of cases, architecture A ranks first, architecture C second, and architecture E third. These three architectures are therefore selected for quantitative analysis.

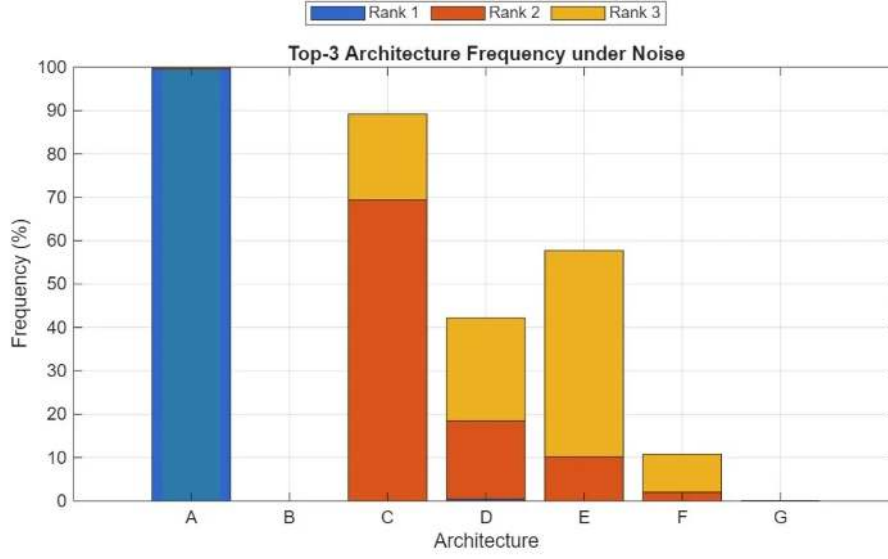


Figure 6: AHP Monte Carlo analysis results.

The three architectures selected from this process are the following:

- A) Mono-Element, Single-Stage:** a single spacecraft acts both as the Carrier of the payload to the Martian orbits and as the Relay. The system uses a single propulsive stage.
- C) Mono-Element, Multi-Stage:** the spacecraft still manages the two functionalities of carrying and relay, but the propulsive system is made of different stages that can be detached once they completed their maneuvers.
- E) Multi-Element, Single-Stage:** the carrier and relay functionalities are separated in two different satellites. The Carrier brings the payload and the Relay satellites to their Martian orbits with a single propulsive stage, and then it is disposed. The Relay has the only functionality to establish the link with the client satellites and with Earth.

It is noted that all architectures performing an Earth return phase were eliminated from the selection process. This exclusion is mainly driven by the significant increase in operational complexity and ΔV associated with executing a return trajectory from Martian orbit.

4.1.3 Quantitative Analysis

A quantitative estimation of the criteria is carried out to make a more reasoned selection of the mission architecture. The assumptions made for the computation of each of the five criteria are presented. The **Payload Mass** is quantified starting from a theoretical launchable mass from the Ariane 64 ^[1], fixed to the same value for all the architectures. From a preliminary ΔV budget, the corresponding mass ratios are used to estimate the structural mass of the three architectures. Given the dry masses, the transportable payload mass is estimated from statistical reports ^[46]. The **Orbital Energy** is estimated as the ΔV that one architecture can perform after the Mars Orbit Insertion (MOI). Fixing the P/L mass to 900 kg (the minimum request from the high level requirements), the corresponding dry mass is estimated from a fitting of data of previous missions. Then the ΔV reachable for the three architectures is computed with information on the estimated propellant mass from the Payload Mass estimation. The **Costs** comprehend the *DDT&E*, launch cost, operational costs (such as personnel, ground stations, and other facilities employed during mission operations) and propellant cost. All

of these components were estimated statistically, linearly fitting the available data of the costs with respect to the spacecraft mass. The **Complexity**, for the best three architectures, comprehends the critical operations, which are only the detachment of the propulsive stage and of the Relay satellite. Finally, the **Building Time** was not quantified because of its minor weight with respect to the others, and the absence of statistical information to estimate it.

Once that the criteria are estimated, they are normalized in a range spanning from 0 to 1 to obtain the quantitative grades for each architecture. The final ranking combines these grades with the criteria weights defined with the AHP. The results of this analysis are presented in Table 4.

Alternative	Payload Mass	Orbital Energy	Costs	Complexity	Building Time	Total
Weights	30.05%	12.03%	27.04%	26.33%	4.54%	100%
Mono-Element Single-Stage (A)	1	0	0	1	0	0.5638
Mono-Element Multi-Stage (C)	0	1	0.7290	0	1	0.3628
Multi-Element Mono-Stage (E)	0.7843	0.6479	1	0	0.5	0.6068

Table 4: Quantitative Analysis Results.

Following the trade-off analysis and preliminary design assessments across MA, PS, and TTMTTC, the **Multi-Element** mission architecture has been selected as the baseline for the MANTIS mission. The reasons motivating this choice are following:

- This architecture avoids the need of an apogee-raise maneuver after aerobraking to reach the relay orbit, preliminarily defined as 10000×300 km of altitude. This is estimated to reduce the ΔV budget for this architecture by 902.05 m/s.
- Because of the reduced amount of propellant needed, the wet mass of the system at launch is reduced significantly, and the overall mass of the PS system is lower.
- The Carrier lifetime does not last until the end of the operative life of the payload, which can be in the order of a few decades. Instead, the Carrier is disposed once all the payloads and the Relay satellite are released, roughly $\sim 13 - 15$ months after launch.
- The SK is needed only for the Relay satellite, as the Carrier is disposed after payload release.

On the other hand, this architecture has some unique criticalities that must be considered. First, it is more complex, having two distinct elements connected with an interface to be designed together with the release system of the Relay. This release also represents an additional operation to be considered in the ConOps. Furthermore, having two separate systems means designing all the necessary subsystems twice, one for the Carrier and one for the Relay. Finally, an additional communication link between the Relay and the Carrier must be designed.

5 Preliminary Mission Definition

5.1 Functional Decomposition

The functional decomposition of the MANTIS mission, presented in Figure 7, identifies the main functions required to fulfill the mission objectives, from launch to disposal. It provides a hierarchical view of the system functionalities, divided into Space and Ground Segments.

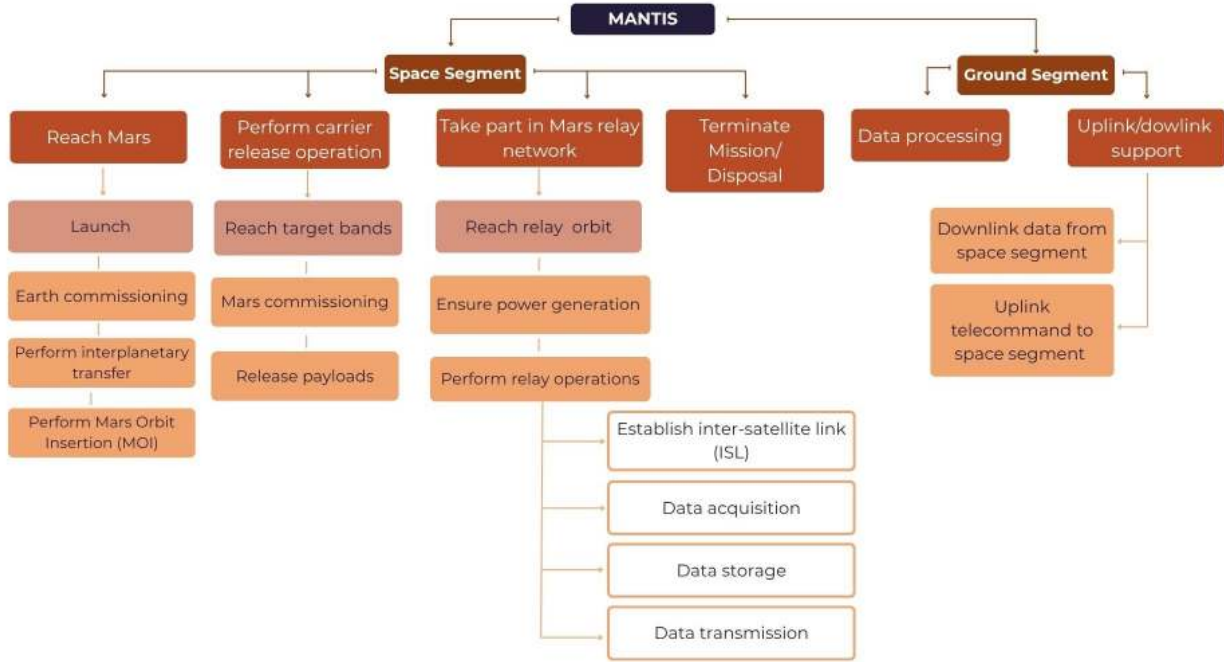


Figure 7: Mission functional decomposition.

The critical functionalities of the MANTIS mission have been identified as those functions that are essential for mission success and involve a significant risk of failure or malfunction.

- **Reach Martian orbit** (Space Segment 1st branch)
 - **MOI:** it is the velocity reduction maneuver to transition from a hyperbolic open trajectory into its first orbit around Mars. The criticality arises from the fact that any failure during this phase would result in a fly-by, causing the spacecraft to miss orbital capture entirely. Additionally, the large distance from Earth prevents real-time control, requiring the spacecraft to execute the injection autonomously, which increases the risks associated with the MOI.
- **Carrier release operations** (Space Segment 2nd branch)
 - **Payloads release:** a correct payload release directly determines the success of both the mission and the provided service. The criticality comes from the need to ensure the accurate positioning and separation of the payload, as well as to minimize the risks associated with potential mechanical failures of the release mechanism, which may operate autonomously.
 - **Reach target bands:** in this functionality the system may need to provide sufficient ΔV to move to a different orbital band to perform the Relay operations. The criticality further increases when multiple orbital bands need to be reached within a single launch.

- **Take part in Mars Relay Network** (Space Segment 3rd branch)
 - **Establish inter-satellite link:** the success of Relay operations defines the fulfillment of the mission objective. The criticality of this function consists in establishing communication links with multiple satellites, both those deployed by the carrier and those already orbiting Mars, while ensuring adequate visibility windows and compatible interfaces.

5.2 Mission Drivers

Three main mission drivers were identified for MANTIS mission.

The first driver addresses **multi-payload adaptability** requested in terms of mass, volume and interface. This is unprecedented in interplanetary missions, which are typically dedicated to a single payload. Our carrier, instead, shall integrate multiple different satellites, possibly from different providers, within a shared structure and thermal environment. This introduces major challenges in interface standardization and mechanical adaptability. It drives the overall system architecture, since both structural design and power data distribution must guarantee this flexibility.

The second driver focuses on the **sequential release capability** according to different orbital targets around Mars. Indeed, the objective moves the mission beyond a standard single-orbit deployment, as all previous Mars missions (e.g., Mars Express ^[47], Mars Reconnaissance Orbiter ^[48]) operated in fixed orbital configurations with at most one release event. The sequential release capability has no Martian heritage and only a partial analogue in ESA's BepiColombo mission ^[49], which will perform a dual-orbiter separation around Mercury; even in this case, however, the payloads were integrated into the design from the outset, not conceived as independent clients.

The last driver is the **payload's environmental protection and survivability** and it originates from the criticalities of the harsh interplanetary environment as cosmic radiation, solar events, micro-meteoroids and extreme thermal swings over several months. The carrier therefore acts as both a transport and protective module, a role rarely implemented in deep-space missions, including ESA's BepiColombo stack configuration ^[49]. Indeed, hosting multiple heterogeneous payloads has been done, which makes MANTIS a first-of-its-kind architecture. This requirement heavily influences the structural layout, thermal control and materials selection above all.

Together, these drivers represent the innovative core and technical risk of the mission: enabling a modular, multi-payload Mars transfer under severe environmental and operational constraints, a capability not yet demonstrated in current interplanetary systems.

5.3 Mission Phases

The mission is structured into distinct phases, each designed to ensure a transition from launch to the achievement of the mission objectives:

1. **Integration and Test operations:** an independent phase due to the complexity and large amount of time that must be allocated for this aim. This phase includes the integration and testing of the Carrier, the accommodation of the client payload within the Carrier and all associated performance verification tests. Particular attention is brought to the interface tests and the assembly validation of the payload into the Carrier. The phase ends once all the integrated elements meet their identified technical and performance requirements.
2. **Launch and Early orbit operations (LEOP):** begins immediately after launch and continues until the spacecraft attains a stable initial attitude. This phase includes the deployment of critical appendages (such as solar arrays) and an initial spacecraft check-out.

3. **Near Earth Commissioning Phase:** its objective is to verify the functionality and performance of all Carrier subsystems. During this phase, the spacecraft undergoes a series of functional tests and system activations to confirm nominal operation before initiating the interplanetary transfer.
4. **Interplanetary cruise and approach:** represents a vast part of the mission lifetime, in which the carrier must ensure its own survival while actively maintaining the health and required operational conditions of the payload set. Two sub-phases are identified:
 - **Cruise:** in which the spacecraft escapes from Earth and transfers towards Mars.
 - **Mars approach:** which is the end of the interplanetary transfer.

The phase culminates as the spacecraft approaches Mars Sphere of Influence.

5. **Mars orbit insertion (MOI):** a critical phase in which one or more maneuvers ensure that the carrier leaves the interplanetary leg and get caught in a Martian orbit.
6. **Mars commissioning phase:** involves verifying the spacecraft's subsystems under martian conditions and validating the functionality of the payload release mechanisms to ensure safe and accurate deployment of hosted satellites.
7. **Carrier release operations:** regardless of the system architecture, the client payloads are deployed into their target operational orbits. These orbits can be located within a single orbital band or split across multiple bands to meet diverse client requirements. The necessary verifications are executed to confirm the successful separation and operational status of the released assets. Different maneuvers are exploited to ensure the correct release of payloads such as placing different payloads in the same orbit with different phasing, change orbit within a band or change the releasing band.
8. **Relay operations:** the spacecraft provides communication services to the newly deployed satellites and, potentially, to other pre-existing assets orbiting Mars developing a constellation. This phase must be maintained for a duration of at least corresponding to the operational lifetime of the served assets. The ability to achieve downlink with Earth for transmitting the relay satellites data is a desirable, non-mandatory capability.
9. **End of mission:** is declared once the operational requirements of the clients have been fully satisfied. At this point, disposal procedures for the spacecraft are formally initiated and executed.

5.4 Concept of Operations (ConOps)

The overall Concept of Operations (ConOps) for the MANTIS mission is illustrated in Figure 8. Because the MANTIS architecture is designed to be repeatable for future launch windows, the specific dates and timeline presented here correspond strictly to the pilot mission, scheduled for launch on April 19, 2033. During this pilot flight, MANTIS will transport and deploy the **PHAROS M** client payload. Crucially, the ConOps divides into two distinct operational timelines following the deployment of the Relay satellite on August 13, 2034, at which point the Carrier and Relay spacecraft assume their independent mission profiles.

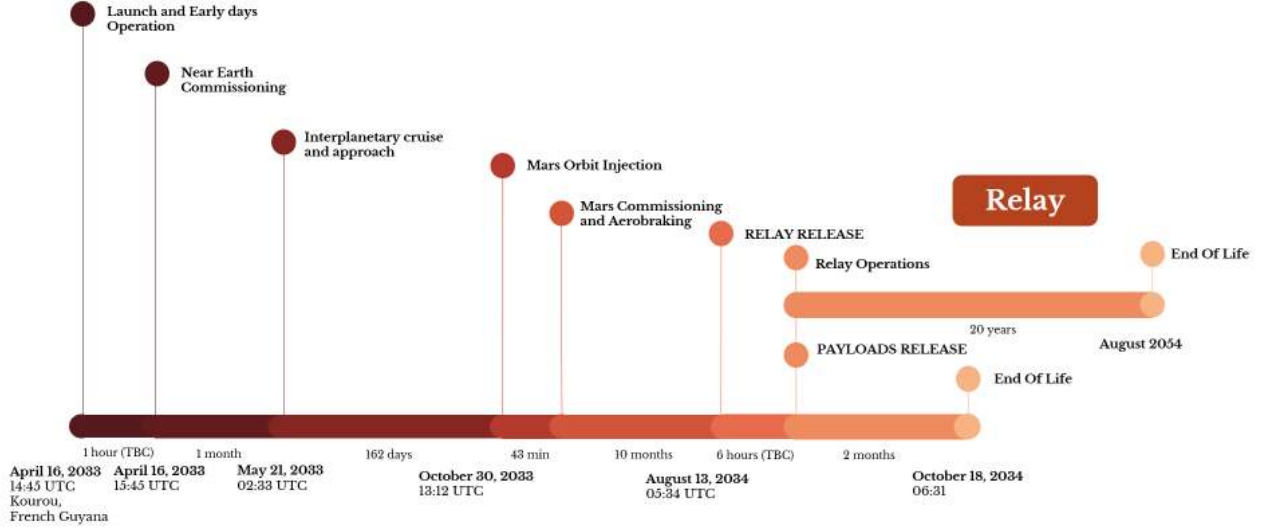


Figure 8: Mission ConOps.

6 Launcher selection

This section presents an analysis of potential launchers for the MANTIS mission, considering dynamic performance, launch site characteristics, fairing envelop, structural loads and preliminary cost per launch estimation. The trade-off study follows the approach defined in the Launchers Selection guidelines, based on three main criteria: **launchable mass as a function of C_3 / v_∞** , to minimize the energy and time required to reach Mars; **cost of a single launch**, since the total mission budget must remain reasonable and multiple launches may be required; **high-level mission requirement** states that the launcher should preferably be European.

6.1 European Launch Vehicle Option: Ariane 64

ESA's new-generation heavy launcher, particularly **Ariane 64** (four P120C boosters and cryogenic Vulcain 2.1/Vinci stages with LOX/LH₂), represents the only fully European vehicle for large interplanetary missions. The launch site is *Guiana Space Centre* in Kourou, French Guiana [5.239° N; 52.768° W]: it benefits from an equatorial velocity boost of ~ 465 m/s, allowing efficient launches to GTO and GEO orbits, accommodating large or clustered payloads within the fairing's envelop of $5.4 \text{ m} \times 20 \text{ m}$. The estimated cost per launch is about €118 M (2018 ^[50]).

The evolution of the payload mass as a function of v_∞ is shown in Figure 9, obtained through interpolation of experimental data ^[51].

In addition to providing full European autonomy, it is currently feasible for MANTIS to be launched on Ariane 64 with the selected architecture, which features a total system mass (Carrier + Relay + 30% system margin ^[52]) of approximately 5418.32 kg. In particular, Ariane 64 is capable of delivering a **hyperbolic excess velocity** v_∞ of **3.58 km/s**.

6.2 Non-European Launch Vehicle Options

American heavy-lift vehicles are also analyzed, as backup launchers, in particular the Falcon Heavy launch vehicle family, whose performances as function of C_3 are shown in Figure 10. At this iteration of the design, all the versions of Falcon Heavy (FH) are capable of launching MANTIS: the *Reusable* configuration; the *Expendable* configuration (no booster recovery, with all fuel used for ascent); the *Expendable + STAR-48* configuration, which includes an additional solid kick stage atop the second

stage to enhance C_3 .

The launch site is *Kennedy Space Center*, Florida [28.6084°N; 80.6043°W], providing a reduced rotational benefit (~ 390 m/s) compared to the previous case. All Falcon Heavy configurations share the same payload fairing, offering an envelope of 5.2 m x 13.1 m [53].

Costs are estimated^[54] at approximately \$100 M (Reusable), \$150 M (Expendable) and \$178 M (Expendable + STAR-48) per launch.

The hyperbolic excess velocity v_∞ provided by each launcher is reported in Table 5. It can be observed that the two *Expendable* versions deliver a v_∞ significantly higher than that required for the MANTIS mission, as will be further discussed in Section 7.4. Conversely, the *Reusable* configuration is currently not capable of launching MANTIS. Therefore, Ariane 64 is confirmed as the most suitable choice.

-	FH Reusable	FH Expendable	FH Expendable + w-STAR48
v_∞ (km/s)	2.85	6.96	7.08

Table 5: Hyperbolic excess velocity provided by Falcon Heavy configurations

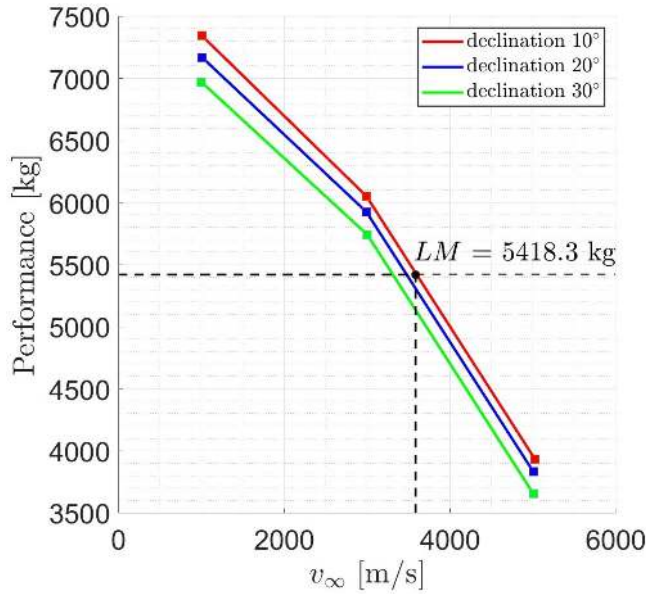


Figure 9: Ariane 64, Payload Mass vs v_∞ .

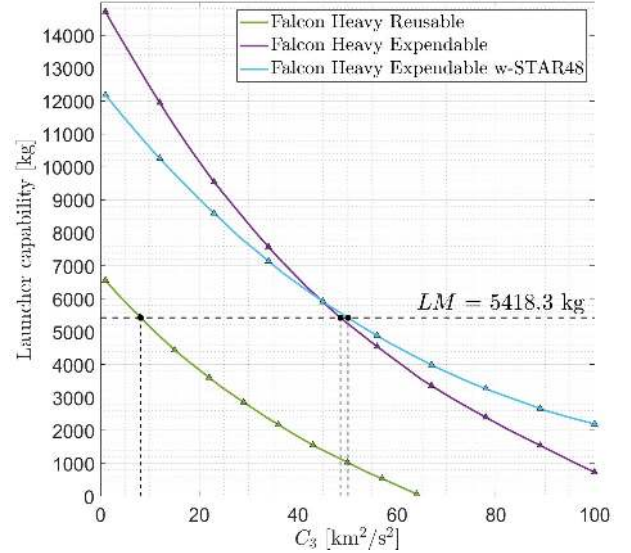


Figure 10: American launch vehicles, Payload Mass vs C_3 .

Despite potential concerns regarding launch frequency, the Ariane 64 schedule is considered sufficient to ensure the repeatability and continuity of the services offered by the MANTIS mission. Moreover, a preliminary structural analysis is performed on both vehicles to verify compliance with all applicable requirements, as will be discussed in detail in Section 17.3.

Finally, a detailed examination of the internal configuration is performed for Ariane 64, specifically considering its extended payload fairing, demonstrating that the system is compliant also with the Falcon Heavy fairing.

7 Mission Analysis

The mission analysis of the MANTIS mission was primarily focused on providing a wide range of payload release orbits to ensure a high level of service and enhance mission attractiveness, while maintaining compliance with the mission’s high-level requirements.

7.1 Orbits of interest

The starting point of the mission analysis is the identification of orbital bands of interest, based on past and planned missions, to support the selection of payload release orbits on which the overall system design is based.

Low Mars Orbits (LMO) Low-altitude Mars orbits typically have altitudes in the range of 100–1500 km, low eccentricity, and inclinations that are predominantly polar or near 60°. These orbits are commonly employed for missions requiring global, high-resolution mapping and scientific observations, enabling precise imaging, topographic measurements, and localized surface analysis.

Medium Mars Orbits (MMO) Medium-altitude Mars orbits generally span altitudes between 1500 and 10,000 km and may be either circular or highly elliptical, as demonstrated by missions such as *Mars Express* and *Mariner 9*. These orbits are particularly suitable for applications including communications relay, regional monitoring, and the development of future navigation infrastructure.

Sun-synchronous Orbits Sun-synchronous low orbits are of particular interest because they maintain consistent illumination conditions by preserving a constant local solar time over observed regions, enabling reliable temporal comparisons. However, they limit simultaneous global coverage, requiring multiple orbital cycles to sample different local times and resulting in discontinuous spatial coverage.

Aerostationary Orbits In the future, areostationary orbits may also be of interest due to the continuous coverage they can provide for both scientific and communication applications. An areostationary orbit is a circular equatorial orbit at approximately 17,000 km altitude in which the spacecraft’s orbital period matches Mars’ rotation, allowing it to remain fixed over a point on the surface.

As a result, it can be concluded that a suitable first orbital band for the MANTIS mission would be Low Mars Orbits. This band meets the immediate needs of stakeholders deploying payloads for scientific observations, especially around the poles. A second band could maintain a high polar inclination while increasing the altitude into the Medium Mars Orbit range, making it suitable for communications and relay support.

7.2 Relevant perturbations

Relevant perturbations in interplanetary space and Mars orbit were reviewed to account for their effects and, where possible, exploit them for orbital changes.

During the interplanetary trajectory, the only perturbation considered is Solar Radiation Pressure (SRP). Its magnitude has been estimated at approximately 10^{-17} km/s², which is considered negligible. During operations within Mars’ sphere of influence (SOI), the following perturbations were analyzed.

- **SRP:** Magnitude similar to interplanetary values and therefore negligible.
- **Magnetic Field:** Negligible, as indicated in Section 2.3.
- **Drag:** Computed for 100, 200, and 300 km in highly elliptical orbits, yielding $a_{\text{Drag}} \approx 10^{-5}$, 10^{-9} , 10^{-11} km/s², respectively. Relevant only at very low altitudes; primarily circularizes orbits, enabling maneuvers such as aerobraking and aerocapture to save ΔV .

- **Third-body:** Sun, Phobos, and Deimos considered for Medium Mars Orbits. Resulting accelerations: $a_{3\text{Body}} \approx 10^{-6}, 10^{-11}, 10^{-13} \text{ km/s}^2$. Phobos and Deimos effects negligible; Sun's effect relevant for long-term propagation.
- **Gravitational:** Dominated by zonal harmonics J_2, J_3, J_4 , with magnitude $\sim 10^{-5} \text{ km/s}^2$ for both orbital bands.

Figure 11a and Figure 11b show the evolution of the orbital elements under the considered perturbations for both the LMO and MMO bands. These effects can be exploited to modify the Argument of Perigee, while the inclination and eccentricity remain close to their nominal values. The variation of the Right Ascension of the Ascending Node (RAAN) strongly depends on inclination; for near-polar orbits, it remains nearly constant, as observed in both figures.

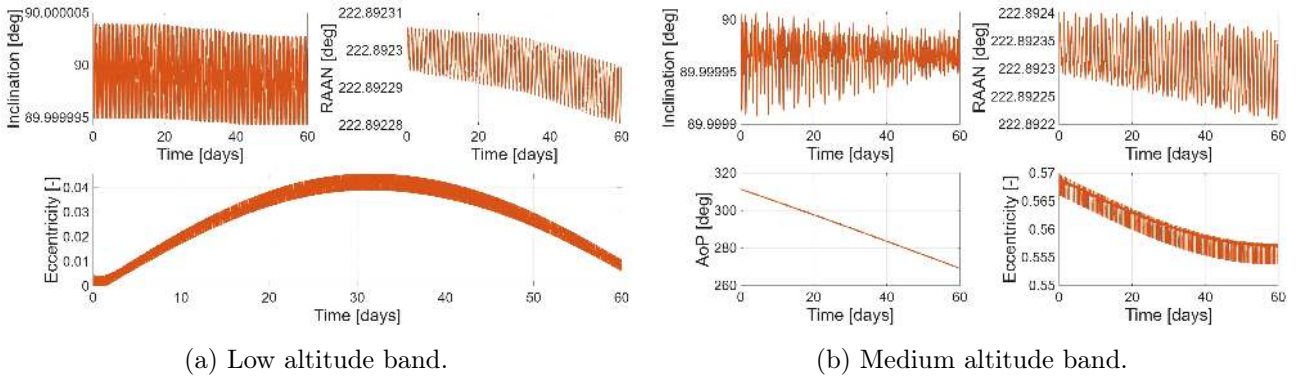


Figure 11: Orbital parameters evolution.

7.3 Trade-off

The MA was conducted by first identifying the mission phases in terms of trajectories and exploring possible solutions for each phase. The baseline was then selected by defining constraints and requirements derived from mission drivers and high-level objectives.

- **ΔV minimization** is prioritized over maneuver duration; due to the high dry mass from the 900 kg payload, reducing propellant consumption is critical.
- The mission shall be **repeatable**; interplanetary trajectories must be feasible at least every Earth–Mars synodic period, avoiding paths that require complex multi-body alignments.
- **Reliability** is essential to safely deliver the client's payload. Tested and validated methods with flight heritage are preferred to minimize mission risk.

7.3.1 Interplanetary transfer

For the interplanetary transfer, two trajectory options were analyzed. The first is a **direct transfer**, which is the most reliable due to its extensive heritage from previous missions and the absence of critical maneuvers. Repeatability is also ensured, as the trajectory depends solely on the synodic period to optimize propellant usage. However, it is the most propellant-intensive option in terms of ΔV . The second option involves using **gravity assists** from other celestial bodies to reduce the required ΔV . Preliminary analysis using Tisserand plots did not identify any trajectory that simultaneously minimized both departure and arrival ΔV . A literature review revealed that the only past mission to attempt such maneuvers was *Nozomi* [55], which planned two lunar flybys and one Earth flyby to save up to 120 m/s. However, this mission ultimately failed to achieve Mars orbit insertion. Given the lack of heritage and repeatability of gravity-assisted trajectories, a direct transfer was selected as the baseline.

7.3.2 Mars Orbit Injection (MOI)

To reduce propellant requirements, Mars SOI insertion is performed by placing the spacecraft into a highly elliptical with an apogee of 60,000 km and a perigee of approximately 100 km. Two insertion techniques were considered: impulsive insertion and aerocapture. **Impulsive insertion** is the most common method, offering high flexibility in the resulting orbit and proven reliability from previous missions. Its main drawback is the high propellant cost, requiring approximately 1 km/s. **Aerocapture** involves a single atmospheric pass that reduces the spacecraft's hyperbolic velocity to achieve an elliptical Martian orbit. However, no past missions have executed aerocapture, and only feasibility studies exist in the literature, making it an unreliable option. For this reason, impulsive insertion was selected as the baseline maneuver.

7.3.3 Orbit Lowering

The orbits identified as most relevant for payload release span a range of altitudes; therefore, an orbit-lowering strategy is required to reach them. Several techniques were assessed. The primary option is **aerobraking**, which consists of multiple atmospheric passes that gradually reduce the semi-major axis and eccentricity of the initial elliptical orbit. Aerobraking has been performed in five Mars missions and, with the exception of *Mars Global Surveyor (MGS)* [56], which experienced operational issues, all demonstrated successful implementation [57]. Notable examples include *ExoMars Trace Gas Orbiter* [58], the largest spacecraft to perform aerobraking to date, and the more recent *Mars Atmosphere and Volatile Evolution (MAVEN)* [59]. The second option consists of using **low-thrust**. The third option is exploiting **impulsive high-thrust**, performing one or several periapsis burns to lower the apoapsis into the target orbit. The Aerobraking has been selected due to its strong state-of-the-art heritage, which demonstrates both reliability and feasibility, but primarily because of the significant ΔV savings it provides.

7.4 Interplanetary trajectory design

The design of the interplanetary transfer begins with the evaluation of potential launch dates to minimize ΔV , using a porkchop plot generated over the 2030–2036 window in Figure 12.

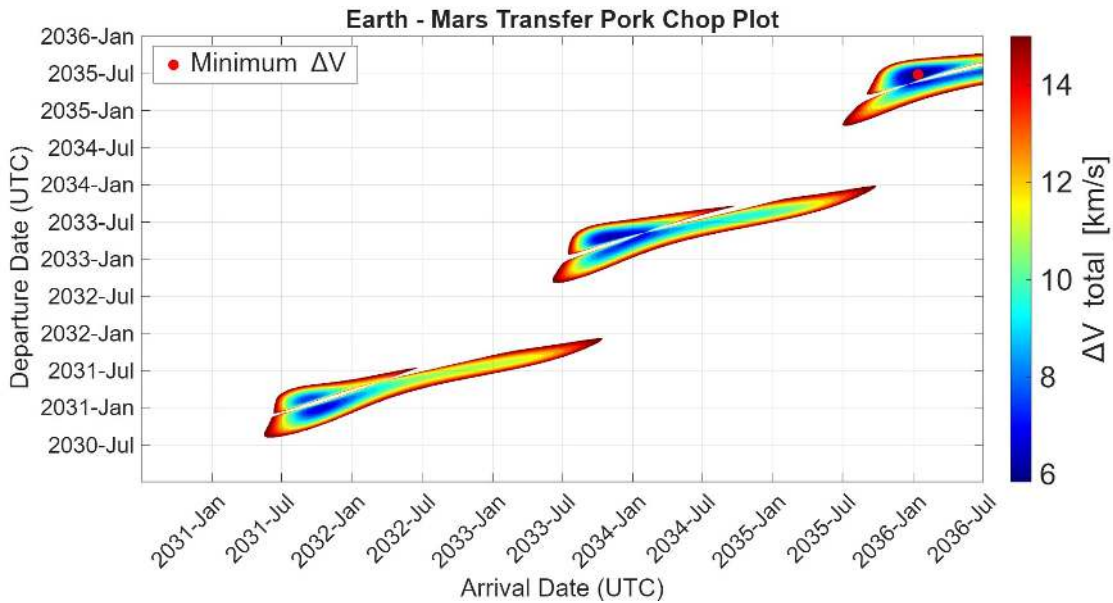


Figure 12: Earth Mars porkchop plot over the time window 2030-2036.

Although the Earth–Mars synodic period is approximately two years, identifying the absolute minimum across the entire window is not meaningful, as the mission must remain repeatable within this interval. Therefore, the baseline sizing is based on the 2033 launch opportunity, which represents a worst-case ΔV scenario while remaining a realistic timeframe for spacecraft design and development. The trajectory parameters are reported in Table 6. The Lambert solution is used as the initial baseline and is subsequently refined through an iterative process to obtain a more realistic trajectory, accounting for perturbations and n-body propagation.

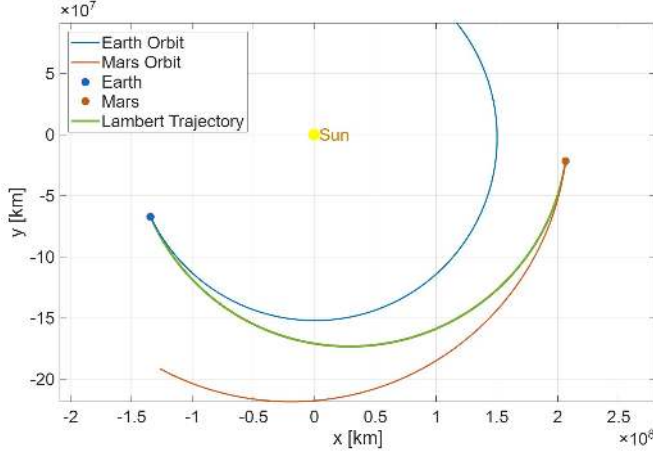


Figure 13: Earth–Mars Lambert transfer in the Sun-centered ecliptic J2000 frame.

Parameter	Value
Departure $ v_{\infty} $	3.0104 km/s
Arrival $ v_{\infty} $	3.3245 km/s
Departure UTC Date	16 Apr 2033
Arrival UTC Date	31 Oct 2033
Time of Flight (ToF)	198 days

Table 6: Lambert solution for Earth–Mars transfer.

7.4.1 Earth Departure

The first step is designing the Earth departure hyperbola, which must match the Lambert departure velocity in both magnitude and direction. Due to the high cost of interplanetary injection, this maneuver is assumed to be performed by the launch vehicle. Following common practice, the pericenter of the departure hyperbola is set to a 500 km low Earth orbit (LEO), and the resulting hyperbolic parameters are computed and reported in Table 7.

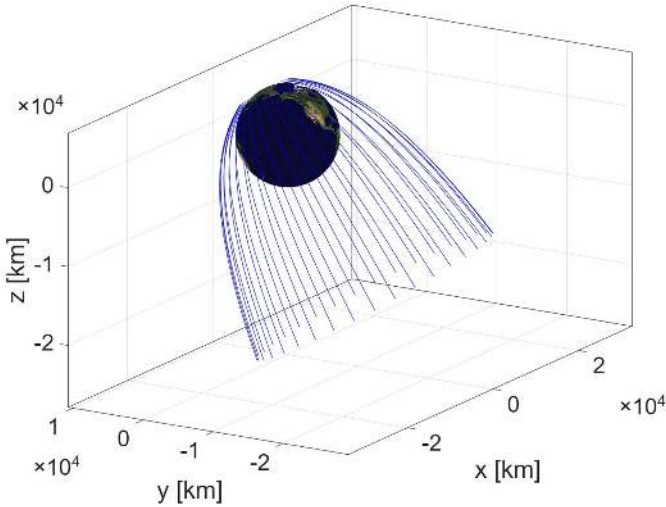


Figure 14: Earth-centered departure hyperbolas in the J2000 reference frame.

Parameter	Value
$ v_{\infty} $	3.0104 km/s
Eccentricity	1.1564
Asymptote pericenter	149.85°
Pericenter radius	6878.1 km
Pericenter velocity	11.178 km/s

Table 7: Earth departure hyperbola relevant parameters.

The computed parameters define asymptotic hyperbolas whose orbital planes contain the Lambert excess velocity vector. The departure orbit inclination is constrained by the launch site. The selected launch vehicle is Ariane 64, departing from Kourou, French Guiana. The minimum achievable inclination corresponds to the launch site latitude (6°), while the theoretical maximum is 90° . In practice, this upper limit is reduced by launcher performance constraints; therefore, a conservative range is adopted and a departure inclination of 80° is selected.

7.4.2 B-Plane Target

The first MANTIS launch is planned to release PHAROS; accordingly, and in line with its mission objectives, a Martian inclination of 90° has been selected. To achieve this orbit, a B-plane targeting approach was carried out. Specifically, the B-plane components required to target Mars' south pole at an altitude of 100 km, which allows immediate aerobraking without additional maneuvers, are computed as $B \cdot R = 0$ km and $B \cdot T = 6246.98$ km. A Trajectory Correction Maneuver (TCM) is then applied using a differential correction scheme in GMAT to target these coordinates. This method accounts for all relevant perturbations, including planetary gravity up to the 8th-order spherical harmonics, n-body effects, solar radiation pressure, and relativistic corrections. The TCM is performed after the spacecraft exits Earth's sphere of influence, following standard practice. To obtain the final trajectory, an iterative process was conducted in which perturbations were progressively included until the solution fully accounted for all effects. The resulting ΔV is 30.4 m/s without margins, fully defining the nominal trajectory.

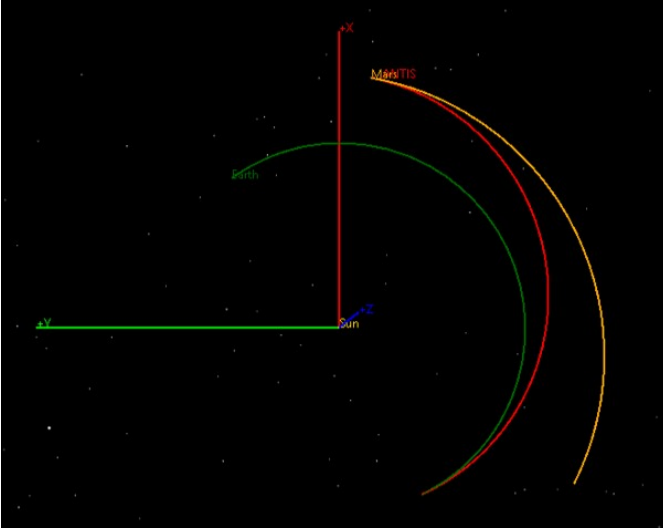


Figure 15: Sun-centered transfer trajectory in the Ecliptic J2000 reference frame.

Parameter	Value
Departure UTC Date	16 Apr 2033
Earth SOI time	3.12 days
TCM UTC Date	19 Apr 2033
Transfer Time	191.37 days
Mars SOI time	2.31 days
Arrival UTC date	30 Oct 2033

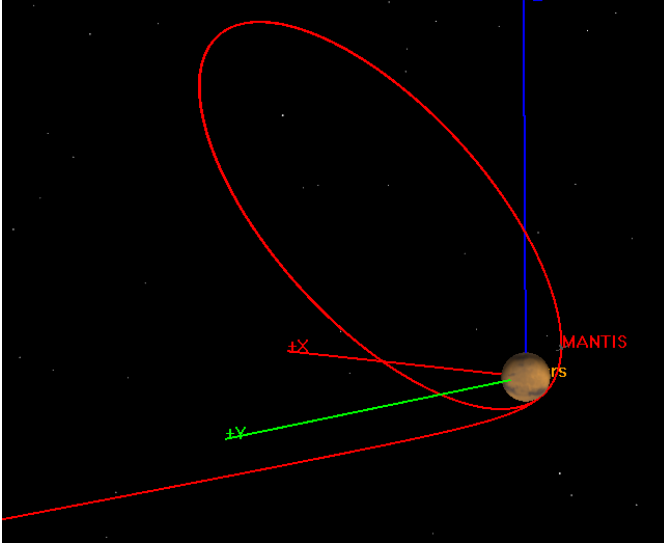
Table 8: Mission relevant dates and phases duration.

Figure 15 shows the final computed trajectory using GMAT's visual representation. Meanwhile, Table 8 lists the dates and durations of the different mission phases. The time of flight differs only slightly from the value obtained using the Lambert approach, while the newly computed departure hyperbolic excess velocity, $\|v_\infty\|$, is 2.975 km/s, again slightly lower than the Lambert value.

7.4.3 Mars Orbit Injection

MANTIS approaches Mars' SOI on 28 October 2033. After this, it continues along its trajectory until the pericenter of its Mars-centered hyperbola is reached. During this phase, further analysis was conducted to ensure that the two Martian moons, Phobos and Deimos, do not influence the trajectory

and that no impacts or close encounters occur. At pericenter, an impulsive burn is performed to insert MANTIS into a $60,000 \times 100$ km orbit, as previously discussed. Figure 16 provides a visual representation of the incoming hyperbolic trajectory and the capture orbit, while Table 9 reports the parameters of the MOI and the resulting capture orbit.



Parameter	Value
MOI $ \Delta V $	1.207 km/s
MOI UTC Date	30 Oct 2033
SMA	33446.19 km
Eccentricity	0.896
Inclination	90°
RAAN (Ω)	311.23°
AOP (ω)	222.89°

Table 9: MOI and achieved orbit relevant parameters.

Figure 16: Mars-centered orbit in the Mars inertial reference frame.

7.4.4 Monte Carlo simulation

A Monte Carlo analysis was conducted to assess Mars arrival performance under uncertainties in both the direction and magnitude of the TCM. A directional uncertainty of 0.1° was adopted, consistent with values reported in the literature and with ADCS capabilities. This value was intentionally not selected to represent the best achievable accuracy, in order to account for additional uncertainties such as mounting errors. A conservative magnitude uncertainty of 0.1 m/s was assumed, consistent with thruster performance. The results were evaluated in terms of arrival altitude and are reported in Figure 17. The analysis shows a high impact probability, with altitude errors ranging within ± 500 km relative to the target altitude.

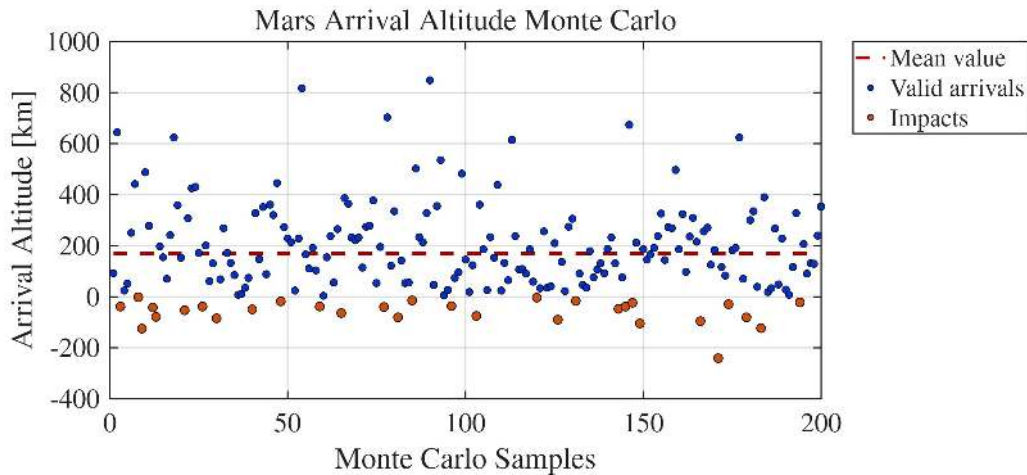


Figure 17: Monte Carlo simulation results under TCM uncertainties.

To mitigate this dispersion, a second TCM is planned and computed, once again using a differential correction scheme to target the previously defined B-plane coordinates. To evaluate the effectiveness of this correction strategy, a complete simulation campaign is performed. The first TCM is perturbed and applied, after which the second TCM is computed to correct the resulting deviation. A perturbation with the same uncertainties described above is then applied to the second TCM. The results, shown in Figure 18, clearly demonstrate that the achieved accuracy is sufficient for the proposed maneuver, with final errors confined within ± 2 km. From Figure 19, it can be observed that the second TCM, when required, is typically small, with values reaching up to 10 m/s.

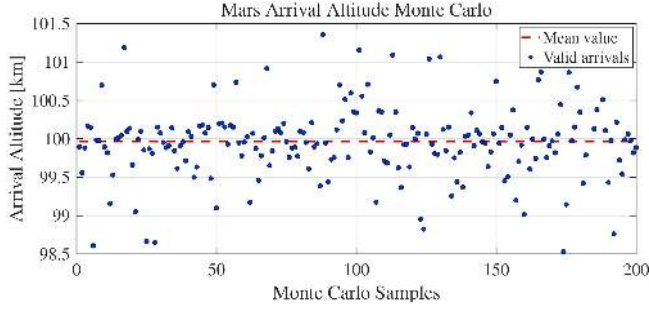


Figure 18: Monte Carlo simulation results under two TCM uncertainties.

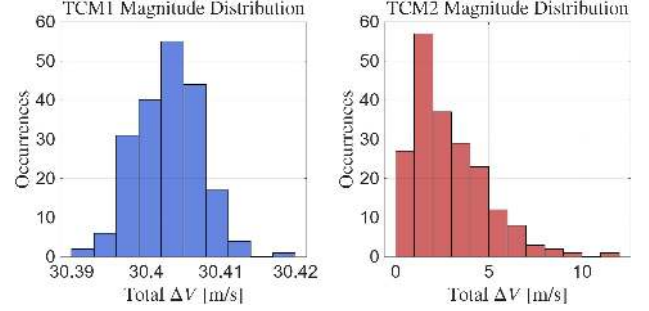


Figure 19: Monte Carlo histograms for TCM magnitude.

7.5 Client Alternatives

The primary objective of the MANTIS mission analysis is to deliver payloads to their designated orbits. Accordingly, it is essential to define the set of feasible release orbits, derived from the orbital bands identified in Section 7.1 and constrained by the propulsion system's maximum achievable ΔV . The design is based on a reference ΔV budget that includes the capability to perform a 10° plane change to reach different relevant inclinations. Ideally, this maneuver would be performed in LMO, where varying inclinations are particularly beneficial for observation purposes. However, due to the high propulsive cost at low altitudes, it is preferable to perform the plane change in MMO and allow the payload to subsequently reduce its altitude. Therefore, a trade-off analysis was conducted to select the optimal altitude for the plane change, ensuring that the client payload can reach the LMO with minimal propellant cost. The ΔV required for a 10° plane change was evaluated at different altitudes, together with the corresponding propellant mass. The results indicate that an apogee altitude of 7000 km provides a satisfactory compromise, minimizing both propellant requirements and release altitude. From this reference ΔV three baseline mission alternatives were defined, each characterized by a different maneuvering strategy. This approach was adopted because the range of possible solutions is effectively unlimited and strongly dependent on client requirements. Consequently, three representative extreme cases are presented, while intermediate configurations remain feasible as long as the maximum maneuvering cost is respected.

Baseline 1 The first baseline proposes an initial MMO band with an apocenter altitude ranging from 10,000 km to 7,000 km and a pericenter of 300 km. A plane change of up to 10° is performed at 7,000 km, followed by insertion into a circular LMO of 300×300 km. A visual representation is shown in Figure 20a.

Baseline 2 The second baseline uses the same initial MMO band but performs no inclination change. Instead, a phasing maneuver is applied in LMO, achieving up to a 44° adjustment in a 300×300 km orbit. A visual representation is shown in Figure 20b.

Baseline 3 The third baseline also starts from the same MMO band. A phasing maneuver is performed, with maximum changes ranging from 135° to 153° depending on the altitude at which it is executed, followed by insertion into a circular LMO. A visual representation is shown in Figure 20c.

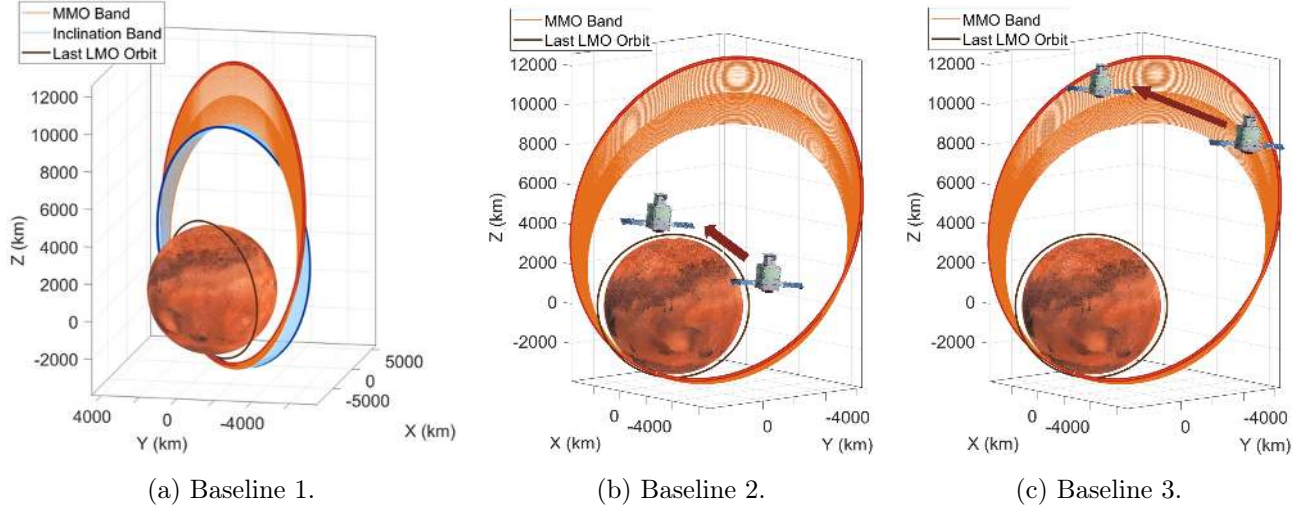


Figure 20: Client Alternatives.

7.6 Aerobraking

To reach the range of altitudes offered to clients while minimizing propellant consumption, an aerobraking strategy was selected, as discussed in Section 7.3.3. The design of this phase prioritizes ΔV minimization, since no time constraints are imposed. Effective orbit circularization requires the pericenter altitude to remain within a corridor where atmospheric effects are significant, defined here as 95 km to 115 km, consistent with navigation prediction capabilities and values reported in the literature. Following these guidelines, the complete aerobraking phase was simulated.

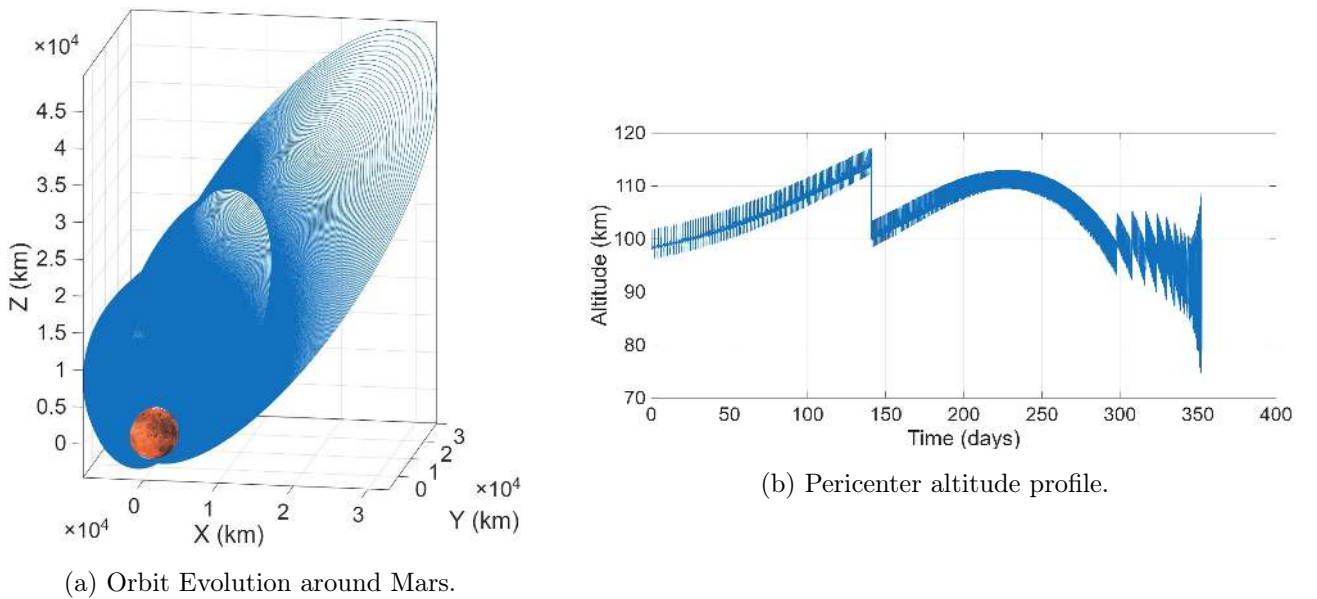


Figure 21: Orbit and pericenter altitude profile around Mars.

The resulting orbit evolution, including pericenter variations, is shown in Figure 21. A total of 20 maneuvers were required, with individual burns ranging from 0.5 m/s to 3 m/s and a cumulative ΔV of approximately 20 m/s. Additional simulations performed at different inclinations produced results consistent with the reference case, confirming the robustness and flexibility of the strategy. As shown in Figure 21b, the total duration of the aerobraking phase is approximately 350 days (about 11 months), which is longer than in comparable missions. Given the relatively small ΔV required, active pericenter maneuvers could be introduced to accelerate circularization. However, because this mission phase is not time-critical, the baseline strategy prioritizes ΔV minimization to enhance versatility in payload release orbits.

7.7 Relay Orbit

This section addresses the selection and long-term maintenance of the relay orbit. First, a trade-off analysis is presented to identify the most suitable orbital configuration based on communication performance, geometry, and mission cost. Then, the station-keeping strategy for the selected orbit is examined to assess its long-term stability and operational sustainability.

7.7.1 Trade-off

The trade-off analysis was performed by combining two complementary approaches. A multi-criteria framework ^[60] was used to quantify geometric and propulsive performance through dedicated figures of merit, while a ground visibility analysis evaluated communication windows between the relay and the ESTRACK ground stations. Three candidate orbits were assessed:

- **Medium Polar Orbit (MPO):** circular, 5000×5000 km, $i = 90^\circ$; a heritage configuration widely used for Mars relay support ^[61].
- **High Elliptic Polar Orbit (HEPO):** 300×10000 km, $i = 90^\circ$, characterized by a low injection ΔV and extended visibility intervals.
- **Areostationary Orbit (ASO):** 17000×17000 km, $i = 0^\circ$; included for completeness but penalized by high ΔV and degraded link geometry.

The comparison relied on five figures of merit:

- **Payload visibility fraction:** portion of the orbit with line-of-sight to the surface payload.
- **Earth visibility fraction:** fraction of time the relay is visible from Earth under ideal alignment.
- **Simultaneous visibility:** overlap between payload and Earth visibility windows.
- **Link-quality proxy:** simplified indicator of received power degradation with range.
- **Cost parameters:** normalized injection and qualitative station-keeping costs.

An overall performance index was defined as a weighted sum of these metrics, prioritizing visibility over propulsive cost to ensure continuous relay capability. Results show that the **HEPO** and the **MPO** provide similar performance, with the **HEPO** slightly outperforming across all metrics while maintaining a comparable injection cost.

A ground visibility analysis was performed for the three candidate relay orbits using the ESA ESTRACK stations (Cebreros, Malargüe, and New Norcia) over a representative 4-day interval. A minimum elevation mask of 10° was applied; shaded regions in Figure 22 indicate periods of valid line-of-sight communication with at least one station. The resulting visibility patterns highlight distinct contact geometries. The **MPO** exhibits regular, symmetric cycles with continuous alternation

among the three ESTRACK sites. The **HEPO** increases the duration of individual contact windows, particularly near apocenter. The **ASO** provides visibility comparable to HEPO, though with different window timing and duration. Overall, the analysis indicates that the **HEPO** configuration provides the most favorable visibility performance among the candidates and is therefore selected as the baseline relay orbit.

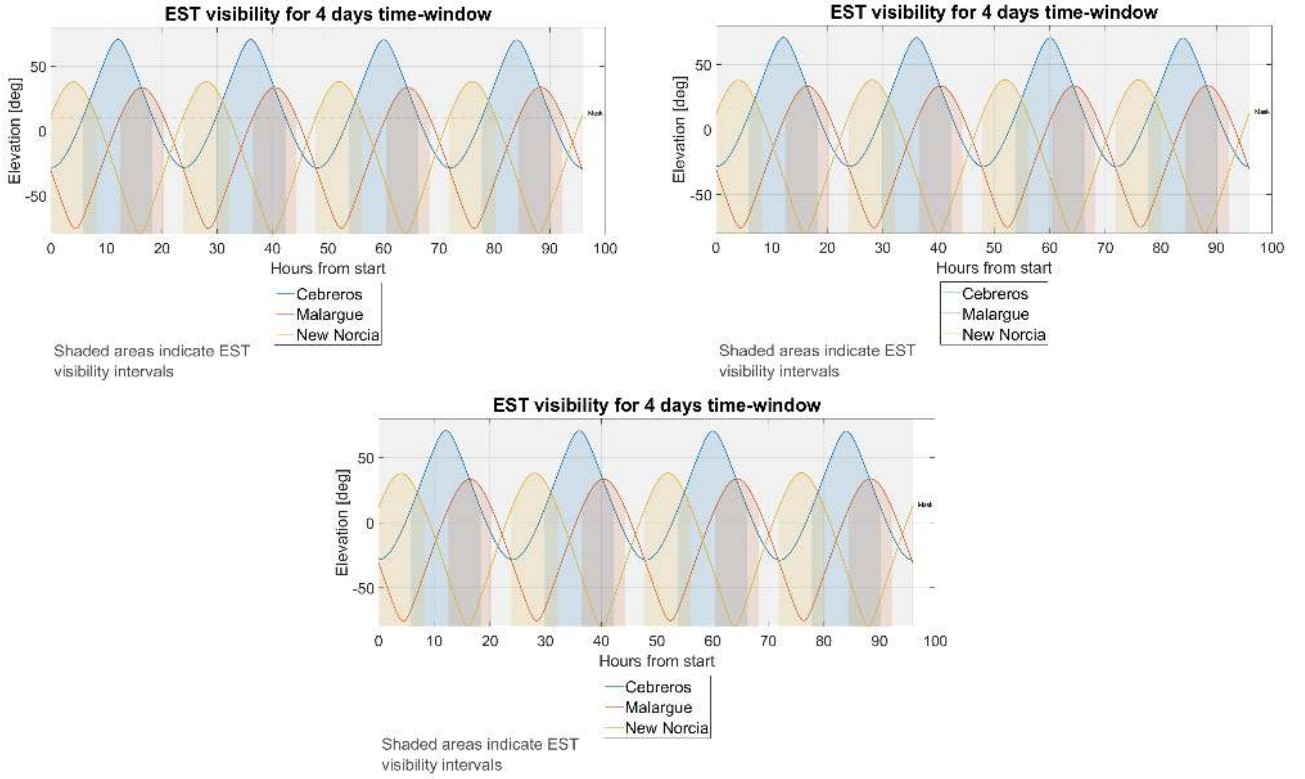


Figure 22: Ground visibility comparison for the three candidate relay orbits.

7.7.2 Station Keeping Strategy

The Carrier continues aerobraking until the apocenter reaches the altitude of 10,000 km. At that point, the pericenter is raised to 300 km to achieve the orbit selected for Relay release. After separation, the Relay begins nominal operations, while the Carrier lowers the pericenter to about 100 km, resumes aerobraking, and starts the payload release phase. The Relay must maintain this orbit for its entire lifetime (~ 20 years). The station-keeping budget is estimated considering Mars' gravity field, solar radiation pressure, atmospheric drag, and third-body perturbations from the Sun, Phobos, and Deimos. Simulations show that the evolution is dominated by Mars' gravity, with other effects negligible over long periods. The orbital plane remains stable, with nearly constant inclination and RAAN. In contrast, the argument of periapsis and periapsis altitude experience significant variations. A control strategy based on radial maneuvers when the argument of periapsis deviates by more than 2° would require 53.9 m/s per correction and interventions every 3–4 days, making it impractical. A periodic oscillation of the semi-major axis causes the periapsis altitude to vary between 300 km and 233 km with a period of about 1.7 years. Restoring the altitude at the minimum would require roughly 8.4 m/s; however, since the natural dynamics restore the nominal value and the minimum remains safely above the atmosphere, passive control is adopted. Potential risks include long eclipse and occultation periods if perturbations shift the apoapsis geometry. A 15-year propagation from the relay release epoch confirms that neither condition occurs during the Relay lifetime, as shown in

Figures 23 and 24.

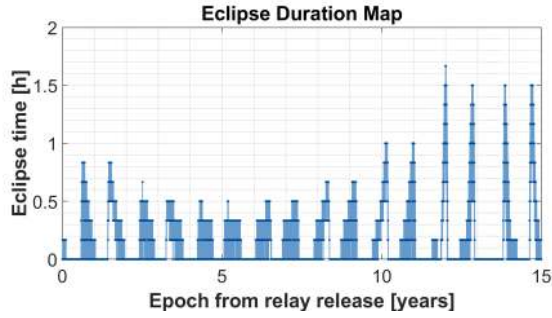


Figure 23: Eclipse Duration Map over 15 years lifetime.

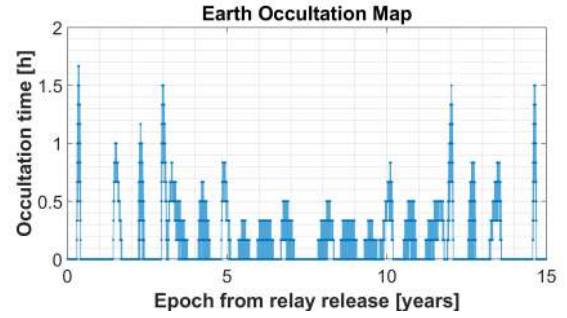


Figure 24: Earth Occultation Duration Map over 15 years lifetime.

7.8 ΔV Budget

This section provides a brief summary of the ΔV budget to give an overall view of the mission cost. A more detailed breakdown for each baseline is provided in the appendix in Sections A.1 to A.3. Margins are computed and applied according to ESA guidelines for Mars missions [52]. Since all maneuvers are evaluated analytically (i.e., deterministic), the applicable margin corresponds to the greater of 5% of the total deterministic ΔV or 10 m/s. In this case, the 5 % criterion governs and is therefore adopted.

Carrier	Estimated ΔV [m/s]	Margin Added	Allocated ΔV [m/s]
TCMs	40.38	+5%	42.40
MOI	1160.00	+5%	1218.00
Aerobraking	18.07	+5%	18.97
Perigee change for relay release	51.06	+5%	53.61
Perigee change after aerobraking	47.66	+5%	50.04
ΔV for payload release	436.22	+5%	458.03
Total	1753.39	-	1841.05
Relay	Estimated ΔV [m/s]	Margin Added	Allocated ΔV [m/s]
Station Keeping	0.00	-	0.00

Table 10: Allocated ΔV budget for Carrier and Relay.

8 Attitude Determination and Control

The design of the Attitude Determination and Control Subsystem (ADCS) is carried out through an iterative process comprising the following steps: definition of control modes; identification of pointing requirements and allocation of the pointing budget per subsystem and per mode; selection of the control architecture for each attitude mode; and selection and sizing of the ADCS hardware (sensors and actuators).

8.1 Modes

ADCS modes are predefined operational configurations that regulate the spacecraft's attitude determination and control strategy according to mission phase and system constraints. Each mode specifies the active sensors, actuators, and control laws required to achieve a particular pointing or stabilization objective. Transitions between modes can be time-based or event-based: time-based scheduling provides the capability to command actions using requests pre-loaded on board the satellite and released at their due time; event-based triggering enables the autonomous execution of predefined actions upon the occurrence of specific on-board events^[62].

The following modes have been defined for the Carrier spacecraft:

- **Acquisition:** Initial determination of the spacecraft's attitude and stabilization to enable ground communication and power generation. This mode may also be used to recover from power upsets or emergency conditions. In this attitude, the spacecraft maintains a power-positive and thermally safe orientation. The entry is typically time-based, while the exit is triggered by the completion of detumble, attitude determination, and panel deployment. If the mode is entered as a consequence of an emergency, the transition is event-triggered^[63].
- **Inertial:** Maintain the spacecraft in an inertially fixed orientation or slew it between two predefined attitudes. This mode can be considered a staging mode between phases^[63].
- **DeltaV:** Activation of the attitude control thrusters and the main engine to provide the needed ΔV for orbit transfer maneuver and the orbit Station Keeping maneuver. Entry and exit are time-based^[64].
- **DeltaH:** Reduction of the angular rate into the acceptable value after dynamical events (separations, releases, burn, disturbances). Entry is triggered by excessive angular rate, while the exit occurs once the exit criteria are satisfied^[64].
- **Release:** Achieve defined attitude and angular rates to correctly release relay and payloads. Entry and exit are time-based.
- **Telecommunication with Earth:** Communication with Earth, slewing into the defined attitude. Entry and exit are time-based.
- **Telecommunication with Relay:** Communication with Relay, slewing into the defined attitude. Entry and exit are time-based.
- **Aerobraking:** Hold the spacecraft in the correct orientation for the aerobraking in terms of solar panels orientation and cross-sectional area. Entry and exit are time-based.
- **Safe mode:** Used in emergencies if regular mode fails or is disabled. Will generally use less power or fewer components to meet minimal power and thermal needs. In this attitude, the spacecraft maintains a power-positive and thermally safe orientation and communication shall be ensured. Entry and exit are event-based or can be commanded from ground^[63].

The Relay spacecraft implements the following modes already defined for the Carrier: Acquisition, Inertial, DeltaV, DeltaH, Telecommunication with Earth, and Safe mode; in addition, a **Telecommunication with Carrier and Payloads** is defined as: communication with Carrier and Payloads, slewing into the defined attitude; entry and exit are time-based.

The state machines associated with the defined ADCS modes are reported in Section B (Figure 106 and Figure 107) for completeness and reference.

8.2 Pointing Budget

The pointing budget, reported in Table 11 for the Carrier and Table 12 for the Relay, is drawn up by compiling the Absolute Performance Error (APE) required by each subsystem or component for every mode and phase, and then by estimating the Absolute Knowledge Error (AKE) as 10% of the corresponding APE as a conservative preliminary assumption, consistent with typical ECSS design practices for early-phase mission analysis. The errors are treated as Gaussian variables, so that the ADCS requirements are defined with a confidence level of 3σ .

	APE	AKE	Notes
Low Gain Antenna (X-band)	12°	1.2°	From TTMTTC sizing
Low Gain Antenna (UHF)	67.47°	6.75°	From TTMTTC sizing
Solar Panels	5°	0.5°	From EPS sizing
Trajectory Correction Manoeuvres	0.1°	0.01°	From MA sizing
Mars Orbit Injection manoeuvre	0.1°	0.01°	From MA sizing
Other manoeuvres	0.1°	0.01°	[65]
	ARPE	ARKE	Notes
Payload release	0.1°/s	0.01°/s	From CONF/MECH sizing

Table 11: Carrier pointing budget.

	APE	AKE	Notes
High Gain Antenna	0.07°	0.007°	From TTMTTC sizing
Low Gain Antenna (X-band)	12°	1.2°	From TTMTTC sizing
Low Gain Antenna (UHF)	53.5°	5.35°	From TTMTTC sizing
Solar Panels	5°	0.5°	From EPS sizing

Table 12: Relay pointing budget.

8.3 Sensors Selection

Based on the pointing and knowledge requirements, the attitude sensor suite was defined. Magnetometers were discarded due to the absence of a global Martian magnetic field, while horizon sensors were excluded as they are not yet fully qualified for Mars applications.

The **Relay** adopts a highly redundant configuration due to its long lifetime, comprising **four Star Trackers** (two nominal and two cold-redundant), **two Inertial Measurement Units (IMUs)** (nominal plus cold-redundant), and **ten Coarse Sun Sensors** (five nominal and five cold-redundant). The number of nominal CSSs is chosen in order to guarantee continuous Sun visibility for attitude

determination, ensuring that at least one sensor is illuminated for any Sun direction around the spacecraft, independently of its attitude. The **Carrier** follows a more compact architecture, with **two Star Trackers** (nominal plus cold-redundant), **two IMUs** (nominal plus cold-redundant), and **ten CSSs** (five nominal and five cold-redundant).

The selected sensor suite ensures full attitude observability in all mission phases. Star Trackers provide high-accuracy absolute attitude knowledge during nominal operations, IMUs provide high-frequency angular rate propagation during slews and in case of temporary star tracker blinding, while Coarse Sun Sensors ensure safe Sun acquisition capability in degraded or contingency conditions.

As shown in Table 13, the selected sensor suite provides full attitude and angular rate knowledge consistent with the pointing budget requirements across all mission phases.

Sensor	Pointing Knowledge	Rate Knowledge	Model	Notes
Star Tracker	0.003°	0.01°/s	Hydra (Sodern)	[66]
CSS	3°	-	CSS (Bradford Space)	[67]
IMU	-	0.001°/s	Astrix 1090 (Airbus)	[68]

Table 13: Sensors selection.

8.4 Relay Architecture

This section details the design and sizing of the Relay spacecraft's ADCS. The following analysis defines the nominal and communication pointing strategies, the sizing of the actuators to handle environmental and maneuver-induced momentum, and the configuration of the sensor suite required to meet the stringent pointing requirements of the high-gain telecommunication links.

8.4.1 Pointing Strategy

The nominal operational mode, **Telecommunication with Carrier & Payloads**, employs a **quasi-inertial** pointing strategy. The spacecraft is oriented with the UHF antenna (aligned along the $+X$ body axis) parallel to the orbit's eccentricity vector, while the solar array drive assembly (Y -axis) is kept perpendicular to the Sun-pointing vector. This strategy provides two main advantages: first, even without dedicated slews, the UHF antenna field of view covers most payload orbits, enabling continuous communication throughout the Relay orbit; second, the quasi-inertial attitude results in a **cyclic gravity-gradient disturbance torque**, which naturally limits reaction wheel momentum accumulation, thereby reducing the frequency and magnitude of wheel desaturation manoeuvres, as the disturbance torque partially self-compensates over each orbit.

During Earth-visibility windows, the spacecraft transitions to a dedicated **Earth-pointing** strategy. In this configuration, the High Gain Antenna (HGA) is steered toward the ground station, while the solar panels maintain their sun-tracking constraint to ensure power consistency.

8.4.2 Actuators Sizing

Actuators Selection To satisfy the stringent pointing requirements, the Relay attitude control system is based on **four Reaction Wheels** (RWs) arranged in a pyramidal configuration with a 35° inclination angle, including one wheel in hot redundancy. This configuration guarantees full three-axis control authority even in the presence of a single wheel failure and improves torque allocation compared to orthogonal configurations. Full three-axis control authority is required to ensure continuous and accurate pointing of the High-Gain Antenna toward Earth, thereby maintaining the communication

link throughout the mission.

Desaturation manoeuvres are performed by **monopropellant thrusters**, selected due to their flight heritage and reliability with respect to other technologies such as bipropellant or electric propulsion.

Sizing Assumptions Actuator sizing was performed under conservative assumptions. Only **Solar Radiation Pressure** (SRP) and **Gravity Gradient** (GG) torques were considered, as they represent the dominant disturbances for the selected orbit: effects from the crustal magnetic field were already proven to be null, while air drag is negligible at orbital altitudes.

SRP was evaluated assuming a worst-case cross-sectional area including the HGA and Solar Arrays, resulting in a constant maximum torque. GG was computed assuming a worst-case 45° inclination of the minimum inertia axis with respect to the local Nadir direction. This condition is representative of transient attitude configurations reached during commanded slew manoeuvres, which may be sustained for time intervals sufficiently long to produce non-negligible angular momentum accumulation in the reaction wheels. The spacecraft inertia was estimated by modelling the Relay as a composite system consisting of the main bus, HGA, and two Solar Arrays, each with **uniformly distributed mass**.

Reaction Wheels Sizing Solar Radiation Pressure (SRP) produces a maximum disturbance torque of 2.85×10^{-6} Nm, corresponding to an accumulated angular momentum of 0.0684 Nms per orbit. Gravity Gradient is the dominant disturbance, with a peak torque of 8.65×10^{-4} Nm and an accumulated angular momentum of 7.98 Nms. On the basis of these results, the selected RWs are **RSI 12-75/60** manufactured by Rockwell Collins^[69], each providing 75 mNm of maximum torque and 12 Nms of storable angular momentum, comfortably exceeding the disturbance requirements. The selected wheels provide a momentum storage capability exceeding the worst-case accumulated disturbance by a factor greater than 1.5, ensuring sufficient operational margin between consecutive desaturation events.

Slew manoeuvres were also verified by limiting the maximum angular rate to $0.5^\circ/\text{s}$. Under this constraint, a 180° reorientation can be performed in approximately 720 s about any axis, requiring a peak torque of 16.3 mNm and storing up to 5.9 Nms of angular momentum, well within wheel capabilities.

Thrusters Sizing Reaction wheel desaturation is performed using **eight MR-103J thrusters** manufactured by Aerojet Rocketdyne^[70], with four units mounted on the $+x$ face and four on the $-x$ face, each inclined by 30° in the y direction: the thrusters located close to the $+z$ face are inclined by 30° toward the $+y$ direction, while those located close to the $-z$ face are inclined by 30° toward the $-y$ direction. This choice was made to limit propellant consumptions during desaturation manoeuvres around x -axis and mitigate plume impingement on the HGA. For desaturation about a given axis, four thrusters are fired simultaneously. Redundancy is not implemented to limit dry mass and system complexity: in case of a single thruster failure, desaturation is performed using the remaining pair, resulting in higher propellant consumption but preserving full three-axis control authority.

Owing to the thruster placement and inclination, the effective force components acting in the yz -plane, required for desaturation about the x -axis, are significantly lower than those available for desaturation about the other axes. Despite this geometric disadvantage, desaturation manoeuvres about all three axes remain feasible within the system constraints. Accounting for the maximum allowable reaction wheel torque and storable angular momentum, a desaturation manoeuvre about the x -axis can be completed in 18.5 s using continuous thrust. For desaturation about the y and z axes, continuous firing would exceed the admissible limits imposed by RWs. Consequently, these manoeuvres are executed using pulsed thrust with **duty-cycled operation**, resulting to be 6.69%

for the y -axis and 6.62% for the z -axis, defined as the fraction of time during a 1 s control cycle in which the thrusters are active. These values are fully compatible with the **minimum impulse bit** specified in the thruster datasheet, ensuring controllability without performance degradation. Under these conditions, desaturation manoeuvres about both the y and z axes have a total duration of approximately 160 s.

The simultaneous firing of four thrusters arranged symmetrically with respect to the spacecraft center ensures a zero net translational force, even in the presence of a displacement between the center of mass and the geometric center. A center-of-mass offset alters the individual lever arms and the torque contribution of each thruster, but the resulting total torque remains unchanged and no parasitic translation is generated.

Finally, based on the expected angular momentum accumulation per orbit and on the maximum allowable reaction wheel angular momentum, desaturation events are required approximately every 3.5 orbits. Under this condition, the total **propellant mass** required over the full mission duration is estimated at 69.03 kg, including a 5.5% margin. This value is driven by the conservative assumption of continuous worst-case gravity gradient torque acting over the entire operational lifetime. The estimate therefore represents a preliminary upper bound: a more accurate assessment would require high-fidelity mission-level simulations accounting for realistic attitude profiles, disturbance variability, operational duty cycles, and optimized desaturation scheduling.

8.4.3 Configuration

Figure 25 illustrates the placement of attitude sensors and actuators on the Relay bus. CSSs are shown in yellow, Star Trackers in blue, and thrusters are indicated by red markers, while Reaction Wheels and IMUs are omitted as they are mounted internally.

Two star trackers (one nominal and one cold-redundant) are mounted on the $+x$ face and inclined by 45° toward the $-z$ direction to prevent Sun intrusion during Earth communications via the X-band LGA, while the remaining two units are placed on the $-z$ face without inclination.

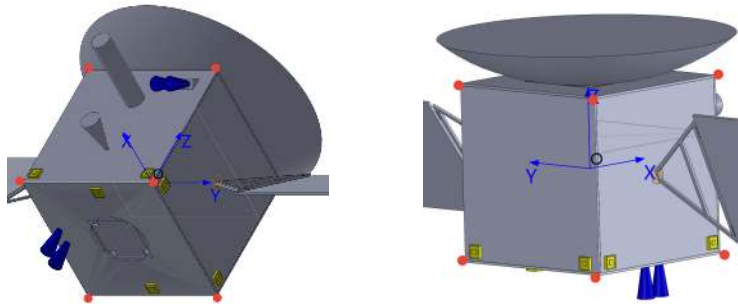


Figure 25: Relay sensors and actuators configuration: CSS in yellow, Star Trackers in blue, thrusters in red.

A specific consideration applies to Coarse Sun Sensors. The presence of the HGA reflector obstructs their Field Of View (FOV), preventing direct Sun detection along that direction. Analysis indicate that at least one CSS remains illuminated for most attitudes; if no Sun detection occurs, the controller holds the last valid attitude in nominal modes or commands a slow spin in safe mode until reacquisition.

8.5 Carrier Architecture

This section presents the design and sizing of the Carrier spacecraft ADCS. The analysis focuses on actuator dimensioning to withstand environmental disturbances, aerobraking-induced aerodynamic torques, and manoeuvre-related momentum accumulation, as well as on the sensor configuration required to support the defined operational modes.

8.5.1 Actuators Sizing

Actuators Selection Similarly to the Relay spacecraft, the Carrier is equipped with **four Reaction Wheels** arranged in a pyramidal configuration with a 35° inclination angle, including one wheel in hot redundancy. This configuration ensures full three-axis control authority and fault tolerance in the presence of a single wheel failure. Three-axis stabilization is required to guarantee accurate pre-release pointing of the Relay X-band Low Gain Antenna toward Earth and to control angular rates within the limits imposed by payload separation requirements. In addition, precise attitude control is necessary during orbit transfer manoeuvres and Mars Orbit Insertion.

Monopropellant thrusters are selected both for reaction wheel desaturation and for attitude control during aerobraking phases. During aerobraking passages, aerodynamic torques can exceed the maximum continuous torque capability of commercially available reaction wheels; therefore, thruster-based control is required to maintain the prescribed aerodynamic attitude and ensure vehicle stability. Due to configuration constraints, four nominal thrusters are mounted at the corners of the bottom face (see Figure 27). Each unit is inclined by 10° in the $\pm x$ directions, enabling the generation of control torques about all three body axes through appropriate thrust combinations.

Sizing Assumptions The actuators are required to provide full control authority throughout the mission, including phases with varying mass and inertia properties due to payload releases. Since the Carrier is designed to accommodate heterogeneous payloads within prescribed mass and envelope limits, the actuation system is sized for **operational robustness** rather than for a single nominal configuration. Three worst-case configurations are considered to bound inertia properties and disturbance torques. In all cases, solar arrays are assumed fully deployed, all components are modeled with uniformly distributed mass, and the system Center of Mass (CoM) height is minimized to maximize the lever arm between CoM and Centre of Pressure (CoP) and thus SRP and air drag torques.

- **Configuration A: Full-stack (Maximum Inertia).** Launch, interplanetary, and early-orbit configuration including the Carrier, the Relay mounted above the payload bay, and a 900-kg payload modeled as a solid block filling the bay (excluding 20 cm lateral and 50 cm vertical margins).
- **Configuration B: Relay-released (Intermediate Inertia).** Same as Configuration A but without the Relay.
- **Configuration C: Payload Worst-case (Maximum Asymmetry).** A single 500-kg payload of dimensions $1 \times 1 \times 1.7 \text{ m}^3$. To simulate a worst-case realistic imbalance, this payload is positioned with a lateral offset of 1.18 m along the interface-plane diagonal.

For each configuration, the inertia tensor and CoM position are computed analytically.

Aerobraking Analysis To identify the optimal flight profile for atmospheric passes, a trade-off analysis was performed between two distinct aerobraking strategies, illustrated in Figure 26, and both exploiting the rear surfaces of the solar arrays to increase effective drag area while shielding solar cells from direct atmospheric flux:

- **Maximum Area case:** $-x$ body axis aligned with the relative velocity vector, maximizing drag area (34.19 m^2 prior to Relay release, 31.94 m^2 after Relay release).
- **Minimum Area case:** $-z$ body axis aligned with the relative velocity vector, minimizing the CoP–CoM lever arm and improving aerodynamic stability (25.70 m^2).

The first option maximizes braking efficiency but produces large aerodynamic torques due to the long CoP–CoM offset, while the second significantly reduces disturbance torques at the expense of braking performance.

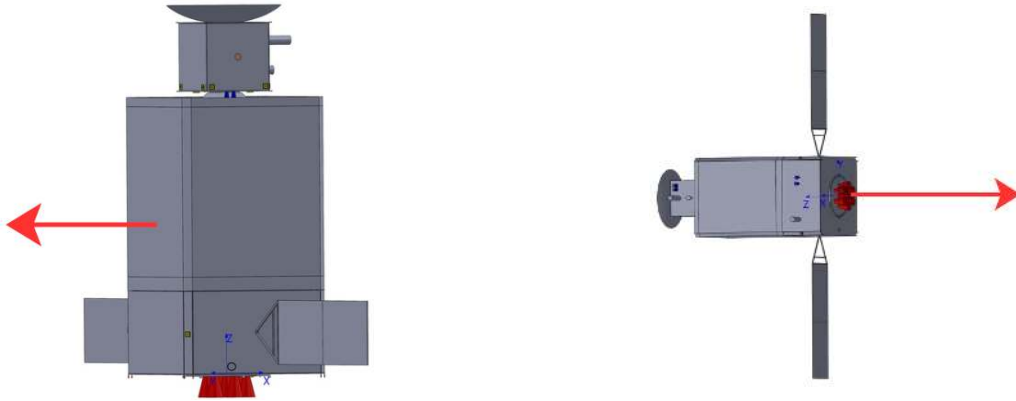


Figure 26: Aerobraking strategies. On the left, maximum area case; on the right, minimum area case.

During atmospheric passes only aerodynamic drag and gravity-gradient torques are considered. Solar Radiation Pressure is neglected, as its contribution is negligible with respect to air drag.

Aerodynamic torques are evaluated assuming atmospheric density $\rho = 5.3 \cdot 10^{-8} \text{ kg/m}^3$, drag coefficient $c_d = 2.2$, cross-sectional areas corresponding to each configuration, and worst-case pericenter velocity for each scenario. For the minimum-area strategy, ideal alignment between CoP and CoM would yield zero torque; a conservative 10° misalignment of the aerodynamic force direction is therefore assumed to account for attitude errors. Gravity Gradient torques are computed assuming a 45° inclination between the minimum-inertia axis and the local Nadir direction. The resulting peak disturbance torques are summarized in Table 14. For the Full-stack and Relay-released configurations, aerodynamic torque acts predominantly along the y -axis due to the vertical alignment of CoM and CoP. In the Payload Worst-case configuration, the CoM offset introduces multi-axis torque components; therefore, only the torque norm is reported.

Configuration	Pericenter velocity [km/s]	Gravity gradient [Nm]	Air drag (max area) [Nm]	Air drag (min area) [Nm]
Full-stack	4.823	0.016	8.088	0.306
Relay-released	4.412	0.006	2.959	0.215
Payload worst-case	4.163	0.003	14.354	9.091

Table 14: Worst-case disturbance torques during aerobraking for the three bounding spacecraft configurations. For each case, the table reports the pericenter velocity and the peak gravity-gradient and aerodynamic drag torques corresponding to both maximum-area and minimum-area strategies.

Reaction Wheels Sizing The disturbance torques during aerobraking (Table 14) exceed by several orders of magnitude the capability of any reaction wheel. Accordingly, RWs are used to counteract

environmental disturbances during exo-atmospheric flight and to damp small attitude perturbations during atmospheric passages.

Torque sizing is performed for the worst exo-atmospheric condition, occurring at the pericenter of the operational orbit (300 km), where gravity-gradient effects are maximum. A conservative 45° inclination between the minimum-inertia axis and the local Nadir direction is assumed, together with normal SRP incidence on the maximum exposed area, including deployed solar arrays. The Full-stack configuration is adopted to maximize inertia and disturbance sensitivity. Including a safety margin of 2, the **required control torque** is 0.027 Nm.

Angular momentum storage is evaluated by integrating the disturbance torques over one orbital period of the final quasi-circular 300 km orbit (in which gravity gradient is maximized), adopting the Payload worst-case configuration, representative of the last operational phase. The resulting accumulated momentum per orbit is 24.956 Nms.

Based on these requirements, the **RSI 30-280/30** reaction wheels by Collins Aerospace are selected, providing 0.28 Nm maximum torque and 30 Nms storage capacity^[71]. This selection provides a torque margin greater than a factor of 10 and a momentum storage margin of approximately 20%, ensuring robustness against modelling uncertainties and long-term disturbance accumulation.

This selection was further validated against **slew manoeuvres**. In the Full-stack configuration, the moments of inertia about the x and y axes are too large to allow reaction-wheel-based slews within acceptable angular momentum and torque limits. Consequently, prior to Relay release, reaction wheels are used for slew manoeuvres only about the z axis, while rotations about the remaining axes are performed using thrusters. After Relay separation, the reduced inertia enables reaction-wheel-based slew manoeuvres about all three axes.

Thrusters Sizing Thruster sizing was driven by the force required to counteract **aerodynamic disturbance torques** during aerobraking. Six cases were analyzed, combining the three mass configurations (Full-stack, Relay-released, Payload worst-case) with the two aerobraking strategies (maximum and minimum cross-sectional area). For both strategies, the worst-case requirement occurs in the Payload worst-case configuration, with a maximum force per thruster of 3.09 N for the maximum-area case and 1.95 N for the minimum-area case. Note that these are the peak values required at the deepest point of the atmospheric passage (100 km), while at higher altitudes the force needed is lower.

To accommodate both strategies with adequate uncertainty margins and to enable slew manoeuvres within acceptable time scales, **MR-111G** thrusters manufactured by Aerojet Rocketdyne were selected, providing a nominal thrust of 4 N and a specific impulse of 225 s^[70]. Since the nominal thrust exceeds the force required during aerobraking, attitude control is achieved through a **variable duty-cycle firing logic**. Thrusters are commanded in pulsed mode, modulating the effective average force by adjusting the pulse duration within each control cycle according to the instantaneous aerodynamic torque. Pulse widths are selected to remain above the minimum impulse bit specified in the datasheet, ensuring predictable thrust delivery and stable control performance while avoiding excessive propellant consumption.

In addition to the four nominal thrusters, four further units are implemented in **hybrid redundancy**. These remain in cold redundancy during nominal operations; however, approximately 30 minutes prior to atmospheric entry, their heaters are activated to precondition the catalytic beds. This strategy ensures immediate availability in the event of a failure of any nominal thruster during aerobraking, preserving full control authority throughout the atmospheric phase.

8.5.2 Propellant Mass Estimation

The propellant mass required by the ADCS thrusters is estimated for each mission phase using the following relations:

$$M_{prop} = \frac{h}{nLI_{sp}g_0}N, \quad M_{prop} = \frac{2t_{acc}nLF_{th}}{I_{sp}g_0} \quad (1)$$

where h is the spacecraft angular momentum to be removed, $n = 2$ is the number of thrusters per manoeuvre, $L = 1.67$ m the lever arm, $F_{th} = 4$ N the nominal thrust, $I_{sp} = 225$ s the specific impulse, $g_0 = 9.81$ m/s², N the number of manoeuvres, and t_{acc} the acceleration (and braking) time.

For detumbling, slew manoeuvres, and reaction wheel desaturations **prior to MOI**, the Full-stack configuration including 1800 kg of MOI propellant is considered and the inertia matrix recomputed. Worst-case separation rates from the Ariane 64 User Manual^[1] (1°/s about x, y and 0.5°/s about z) lead to a detumbling propellant of 0.079 kg. Ten conservative slew manoeuvres prior to MOI require 0.180 kg. Reaction wheel desaturation during the interplanetary cruise, considering worst-case constant SRP torque at maximum area and normal incidence, requires 0.010 kg.

Aerobraking propellant is computed from the **number of atmospheric passages** provided by Mission Analysis for both maximum- and minimum-area strategies. The aerodynamic torque is modeled with a triangular time history over a mean atmospheric flight time of 5 minutes per passage before Relay release and 15 minutes per passage after Relay release (according to Mission Analysis computations). The resulting propellant budgets are reported in Table 15.

	Number of passages Pre-Relay release	Number of passages Post-Relay release	Propellant [kg]
Max area case	460	440	109.929
Min area case	525	635	8.475

Table 15: Estimated number of aerobraking passages and corresponding ADCS propellant consumption for the two investigated aerobraking strategies.

Using the same number of orbits, the propellant for **post-MOI** slew manoeuvres prior to Relay release is estimated assuming two manoeuvres per orbit, yielding 13.462 kg for the maximum-area case and 15.364 kg for the minimum-area case. To reduce propellant consumption, slews are performed at a maximum angular rate of 0.1°/s: a 180° rotation therefore requires approximately 30 minutes.

Propellant for reaction wheel desaturation during Mars orbit operations is computed considering constant worst-case SRP and gravity-gradient torques, the latter conservatively taken as the mean between the values at pericenter (300 km) and first apocenter (60000 km). The number of desaturations is obtained from the ratio between the total accumulated angular momentum and the wheel storage capability, resulting in 0.500 kg.

Summing all contributions and applying a 5.5% margin, the total ADCS propellant mass is 131.264 kg for the maximum-area strategy and 31.238 kg for the minimum-area strategy.

8.5.3 Aerobraking Strategy Selection

The quantitative assessment of the ADCS propellant budget shows a significant difference between the two configurations. The higher consumption in the maximum-area case is directly linked to the larger peak aerodynamic torques generated at each atmospheric passage, which demand stronger and more frequent control actions. In contrast, the minimum-area configuration reduces instantaneous

aerodynamic loads and therefore lowers the cumulative propellant expenditure associated with attitude control during aerobraking.

Despite this substantial difference in propellant mass, the maximum-area strategy is selected as baseline due to its operational and structural advantages. By exposing the spacecraft with the $-x$ direction facing the flow, the Carrier and Relay Low-Gain Antennas located on the $+x$ faces are shielded from direct aerodynamic loading. This configuration reduces mechanical stress and mitigates thermo-structural risks on critical communication hardware.

Moreover, the increased drag in the maximum-area configuration enhances orbital energy dissipation per passage, significantly reducing the total number of aerobraking revolutions required to reach the target orbit. The resulting shorter aerobraking phase decreases cumulative exposure to atmospheric uncertainties and reduces overall mission duration, leading to improved operational robustness and lower campaign risk.

For these structural and programmatic reasons, the **maximum-area aerobraking strategy** is adopted as the baseline configuration, accepting the higher propellant expenditure in exchange for reduced mission duration and improved hardware protection.

8.5.4 Configuration

Figure 27 illustrates the placement of attitude sensors and actuators on the Carrier bus. CSSs are shown in yellow, Star Trackers in blue, and thrusters are indicated by red markers, while Reaction Wheels and IMUs are omitted as they are mounted internally.

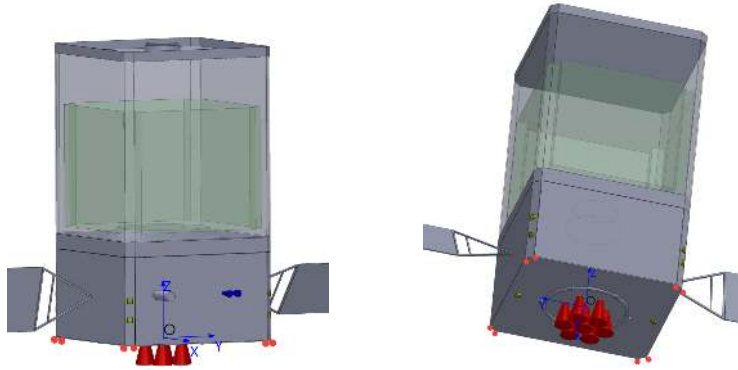


Figure 27: Carrier sensors and actuators: CSS in yellow, star trackers in blue, thrusters in red.

Similarly to the Relay, the presence of the payload bay obstruct the FOV of CSSs, preventing Sun detection along that direction. Analysis indicate that at least one CSS remains illuminated for most attitudes; if no Sun detection occurs, the controller holds the last valid attitude in nominal modes or commands a slow spin in safe mode until reacquisition.

8.6 Performance

Two nonlinear dynamic simulators were developed in Simulink to verify the performance of the selected ADC architectures. The first model assesses the Carrier attitude control performance during aerobraking, where aerodynamic disturbance torques dominate the control design. The second model evaluates the Relay attitude control performance during Earth communication phases, representing the most stringent condition in terms of pointing accuracy. A detailed description of the simulators architectures are provided in Section B.3 and Section B.4.

8.6.1 Aerobraking Performance

Robustness during the aerobraking phase was assessed through a Monte Carlo campaign (100 runs) including simultaneous uncertainties in initial conditions (attitude error up to 3° and angular velocity up to $0.03^\circ/\text{s}$) and propulsion system (thrust magnitude dispersion of $\pm 5\%$ and thruster misalignment (cant angle) errors of $\pm 1^\circ$). The results are reported in Figure 28. All cases converge within approximately 10 minutes from maneuver start. The right-hand plot focuses on the most critical altitude (approximately 100 km), corresponding to peak aerodynamic torque. Even under maximum disturbance conditions and parametric uncertainties, the controller maintains the pointing error well below 0.1° , confirming the efficiency of the architecture and the feasibility of the maneuver despite the unusual aerobraking attitude.

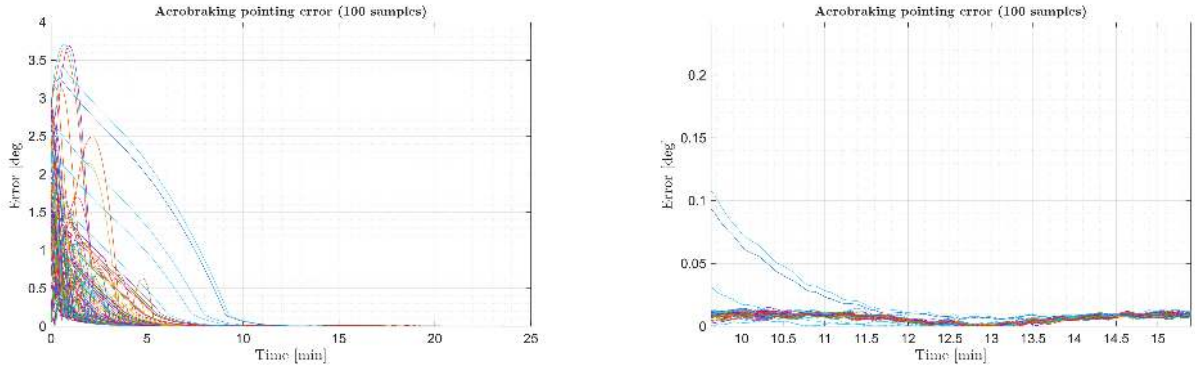


Figure 28: Monte Carlo analysis of the aerobraking phase. Left: full transient response. Right: zoom at peak aerodynamic loading (100 km altitude).

8.6.2 Relay Pointing Performance

Robustness during the telecommunication-with-Earth phase was assessed through a Monte Carlo campaign (100 runs) including simultaneous uncertainties in initial conditions (attitude error up to 180° and angular velocity up to $0.5^\circ/\text{s}$). Results are reported in Figure 29. All samples converge to the target Earth-pointing attitude within approximately 40 minutes, demonstrating global stability of the control architecture even under large initial misalignments.

The right-hand plot provides a steady-state zoom after convergence. The pointing error remains consistently below 0.005° , well within the 0.07° APE requirement defined in the pointing budget. The residual oscillations are dominated by sensor noise and reaction wheel torque quantization, indicating that performance is measurement-limited rather than control-limited.

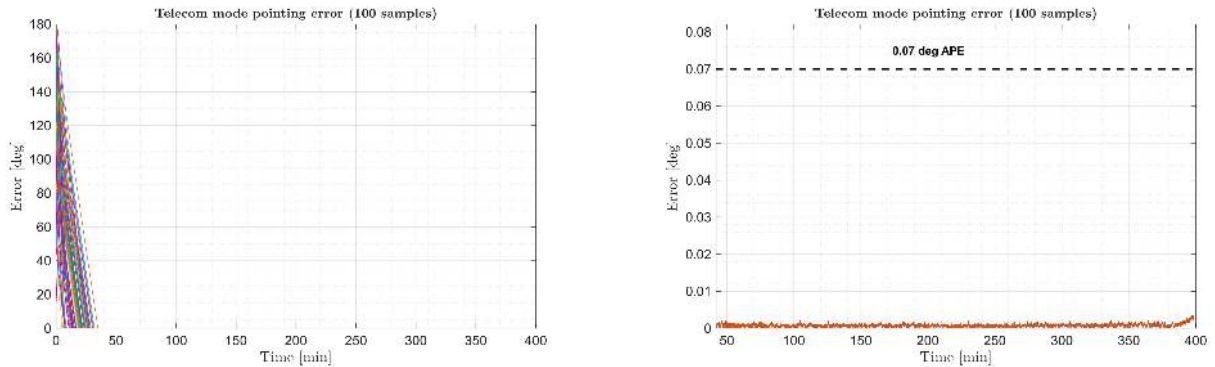


Figure 29: Monte Carlo analysis of the telecommunication phase. Left: acquisition from large initial errors. Right: steady-state pointing accuracy after convergence.

9 Propulsion Subsystem

In this Chapter, the propulsion systems (PS) of the Carrier and the Relay will be analyzed in terms of propellant mass and volume, as well as the chosen architecture (thrusters, feeding systems and tanks).

9.1 Carrier Propulsion Subsystem

9.1.1 Criteria & Trade-off

The design input is the ΔV budget presented in Section 7.8, also considering the constraint that, according to Mission Analysis, the ΔV -MOI must be performed as an impulsive maneuver, thus always relying on chemical propulsion. The selection of the Carrier PS architecture began with the evaluation of several potential candidates, including Dual-mode with N_2H_4 /MON-3, Bipropellant with MMH/MON-3, and a Hybrid configuration combining N_2H_4 /MON-3 with resistojets. To identify the final PS architecture among these three alternatives, a trade-off analysis was conducted based on the following criteria:

1. **Total propellant mass** $M_{p,tot}$ (both fuel and oxidizer), since a lower mass would also reduce the corresponding inert masses of tanks, feeding lines and the pressurization system;
2. **Maximum power** required among the maneuvers;
3. **Maximum value of the product between power and time** (Power $\cdot$ Time) among the maneuvers, to capture the effect of maneuver duration and not only peak power; this criterion, together with the second one, characterizes the constraints that the PS imposes on EPS;
4. **Operational flexibility**, highlighting a propulsion system capable of adapting to both large-scale and highly precise maneuvers, a particularly valuable feature for the MANTIS mission profile;
5. **Reliability**, understood as proven heritage, i.e., whether the corresponding propulsion architecture has already been employed in at least one interplanetary mission to Mars.

The weights for each criterion are computed using the *AHP* method^[72]: 33.47% $M_{p,tot}$, 25.37% (Power $\cdot$ Time), 19.23% Flexibility, 14.57 % Power and 7.36% Reliability. The final ranking places the **Dual-mode** alternative in first position (Score: 0.665), the Bipropellant alternative in second position (Score: 0.624) and the Hybrid alternative in last position (Score: 0.335).

9.1.2 Modeling & Assumptions

The Carrier propulsion system must be capable of fulfilling any baseline selected by Mission Analysis. Consequently, the sizing was performed by identifying the most demanding baseline in terms of required propellant mass, while maintaining the same initial assumptions:

- dry mass of the Carrier equal to $\sim 1425 \text{ kg}$ ¹ and of the Relay equal to $\sim 385 \text{ kg}$;
- for all baselines, the dry mass evolution was evaluated assuming the release of a single 900 kg payload at the last release opportunity (most critical condition in terms of propellant demand);
- the duration required for each maneuver is verified to remain below the maximum continuous burn time and above the minimum impulse bit time (valve opening time) of the thrusters considered;
- the accelerations are computed using the mass immediately before performing each Δv (no time-varying mass evolution is considered).

The propellant mass was calculated iteratively using the *Tsiolkovsky* equation, starting from the last maneuver and working backwards to update the dry mass with the newly computed propellant portion,

¹It should be noted that the sizing of the propulsion system is an iterative process, as it takes as input the dry mass of the system, which in turn includes the dry mass of the propulsion system itself. Therefore, multiple design iterations are required to achieve convergence.

and, eventually, adding the mass of the payload and the relay to simulate their release.

To properly size the system, margins shall be added to the propellant mass: 3% to account for the ullage, 2% for propellant residuals that could remain in the tank (MAR-MAS-080^[52]) and 0.5% for the loading uncertainties. Subsequently, the corresponding volumes of N_2H_4 and MON-3 are computed using $\rho_{N_2H_4} = 1010 \text{ kg/m}^3$ and $\rho_{MON-3} = 1435 \text{ kg/m}^3$ and, to take into account the unusable volume, a 10% margin must be added (MAR-CP-010^[52]).

Baseline 3 is the most demanding in terms of propellant mass, with a total of 2496.84 kg with 5.5% margin ^[52], including the hydrazine mass required by the ADCS. The propellant consumed for each individual maneuver, as well as the maneuver time and acceleration, are reported in Table 16. In particular, the N_2H_4 mass is 1472.30 kg, while the MON-3 mass is 1024.54 kg. Looking at the distribution of propellant mass throughout the mission, 66.63 kg is used for the TCMs, 1599.12 kg for the MOI, and 576.75 kg for all maneuvers following the MOI. It can be observed that nearly 65% of the onboard propellant is consumed solely for the MOI, a value perfectly consistent with the literature (e.g. MRO mission).

ΔV [m/s]	Prop mass [kg]	Thrusters	Maneuver time [min]	Acceleration [m/s ²]
31.5	50.06	3 x Biprop	2.255	0.233
10.5	16.58	3 x Biprop	0.744	0.235
1218	1599.12	6 x Biprop	43.029	0.472
0.525	0.83	3 x Monoprop	0.128	0.069
0.525	0.829	3 x Monoprop	0.128	0.069
0.525	0.829	3 x Monoprop	0.128	0.069
0.525	0.829	3 x Monoprop	0.128	0.069
0.525	0.829	3 x Monoprop	0.127	0.069
0.525	0.829	3 x Monoprop	0.127	0.069
26.775	42.00	3 x Monoprop	6.422	0.069
26.775	35.04	3 x Monoprop	5.358	0.083
1.05	1.365	3 x Monoprop	0.210	0.083
30.45	39.31	3 x Monoprop	6.007	0.084
165.9	141.80	3 x Biprop	6.194	0.446
165.9	134.70	3 x Biprop	5.884	0.470
30.45	34.99	3 x Monoprop	5.346	0.095
30.45	34.51	3 x Monoprop	5.273	0.096
30.45	34.03	3 x Monoprop	5.200	0.098
1.05	1.165	3 x Monoprop	0.179	0.098
1.05	1.165	3 x Monoprop	0.179	0.098
1.05	1.164	3 x Monoprop	0.179	0.098
1.05	1.164	3 x Monoprop	0.179	0.098
1.05	1.163	3 x Monoprop	0.179	0.098
1.05	1.162	3 x Monoprop	0.179	0.098
1.05	1.162	3 x Monoprop	0.179	0.098
1.05	1.161	3 x Monoprop	0.179	0.098
1.05	1.161	3 x Monoprop	0.179	0.098
2.1	2.320	3 x Monoprop	0.357	0.098
2.1	2.318	3 x Monoprop	0.356	0.098
4.2	4.629	3 x Monoprop	0.712	0.098
49.875	54.301	3 x Monoprop	8.261	0.101

Table 16: Propellant mass (without 5.5% margin), Maneuver time and acceleration for Baseline 3.

Equivalent tables for Baseline 1 (total propellant mass of 2496.49 kg, considering ADCS propellant mass and 5.5%) and Baseline 2 (total propellant mass of 2454.87 kg, considering ADCS propellant mass and 5.5%) can be found in Section C.

9.1.3 Dual-mode Architecture

In this Section, the **Dual-mode architecture**, selected as the winning solution from the trade-off analysis, is examined with respect to the number of thrusters and their required placement within the configuration, as well as the definition of the number and sizing of the tanks. In particular, the selected thrusters and their performance are presented in Table 17, where [N] denotes the nominal thrusters and [R] denotes the thrusters in warm redundancy. The high TRL of the thrusters is emphasized, as they are qualified and flight-proven, thus exhibiting a high level of maturity.

Property	Main Thrusters	Medium Thrusters
Thruster	R-4D-15 HiPAT Rocket Engine <i>Aerojet Rocketdyne</i> ^[73]	MR-107T Rocket Engine Assembly <i>Aerojet Rocketdyne</i> ^[74]
Propellant	N ₂ H ₄ / MON-3 (O/F = 1)	N ₂ H ₄
Number	6 [N] + 1 [R]	3 [N] + 3 [R]
Thrust (steady state) [N]	329 - 556 selected: 400	54 - 125 selected: 80
Specific Impulse [s]	329	225
Power [W]	46 (valve) + 15 (feed)	34.8 (valve) + 4 (feed) + 13.2 (bed)
Inlet Pressure [bar]	21.4 - 15.2	35 - 7
Nozzle exit diameter [cm]	36.195	6.604
Mass [kg]	5.44	1.01
TRL	8/9	8/9

Table 17: PS Carrier's thruster selection.

The PS features a hierarchical thruster architecture, in terms of thrust level, that enables the full mission profile to be accomplished. The *main* thrusters (T1-T7 in Figure 37) are bipropellant engines, used as **primary propulsion** for TCMs, MOI, and the most demanding maneuvers following MOI. Although TCMs are small corrective maneuvers, they are performed at the beginning of the mission during the interplanetary leg, when the spacecraft mass is at its maximum (including the complete payload, the relay element, and all propellant required for subsequent maneuvers), exceeding 5359 kg. Consequently, such corrective maneuvers are carried out using the main propulsion system.

The *medium* thrusters (T8-T13 in Figure 37) are monopropellant engines, considered **secondary propulsion** and used for aerobraking and perigee lowering/raising maneuvers.

Finally, although not listed in the table, the system also includes eight *small* thrusters, also monopropellant, shared with the ADCS for attitude maneuvers and/or reaction wheel desaturation, as presented in Section 8.5.1.

9.1.4 Thrusters configuration

An important issue in the use of any spacecraft PS is the assessment and mitigation of effects caused by interactions between thruster plumes themselves and with spacecraft surfaces. Plume interference can alter nozzle back-pressure conditions, affecting thrust performance and may also cause localized over-heating. Direct impingement of a thruster plume on spacecraft surfaces can generate undesired torques, localized heating and surface contamination, primarily impacting the ADCS and TCS. Therefore, the geometric arrangement of the thrusters was defined based on a **plume impingement analysis**.

The minimum spacing between thrusters was calculated assuming a plume cone half-angle of 20° for the monopropellant thrusters and 15° for the N₂H₄/MON-3 thrusters (values consistent with studies in literature ^{[75][76]}). A cone length $L = 0.8$ m was chosen for both the *main* and *medium* thrusters, ensuring no plume impingement up to a distance L from the Carrier base.

Therefore, the minimum spacing between²:

- bipropellant–bipropellant thrusters is $2 \cdot L \cdot \tan(15^\circ) = 0.429$ m;
- monopropellant–monopropellant thrusters is $2 \cdot L \cdot \tan(20^\circ) = 0.582$ m;
- bipropellant–monopropellant thrusters is $L \cdot (\tan 20^\circ + \tan 15^\circ) = 0.506$ m.

The diameter of the *main* thruster ring is 0.86 m, whereas the diameter of the *medium* thruster ring is 1.87 m.

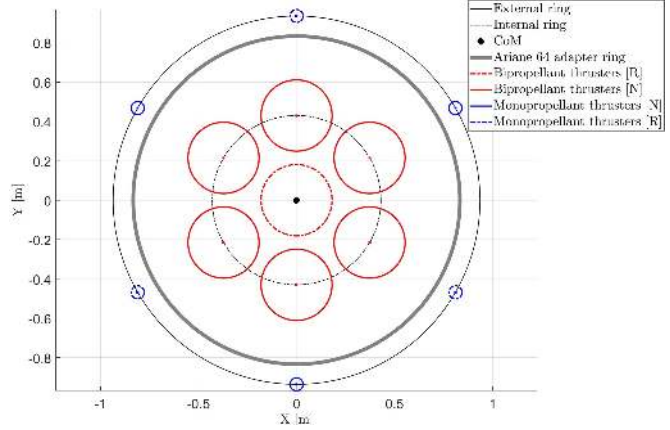


Figure 30: Thrusters configuration geometry.

These distances defined the thrusters' geometry on the Carrier base ($X - Y$ plane), as illustrated in Figure 30. Both nominal [N] and redundant [R] thrusters are depicted together with the 1.666 m diameter Ariane 64 adapter. Positioning the monopropellant thrusters outside the adapter ensures layout flexibility and avoids physical interference with umbilical connectors, push springs and internal separation sensors. Additionally, this configuration facilitates technician access during launch pad integration and reduces wiring complexity. The bipropellant thrusters are positioned symmetrically with respect to the carrier's Z -axis, along which its center of mass lies; concentrating these heavy components near the central Z -axis preserves the spacecraft's balance. Furthermore, since acoustic loads and vibrations are more intense at the periphery during launch, placing the larger engines within the adapter's envelope provides superior protection compared to a peripheral location. Figure 31 shows a 3D visualization of the plume cones of *main* and *medium* thrusters.

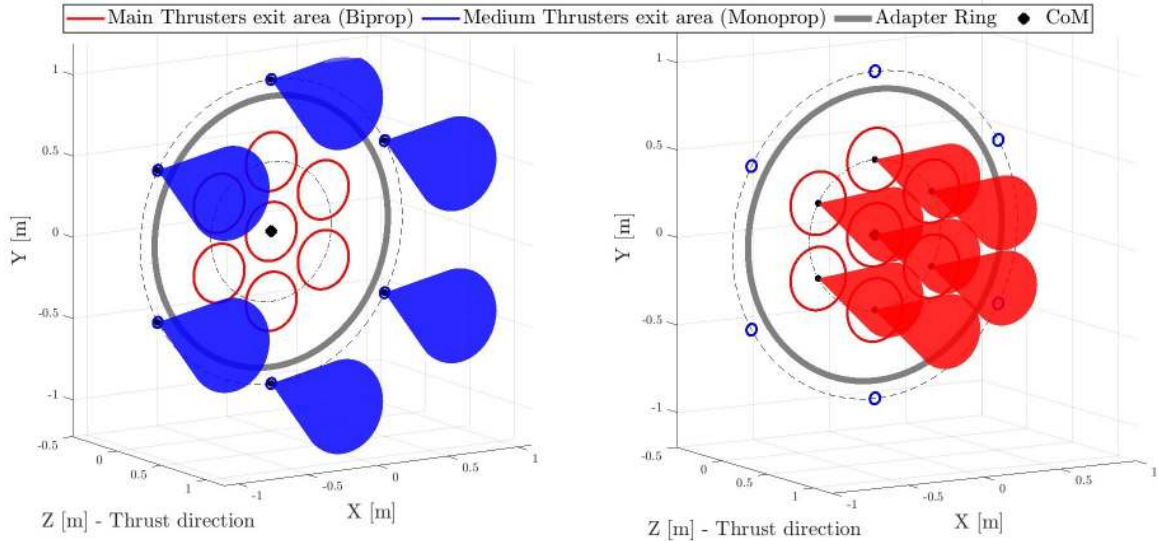


Figure 31: 3D Plume impingement visualization.

It should be emphasized that this geometry is highly robust: the monopropellant thrusters pose no risk of plume impingement on the adapter, as they are activated only well after separation from the launch vehicle; the required monopropellant–bipropellant spacing is strictly maintained, despite these thrusters never being fired simultaneously.

²Note that the minimum distances are computed with respect to the centers of the exit sections of each thruster.

9.1.5 Number Selection and Redundancy Analysis

To determine the number of *main* (T1-T7) and *medium* (T8-T13) thrusters shown in Table 17, a **single Engine failure analysis** was conducted over the entire mission envelope. An important constraint exists: for maneuvers nominally assigned to bipropellant thrusters, the most straightforward failure management involves maintaining the same operational mode. Indeed, switching to monopropellant would require more hydrazine than is currently stored on board for that specific maneuver; conversely, the same rationale applies to maneuvers nominally assigned to the monopropellant thrusters.

The results are described below, with Figure 32, Figure 34, Figure 33 illustrating the activation scheme.

- **TCMs:** nominally, three bipropellant thrusters arranged in a symmetric triangular configuration are active (total thrust of 1200 N); in the event of a failure of any one of the three thrusters, the corresponding redundant symmetric set of three thrusters is activated, enabling the maneuver to be performed with the same nominal total thrust;
- **MOI:** is nominally performed with six bipropellant thrusters (total thrust of 2400 N); depending on which thruster fails, the maneuver can be completed using five thrusters at 400 N each or four thrusters at 556 N each by activating the central thruster, while always maintaining a configuration symmetric with respect to the Z-axis and a total thrust close to the nominal value;
- **Post-MOI:** if the maneuver is performed using three bipropellant thrusters (total thrust of 1200 N), the failure analysis is the same as for the TCMs. If the maneuver is performed using three monopropellant thrusters (total thrust of 240 N), the failure of one thruster triggers the activation of the second symmetric branch of three monopropellant thrusters.

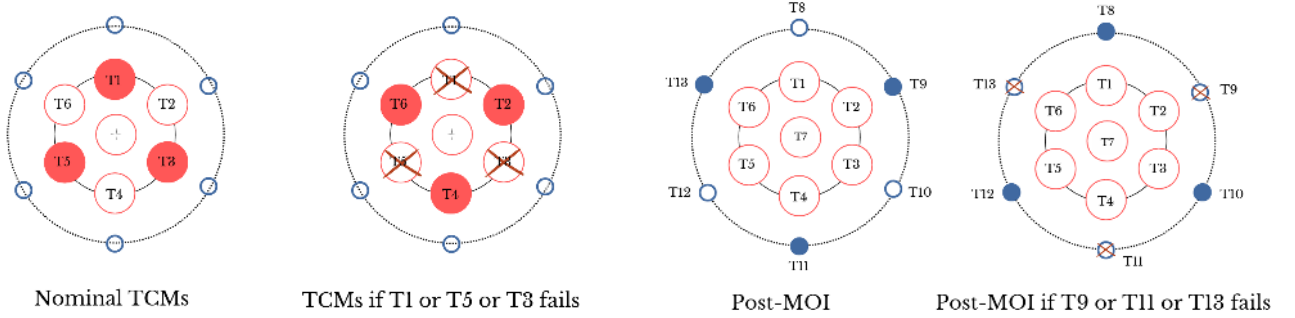


Figure 32: TCMs single failure management.

Figure 33: Post-MOI single failure management.

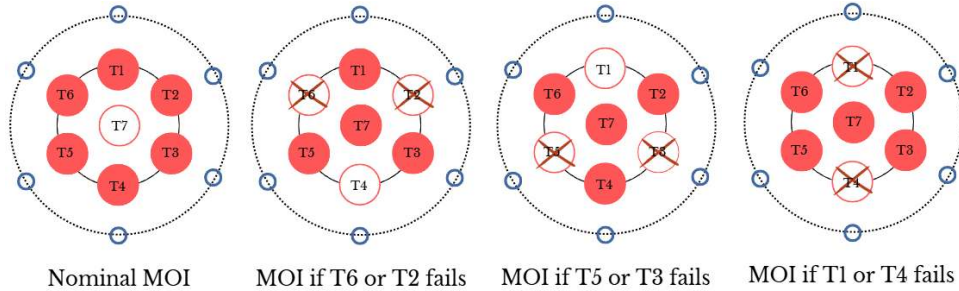


Figure 34: MOI single failure management.

It can be observed that the three monopropellant thrusters (T8, T10, T12) and the central bipropellant thruster (T7) are in *warm redundancy*, as they are kept operationally ready during maneuvers, receiving power for thermal maintenance via heaters (and, eventually the catalytic bed); in this way, they can be promptly used in the event of a failure. In particular, this choice is driven by the mission profile: T7 ensures the success of the MOI, while T8, T10, and T12 enable the proper execution of all aerobraking and payload release maneuvers, which are essential to achieve MANTIS mission goals.

9.1.6 Feeding Strategy

Tanks Layout The propellant tanks are made of Ti6Al4V ($\sigma = 950$ MPa, $\rho = 4429$ kg/m³)^[77], as it is chemically compatible with pure hydrazine and resistant to MON-3-induced corrosion. The propellant tanks are cylindrical with hemispherical end caps, while the helium tank is spherical. The tank layout exploits geometric symmetry (Figure 35).

Two MON-3 tanks are placed 180° apart on the X-axis, the four N_2H_4 tanks are arranged in symmetric pairs at 50°, 120°, 240° and 300° and the spherical Helium tank centered at (X,Y) = (0,0). The resulting geometry of each tank is reported in Table 18; a wall thickness of 1 mm was imposed whenever the resulting value was lower, ensuring manufacturability. For the propellant tanks, the *E3000 Family – 124 L / 590 L Bi-propellant PMD Tank*^[78] (Category A^[52]) is considered, as it is compatible with both the required volumes and the Maximum Expected Operating Pressure (MEOP) of 24 bar. The helium tank, instead, is considered customized (Category D^[52]), designed as a COPV with an internal Ti liner (1 mm thickness), using $\sigma_{\text{carbon}} = 1800 \times 10^6$ Pa and $\rho_{\text{carbon}} = 1800$ kg/m³, in order to save mass while ensuring a MEOP of at least 235 bar (assuming a safety factor SF equal to 1.5).

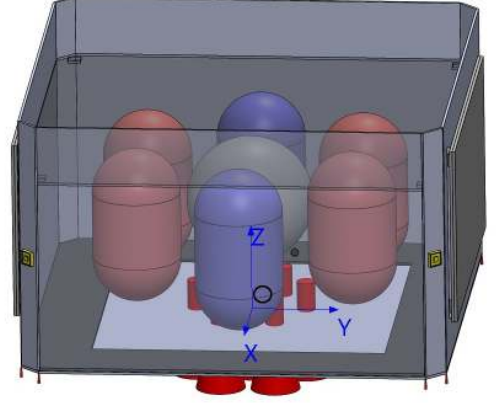


Figure 35: Tanks' layout.

N_2H_4 tank [x4]	MON-3 tank [x2]	Helium tank [x1]
$R_{\text{tank}} = 0.345$ m	$R_{\text{tank}} = 0.345$ m	$R_{\text{tank}} = 0.49$ m
$h_{\text{cyl}} = 0.511$ m	$h_{\text{cyl}} = 0.585$ m	/
$h_{\text{tot}} = 1.202$ m	$h_{\text{tot}} = 1.276$ m	$h_{\text{tot}} = 0.98$ m
Tank Volume ~ 0.365 m³	Tank Volume ~ 0.392 m³	Tank Volume ~ 0.500 m³
Tank Mass* ~ 19 kg	Tank Mass* ~ 20 kg	Tank Mass ~ 40 kg

Table 18: Tanks geometric specifications.

* the mass has been increased by 60% in order to account for heavy internal Propellant Management Devices (PMD), reinforced polar mounting interfaces and weld-zone thickness compensations.

Common Manifold System To ensure that the Carrier CoM remains aligned with its Z-axis throughout the entire mission, the PS adopts a strategy of simultaneous and balanced tank draining: propellant is always withdrawn in parallel from symmetric tank pairs with the same mass flow rate.

- *Uniform Pressure Distribution*: the central high-pressure Helium tank provides the driving force, ensuring that pressure is distributed uniformly across all propellant tanks.
- *Flow Convergence*: The outlet lines from the four hydrazine tanks converge at a single common point within the manifold, creating a shared supply line for the thrusters. An analogous configuration is adopted for the two MON-3 tanks.
- *Bipropellant O/F Ratio Management*: operating the bipropellant system at O/F ratio of 1.0 further simplifies the mass balance logic:
 - Symmetric Mass Loss: when the system consumes 1 kg of MON-3 for every 1 kg of N_2H_4 , mass is removed symmetrically from both sides of the satellite's structure simultaneously.

- Torque Mitigation: this 1:1 consumption ratio ensures that one side of the Carrier does not become significantly lighter than the other. As a result, the CoM remains stable along the Z axis, preserving the pointing accuracy required for critical maneuvers such as MOI. This minimizes disturbing torques, significantly reducing the workload on the ADCS.

Pressure regulated architecture The *medium* thrusters^[74] have a very wide operating feed pressure range (35–7 bar), which suggests that they are designed to perform well even in *blow-down* mode. However, the *main* thrusters^[73] operate over a much narrower range (21.4–15.2 bar), therefore represent the critical constraint: if the pressure drops below 15.2 bar, the main engines may experience a loss of efficiency or combustion stability. This condition is particularly critical since, regardless of the baseline adopted by the Mission Analysis, the bipropellant *main* engines are used from the very beginning of the mission for the 1st Trajectory Correction Maneuver.

As discussed previously in Section 9.1.5, the PS is designed to guarantee the successful execution of MOI, which is nominally performed by 6 *main* thrusters delivering 400 N each, even in the presence of only 4 active *main* thrusters. In particular, the MOI can be completed with either the thruster set [T1, T3, T5, T7] or [T2, T4, T6, T7], arranged in a symmetric configuration with respect to the Carrier Z -axis, with each thruster operating at 556 N. Therefore, in order to manage the dual thrust mode (400 N nominal / 556 N failure) and to ensure a high level of system redundancy, a **two-branch parallel pressurization architecture** was selected:

- **Primary Nominal Branch A:** (in *green* in Figure 37) regulated to a nominal pressure P_{nom} of 17.1 bar ($+\Delta P$ losses), allowing the *main thrusters* to operate in steady-state conditions at 400 N.
- **Secondary Failure Branch B:** (in *orange* in Figure 37) when activated, this branch provides a pressure P_{high} of 21.4 bar ($+\Delta P$ losses³), enabling the engines to reach their maximum qualified thrust of 556 N.

The failures that this design is able to handle are the following: **Pressurization Failure**, the secondary branch replaces the primary line if the pressure regulator fails in a closed or open state (in the latter case, the faulty line is isolated by means of latch and pyrotechnic valves); **Single Thruster Failure**, if a *main* engine (T2 or T3 or T5 or T6) fails to ignite (or if its performance is not stable), the high-pressure secondary branch is triggered to increase the thrust of the remaining operational engines to 556 N. It is important to note that the sizing performed to compute the required helium volume V_{He} and mass M_{He} is carried out considering the high-pressure failure condition.

Three-stage pyrotechnic ladders The implemented design employs synchronized three-stage pyrotechnic ladders at both the high-pressure helium manifold and the propellant tank inlets (Figure 36). This dual-barrier approach, based on *Normally Closed* (NC) and *Normally Open* (NO) valves, ensures complete hermetic sealing. It protects sensitive downstream components, such as regulators and check valves, from both pneumatic loads and corrosive vapor migration, while guaranteeing zero leakage during extended periods of inactivity. The sequence of valve closures and openings is as follows:

- **1st Pyrotechnic Ladder, from Launch to TCM-1**
 - Launch State: the system is launched in a hermetically sealed configuration. All NC valves are closed to protect the regulators from launch vibrations and to prevent any leakage;

³**Liquid Section (Post- Propellant Tank):** To achieve a pressure of 21.4 bar at the thruster inlet, the propellant tank pressure must be set to 22.2 bar. This value accounts for a pressure drop of 0.45 bar across the propellant filter and a cumulative loss of 0.35 bar due to latch valves and manifold friction. **Gas Section (Pre-Propellant Tank):** The pressure regulator must maintain a static set-point of 23.4 bar, compensating for a pressure drop of 0.4 bar across the quad check valves and gas lines, as well as an additional dynamic pressure droop of 0.8 bar occurring during active flow conditions.

- Activation (NC-1): on 19/04/2033, the first NC valve is fired to pressurize the tanks at P_{nom} for TCM-1;
- Isolation (NO-1): once the TCM-1 is complete, the first NO valve is fired to seal the helium tank, ensuring the safety and sealing of the system during the six-month deep-space cruise phase.
- **2nd Pyrotechnic Ladder, from TCM-1 to TCM-2**
 - Activation (NC-2): after approximately 6 months of interplanetary cruise, the second NC valve is fired to re-pressurize the system for TCM-2 and subsequently for the MOI. *Note: at this stage, Branch B becomes available as a backup in case Branch A fails;*
 - Isolation (NO-2): after MOI, the second NO valve is fired. Since the next maneuver is scheduled approximately two months later, this action prevents propellant vapors from migrating back into the gas lines and avoids helium loss.
- **3rd Pyrotechnic Ladder, from MOI to Carrier release operations Phase**
 - Activation (NC-3): two months after MOI, the third NC valve is fired to enable the maneuvers required for aerobraking and payload release.

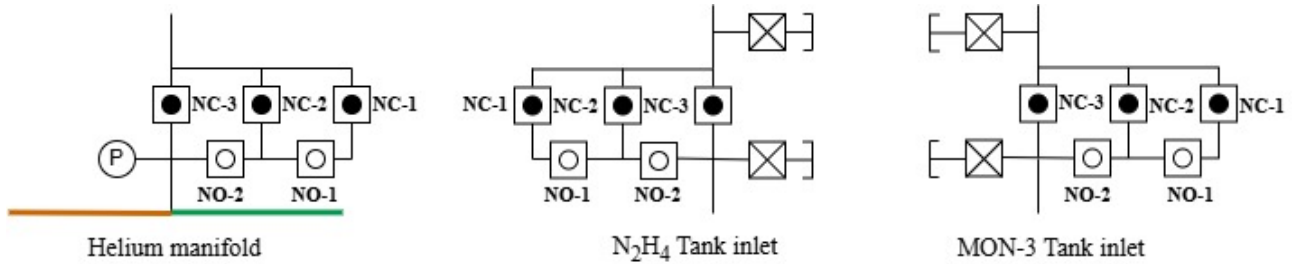


Figure 36: Three-stage pyrotechnic ladders activation scheme.

Pyrotechnic Isolation The two pyrotechnic valves on the Failure Branch serve as high-integrity hermetic seals throughout the nominal mission phases. Their primary function is to isolate the high-pressure regulation circuit from the rest of the manifold. In the event of a failure, these valves are commanded to open by the onboard computer, enabling rapid system reconfiguration.

Downstream Pressure Transducer The pressure transducer serves as the primary diagnostic tool, providing the real-time feedback necessary to verify that the active regulator is maintaining the setpoint pressure — either P_{nom} for nominal operations or P_{high} during failure recovery. A dual-transducer configuration is implemented in the connection line between the primary and secondary branches in order to eliminate a Single Point of Failure. This arrangement allows the onboard software to employ majority voting logic, distinguishing between an actual regulator failure and a faulty sensor reading. Without this redundancy, a single erroneous low-pressure measurement could prematurely trigger the irreversible firing of the orange failure branch pyrotechnics, wasting valuable system reserves.

Other components: *Quad check valves* prevent propellant vapor backflow into the gas lines through a redundant two-parallel-by-two-series configuration, ensuring protection against single-point failures such as valve leakage or failure to open.

Service valves are manually operated ground-interface ports used for propellant loading, helium pressurization, and system purging.

Calibration orifices are fixed-diameter restrictions employed to accurately balance flow rates and impose controlled pressure drops.

Latch valves are bistable electromagnetic valves that retain their last commanded position without power, providing subsystem isolation while minimizing electrical consumption.

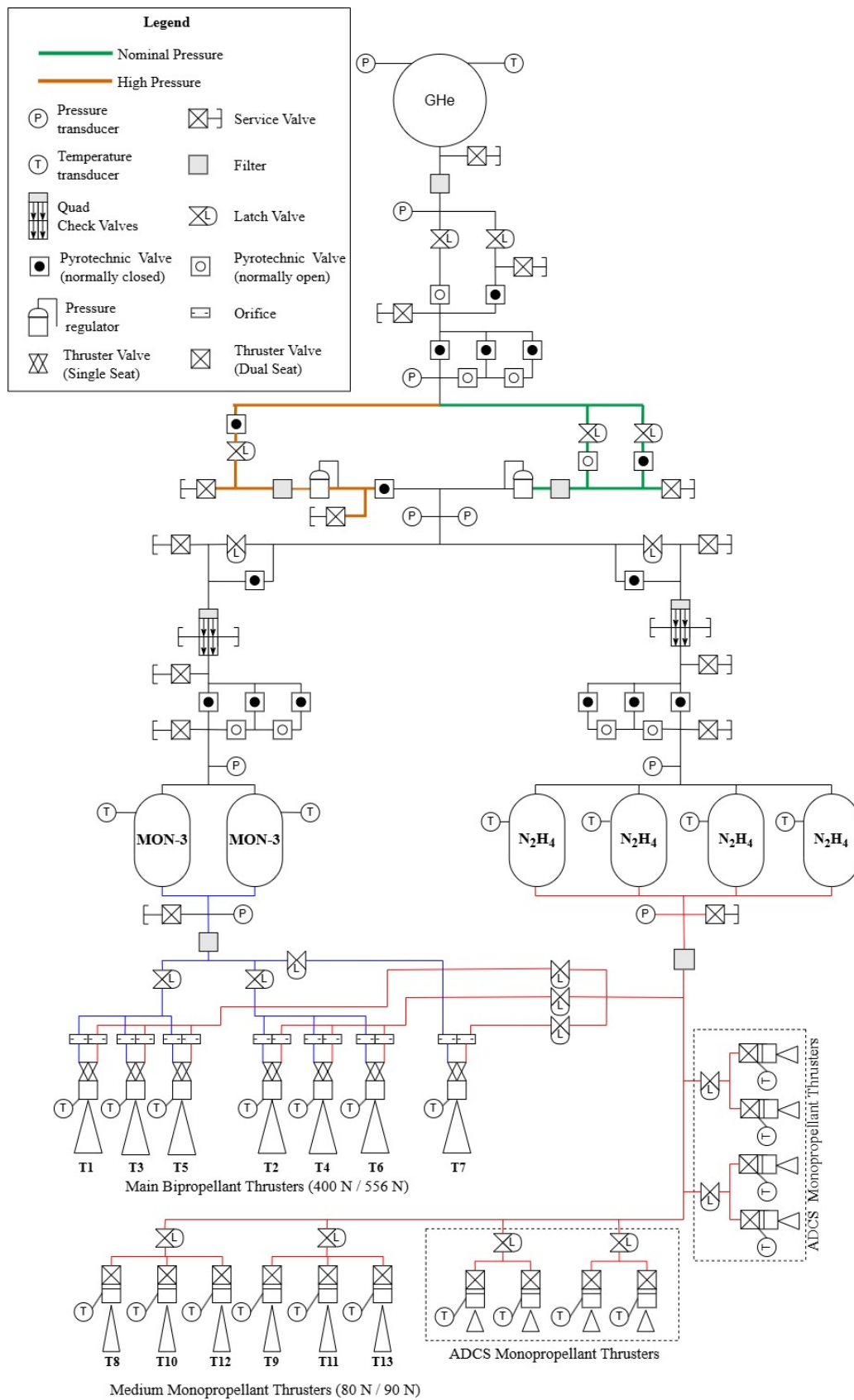


Figure 37: Carrier Propulsion Schematics.

9.1.7 Effects of Carrier PS on s/s

The propulsion subsystem affects several other subsystems through both requirements and constraints, making it one of the main drivers in the overall spacecraft design.

In terms of requirements, the most impacted subsystem is the EPS. The power demand of the PS arises primarily from the latch control valves, the catalytic bed heater and its associated valve. In particular, the power consumption exceeds 400 W during the MOI phase, representing one of the peak loads of the mission, as shown in Table 33.

Moreover, the TCS is strongly influenced by the need to maintain critical PS components within their allowable temperature ranges. Special attention must be given to propellant thermal control, especially to ensure that hydrazine remains above its freezing point of 1.4°C, thus preventing line blockage or performance degradation.

The PS also affects the ADCS: thruster firings generate torques that disturb spacecraft pointing, while the progressive consumption of propellant causes a shift in the center of mass.⁴ These aspects impact the management of impulsive torques, overall pointing accuracy and thruster placement, particularly since some thrusters are shared between maneuvering and attitude control functions.

Finally, the PS has a significant impact on the CNF/STR subsystem in terms of tank number and geometry, mounting interfaces, feed lines routing, valve accommodation, thruster layout and structural supports. These effects translate into constraints on the inertia matrix management, structural stiffness and the avoidance of plume impingement on solar arrays, optical surfaces and payloads.

9.2 Relay Propulsion Subsystem

A **Monopropellant Blowdown** system offers the best balance of low dry mass, system heritage, and power availability for the Relay system, whose results are presented in Table 19. It is important to note that the sizing was performed considering only the hydrazine mass required by ADCS, as no station-keeping ΔV was included in the ΔV budget defined by the Mission Analysis in Table 10.

Parameter	Value
Propellant (N_2H_4) Mass (+ 5.5% MAR-MAS080 ^[52])	69.03 kg
Propellant (N_2H_4) Volume (+ 10% MAR-CP-010 ^[52])	75.18 L
Initial Helium Pressure (MEOP)	24.6 bar
Final Helium Pressure	5.78 bar
Blowdown Ratio (B)	4.26
Helium Mass (+ 20% MAR-MAS-090 ^[52])	0.112 kg
Helium Volume	23.09 L
Estimated Total Volume	99.25 L
Tank Model	ArianeGroup Surface Tension Tank OST 31/0 ^[79]
Tank Volume (78 L for N_2H_4)	104 L
Tank Mass	6.85 kg
PS Total Dry Mass* (without Design Maturity Margin)	14.02 kg
*Includes: Helium tank, Helium mass, 8 MR-103J thrusters, valves + 10% cables, transducers & harness.	

Table 19: Propulsion System Specifications and Tank Data

⁴In this phase of the design, propellant sloshing has been neglected as an internal disturbance.

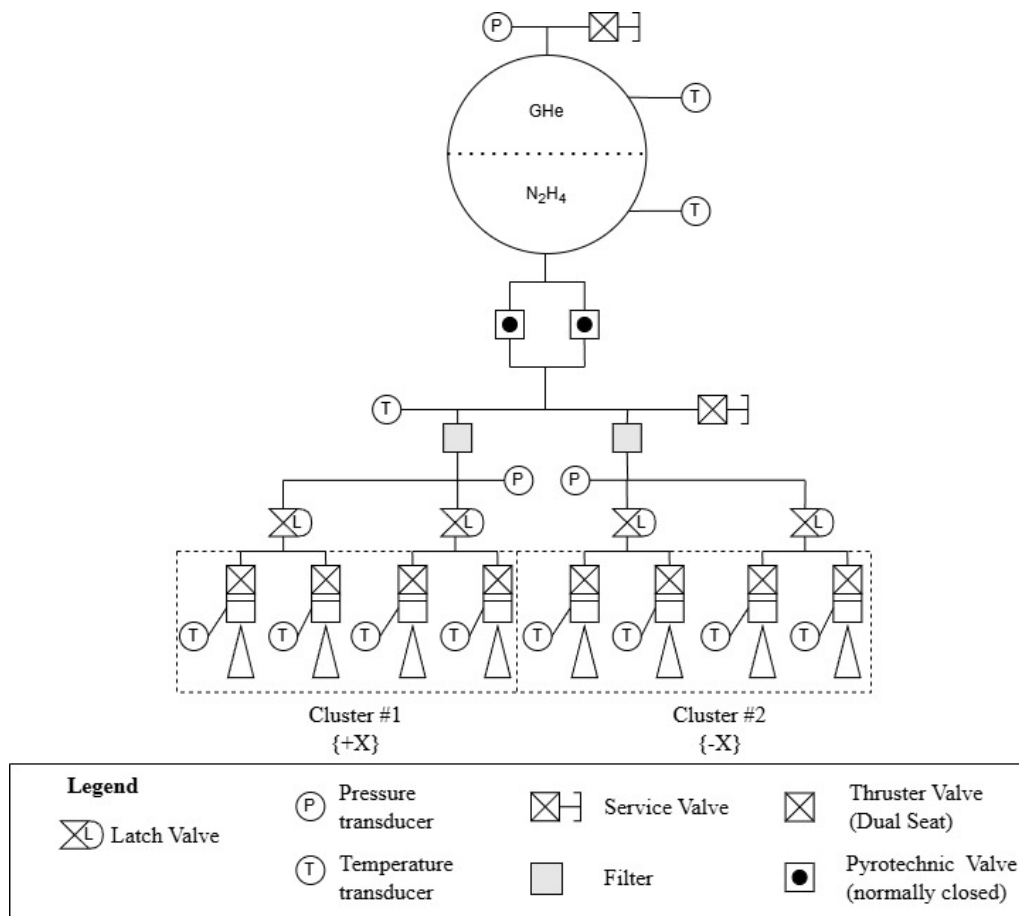


Figure 38: Relay Propulsion Schematics.

The complete schematic is shown in Figure 38, in which only the ADCS thrusters are represented. In this configuration, the pyrotechnic valves ensure that the thruster valves and catalyst beds remain unpressurized and completely free from propellant exposure throughout the entire interplanetary cruise and part of the aerobraking phase (approximately 16 months). Although latch valves provide operational isolation, they are not, by themselves, sufficient to meet the stringent zero-leakage requirements imposed by long-term storage and launch constraints.

10 Tracking, Telemetry and Telecommand

This chapter presents the design and analysis of the Tracking, Telemetry and Telecommand (TTMTC) subsystem for both Carrier and Relay spacecraft. The subsystem has been assessed in terms of communication architecture, link budget analysis, and tracking strategy, ensuring compliance with mission objectives and high level requirements.

10.1 Preliminary Identification of Data Budgets to Download

In order to define the communication requirements, a quantitative analysis has been performed to estimate the amount of data to be downloaded during the mission. This assessment includes the characterization of telemetry and science data. Firstly, the data to be downloaded is identified:

- **Carrier and Relay telemetry:** to ensure the correct operations. In the mono-element architecture, these data coincide, while in the multi-element they determine two different data budgets;
- **Payload telemetry:** the mission has to ensure the survivability of the payloads that are accommodated inside and, after the release, ensure the relay service for one to four payload;
- **Payload data science:** the relay service ensures the transmission of scientific data from the payloads through our ground link. By relying on this service, user spacecraft can significantly reduce the mass and complexity of their own TMTTC subsystem, enabling mission designers to either accommodate additional scientific instruments, improve data quality through enhanced payload performance, or simply decrease the overall spacecraft mass, leading to more efficient mission architectures. Similar data for MAVEN and ExoMars TGO were not found.

To estimate the total amount of data expected to be downlinked during the mission, reference values from the Mars Relay Network were considered. The estimated data rates are summarized in Table 20. Some assumptions were made to compute data rates of past missions, as for *Payload Data Science* corresponds to the average data volume transmitted by the Mars Relay Network^[80], assuming a 16-hour daily communication window for MRO^[81] and 10-hour windows for the other orbiters; *Payload Telemetry* is estimated as 1% of *Payload Data science*; *Carrier Telemetry* represents the total data rate of the orbiters: for MRO, between November 2006 and November 2013, approximately 200 Tb were transmitted, 99.9% of which correspond to science data^[82], whereas the 2001 Mars Odyssey orbiter transmitted 17.1 Tb over 23 years, assuming the same 99.9% proportion of science data^[83].

	MRO	Odyssey	MAVEN	ExoMars TGO
Carrier Telemetry [Mb/h]	50	2	-	-
Payload Telemetry [Mb/h]	179.72	68.44	360.44	627.59
Payload Data science [Mb/h]	1797.2	684.4	3604.4	6275.9

Table 20: Mars Relay Network telemetry data rate estimation.

Provided in the Appendix are the overall preliminary data budget considered through the link budget analysis.

10.2 Communication Strategy

The design and selection of the communication strategy are crucial to ensure reliability and continuous data exchange to fulfill the mission objective. The selection of ground segment infrastructure, including ground stations, frequency bands and communication protocols, is crucial and is conducted

to guarantee adequate coverage, compliance with international standards, and efficient utilization of communication resources.

Ground Stations (GS) selection is inherently constrained by international and national regulations governing the use of radio frequencies for space communications. Frequency allocation and coordination requirements are established by International Telecommunication Union (ITU) and national space agencies. These constraints must be taken into account from early design phases to ensure compliance, avoid interference, and guarantee the authorization of communication links throughout the lifetime. For deep space communication the available ground stations are ESA ESTRACK and NASA Deep Space Network (DSN). Since MANTIS is an European mission, the preferred GS is ESTRACK with the possibility for a combined strategy, already been used for ExoMars, TGO ^[84] and MAVEN, using both DSN and ESA ESTRACK.

Frequency Bands selection has been carried out evaluating performance, heritage, and applicability. Due to regulation, X band is chosen for its robustness and reliability, supported by extensive operational heritage in deep space. Mature technology, low propagation losses in atmosphere, and broad compatibility with existing ground stations make it the baseline for both telecommand and telemetry. In addition, to accomodate high data science downlink, the Ka band has been considered in the study, as in the last decade lots of scientific missions have adopted it due to its ability to support significantly higher rate compared to X band. However, it is susceptible to atmospheric attenuation due to a higher wavelength, requires stringent pointing requirements and introduces a higher subsystem complexity. The trade off analysis is reported in Section 10.3.1.

Lastly, for the relay - payload link, it has been considered the UHF band as primary option due to its high heritage in Martian missions where it has been widely employed for relay of landers or vehicles on Mars' surface. Its reliability, good propagation capability and compatibility with low gain antennas decreases the size of payloads that exploit our relay service, make UHF usually a robust baseline for short-range intersatellite link. As an option in Proximity Link-1 CCSDS, the S band is taken into account, then discarded due to higher power demand and less heritage, as reported in Section 10.4.2. [61] [85] [86] [87] [88]

The TMTC subsystem is designed to support a set of **Operational modes**, each associated with specific active communication links. The activation of links and data rates is driven by the mission phase and the required level of autonomy, shown in Section D.1. Each mission phase incorporates the following operational modes:

- **Initialisation mode:** active after a system reset or startup or when a new RF chain is engaged. In this mode the subsystem operates at a minimal level, accepting only critical telemetry and telecommand signals;
- **Safe mode:** survival oriented focused on essential housekeeping. Communication occurs at a low rate to ensure subsystem preservation under fault conditions;
- **Nominal mode - relay:** provides nominal rate communication with ground supporting routine operations;
- **Nominal mode - carrier:** enables nominal rate communication directly with the relay, maintaining standard operational connectivity;
- **High rate downlink - relay:** supports high datarate transmission, typically used for science or payload data, via relay links;
- **Proximity link mode:** offers robust medium rate communication without strict pointing requirement;
- **Standby mode:** maintains system readiness with minimal activity, conserving resources while awaiting the next operational command or phase.

The selection of a mode governs which communication links are active and the associated data rates,

balancing mission objectives and communication performance. During the mission, the subsystem transitions between modes as required by the evolving operational context.

Near Earth Commissioning: during the early commissioning phase in Earth proximity, the UHF antenna on the carrier is activated. This choice is driven by the limited pointing accuracy available at this stage and the need to manage lower transmit power. Only telemetry data are transmitted in this phase. The Carrier operates in Initialization, using UHF links to uplink commands and downlink extended housekeeping telemetry with a target BER of 10^{-5} , and Standby. These modes support RF channel initialization, basic health monitoring, and recovery from potential early anomalies.

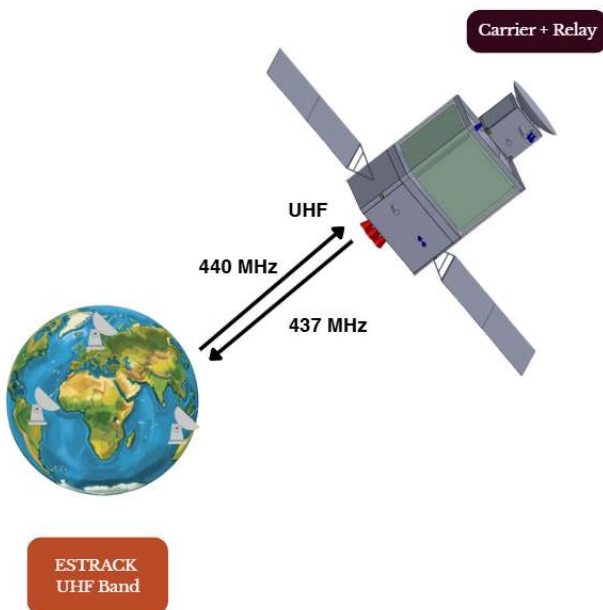


Figure 39: Communication strategy: Near Earth Commissioning.

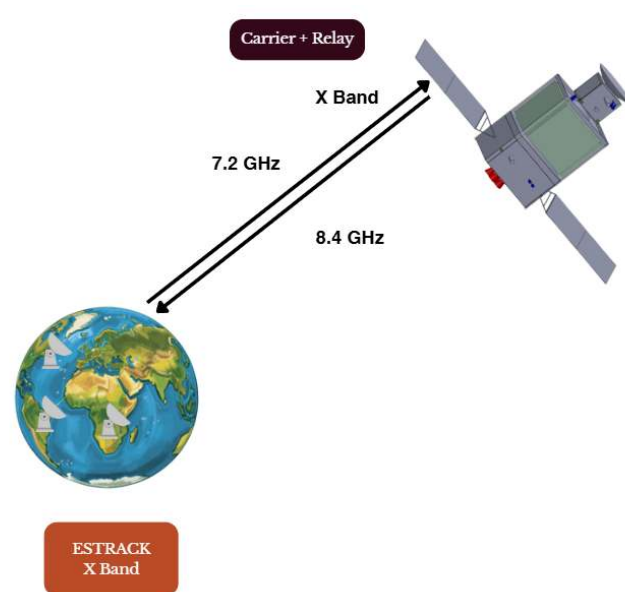


Figure 40: Communication strategy: Interplanetary and MOI.

Interplanetary Leg: throughout the cruise phase, the X-band horn antenna is activated, while the UHF antenna is in Standby mode. During this period, the relay's TMTC subsystem remains continuously operational, ensuring communication with Earth. The Relay operates primarily in Initialization, Nominal and Standby modes through the X-band horn Antenna while the parabolic antenna is still switched off. This configuration enables command reception and housekeeping telemetry downlink at low data rates, without imposing stringent pointing requirements.

Mars Orbit Insertion and Aerobraking: during Mars Orbit Insertion and the subsequent aerobraking passes, the spacecraft operates in a highly autonomous configuration dominated by Standby and Safe modes. Continuous communications are intentionally interrupted due to the loss of Earth pointing during atmospheric drag passes. Prior to each drag pass, the X-band downlink is terminated and the uplink is reduced to a minimum rate to guarantee post aerobraking reacquisition. A temporary loss of the Earth communication link occurs for approximately 10 – 15 minutes, while the subsystem remains in a standby condition. After this period, Earth is reacquired using the relay X-band horn antenna, followed by a gradual restoration of nominal telemetry. This operational concept follows established Mars aerobraking heritage and prioritizes robustness over continuous connectivity.

Relay Release: both the X-band horn antenna and UHF antennas on the relay are activated, supporting the downlink of telemetry only during the separation process. The carrier TMTC subsystem exit the standby mode of UHF antenna transmitting only telemetry of its system and allocated pay-

loads. Both spacecraft operate in a dedicated proximity link and Standby mode. Short UHF contacts are established between the Carrier and the Relay to confirm successful separation and downlink critical telemetry and event logs. Conservative data rates and limited contact windows are adopted to maximize reliability.

Carrier Payload Release: during payload deployment, the Carrier operates in a dedicated Carrier nominal mode while the Relay remains fully operative and provides relay services to the Carrier. Short-range UHF proximity links are used to transmit Carrier telemetry and deployment confirmations to the Relay, which then forwards the data to Earth via X-band link.

Relay Operations: once the carrier is disposed only the relay remains operational for a 20 year lifetime. High data rate operations rely on the X-band High-Gain Antenna, with the spacecraft operating in High-Rate Downlink modes while for nominal relay operations the signal switch to the horn antenna. Telecommands are uplinked from Earth, while housekeeping and payload science data are downlinked with a target BER of 10^{-7} . High-rate downlink modes are activated during scheduled Earth visibility windows near pericenter of relay orbit. Complementary, as shown in ADCS pointing strategy, the Relay operates in Proximity Link mode on UHF to communicate with payloads, collect telemetry and science data, and implement a store-and-forward strategy towards Earth. During blackout windows and waiting periods, the Relay transitions to Standby modes.

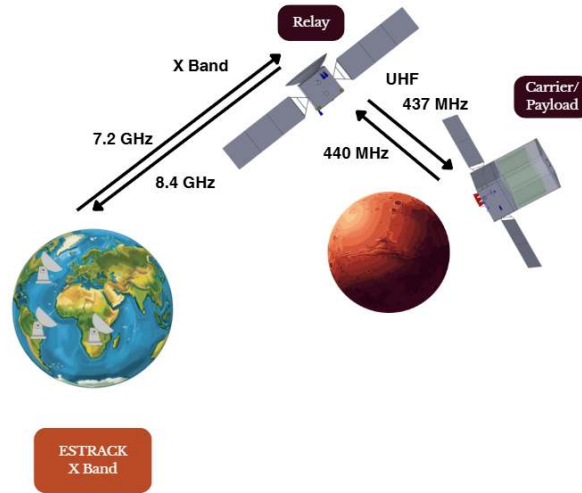


Figure 41: Communication strategy: Relay Operations.

10.3 Relay

10.3.1 Frequency Trade-Off: X-Band vs Ka-Band

A preliminary link budget assessment was carried out considering both X-band and Ka-band solutions for the Earth communication link. The analysis was performed under the conservative assumptions reported in Table 54, and the resulting preliminary budgets are summarized in Section D.2. To generalize the design space, parameterized equations were implemented in MATLAB in order to determine the minimum transmission power required to guarantee a 3 dB margin on both E_b/N_0 and SNR. The analysis was conducted as a function of antenna diameter and pointing accuracy, spanning diameters between 0.5 m and 5 m and pointing errors between 0.001° and 0.12° . Although Ka-band provides higher antenna gains and reduced required transmission power for equivalent data rates, the analysis

highlighted its strong sensitivity to pointing accuracy and atmospheric attenuation. Due to the stringent pointing requirements and increased vulnerability to atmospheric impairments, system reliability was prioritised over peak performance. Consequently, X-band was selected as the baseline frequency for the Earth communication link.

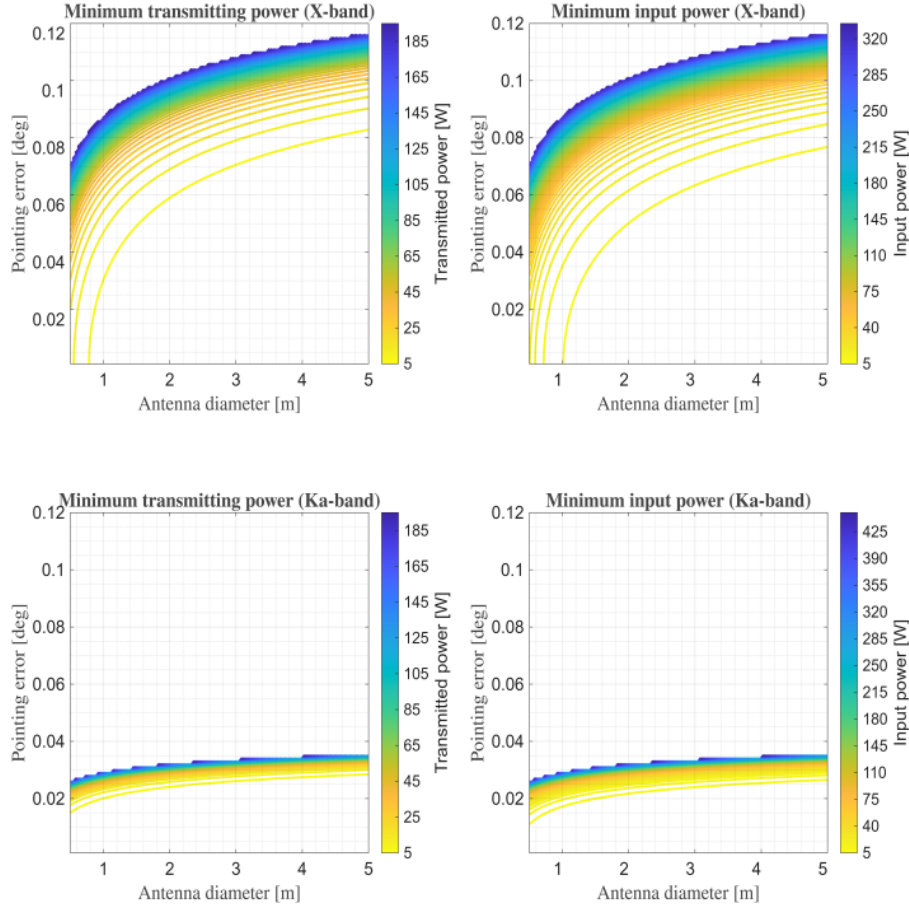


Figure 42: Minimum input and transmitting power required to overcome system margins, shown as a functions of antenna diameter and pointing error for X-band and Ka-band.

10.3.2 Parabolic Antenna Sizing for Scientific Data Downlink

Once X-band was selected as the baseline solution, the antenna diameter was sized to guarantee the required scientific data rate under worst-case operational conditions. The selected configuration (2.5 m parabolic reflector with 0.55 efficiency) ensures closure of both uplink and downlink with comfortable margins for BER levels of 10^{-5} (telemetry) and 10^{-7} (scientific data), as reported in Table 55. The adopted diameter represents a compromise between achievable data rate, transmission power, structural accommodation, and mass constraints. A larger reflector would increase link margin but introduce structural penalties and tighter pointing requirements, while a smaller antenna would require significantly higher RF power. Therefore, the chosen diameter guarantees the required data throughput while limiting spacecraft accommodation impact.

10.3.3 Ground Segment Verification with ESA ESTRACK

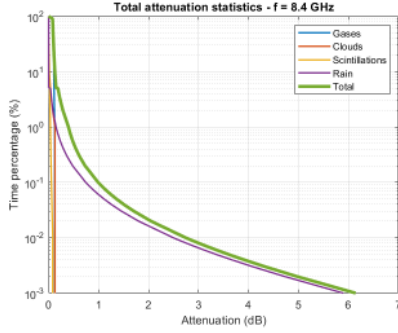
The X-band link budget was subsequently verified considering the ESA ESTRACK 35 m ground stations under conservative operational conditions.

Two values of the allowable bit-error rate (BER) have been considered in the analysis: $\text{BER} = 10^{-5}$, for standard telemetry transmission, and $\text{BER} = 10^{-7}$ for high-quality scientific data downlink, considered the critical one and reported in Table 21.

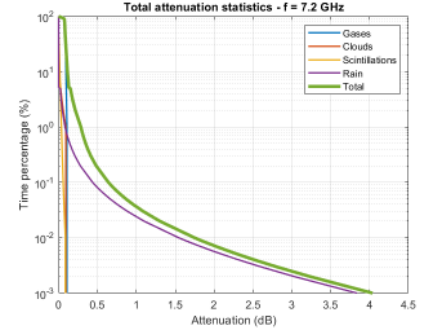
	Cebreros	Malargüe	New Norcia
Antenna efficiency	0.55	0.55	0.55
GS diameter	35 m	35 m	35 m
Latitude	40.452°	-35.776°	-31.048°
Longitude	-4.367°	-69.398°	116.191°
Altitude	794e – 3 km	1550e – 3 km	252e – 3 km
Elevation	30°	30°	30°
Polarization	0°	0°	0°
X-band Downlink			
Frequency	8.4e9 Hz	8.4e9 Hz	8.4e9 Hz
BER	1e – 7	1e – 7	1e – 7
EbN0 min	5.6 + 3 dB	5.6 + 3 dB	5.6 + 3 dB
Free space losses	- 289.22 dB	- 289.22 dB	- 289.22 dB
Pointing accuracy	0.07°	0.07°	0.07°
Pointing losses	-8.53 dB	-8.53 dB	-8.53 dB
Cables losses	-2 dB	-2 dB	-2 dB
Atmosphere losses	-2.63 dB	-1.83 dB	- 3.6 dB
X-band Uplink			
Frequency	7.2e9 Hz	7.2e9 Hz	7.2e9 Hz
BER	1e – 5	1e – 5	1e – 5
EbN0 min	4.2 + 3 dB	4.2 + 3 dB	4.2 + 3 dB
Free space losses	- 287.89 dB	- 287.89 dB	- 287.89 dB
Pointing accuracy GS	0.01°	0.01°	0.01°
Pointing losses	-0.5 dB	-0.5 dB	-0.5 dB
Cables losses	-2 dB	-2 dB	-2 dB
Atmosphere losses	-1.6 dB	-1.08 dB	-2.31 dB

Table 21: X band link budget considering ESA ESTRACK ground station and data science in downlink.

For both cases, the calculations were performed assuming the worst-case distance between the Relay and the ground station so distance $4e-11$ m. Atmospheric attenuation has been modelled by incorporating, for each ground station, the site-specific atmospheric losses due to rain, clouds, gasses and scintillation, as regulated in ITU Figure 43. Additional losses have been included as fixed cable losses $L_{cables} = -2$ dB, pointing losses, computed using a pointing accuracy of 0.07° for the spacecraft antenna and free-space path loss. A double for-loop has been implemented to iterate over combinations of transmit power and antenna diameter, firstly to verify the link closure then only physically realistic and implementable configurations have been retained where the closure has been verified for each scenario allowing the determination of the maximum achievable data rate associated with the extreme feasible couple of power - diameter.

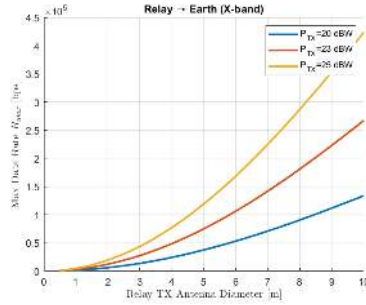


(a) Downlink of data science in New Norcia.

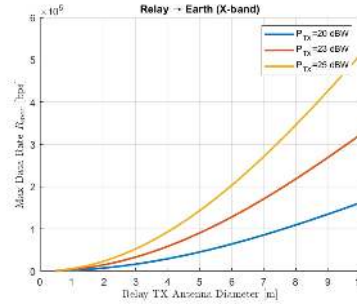


(b) Uplink of data science in New Norcia.

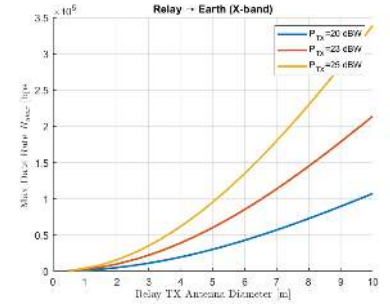
Figure 43: Atmospheric losses X band: worst case.



(a) Downlink Cebreneros.



(b) Downlink Malargüe.



(c) Downlink New Norcia.

Figure 44: Data rate dependency on the power transmitted and diameter of the antenna.

10.3.4 Communication Architecture

The final communication architecture of the Relay spacecraft is based on different antenna systems allocated according to mission phase and data rate requirements. High-rate payload release and relay operations rely on a lightweight X band parabolic antenna, sized to ensure the required scientific data downlink while minimizing structural and accommodation penalties. Since not all mission phases require high data rates, communication robustness during aerobraking, early mission phases, and contingency scenarios is ensured through a redundant X band horn antenna. This antenna provides wider beamwidth coverage at reduced data rate, lower power demand, guaranteeing link availability even in case of degraded pointing performance. Both antenna sizes have been selected to ensure compliance with performance requirements while minimizing volume, mass, and power impact on the overall configuration. The link budget has been computed for the worst-case conditions of each mission phase. Detailed results for all antenna configurations and operational scenarios are reported in Section D.3.

10.4 Carrier

After consolidating the preferred architecture choice of multi-element system, a dedicated trade-off was carried out to define where to locate the interplanetary link complexity. In the next paragraph, two architectures are considered. The trade-off focuses on the Carrier antenna type and associated RF chain, with three criteria: antenna properties, required RF transmit power, and Relay–Carrier

visibility during the time window when the Carrier starts to deploy the payloads. The sizing of the Carrier depends on its antenna and operative band.

Option A: UHF LGA

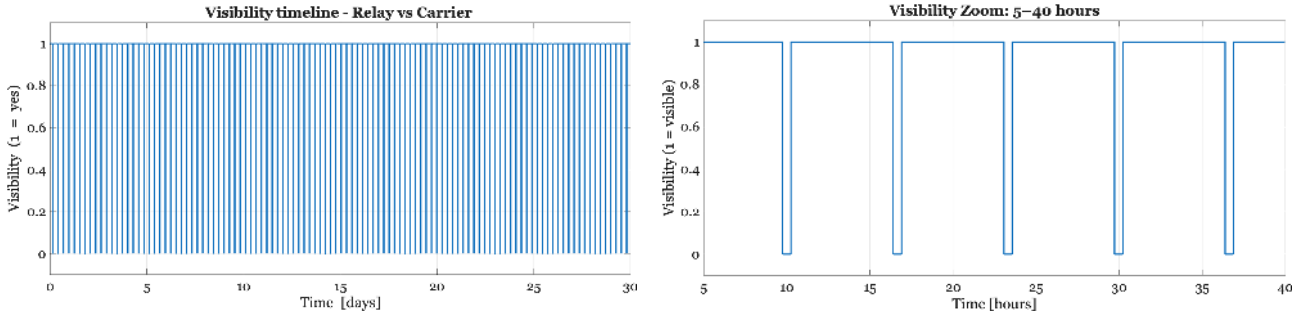
The UHF solution adopts a right hand circularly polarized helical Low Gain Antenna, omnidirectional. The diameter was obtained through a parametrization in which the transmitted power (1–15 W) was swept and the corresponding feasible antenna diameters were derived from the link budget closure. An housekeeping datarate for the UHF link was set to $R_{\text{UHF}} = 5\text{ kbps}$. This produced a feasible range of $D_{\text{UHF}} = 0.12\text{--}0.18\text{ m}$. For the most realistic reference value of $P_{\text{TX}}^{\text{UHF}} = 5\text{ W}$ corresponding to an EPS input power of $P_{\text{in}} = 14.3\text{ W}$ considering a heritage efficiency of $\mu = 0.07$. Since the LGA has no pointing requirement, the TTMTTC subsystem does not require the Carrier to be three axis stabilized during payload release, compliant with the drag driven attitude dynamics of the MOI–aerobraking sequence.

Option B: X-band HGA

For the X-band the required HGA diameter is obtained by parametrically sweeping the available transmitted power P_{TX}^{X} varied over the range 30–120 W and the minimum diameter required to close the link is computed for each feasible power level. The sweep produces a corresponding diameter range of $D_{\text{X}} = 0.3\text{--}0.94\text{ m}$, with smaller apertures enabled only at higher transmitted powers. The link is evaluated in the worst-case geometry, when the Relay is at the apocentre of its HEPO orbit while the Carrier has already descended into the lowest deployment band for payload release. The analysis assumes a system noise temperature of $T_{\text{sys}} = 20\text{ K}$, a gross data-rate of $R_{\text{gross}} = 5\text{ kbps}$, a required energy-per-bit to noise density ratio $(E_b/N_0)_{\text{req}} = 5\text{ dB}$ (extracted using MATLAB BER tool), and a design margin of 3dB. Representative losses are included as $L_{\text{cab}} = -2\text{ dB}$, and $L_{\text{point}} = -1\text{ dB}$. The antenna diameter derived that ensures link closure at the most representative transmission level, namely $P_{\text{TX}}^{\text{X}} = 40\text{ W}$, is found to be $D_{\text{X}} = 0.847\text{ m}$. This configuration requires a two-axis gimbal to maintain continuous Earth-pointing, 30–100 W of transmitted power (corresponding to an EPS input power of $P_{\text{in}} \approx 110\text{ W}$), and high stability three-axis stability to meet the 0.5 pointing requirement.

10.4.1 Relay–Carrier Visibility During Payloads Release

An important factor in assessing whether the Carrier truly requires its own Direct-To-Earth (DTE) capability is the analysis of the Carrier–Relay visibility windows. The fraction of time in which the two spacecraft maintain line of sight it's been computed, also evaluating the duration and continuity of these windows. A dedicated analytical visibility study was performed using ^[89], computing rise set times by constructing the visibility function through a parabolic-blending interpolation of uniformly sampled line of sight. The two spacecraft maintain line-of-sight for approximately 77.35% of the time. Figure 45a shows the resulting visibility timeline, which exhibits a strongly intermittent pattern characterised by short occultation gaps and longer visibility arcs. This behaviour arises from the natural phasing between the Relay's highly elliptical orbit and the Carrier's aerobraking driven deployment trajectory. As visible in Figure 45b, each orbital cycle produces a long, uninterrupted visibility interval of approximately 9–10 hours, followed by a short ~ 1 hour occultation when Mars intercepts the line of sight. This pattern repeats with near constant periodicity across each ~ 5 -day window. As the Carrier undergoes the sequence of orbit lowering maneuvers and transitions across the defined orbital bands, its relative geometry with respect to the Relay evolves periodically, producing the observed alternation of visibility and blackout windows.



(a) Visibility window between Carrier and Relay during release payload phase.

(b) Relay–Carrier visibility over a 5-day window showing ~ 10 h visibility separated by ~ 1 h occultations.

Figure 45: Relay – Carrier Visibility Analysis

As visible in Figure 45b each Relay–Carrier orbital cycle exhibits a long, uninterrupted visibility interval of approximately 9–10 hours, followed by a short occultation gap of about 1 hour when Mars intercepts the line of sight. This pattern repeats with near constant periodicity across each 5-day window.

10.4.2 Frequency Trade Off: UHF vs S-Band

A preliminary trade off analysis was conducted to assess the suitability of a UHF based intersatellite link for the Relay–Payload communication. UHF was initially considered due to its heritage in proximity operations, its relaxed pointing requirements, and its intrinsic robustness. At lower frequencies, the link budget benefits from reduced free space loss and wider beamwidth antennas, enabling quasi-omnidirectional coverage with helical antennas. The analysis was performed assuming heritage Proximity-1 BPSK/QPSK modulation with Reed Solomon convolutional coding, ensuring compatibility with existing space-qualified communication standards. For each pair (P_{TX}, D) , the instantaneous maximum achievable data rate R_{max} was evaluated at the worst-case separation distance defined by the mission geometry.

UHF band downlink	
Assumption	
Frequency	440e6 Hz
Antenna efficiency	0.55
Antenna noise temperature	500 K
G_{tx} patch antenna	3 dBi
EbN0 required	7 + 3 dB
Distance (worst case)	3000e3 m
Power transmitted range	0 - 10 dBW
$\frac{C}{\lambda}$	0.8 - 1.2
Characteristic length	0.1 - 2.5 m
Cable losses	- 2 dB
Pointing losses	- 1.5 dB
Computed	
Free space losses	-161.16 dB

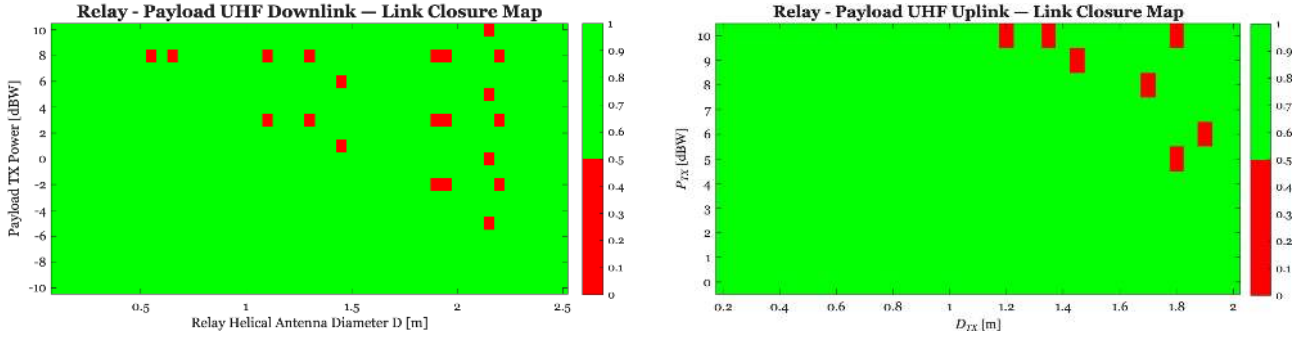
Table 22: Link budget design of UHF band downlink.

UHF band uplink	
Assumption	
Frequency	437e6 Hz
Antenna efficiency	0.55
Antenna noise temperature	500 K
G_{rx} patch antenna	3 dBi
EbN0 required	5.5 + 3 dB
Distance (worst case)	3000e3 m
Power transmitted range	0 - 10 dBW
$\frac{C}{\lambda}$	0.8 - 1.2
Characteristic length	0.1 - 2.5 m
Cable losses	- 2 dB
Pointing losses	- 1.5 dB
Computed	
Free space losses	-154.8 dB

Table 23: Link budget design of UHF band uplink.

Results yield to a downlink capacities of 10^3 – 10^4 bps and uplink capacities of 10^4 – 10^5 bps. The feasible

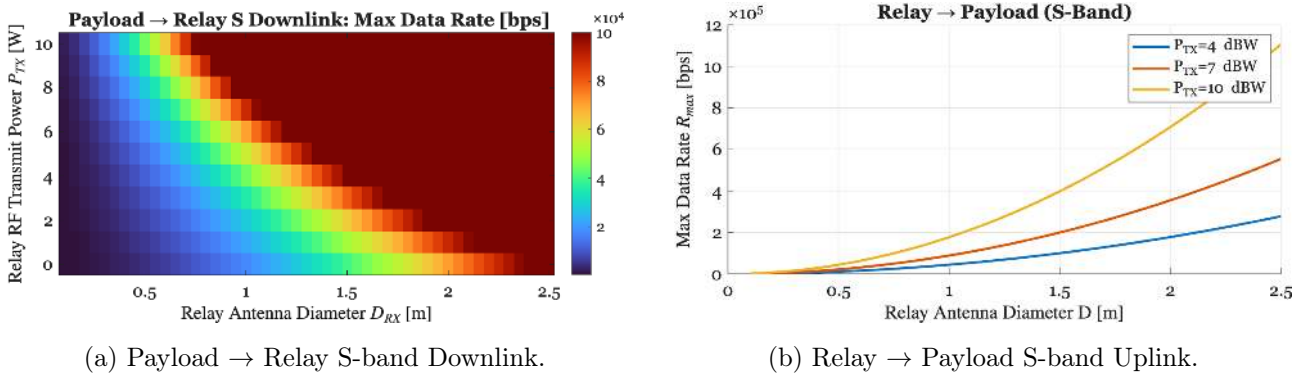
UHF envelope amply supports the mission housekeeping requirement $R_{HK} = 4kbps$.



(a) Relay-Payload UHF Downlink Link Closure Map. (b) Relay-Payload UHF Uplink Link Closure Map.

Figure 46: Parametric link-closure analysis as a function of payload TX power P_{TX} and relay helical antenna diameter D . Green regions indicate closed links ($R_{max} \geq R_{HK}$), while red points represent configurations that fail to meet the required (E_b/N_0) margin.

A second realistic and technically feasible alternative was investigated at first. Based on findings from the literature, as referenced in the cited article [90] as a result from the market analysis, the S and K band emerges as potential users in the decade of interest (2020-2040). The S-band has been analysed using the same methodology used for UHF, but modeling the transmitting and receiving elements as parabolic antennas.



(a) Payload → Relay S-band Downlink.

(b) Relay → Payload S-band Uplink.

The S-band heatmap (a) shows the maximum downlink data rate as a function of the relay antenna diameter and the payload transmit power; the boundary between the green and brown regions highlights the minimum combinations required to achieve link closure. The uplink plot (b) displays the corresponding $R_{max}(D)$ curves for three representative relay transmit powers, confirming that larger antenna diameters yield rapidly increasing throughput. Together, the two views validate S-band antenna sizing for both directions of the Relay-Payload link.

S band downlink	
Assumption	
Frequency	4e9 Hz
Antenna efficiency	0.55
Antenna noise temperature	500 K
G_{tx} patch antenna	3 dBi
EbN0 minimum	7 + 3 dB
Distance (worst case)	3000e3 m
Power transmitted range	0 - 10 dBW
C/λ	0.8 - 1.2
Characteristic diameter	0.1 - 2.5 m
Cable losses	- 2 dB
Pointing losses	-0.45 dB
Computed	
Free space losses	-174.02 dB

Table 24: Link budget of S band downlink.

S band uplink	
Assumption	
Frequency	2.4e9 Hz
Antenna efficiency	0.55
Antenna noise temperature	500 K
G_{rx} patch antenna	3 dBi
EbN0 minimum	5.5 + 3 dB
Distance (worst case)	3000e3 m
Power transmitted range	0 - 10 dBW
C/λ	0.8 - 1.2
Characteristic length	0.1 - 2.5 m
Cable losses	- 2 dB
Pointing losses	- 1.5 dB
Computed	
Free space losses	-169.6 dB

Table 25: Link budget of S band uplink.

While S-band allows significantly higher achievable data rates this advantage comes at the cost of stricter pointing accuracy requirements and increased mechanical complexity. In addition, the narrower beamwidth makes the link more sensitive to misalignment. On the other hand, UHF operates with lower freespace loss and much wider beamwidths, enabling relaxed pointing constraints and mechanically simpler antenna. This translates into a more tolerant and robust architecture, therefore UHF has been selected as the baseline solution.

10.5 Architecture

The TTMTTC subsystem architecture builds upon the previously consolidated selection of antennas and necessity of tracking and navigation strategy.

10.5.1 X-Band RF Chain

The **X-band** RF chain starts from the OBDH, which provides telemetry and telecommand data to the Small Deep Space Transponder (SDST). The SDST implements digital processing, modulation/demodulation, and IF/RF conversion, acting as the core element of the deep-space communication link. The SDST is selected for its strong flight heritage (TRL 9), reduced interface complexity compared to a distributed RF architecture, compliance with MIL-STD-1553 bus and native support of coherent Doppler and ranging observables required by the GNC subsystem. The highrate downlink relies on a parabolic High Gain Antenna (HGA) with a diameter of 2.5 m and efficiency of 0.55, providing a gain of 44.25 dBi at 8.4 GHz. The corresponding halfpower beamwidth is approximately 1° , enabling high directivity and efficient transmission toward Earth. A horn Medium Gain Antenna (MGA) is added to ensure communication robustness during cruise, aerobraking, and nominal modes, when precise pointing cannot be guaranteed or high datarate not needed. The horn antenna has a diameter of 0.2 m and efficiency of 0.52, providing a gain of 22.1 dBi at 8.4 GHz. The corresponding halfpower beamwidth is less stringent with respect to HGA, approximately 13° . In transmission, the RF signal is amplified by a Traveling Wave Tube Amplifier (TWTA) and routed through a diplexer and RF switch network to either the HGA for highrate relay operations or the MGA for wider beam coverage. In reception, the signal follows the reverse path through the Low Noise Amplifier then coherent processing inside the SDST. Concerning modulation and channel encoding, the X band RF chain adopts BPSK modulation combined with LDPC channel encoding. This selection results from a trade off between spectral efficiency, robustness, and technological maturity. BPSK enables high data rate transmission with good power efficiency and ranging, while LDPC coding ensures strong forward error

correction performance, particularly suitable for nominal and scientific highrate downlink operations [91].

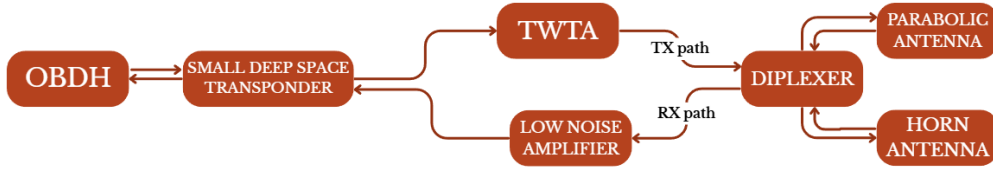


Figure 48: RF chain: X band.

10.5.2 UHF RF chain

The **UHF proximity link** on the Relay and Carrier spacecraft supports short range communications with the Carrier, during early mission phases, and with the payloads, during relay operations. The UHF RF chain is identical for both spacecraft with the difference on the size of helix antenna. It starts from the OBDH and is based on a coherent transponder performing digital processing and IF/RF conversion. The antenna solution consists of a helical Medium Gain Antenna, for relay, and helical Low Gain Antenna, for carrier. The Relay MGA has a characteristic length of 0.8 m providing a gain of approximately 10.1 dBi. The radiation pattern is quasi-omnidirectional, ensuring wide angular coverage and relaxed pointing constraints, which are essential during proximity operations.

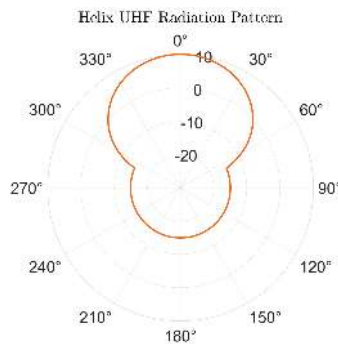


Figure 49: Radiation pattern: Relay UHF helix antenna.

The Carrier LGA has a characteristic length of 0.5 m providing a gain of approximately 3.3 dBi. This difference of length with respect to the Relay antenna stays in its function, the carrier antenna doesn't have a highrate mode and also staying onto the carrier in the launcher has stricter configuration constraints. In transmission, the signal is amplified by a solid state power amplifier and routed through an RF switch to the helical antenna. In reception, the signal is amplified by a Low Noise Amplifier and downconverted before demodulation inside the transponder. For what concerns modulation and channel encoding, the UHF RF chain adopts BPSK modulation combined with concatenated Viterbi and Reed–Solomon convolutional coding (rate 1/2).

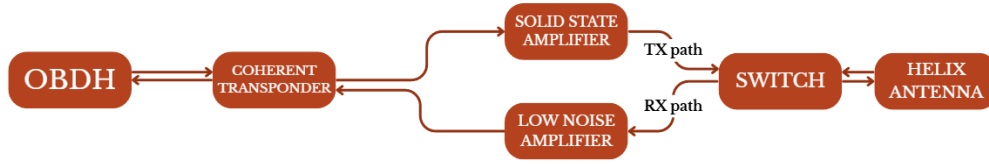


Figure 50: RF chain: UHF band.

10.6 Tracking Strategy

The relay spacecraft not only provides data forwarding capabilities, but also delivers tracking services. The tracking architecture is therefore structured into two distinct segments: Relay–Earth and Relay–Payload, each tailored to its specific operational objective.

10.6.1 Relay-Earth

The Relay–Earth link supports absolute navigation through coherent carrier-based tracking performed at the ground segment. The spacecraft is equipped with a coherent Small Deep Space Transponder (SDST), enabling Earth coherent uplink-downlink frequency conversion. The link adopts BPSK modulation with residual carrier, allowing stable carrier phase tracking at the ground station. Tracking functions are implemented on ground via carrier tracking using a Phase-Locked Loop (PLL), PN ranging and multi station interferometry. The resulting observables are Range-rate, Range and ΔDOR generated at the ground segment and support absolute navigation and orbit determination. The modulation index adopted for tracking purposes is $m = 1.167$, selected through a trade-off between E_b/N_0 margin and data power allocation. The chosen value lies within the typical operational range (0.8–1.2) and ensures a balanced distribution of power between carrier and sidebands, preserving both tracking stability and communication performance.

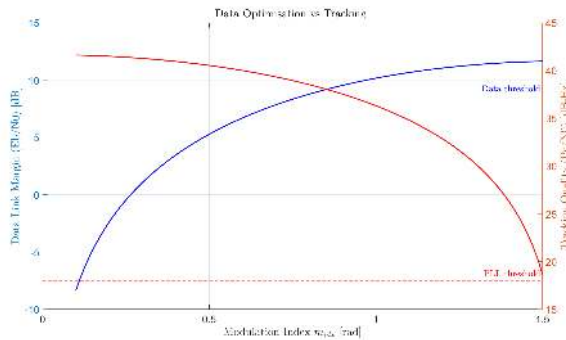


Figure 51: Modulation index behaviour.

The Phase-Locked Loop (PLL) operates with a Carrier-To-Noise density ratio P_c/N_0 of 33.55 dB-Hz, ensuring stable carrier phase tracking at the ground station. Under these conditions, the resulting tracking performance guarantees a range observable precision on the order of 1 m.

10.6.2 Relay-Payload

The Relay–Payload link supports navigation during proximity operations. The UHF link operates with BPSK modulation and suppressed carrier. Two complementary techniques are implemented

onboard the relay: Regenerative ranging for range estimation, which removes uplink noise effects and provides accurate time-of-flight measurements, and carrier tracking via a Costas loop for Doppler estimation. Since the carrier is suppressed, a BPSK Costas Loop reconstructs a coherent carrier replica, enabling frequency tracking and range-rate estimation. The observables generated are Range and Range-rate. These measurements are provided to the GNC subsystem and processed through a UKF-based relative navigation filter. Local coherence between receiver and transmitter onboard the relay is required. Although this solution is technically viable and compatible with low complexity UHF architectures, it may not represent the optimal strategy for high precision proximity. Other options will be considered in future design iterations as potential enhancements to the proximity navigation.

11 Guidance, Navigation, and Control

The Guidance, Navigation, and Control (GNC) architecture for the MANTIS mission focuses on the definition of a robust Absolute Navigation strategy, which must be designed to be compliant with the TMTC subsystem while leveraging the capabilities of the Relay satellite to perform precise Orbit Determination (OD). The navigation solution must provide sufficient accuracy to estimate the pericenter altitude within the strict aerobraking corridor defined in Section 7.6. Furthermore, it must ensure that orbital injection errors for the client payloads remain within acceptable thresholds.

11.1 MOI Uncertainty Analysis

Before detailing the navigation architecture, an uncertainty analysis of the Mars Orbit Insertion (MOI) maneuver was performed. As the MOI represents the most time-critical and ΔV -intensive maneuver of the mission, evaluating its execution errors is important to quantify the resulting dispersions of the Martian capture orbit. To this end, a 1,000-sample Monte Carlo simulation was conducted. The nominal maneuver was perturbed using standard execution uncertainties: a vector magnitude error of $\sigma_{mag} = 0.001\Delta V_{MOI}$ ($\sim 1.2m/s$) and a pointing direction error of $\sigma_{dir} = 0.1^\circ$. The outputs evaluated were the spatial dispersion of the initial apoapsis, alongside key Keplerian elements, specifically inclination, argument of periapsis (AOP), and Right Ascension of the Ascending Node (RAAN). The dispersion of the capture orbit apoapsis position is presented in Figure 52.

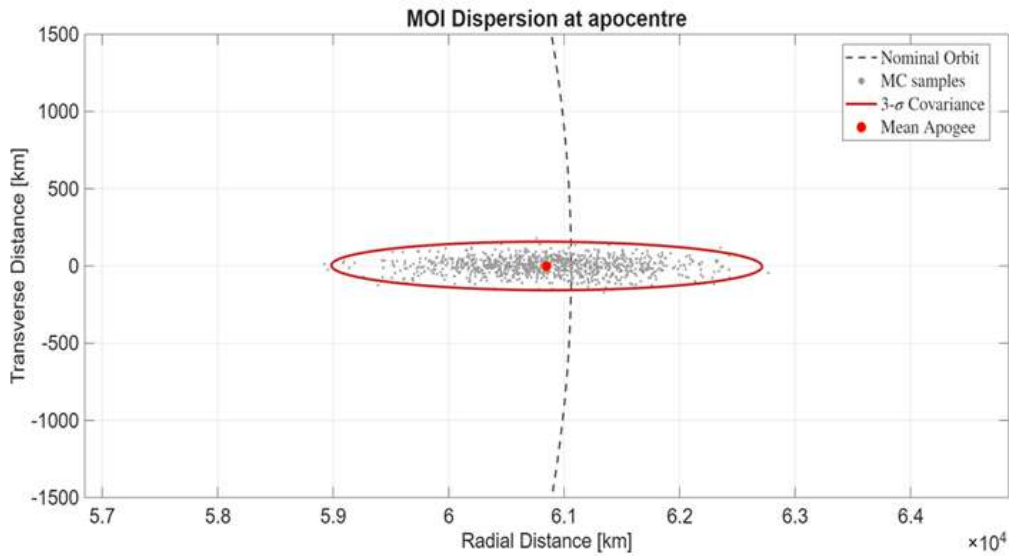


Figure 52: Monte Carlo Apoapsis Dispersion.

The analysis verifies that the majority of the apoapsis dispersion is present along the radial direction ($\approx 4,000$ km). However, this uncertainty is not a critical driver for the mission design. Its primary effect is a minor variation in the overall duration of the aerobraking campaign, extending or reducing the operational timeline by only a few days.

On the other hand, the dispersion of the key Keplerian elements is shown in Figure 53. Variations in these parameters are significantly more critical to mission success; they must closely match to the nominal target values dictated by the client payload requirements and the overall pilot mission architecture, with inclination being a particularly important constraint. The Monte Carlo simulation confirms that the dispersion of these angular parameters remains tightly bounded, yielding $3\sigma_i \approx 0.05^\circ$, $3\sigma_\omega \approx 0.15^\circ$, and $3\sigma_\Omega \approx 0.08^\circ$.

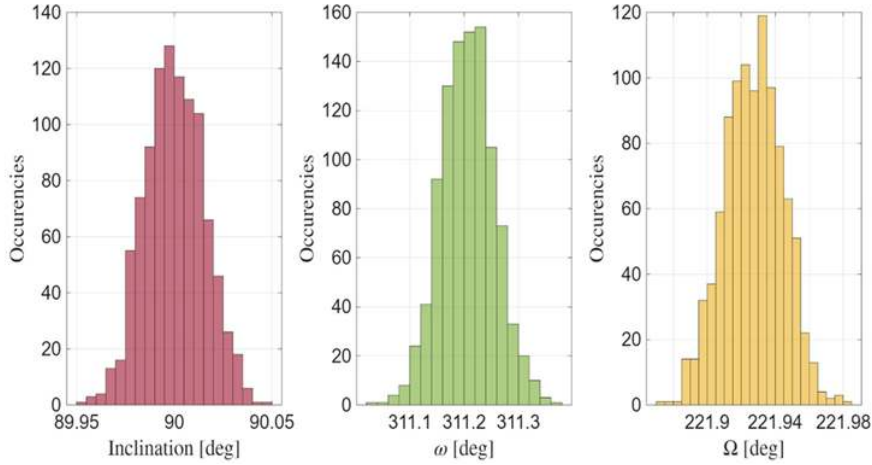


Figure 53: Inclination, AOP and RAAN Monte Carlo Dispersions.

The execution uncertainties associated with the MOI maneuver revealed no critical risks to the overall mission design. Also, the resulting apoapsis dispersions establish a preliminary order of magnitude for the initial Martian state error, serving as a baseline for the GNC subsystem. Future higher-fidelity analyses could fully integrate these initial state dispersions into the GNC architecture, allowing for the development of dedicated trajectory correction maneuvers to restore the orbit to its nominal parameters. However at the current stage, given the non-criticality of these dispersions, no correction maneuvers are considered.

11.2 Navigation Strategy

This section details the Navigation strategy for the two MANTIS spacecraft. To ensure a highly reliable, state-of-the-art approach, the OD architecture is fundamentally ground-based. The baseline tracking data types utilized for absolute OD are **2-way range**, **2-way Doppler (range-rate)** and **Delta-Differential One-way Ranging (Δ DOR)**. The acquired radiometric telemetry is downlinked to Earth, where spacecraft state estimation is performed utilizing an **Unscented Kalman Filter (UKF)**. The set of radiometric data, combined with a centralized, ground-based processing scheme, was chosen based on the extensive flight heritage of comparable Martian orbiters, such as the Mars Reconnaissance Orbiter (MRO), Mars Odyssey, and the Trace Gas Orbiter (TGO) [92–94]. Furthermore, the objective of the navigation architecture was to meet the OD requirements utilizing only the TMTC hardware, avoiding the mass and complexity addition of dedicated sensors such as cameras or LiDAR. While these are highly effective for missions with critical relative navigation phases, MANTIS does not require these capabilities. Vision-based absolute navigation was initially evaluated but then discarded once radiometric tracking alone demonstrated sufficient OD accuracy. This decision aligns with past Martian missions, where optical data was typically utilized only as an auxiliary source during critical events (e.g., MOI) rather than as primary tracking, often exploiting scientific imaging payloads already present on the spacecraft rather than dedicated navigation cameras.

As a consequence of the MANTIS multi-element configuration and the chosen TMTC architecture, the Carrier's navigation strategy differs between the pre-deployment and post-deployment phases of the Relay satellite, as detailed in the subsequent sections.

11.2.1 Pre-Relay Release Navigation

Prior to the deployment of the Relay, the MANTIS mission encounters the following navigation phases: LEOP, the interplanetary transfer, MOI, and the initial aerobraking sequence. During this first aerobraking phase, the apoapsis altitude is reduced from 60,000 km to 10,000 km, at which point the Relay separation is executed. Throughout these operations, the Carrier and Relay remain in a stacked configuration, and a **2-Way Direct-to-Earth link** with the GS is maintained utilizing the parabolic HGA mounted on the Relay. Furthermore, the integration of the X-band Small Deep Space Transponder (SDST) enables the collection of the following high-precision radiometric tracking data:

- **2-way X-band range** measurements
- **2-way X-band Doppler** measurements
- (Δ DOR)

Each communication window is constrained to a duration of approximately 20 minutes, driven by the thermal limitations of the TMTC antenna. For this preliminary analysis, the number of contacts per aerobraking orbit was deliberately restricted to reduce overheating risks and to alleviate the processing and operational loads on the TMTC and OBDH subsystems. However, subsequent analyses may explore the feasibility of extending these contact durations. The navigation approach for the Carrier-Relay stacked configuration is summarized in table 26.

Phase	Navigation	Notes
Interplanetary Arc	Ground Nav	2–3 contacts per week, ~ daily contact after TCMs
MOI	Before: Ground Nav During: No Nav After: Ground Nav	Before: 48–24 h before the MOI near continuous ground contact During: On-board execution After: Contact re-established
Aerobraking	Ground Nav	~ 6 contacts per revolution

Table 26: Carrier navigation approach pre-Relay release.

During the interplanetary transfer phase, the navigation strategy relies on a sparse tracking schedule of a few contacts per week, consistent with the heritage of previous Martian missions. In the 24 to 48 hours preceding the MOI maneuver, ground contact becomes continuous to ensure a highly accurate state estimation and trajectory reconstruction prior to the burn. The MOI maneuver itself represents an expected communications blackout; the spacecraft attitude required to align the Δv thrust vector is fundamentally incompatible with pointing the HGA toward Earth. Upon completion of the burn, the telemetry link is re-established at the earliest opportunity to perform preliminary OD and verify a nominal capture trajectory.

Results: The UKF-based OD strategy was simulated specifically for the aerobraking phase, representing the most critical navigational segment of the mission. During this phase, highly accurate estimation of the periapsis is mandatory to ensure compliance with the predefined aerobraking safety corridor. Furthermore, precise state estimation is required following each atmospheric transit to verify that the apoapsis decay is proceeding nominally. The plots of the actual orbit path and the UKF estimation are shown in Figure 54.

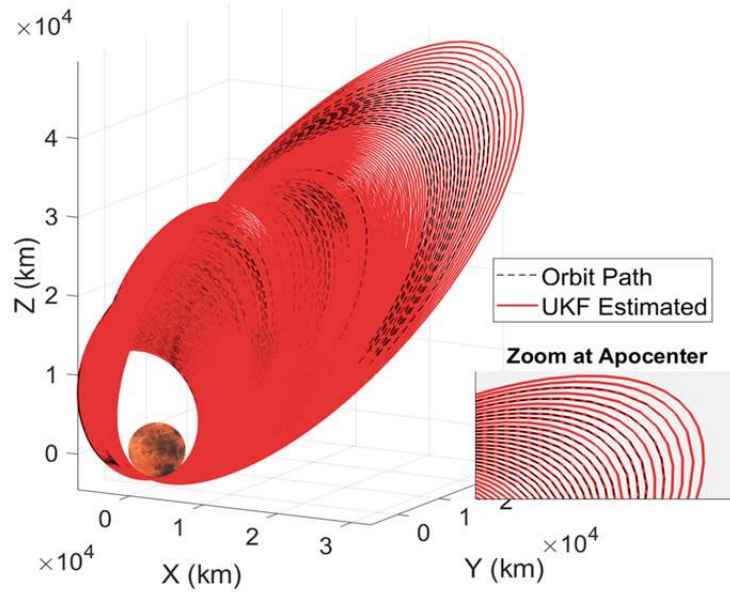


Figure 54: Carrier OD Pre-Relay Release.

The maximum absolute and relative errors at the apsis encountered in this phase are shown in Table 27.

Apsis	Absolute Error	Relative Error
Pericentre	< 5 km	< 0.41%
Apocentre	< 20 km	< 1.07%

Table 27: Carrier OD position errors pre-Relay release.

The OD position errors at pericenter consistently remain below 5 km. This performance is satisfactory enough, providing a substantial safety margin relative to the ~ 20 km width of the aerobraking corridor.

11.2.2 Relay Navigation

Following deployment into its nominal operational orbit, the Relay must maintain precise absolute OD to serve as a reliable navigational anchor for both the Carrier and the client payloads. To achieve this, the Relay's navigation relies on a single 20-minute Earth-contact window executed at periapsis. This operational constraint is dictated by the Relay's primary mission architecture: the majority of its orbital period is dedicated to establishing proximity links with the client satellites to aggregate scientific, telemetry, and tracking data. Instead, the periapsis region is exclusively reserved for Earth communications. During this brief window, the Relay downlinks the collected data and concurrently provides the 2-way X-band radiometric tracking measurements necessary for its own OD to the GS.

Phase	Navigation	Notes
Relay Operations	Ground Nav	1 contact per revolution

Table 28: Relay Navigation approach.

Results: The plots of the actual orbit path and the UKF estimation are shown in Figure 55.

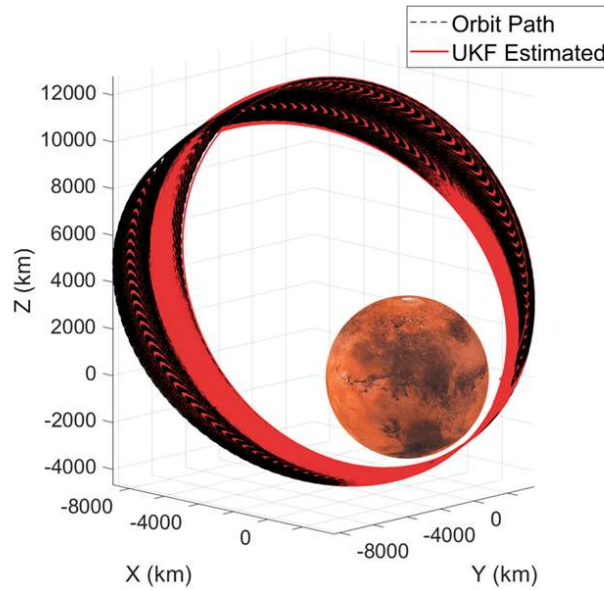


Figure 55: Relay OD.

The maximum absolute and relative errors at the apsis encountered in this phase are shown in Table 29.

Apsis	Absolute Error	Relative Error
Pericentre	< 10 km	< 0.29%
Apocentre	< 25 km	< 0.19%

Table 29: Relay OD position errors.

Relying on only a single communication window at pericenter inherently limits the OD performance of the Relay. Should future design iterations necessitate higher navigational accuracy, the implementation of additional tracking passes, coupled with a different tuning of the UKF parameters, will be investigated. These modifications will be evaluated to ensure compliance with the operational limits of the ADCS and TTMTTC subsystems.

11.2.3 Post-Relay Release

Following the deployment of the Relay, the Carrier continues its aerobraking campaign, reducing its apoapsis from 10,000 km down to 300 km, and concurrently executing the deployment of the client payloads. This period represents the most critical navigational phase of the mission. The complexity is driven not only by the stringent safety constraints of the aerobraking corridor and the uncertainties of the Martian atmosphere, but also by the strict OD accuracy required for precise insertion of the client payload. Because the Carrier lacks a Direct-to-Earth communication capability, its navigation relies entirely on the Relay. This is achieved by establishing a 2-way inter-satellite UHF proximity link, enabling the Relay to acquire relative radiometric tracking data of the Carrier. The specific data types collected are:

- **2-way UHF-band range** measurements
- **2-way UHF-band Doppler** measurements

The acquisition of these relative radiometric measurements is made possible by the dedicated, coherent UHF transponders equipped on both the Carrier and Relay spacecraft.

To achieve the requisite OD accuracy during this critical phase, the preliminary analysis assumes six 20-minute Relay-Carrier communication windows per orbit. This high-cadence tracking schedule is possible due to the Relay’s operational profile, which dedicates most of its orbital period to proximity link operations with the Carrier. Furthermore, the Carrier is only constrained to its specific aerobraking attitude during the periapsis drag pass; throughout the rest of its trajectory, the spacecraft is free to reorient and actively point toward the Relay to close the communication link. However, further higher-fidelity mission analysis iterations may optimize both the duration and number of these contact periods.

Phase	Navigation	Notes
Aerobraking & Payload release	Ground Nav	~6 contacts per revolution
Disposal	Ground Nav	

Table 30: Carrier navigation approach post-Relay release.

Results: The plots of the actual orbit path and the UKF estimation are shown in Figure 56.

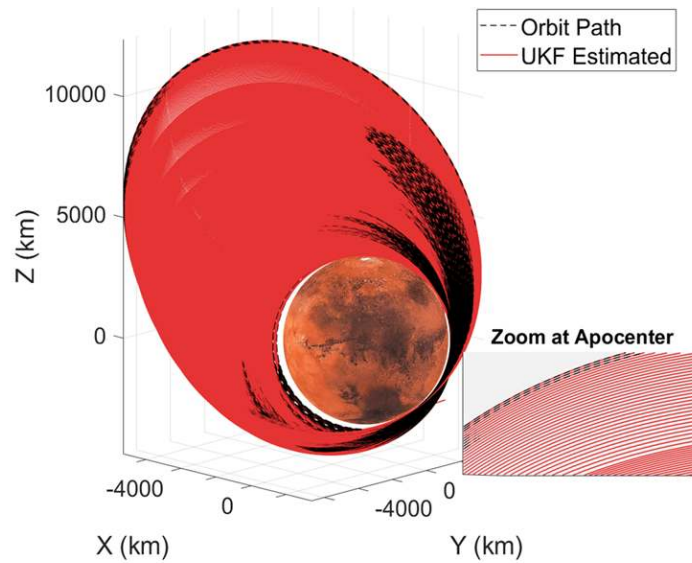


Figure 56: Carrier OD Post-Relay Release.

The maximum absolute and relative errors at the apsis encountered in this phase are shown in Table 31.

Apsis	Absolute Error	Relative Error
Pericentre	< 5 km	< 0.23%
Apocentre	< 20 km	< 0.25%

Table 31: Carrier OD position errors post-Relay release.

The inter-satellite navigation strategy demonstrated satisfactory performance. Because the Carrier’s apoapsis is specifically designated for payload deployment operations, this preliminary analysis indicates that the MANTIS architecture can guarantee a client satellite injection accuracy of better than 20 km. While this baseline performance satisfies initial mission requirements, further higher-fidelity analyses are expected to refine and potentially improve these deployment margins.

12 On Board Data Handling

This section introduces the On-Board Data Handling subsystem preliminary analysis for MANTIS developed starting from the mission data and power budget that were derived in accordance with the constraints and needs of the other spacecraft subsystems. Therefore, the work addresses the definition of OBDH scheme for the Carrier and Relay spacecraft, the selection of a suitable architecture, the allocation of functions to hardware components, and the modelling of operational states and autonomy logic. Then, a preliminary sizing of computational resources and mass memory is performed to verify the feasibility of the proposed solution under worst-case scenario.

12.1 Design Drivers and Component Selection Rationale

The selection of the OBDH architecture and hardware components has been driven by two primary design constraints: performance requirements and environmental conditions. From a performance perspective, the Solid State Recorder (SSR) shall provide at least 96 GB of storage capacity to tolerate prolonged Earth visibility blackouts as computed and reported in the following section. A high-rate data path of 1 Mbps is required for relay operations, while a minimum of 256 MB of non-volatile MRAM memory is allocated for configuration data, housekeeping storage, and FDIR tables. These requirements directly drove the adoption of a dual RAD750 architecture, a dedicated rad-tolerant FPGA for data-path offloading, and a modular rad-hard NAND Flash SSR. Environmental constraints represent the second fundamental driver. The OBDH must operate in a deep-space radiation environment corresponding to Total Ionizing Dose (TID) levels exceeding 100 krad(Si) and stringent Single Event Upset (SEU) rate requirements. Therefore, rad-hard and rad-tolerant components (RAD750 processor and RTG4 FPGA) have been selected. Error Detection and Correction (EDAC), subsystem redundancy, and hierarchical FDIR mechanisms are implemented to mitigate radiation-induced faults and guarantee long-term reliability.

12.2 Relay OBDH Architecture

The Relay spacecraft OBDH architecture is defined to support on-board autonomous operations, high-rate payload data handling, and fault management in a Deep-space environment.

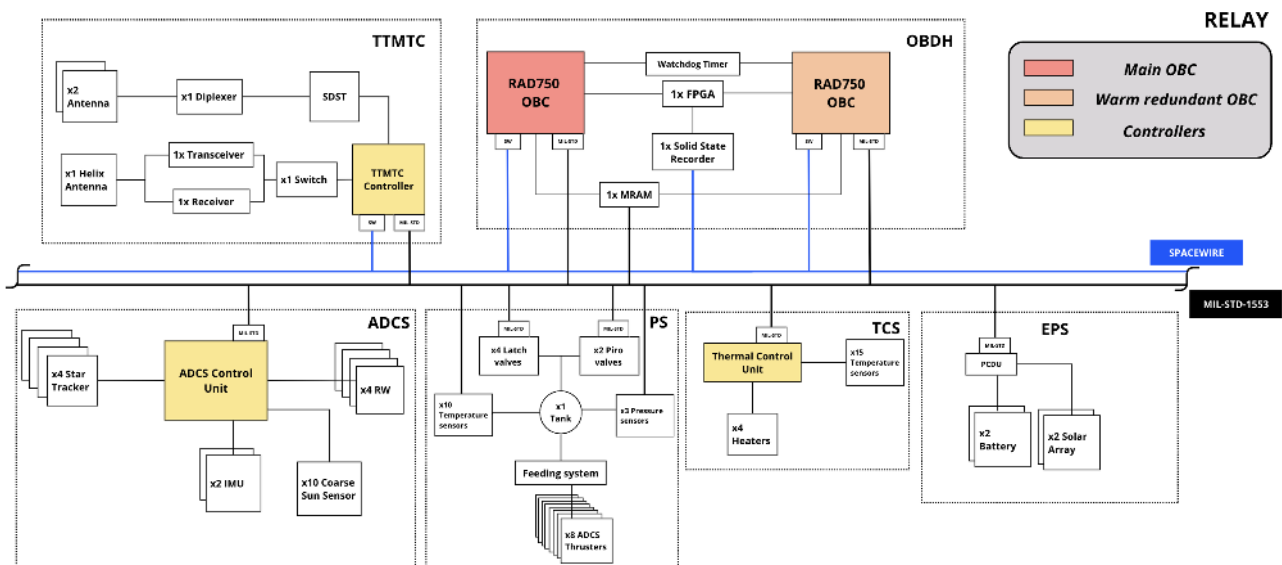


Figure 57: Relay OBDH architecture.

The proposed architecture is derived from the functional roles and operational drivers identified at system level. The architecture adopts a federated approach, in which control, monitoring, and system-level functions are logically separated from high-throughput payload data handling. This separation is reflected in using two on-board buses: a MIL-STD-1553B main bus for command, control, and housekeeping functions, and SpaceWire network dedicated to payload data routing and mass memory access.

At the core of the OBDH, two radiation-hardened RAD750 On-Board Computers are implemented in a supervised warm-redundant configuration. The main OBC acts as the primary spacecraft controller, managing system modes, telecommand execution, telemetry acquisition, FDIR functions, and bus arbitration. The redundant OBC remains powered and connected to both communication buses, operating in a monitoring role to ensure rapid takeover capability in case of anomaly. A dedicated watchdog timer and a rad-tolerant FPGA provide supervision and reconfiguration logic, enabling autonomous failover between the two processing chains. The architecture clearly separates low-rate platform control functions from high-rate payload data handling through the adoption of two distinct communication buses. The MIL-STD-1553B bus constitutes the deterministic command and control backbone of the spacecraft, interfacing the OBDH with ADCS, Propulsion Subsystem (PS), Thermal Control Subsystem (TCS), Electrical Power Subsystem (EPS), and the TTMTTC controller. Through this bus, the OBDH manages sensors, actuators, propulsion valves, thermal units, and the SDST transponder. In parallel, a SpaceWire network is implemented for high-throughput data transfer. Both OBCs are connected to the SpaceWire backbone, which links them to the Solid State Recorder (SSR) and to the FPGA used for payload data buffering, formatting, and packet manipulation. This architectural decoupling ensures that high-rate relay data from payload users does not interfere with critical spacecraft command and monitoring activities on the MIL-STD-1553B bus.

Two memory layers are implemented. A rad-hard MRAM, connected to the main system bus, stores non-volatile spacecraft data and housekeeping information. Separately, the SSR provides mass memory dedicated to high-rate payload data storage prior to downlink. This separation increases robustness and protects platform data from payload traffic saturation. Power distribution to all OBDH elements is provided by the Power Control and Distribution Unit (PCDU) within the EPS.

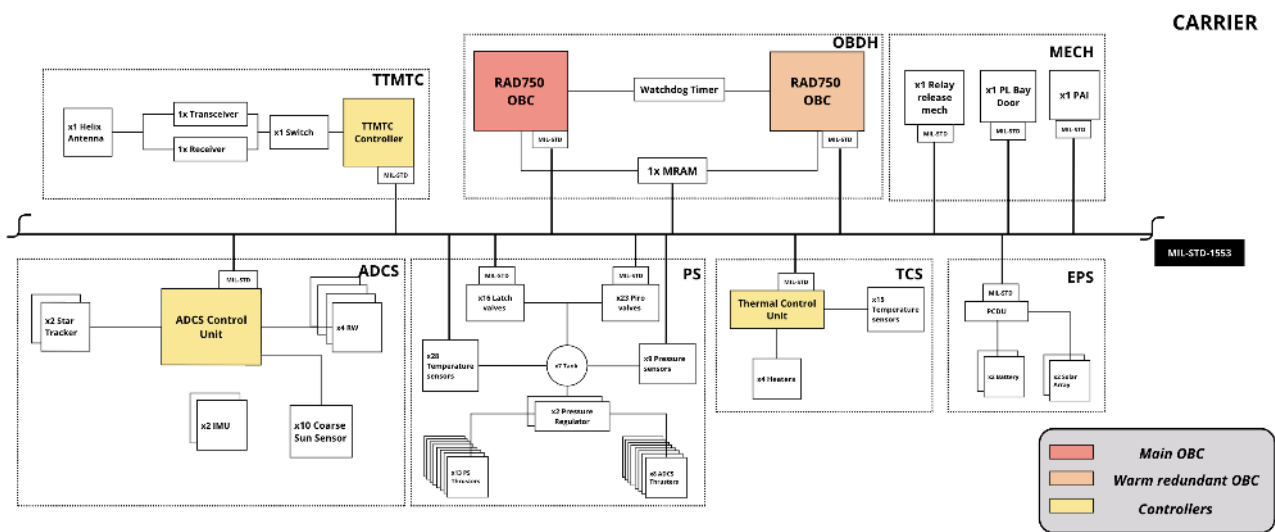


Figure 58: Carrier OBDH architecture.

The Carrier OBDH architecture, shown in fig. 58, is designed around a simplified and robust command-

and-control structure based exclusively on the MIL-STD-1553B bus.

At its core, two RAD750 On-Board Computers are implemented in a supervised warm-redundant configuration. Unlike the Relay architecture, the Carrier does not include a high-rate SpaceWire network, as no continuous payload data handling is required. Particular emphasis is placed on the mechanism control logic. The OBDH directly interfaces, via MIL-STD-1553B, with the relay release mechanism, payload bay door actuators, and Payload Adapter Interface (PAI). Dedicated control logic within the OBC software ensures sequencing, interlocks, and safety monitoring during deployment operations, which represent the most critical mission phase for the Carrier element.

12.3 Component Selection

Based on the functional allocation architecture, a preliminary selection of the main OBDH hardware components for the Relay spacecraft is performed. The selected components are flight-proven and space-qualified technologies, and are consistent with the mission lifetime and above all with the radiation environment at Mars. The selected solutions provide also sufficient performance margins to support high-rate payload data handling.

Relay	CPU – RAD750	Mass Memory – SSR	FPGA – RTG4
Technology / Architecture	Single-core rad-hard PowerPC	Rad-hard NAND Flash	Flash-based rad-tolerant FPGA
Role in OBDH	Central OBDH computer	Payload data storage	Data handling / compression
Max Frequency / Speed	200 MHz	≤ 10 Gbit/s per slice	~ 300 MHz
Capacity / Resources	400 MIPS processing capability	780 Gbit (96 GB) usable memory (4 slices)	> 500 kLUT, integrated DMA
Fault Tolerance	EDAC, watchdogs, warm redundancy	N-out-of-(N+1)	SEU-immune, EDAC
Power Consumption	< 10 W	~ 15 W	~ 5 – 10 W
Mass	~ 1 – 1.2 kg	~ 7 kg	~ 20 – 40 g
Operating Temperature	-55°C to $+125^{\circ}\text{C}$	-25°C to $+60^{\circ}\text{C}$	-55°C to $+125^{\circ}\text{C}$
Radiation Hardness	TID 1 Mrad(Si), latch-up immune	TID ≥ 100 krad(Si)	TID ≥ 100 krad(Si)

Table 32: Selected OBDH components for the Relay spacecraft.

12.3.1 FPGA adoption rationale

The inclusion of a dedicated FPGA is motivated by the need to support payload data handling functions characterized by high throughput and a deterministic timing. During the preliminary architectural analysis, a CPU-only solution based on the RAD750 was considered; however, this option would concentrate both system management and high-rate data-path functions on the same processing unit. The FPGA is therefore introduced as a dedicated data-path support element, intended to offload repetitive operations such as payload data framing, buffering, integrity checks, and optional pre-processing. This separation allows the central OBDH computer to remain focused on system-level functions, autonomy logic and to preserve good performance during peak data acquisition phases. In addition, the presence of a reconfigurable hardware element enables the accommodation of multiple payloads with different data characteristics within a unified OBDH framework. The FPGA provides architectural flexibility to adapt data handling chains to different payload interfaces, data rates, and operational timelines, without impacting the core system management functions executed by the OBC. Indeed, at this level, the FPGA is treated as a configurable and adaptable resource. The detailed definition

of internal protocols (including firmware partitioning and specific processing chains) is intentionally deferred to later design phases, once payload interfaces and data formats are fully consolidated.

12.3.2 MRAM adoption rationale

A radiation-hardened MRAM device was considered on the OBDH main bus to provide non-volatile system memory. MRAM enables fast, SRAM-like access to boot code, configuration data, and FDIR tables while guaranteeing data retention without refresh and intrinsic robustness against SEEs. Having a memory on the main bus avoids relying on the SSR as the only non-volatile memory resource. Without it, system data would be forced to traverse the MIL-STD-1553 bus, be translated onto SpaceWire, and then stored on the SSR. MRAM therefore decouples system-level memory management from the high-rate payload data path.

12.4 Relay State Diagram

The overall spacecraft operational behaviour is described by the state diagram shown in fig. 59, which separates the Relay Operation Phase and the Carrier Operation Phase, while also including pre-launch and safe-mode logic.

The sequence starts from the **OFF** state. Upon power application, the system transitions to **PRE-LAUNCH BOOT**, and, if successful (OK transition), to **LAUNCH STAND-BY**. From this state, the operational branch splits depending on the active spacecraft element.

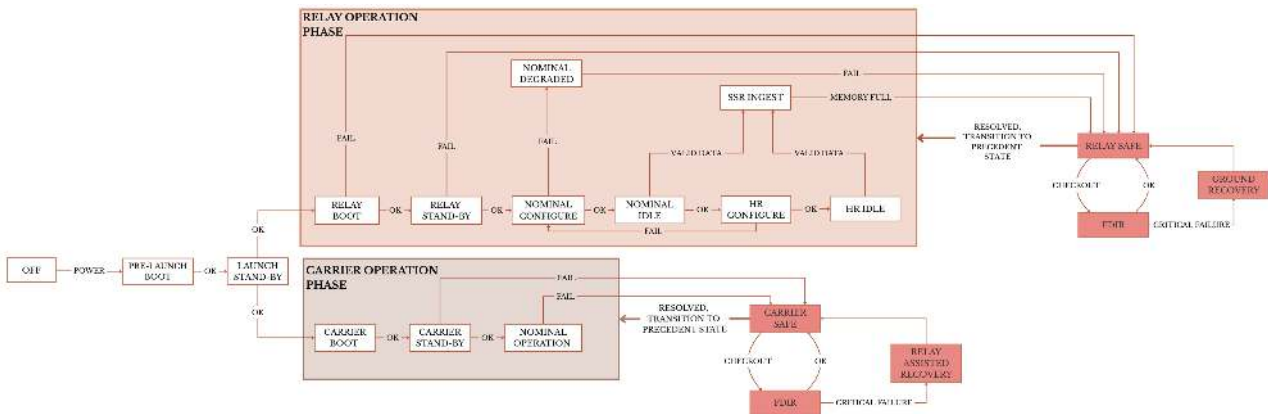


Figure 59: OBDH State Diagram.

12.5 Relay Operation

Once the relay element is activated, the OBDH enters **RELAY BOOT**. After successful initialization (OK), the system transitions to **RELAY STAND-BY**. From **RELAY STAND-BY**, an OK transition leads to **NOMINAL CONFIGURE**, followed by **NOMINAL IDLE**. These states represent the nominal operational chain. When high-rate relay operations are required, the system transitions from **NOMINAL IDLE** to **HR CONFIGURE**, and upon successful configuration (OK), to **HR IDLE**. The high-rate relay operations consist in receiving data from the served payload spacecraft via the proximity link, storing the acquired data in the Solid State Recorder (SSR), and subsequently forwarding them to the TMTC chain for transmission to ground. Indeed, in the **HR IDLE** state, whenever valid data have been collected, the system transitions to **SSR INGEST**. After successful ingestion of valid data, the state returns to **HR IDLE**. If the SSR memory becomes full, a transition toward the fault-management branch is triggered.

At multiple points in the nominal chain, a detected failure (FAIL transition) leads upward to **NOMINAL DEGRADED**. This state represents a degraded but still operational configuration. In this state, the system

remains operational but with reduced functionality, only vital housekeeping and critical mission data are stored and preserved. If failures persist or escalate, the system transitions to **RELAY SAFE**. In this state, relay services are suspended. From **RELAY SAFE**, the FDIR block performs checkout procedures. If the anomaly is resolved on-board (OK), the system transitions back to the preceding operational state. In case of a critical failure, control is handed over to **GROUND RECOVERY**, requiring ground intervention before returning to nominal conditions.

12.6 Carrier Operation

In parallel, the Carrier Operation Phase is represented by a simpler chain. From **LAUNCH STAND-BY**, the system transitions to **CARRIER BOOT**, then to **CARRIER STAND-BY**, and finally to **NOMINAL OPERATION** under OK conditions. Any failure occurring in these states leads to **CARRIER SAFE**. Similarly to the relay branch, the **CARRIER SAFE** state is supervised by FDIR checkout logic. If the anomaly is resolved, the system transitions back to the preceding state. In case of critical failure, a transition to **RELAY ASSISTED RECOVERY** is performed, indicating the possibility of recovery support from the relay segment.

12.7 Relay Solid State Recorder (SSR) sizing

The sizing of the Relay Solid State Recorder (SSR) is derived from the worst-case operational conditions identified in the mission data budget that has been refined in the previous analysis. The design constraint is given by the High-Rate Relay phase around Mars. The worst-case downlink condition corresponds to a high-rate X-band link with a net data rate of $R_{\text{net}} = 500$ kbps. The Relay spacecraft is required to tolerate prolonged loss of Earth visibility. Based on the analysis of the Relay orbit and on the identification of the visibility windows available between the Relay and Earth, considering data from multiple payload operational scenarios, the maximum continuous communication blackout duration was derived from the worst-case occultation conditions and was found to be $T_{\text{occ}} = 3$ days (259 200 s). Under these conditions, the maximum accumulated data volume per single payload is:

$$V = R_{\text{net}} \cdot T_{\text{occ}} = 129.6 \text{ Gbit} = 16.2 \text{ GB}.$$

A global margin of 100% is applied to account for uncertainties in payload data generation profiles, compression efficiency, and scheduling constraints. Considering the full mission payload set (x3), the resulting SSR sizing requirement amounts to approximately 780 Gbit (96 GB).

The SSR is operated according to a controlled store-and-forward policy, fully managed by the OBDH subsystem. Data generated during blackout intervals are classified, prioritized, and stored based on predefined priority levels and memory allocation rules. Upon restoration of Earth visibility, the OBDH automatically schedules downlink activities at the earliest available contact window, ensuring timely depletion of stored data. Priority-based downlink management guarantees that time-critical and high-value data are transmitted first, and this ensures that the SSR operates within a bounded data flow, to avoid uncontrolled long-term accumulation and maintain compliance with mission autonomy and reliability requirements over the entire Relay lifetime.

13 Electric Power Subsystem

13.1 Introduction

The Electrical Power System (EPS) of the MANTIS mission is designed to ensure reliable power generation, storage and distribution for the Carrier and Relay spacecraft throughout all mission phases, from launch to End-of-Life (EOL). The system is sized according to mission requirements, operational power requests and orbital constraints. Aligned with the mission's dual spacecraft architecture, the design features two independent electrical power systems, delivering tailored solutions for the distinct operational profiles of each module:

- **Carrier:** functions as the primary hub during the mechanically mated phases (interplanetary cruise, Mars Orbit Insertion, and aerobraking). In this configuration, the Carrier EPS powers the entire stack. This load includes all platform functions (PS, ADCS, TCS, OBDH), the Relay's X-band horn antenna, which serves as the communication asset while the vehicles are mated, and continuous keep-alive power for payloads such as PHAROS.
- **Relay:** designed for long term survivability in Martian environment, prioritizing component degradation margins and radiation hardness. Once the target orbit is reached and separation occurs, the Relay activates its own EPS. From this point onward, it relies on its own primary and secondary power source to support high power X-band downlink to Earth and UHF reception from the payloads. Conversely, the Carrier remains active for the payload delivery phase, utilizing a low power UHF link to report its status to the Relay spacecraft.

13.2 Power Analysis Methodology

A fundamental step in the sizing of the EPS is the formulation of a comprehensive **power analysis**, which maps the energy demand of each subsystem across all defined mission phases. The power consumption analysis is performed by evaluating the active equipment required for each specific operational phase. For every subsystem, the power profile is constructed by defining the maximum power request of each individual unit (derived from manufacturer data sheets) and the number of active units per phase, alongside the operational duty cycle. This duty cycle, expressed as a percentage from 0 % to 100 %, represents the fraction of maximum power utilized during a specific phase to compute both average and peak demands. A rigorous distinction between peak and average power is essential for the EPS architecture. Peak power represents the absolute maximum draw when a component operates at its highest capacity. Conversely, average power reflects the mean consumption over a complete operational cycle, accounting for both active and quiescent phases. The generation and storage devices are sized around the average power, as it reflects the typical power needs of the satellite over an operational phase.

13.2.1 Margin Philosophy

To mitigate developmental risks and account for uncertainties inherent in preliminary design, power margins are applied in accordance with ECSS standards ^[52]. A **design maturity power margin** is applied at equipment level:

- 5 % for off-the-shelf products without modifications.
- 10 % for off-the-shelf products with modifications.
- 20 % for newly designed and developed products.

Furthermore, a 30 % **system level margin**, typical for Martian missions, is superimposed, given that the MANTIS mission is currently in Phase A level of development. Adopting this conservative margin philosophy is a standard engineering practice for preliminary studies, safeguarding power availability against the inherent growth in system mass and complexity during subsequent phases.

13.3 Power Budget Breakdown

Applying the established methodology and margins, the power consumption profiles are computed for both spacecraft. Specifically, for the **Carrier** spacecraft, the Mars Orbit Insertion (MOI) phase is identified as the worst-case, exhibiting a margined average load of about 1160 W. This elevated demand is primarily dominated by the Propulsion Subsystem, which requires continuous power (~ 400 W) to sustain the bipropellant thruster operations, complemented by substantial Thermal Control loads dedicated to propellant tank heating. Figure 60 reports the subsystem power consumptions per each phase of the mission.

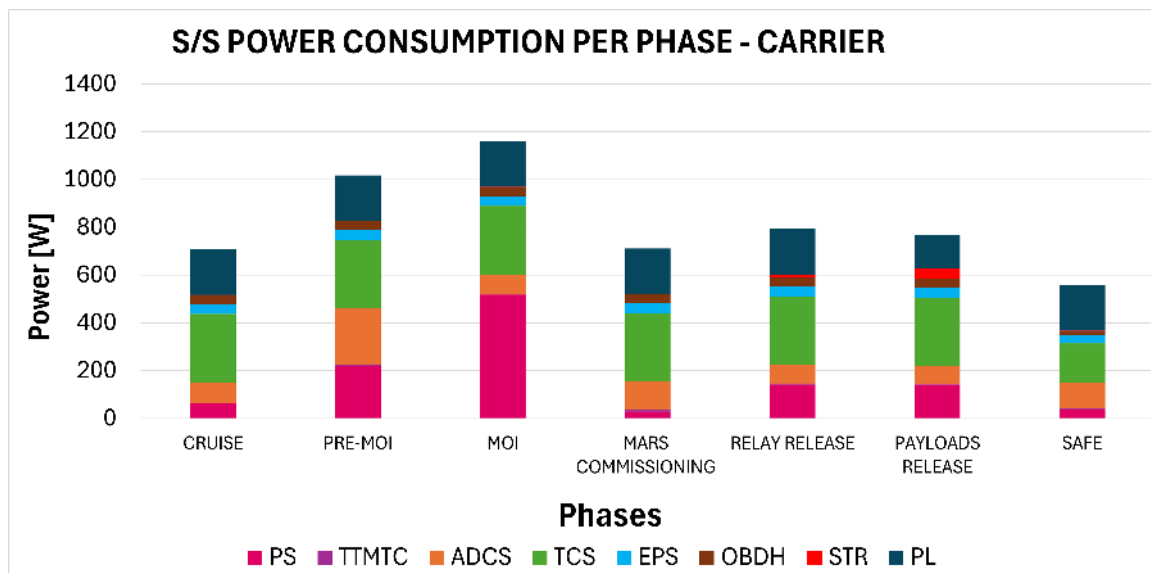


Figure 60: Subsystem Power Consumption per Phase (Carrier).

To provide a deeper insight into the dimensioning case (MOI phase), Table 33 details the specific power distribution percentage among the subsystems.

Subsystem	PS	ADCS	TCS	EPS	OBDH	STR	PL
Percentage	44.6 %	7.3 %	24.7 %	3.5 %	3.3 %	0.1 %	16.5 %

Table 33: Subsystem Power Distribution during MOI.

Conversely, the **Relay** EPS sizing is primarily driven by the High-Rate Operations phase, which constitutes the mission's highest power demand. Characterized by a margined average load of about 580 W, this phase represents the most stringent requirement for the primary power source. The consumption profile is dominated by the TTMTTC subsystem, specifically the activation of the high-power X-band parabolic antenna. This unit requires significant power for high-bandwidth data transmission to Earth while operating concurrently with the ADCS, which must maintain high precision pointing

throughout the downlink window. The Relay power demands divided by subsystem are shown in Figure 61.

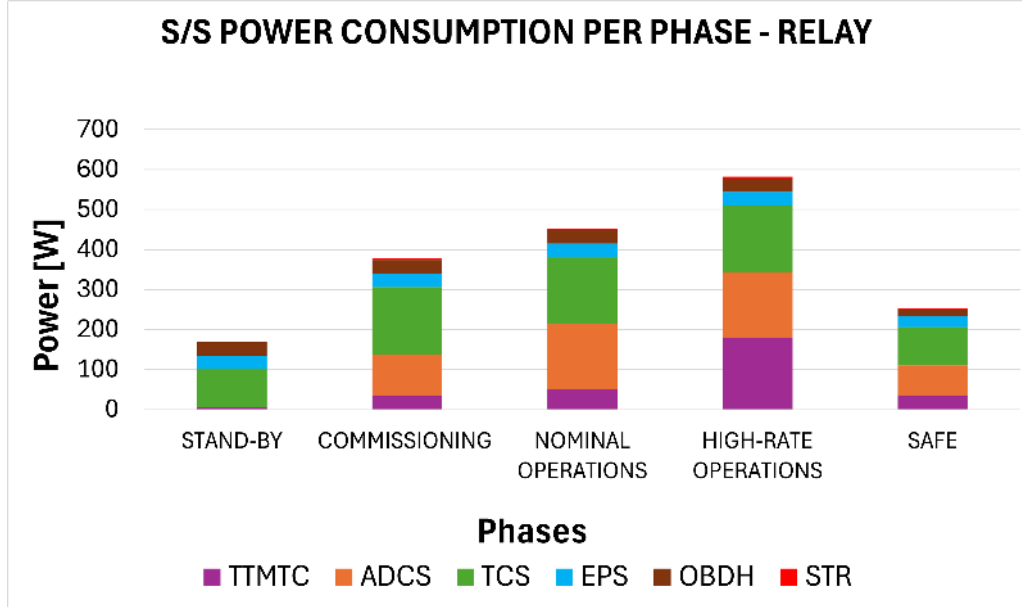


Figure 61: Subsystem Power Consumption per Phase (Relay).

Similarly, the power allocation for the Relay's most critical phase (High-Rate Operations) is analyzed in Table 34, highlighting the predominance of the communication payload.

Subsystem	TTMTC	ADCS	TCS	EPS	OBDH	STR
Percentage	31.0 %	27.9 %	28.7 %	5.9 %	5.8 %	0.7 %

Table 34: Subsystem Power Distribution during High-Rate Operations.

13.4 Primary Power Source

A trade-off analysis is conducted to identify the optimal primary power architecture. Evaluating all the different options, Radioisotope Thermoelectric Generators (RTGs) are discarded because their low specific power necessitates numerous units, leading to a prohibitive mass penalty for the mission. The solar arrays selection is dictated by the high peak power demand and the extended mission duration. Moreover, body-mounted arrays are fundamentally incompatible with the mission's power requests, as their limited sun exposure would require a three to four area increase to meet power demands with respect to deployable arrays, causing a detrimental impact on the spacecraft's total mass budget. Consequently, **deployable solar arrays** utilizing triple junction solar cells (InGaP/GaAs/Ge, CTJ30) [95] are selected for both spacecraft to provide high power to weight ratios using lightweight substrates. By integrating Solar Array Drive Mechanism (SADM), these arrays maintain near normal sun incidence and maximum efficiency, unlike fixed body-mounted solutions. Furthermore, a symmetric dual-wing layout is adopted for both the Carrier and the Relay; this symmetry is fundamental for balancing the center of mass and to guarantee that disturbance torques arising from solar radiation pressure and aerodynamic drag during the aerobraking phase remain manageable for the ADCS.

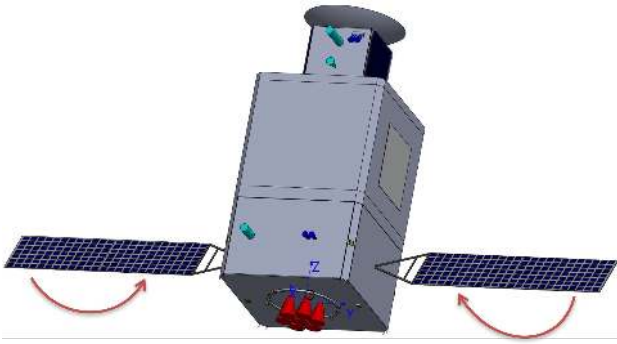


Figure 62: Carrier Solar Arrays Configuration.

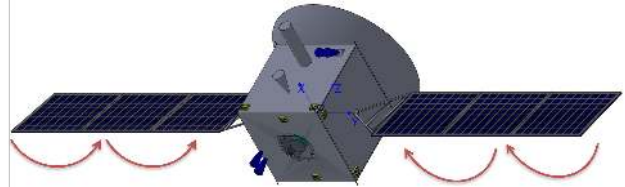


Figure 63: Relay Solar Arrays Configuration.

The specific mechanical configuration of the solar arrays is driven by the launch vehicle fairing envelope constraints, which lead to the adoption of a **Z-fold** deployment strategy for both spacecraft modules. For the Relay, the stringent volumetric limitations necessitated a three panel Z-fold architecture; this specific design is selected to pack the required photovoltaic surface area into a compact stowed footprint that fits within the bus face dimensions, while simultaneously preserving sufficient lateral margins for hinge mechanisms. In contrast, the Carrier benefits from a more generous volume allowance, adopting a two panel Z-fold configuration that simplifies the deployment kinematics. The deployment is actuated by spring-loaded hinges located at inter-panel lines, with the deployment rate controlled by viscous dampers to minimize latching shocks. To guarantee structural integrity during the launch phase, both systems rely on Hold Down and release Mechanisms (HDRM) to secure the stowed pack.

From a dimensioning perspective, the total active area was governed by the sizing driver: the highest $T_{Eclipse}/T_{Sunlight}$ ratio encountered during the mission. This critical condition dictates the maximum energy generation required during the limited sunlight window to recharge batteries while sustaining bus loads. Moreover, the arrays area computation is refined through a detailed environmental analysis performed using **ESA-SPENVIS** software [96]. This simulation modeled the specific radiation environment that both Carrier and Relay will endure throughout their respective lifetimes. The analysis quantifies the total efficiency loss of the solar cells, resulting in a retained power capability (P_{EOL}/P_{BOL}) of about 85.4 % for the Carrier and 75.4 % for the Relay. Consequently, the final sizing incorporates a precise degradation margin, verifying that the End-of-Life power output remains sufficient despite the harsh Martian Environment.

The solar panel dimensions are determined using a geometric procedure, which translates the required photovoltaic area, augmented by a 10 % design maturity margin [52], into a physical layout. The surface is discretized into individual solar cells, explicitly accounting for inter-cell gaps and additional mechanical margins for handling edges and hinges. The arrangement adopts a **String Folding** filling strategy to optimize the layout. The maximum panel height is constrained by the available area on the spacecraft bus face. The method maximizes the number of cell rows within this height and then extends the panel width to accommodate the total number of cells. This geometric approach decouples the mechanical sizing from electrical string constraints, maximizing the packing factor and eliminating dead zones. Vital for mission assurance, the electrical design also implements a robust redundancy scheme: the string layout includes an extra string, so that the loss of a string does not compromise the mission's overall power budget. Incorporating these conservative factors, the final design yields a robust energy margin: specifically, the design confirms a power produced at EOL at Mars of approximately 1490 W for the Carrier and 800 W for the Relay, fully satisfying the worst-case load requirements. In Table 35 the specific dimensions for both spacecraft are reported.

Parameter	Carrier	Relay
Configuration	2 Wings	2 Wings
Panels per Wing	2	3
Single Panel Dimensions	1.55x2.35m	1.30x1.32m
Total Wing Dimensions	1.55x4.69m	1.30x3.96m
Panel Thickness	22 mm	22 mm
Stowed Panels Thickness	56 mm	90 mm
Total Number of Solar Cells	4251	2925
Total Area (including 10 % margin) ^[52]	14.55 m ²	10.30 m ²
Total Mass (including 10 % margin) ^[52]	56.37 kg	39.93 kg

Table 35: Solar arrays design parameters.

13.5 Secondary Power Source

Complementing the primary generation system, **Lithium-ion (Li-ion) batteries** are selected for energy storage, as this chemistry is preferred over traditional Nickel-Cadmium (NiCd) or Nickel-Hydrogen (NiH₂) alternatives. Specifically utilizing the Yardney NCP 25-1 cells, the design leverages a solution with proven spaceflight heritage to minimize technological risk. This technology offers a superior mass efficiency and volumetric compactness, boasting a specific energy of 164 W h kg⁻¹ and a high energy density of 310 W h dm⁻³. These characteristics allow for a drastic reduction in the overall structural mass and volume budget while maintaining high performance. Functionally, the storage system is essential for maintaining operational continuity during Martian eclipses and peak demand transients, providing the necessary capacity to sustain critical loads when solar generation is absent or insufficient. The battery sizing is defined by the maximum eclipse duration, which is around 1.11 h for the Carrier and about 1.68 h for the Relay (see Figure 23). Design reliability is maximized through a redundant architecture: both spacecraft carry 2 independent battery packs. This configuration establishes a fault-tolerant architecture by eliminating single points of failure; in the event of a battery pack malfunction, the redundant unit provides sufficient capacity to maintain mission requirements. The obtained sizing parameters are summarized in Table 36.

Parameter	Carrier	Relay
Number of Battery Packs	2	2
Total Mass (including 10 % margin) ^[52]	46.36 kg	30.91 kg
Total Volume (including 10 % margin) ^[52]	0.0245 m ³	0.0164 m ³

Table 36: Batteries design parameters.

To validate the sizing of the energy storage system, **dynamical power simulations** are performed for both the spacecraft using Simulink. The analysis is conducted under the worst-case conditions, characterised by the maximum eclipse duration and peak operational loads. The simulation tracks the State of Charge (SoC) of the batteries over multiple orbital revolutions, incorporating the specific electrochemical characteristics of the selected Li-ion cells.

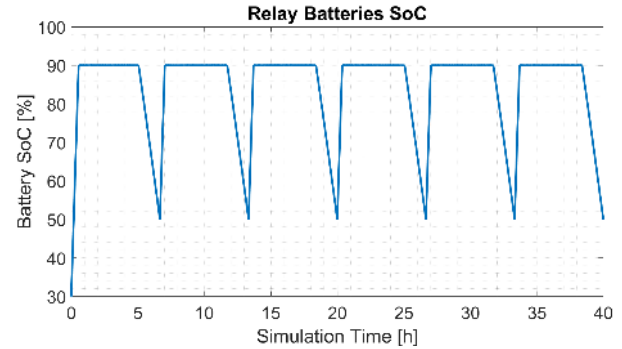
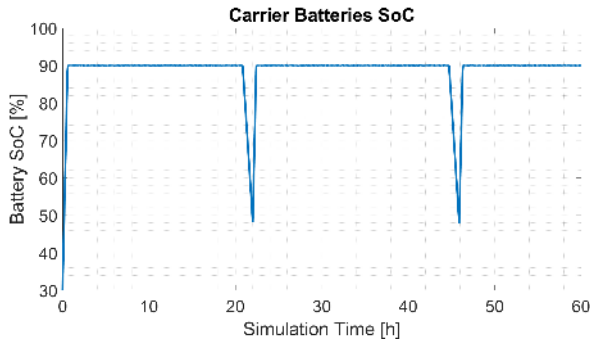


Figure 64: Carrier Batteries Dynamical Simulation. Figure 65: Relay Batteries Dynamical Simulation.

The system shows a stable and predictable cyclic behaviour. During the eclipse phases, the SoC decreases linearly as the batteries sustain the bus loads; the simulation confirms that the minimum charge level remains well above a safety threshold, maintaining a Depth of Discharge (DoD) of about 40 %. This conservative margin is vital to prevent deep discharge stresses, thereby mitigating degradation and preserving battery health for high-cycle operations over the extended mission lifetime. Conversely, upon entering the sunlight phase, the curve displays a steep charging slope. The rapid recovery confirms that the solar arrays generate a significant power margin, allowing the batteries to fully replenish the dissipated energy before the onset of the next eclipse. The consistent return to the maximum charge level at every cycle demonstrates a guaranteed positive energy balance and confirms the robust and reliable sizing of the entire power subsystem.

13.6 EPS Architecture

The Electrical Power System architecture, illustrated in Figure 66, establishes a fully regulated 28V bus as the common baseline for both spacecraft. This voltage standard is selected as the optimal compromise for power demands below 2kW. Central to this regulation strategy is the Battery Charge/Discharge Unit (BCDU). This bidirectional converter manages the energy flow: it controls the battery charging profile during sunlight and regulates the discharge voltage to maintain a stable 28V on the bus during eclipses, regardless of the battery's state of charge. Integrated within the battery assembly, a battery management system performs continuous cell voltage monitoring and active balancing. This ensures that no single cell exceeds its voltage limits during charge, maximizing the pack's operational life.

While the bus voltage is standardized, a distinct control strategy is tailored to the specific mission profile of each spacecraft to optimize efficiency and reliability:

- **Carrier - Peak Power Tracking (PPT):** the Carrier employs a PPT topology that electrically decouples the solar arrays from the main bus. This enables the system to continuously adjust the operating voltage to track the maximum power point on the I-V curve. This capability is necessary to maximize power extraction and to manage the drastic solar flux variations and temperature shifts experienced during the transfer from Earth to Mars.
- **Relay - Direct Energy Transfer (DET):** the Relay utilizes a DET topology equipped with a sequential shunt regulator. In this configuration, the arrays are directly connected to the bus, and regulation is achieved by shunting excess current to external dissipators when power generation exceeds demand. Since the spacecraft operates in a stable orbit with predictable flux, this architecture offers a simpler, more robust solution, optimizing reliability for the 20-year mission lifetime.

The architecture scheme of the electrical subsystem is reported in Figure 66.

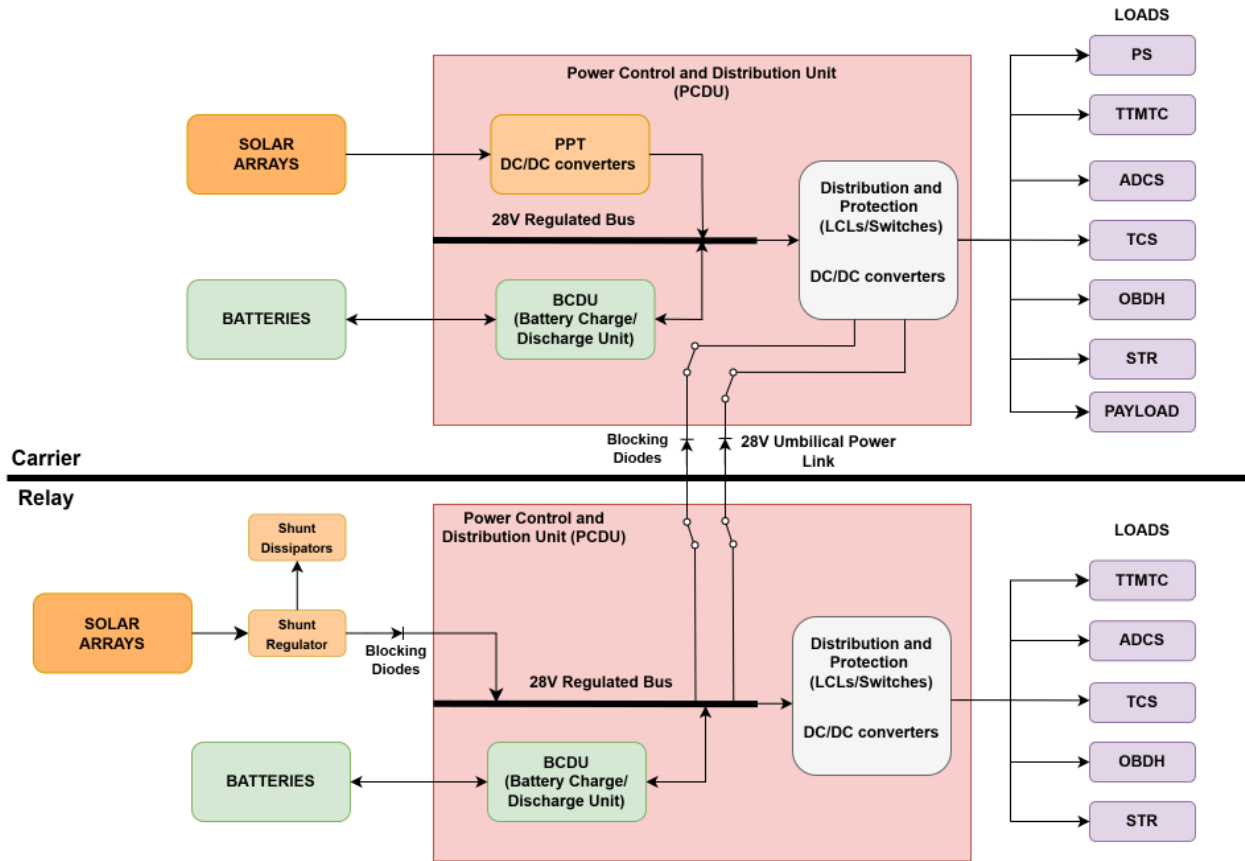


Figure 66: EPS Architecture.

Downstream of the main bus, the Power Control and Distribution Unit (PCDU) serves as the centralized node for power routing. Consistent with a centralized architecture, secondary voltage conversion is performed internally within the PCDU; dedicated DC/DC converters adapt the 28V bus voltage to the specific levels required by individual payloads, distributing regulated lines directly to the subsystems. For protection, the PCDU incorporates Latching Current Limiters (LCLs) to monitor consumption and isolate specific faults. If a payload short-circuits, the LCL cuts the line immediately, preventing failure propagation to the main bus. Physically, the PCDU has a mass of about 20 kg for the Carrier and 13 kg for the Relay. Additionally, the total harness and cabling mass budget, which accounts for the distributed nature of the loads and interplanetary interface, is set to be 15 % of the total EPS mass.

A critical feature of the design is the **Umbilical Power Link** that connects the two modules. During the cruise phase, when the Relay's arrays are stowed, this link allows the Carrier to power the Relay's critical loads. The physical disconnection occurs prior to the Relay release, effectively decoupling the two power buses. From this moment onward, the Relay operates as a self-sustaining system, independent from the Carrier's resources. To ensure safety, a blocking diode is installed on the umbilical line; this protection mechanism inhibits hazardous reverse currents, preventing potential voltage differences between the two independently regulated buses from compromising system safety during the separation sequence.

13.7 Effects and Constraints

The EPS acts as a primary design driver, imposing strict physical and operational constraints on the entire spacecraft architecture. System interoperability is managed via three distinct channels: mechanical, thermal, and electromagnetic interfaces. The following points outline the major system level interfaces and constraints dictated by the electrical system configuration and its operational profiles:

- **Thermal Control Subsystem (TCS):** the EPS imposes among the most stringent thermal requirements of the mission. Specifically, the Li-ion batteries are chemically sensitive to temperature fluctuations and must be maintained within a narrow operating window (nominally -20°C to 40°C) to prevent capacity loss and safeguard battery health. Similarly, the PCDU requires precise thermal regulation. Furthermore, the Relay's DET architecture introduces a significant heat rejection requirement, necessitating dedicated radiator surfaces to dissipate the excess power diverted by the shunt regulators when the batteries are fully charged.
- **Attitude Determination and Control Subsystem (ADCS) and Structures:** the large surface area of the deployable solar arrays is a primary source of external disturbance torques; these disturbances stem from solar radiation pressure and, critically for the Carrier, atmospheric drag during aerobraking, requiring active ADCS compensation to maintain pointing accuracy. Simultaneously, the internal configuration is influenced by the battery packs; as one of the densest components on board, their placement acts as a constraint for the spacecraft's inertial profile. By defining the center of mass and moments of inertia, they directly dictate the structural stability and control effort required by the ADCS to execute precise manoeuvres.
- **Payload and On Board Data Handling (OBDH):** the EPS sizing establishes the limit for the payload's operational duty cycle. The available energy budget determines the maximum allowable operating time for high power instruments and data transmission systems, requiring the OBDH to manage activation windows to uphold a positive energy balance across the orbit.
- **Electromagnetic Compatibility:** the Carrier's PPT regulators introduce high frequency switching noise. To prevent these conducted emissions from interfering with the Relay's electronics during the cruise phase, rigorous input/output filtering is implemented. Maintaining a low electromagnetic noise floor is paramount to safeguard the operational sensitivity of both the UHF transceiver and the X-band receiver. By minimizing potential interference, this architecture optimizes the Signal to Noise Ratio (SNR), ensuring high fidelity data recovery during critical communication windows.

14 Thermal Control

The MANTIS Thermal Control Subsystem (TCS) has been designed to ensure reliable operation throughout all mission phases, from interplanetary transfer to the long-term operational phase of the Relay spacecraft in Mars orbit. The primary objective of the TCS is to maintain every spacecraft component within its allowable temperature ranges for the entire mission duration. Consequently, a rigorous definition of temperature requirements represents the starting point of the entire thermal analysis process.

14.1 Requirement Temperatures

The first step for thermal analysis concerns defining the temperature requirements for each component installed on board of both Carrier and Relay spacecraft. For every component, two distinct temperature intervals shall be identified:

- the **Operational Temperature** range, within which the unit shall operate nominally and meet all performance specifications;
- the **Non-Operational Temperature** range, defining the temperature limits within which the unit may remain powered off without incurring permanent degradation or damage.

An overview of the collected data is presented in Table 37, 38. The reference temperature ranges were primarily derived from the dedicated component datasheet. For components without reference temperatures reported in the respective datasheet, the required values were established based on analogous space-qualified components, following a careful assessment of technological similarities, materials, functional roles, and heritage.

Unit	Oper. Min (°C)	Oper. Max (°C)	Non-Oper. Min (°C)	Non-Oper. Max (°C)	Ref.
Star Tracker	-30	+60	-40	+70	[66]
Sun Sensor	-70	+110	-80	+120	[67]
IMU	-25	+60	-30	+65	[68]
RW	-20	+70	-40	+75	[69]
Solar Panels	-150	+110	-200	+130	[91]
Battery	-20	+40	-30	+50	[97]
PCDU	-30	+45	-50	+60	[98]
SADM	-40	+80	-50	+90	[91]
HDMR	-65	+75	-150	+150	[99]
Hydrazine Tank	+15	+40	+5	+50	[100]
MON-3 Tank	+5	+55	-5	+65	[101]
Helium Tank	-10	+50	-30	+60	[102]
CPU	-55	+125	-65	+135	[103]
UHF antenna	-55	+125	-65	+150	[91]
UHF transponder	-35	+80	-35	+80	[104]
Low noise amplifier	-55	+125	-65	+135	[105]

Table 37: Carrier Temperature Range.

Unit	Oper. Min (°C)	Oper. Max (°C)	Non-Oper. Min (°C)	Non-Oper. Max (°C)	Ref.
Star Tracker	-30	+60	-40	+70	[66]
Sun Sensor	-70	+110	-80	+120	[67]
IMU	-25	+60	-30	+65	[68]
Reaction Wheel	-20	+70	-40	+75	[69]
Solar Panels	-150	+110	-150	+130	[91]
Battery	-20	+40	-30	+50	[97]
PCDU	-30	+45	-50	+60	[98]
SADM	-40	+80	-50	+90	[91]
HDMMR	-65	+75	-150	+150	[99]
Tank	+15	+40	+5	+50	[100]
CPU	-55	+125	-65	+150	[103]
SSR	-25	+60	-35	+70	[106]
FPGA	-55	+125	-65	+150	[107]
UHF antenna	-55	+125	-65	+150	[91]
UHF transponder	-35	+80	-35	+80	[104]
Horn antenna	-55	+125	-65	+150	[91]
HGA	-140	+90	-140	+120	[91]
Deep Space transponder	-40	+60	-50	+70	[108]
Diplexer	-40	+85	-65	+150	[109]
TWTA	-40	+70	-65	+125	[110]
SSA	-30	+60	-40	+70	[111]
Low noise amplifier	-55	+125	-65	+150	[105]

Table 38: Relay Temperature Range.

14.2 Internal heat dissipation

The internal heat dissipation of the Carrier and Relay spacecraft is determined for each operational mode by evaluating the power consumed by all subsystems. The results, presented in Tables 40 and 39, are organized by subsystem and were derived from the EPS average power budget, assuming full conversion of electrical power into thermal energy. For the spacecraft solar cells, heat dissipation is determined taking into account the instantaneous solar flux incident on each cell and their conversion efficiency at each moment. This method provides a realistic, time-resolved evaluation of thermal loads under varying illumination conditions throughout the mission. The spacecraft battery's heat contribution was calculated considering both charge/discharge efficiency and Joule heating effects. This allowed for an accurate estimation of the fraction of power actually released as heat.

Subsystem	Cruise [W]	Pre-MOI [W]	MOI [W]	Mars Commissioning [W]	Payloads Release [W]	Relay Release [W]	Safe [W]
PS	62.24	218.27	518.02	28.34	141.69	141.69	35.63
TMTC	1.81	7.25	0.00	7.25	4.54	0.72	6.29
ADCS	86.54	235.05	84.68	118.76	77.81	77.81	109.07
EPS	40.95	40.95	40.95	40.95	40.95	40.95	32.76
OBDH	37.78	37.78	37.78	37.78	37.78	37.78	18.89
STR	0.96	0.96	0.96	0.96	13.12	43.96	0.96
Payload	191.10	191.10	191.10	191.10	191.10	136.50	191.10
Total	421.38	731.37	873.50	425.14	506.97	479.30	394.71

Table 39: Carrier internal heat loads.

Subsystem	Stand-By [W]	Commissioning [W]	Nominal Operations [W]	High-Rate Operations [W]	Safe [W]
TMTC	5.72	34.35	51.53	180.43	31.72
ADCS	0.00	104.43	162.31	162.31	76.31
EPS	34.13	34.13	34.13	34.13	27.30
OBDH	34.10	34.10	34.10	34.10	17.04
STR	0.00	3.83	3.83	3.83	3.83
Total	73.94	210.82	285.88	414.78	156.19

Table 40: Relay internal heat loads.

14.3 Analysis cases

In order to identify the hot and cold design cases, a comprehensive analysis of the thermal environmental conditions was carried out. The MANTIS spacecraft encounters significantly different external heat fluxes and operational states throughout the various phases of its mission. For this reason, the analysis was structured by mission phase, ensuring that all relevant external thermal loads were properly assessed for each spacecraft configuration.

14.3.1 Interplanetary Cruise

During this phase, the spacecraft thermal environment is primarily determined by its heliocentric distance. The dominant external heat source is the direct solar flux, which decreases with increasing distance from the Sun, varying from approximately 1367.5 W m^{-2} to 589 W m^{-2} at Mars approach. In this phase, the Carrier and the Relay operate in a stacked configuration. The carrier is in Cruise Mode, while the relay remains in Stand-By Mode. This condition represents a relatively stable environment and although not dimensioning, it verifies the correct thermal behavior of the spacecraft during transfer to Mars.

14.3.2 Release Operations

This phase represents the most thermally demanding segment of the mission. During aerobraking, the spacecraft experiences significant aerodynamic heating. As the aerobraking sequence progresses, the orbital period decreases, resulting in longer fractions of each orbit being exposed to aerodynamic heating as illustrated in Figure 67a and 67b. Simultaneously, eclipse periods introduce severe cold sinks, further amplifying the thermal challenge. Figure 67c summarizes the main external fluxes acting on the spacecraft during the longest eclipse periods ($\approx 1.1 \text{ h}$). Together, these effects make this phase critical from a thermal design perspective.

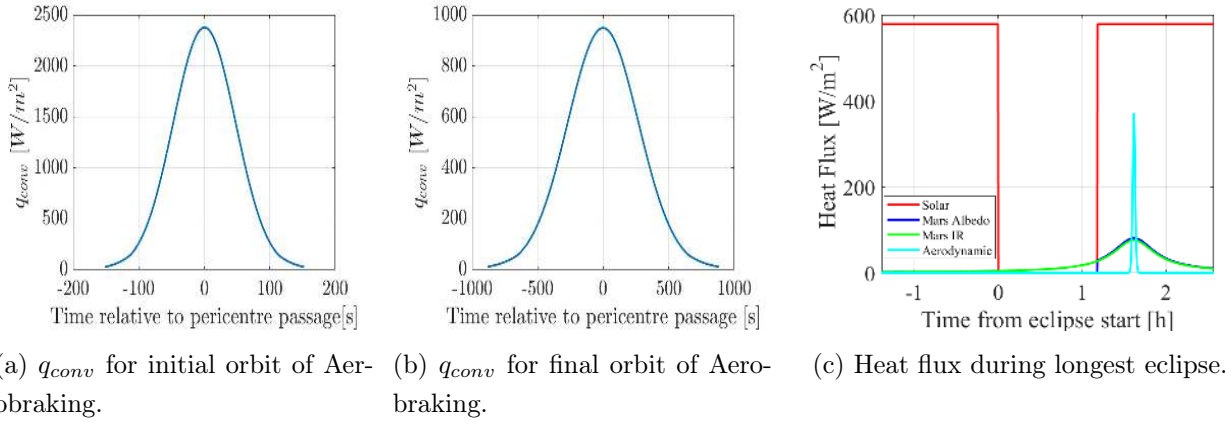


Figure 67: Aerodynamic heat flux evolution during the Release Operation (left) and environmental heat flux during the longest eclipse period (right).

14.3.3 Relay nominal orbit

The thermal environment of the Relay in its nominal Mars orbit has been analyzed by considering the most extreme orbital conditions expected during long-term operations, as shown in Figure 68. The most critical low-temperature condition occurs during the time interval characterized by the longest eclipse durations (≈ 1.7 h), when the spacecraft operates in Safe Mode. In this configuration, the absence of solar input combined with reduced internal power dissipation leads to minimum temperature levels. The most demanding high-temperature condition was evaluated during periods of continuous sunlight, when the spacecraft is continuously exposed to direct solar radiation and operates in High-Rate Operations Mode, corresponding to the maximum internal power dissipation.

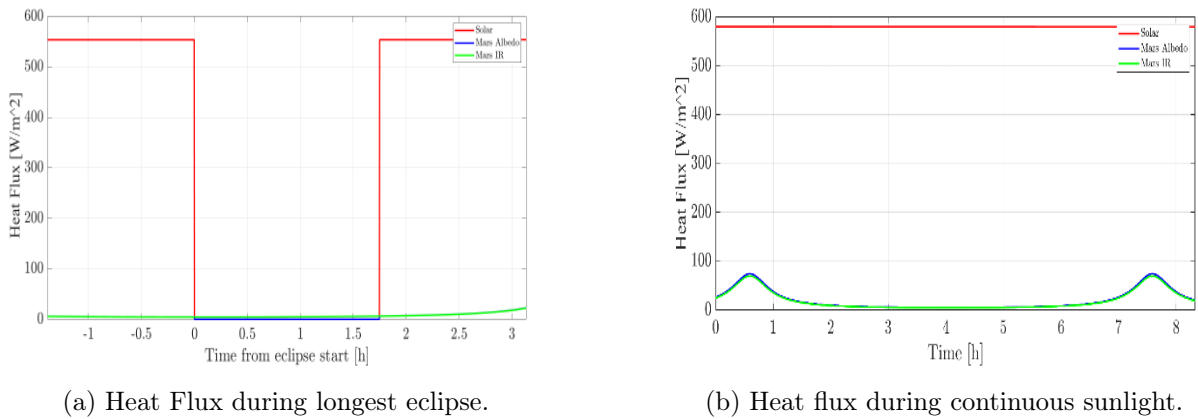


Figure 68: Evolution of the environmental fluxes during worst cases in Nominal Orbit.

14.3.4 Identification of Design Cases

Following the assessment of all mission phases and configurations, the design cases are identified as follows:

- **Carrier Hot Case:** occurs during aerobraking in lower orbits during Payloads Release phase, due to the combined effect of aerodynamic heating and external radiative fluxes;

- **Carrier Cold Case:** takes place during longest eclipse periods in the Release Operation phase when operating in Safe Mode;
- **Relay Hot Case:** arises in Relay nominal Mars orbit during continuous sunlight exposure while operating in High-Rate Operations Mode;
- **Relay Cold Case:** is observed during longest eclipse periods in Release Operations phase while operating in Stand-By Mode in stacked configuration.

14.4 Multinodal thermal analysis

In order to achieve a realistic representation of the spacecraft thermal behavior and to accurately monitor the temperature of the most thermally critical components, a transient multi-node thermal analysis was performed for all mission configurations. The analysis was implemented through a *Simulink/Simscape* based lumped parameter thermal network model, capable of simulating the time dependent thermal response of the system. The simulations account for orbital evolution, spacecraft attitude and pointing strategy, operational modes and the corresponding internal power dissipation profiles. This integrated approach ensures consistent coupling between environmental boundary conditions and internal heat loads, providing reliable temperature predictions throughout all mission phases.

14.4.1 Stacked configuration Nodalization Scheme

For the stacked configuration, a 29-node thermal model was developed to represent the principal external surfaces and thermally relevant subsystems of the integrated Carrier–Relay system.

The **external nodes** include:

- one node for each external face of the Carrier bus, Payload bay and Relay;
- one node for each face of the Carrier solar array;
- one node for each Relay folded solar panels;
- one node for the CPDI;
- one node for the HGA.

The **internal nodes** represent the most thermally relevant components, including:

- Carrier propellant tanks;
- Carrier EPS module (including batteries and PCDU);
- Payload;
- Relay EPS module;
- Relay tank.

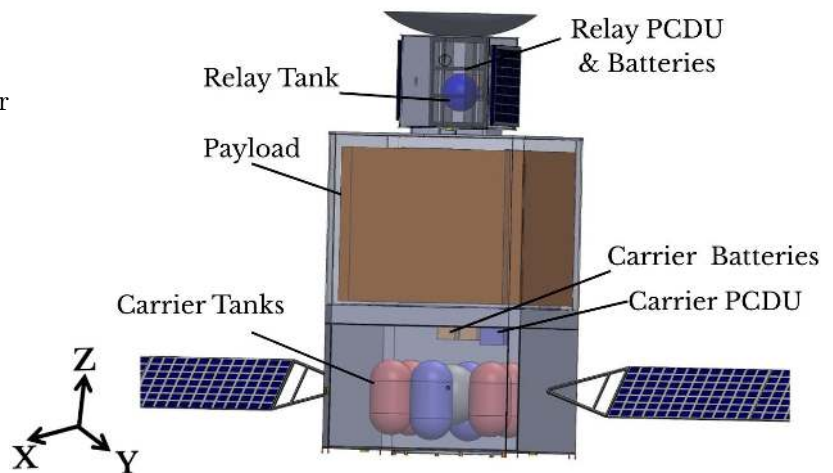


Figure 69: Stacked configuration internal thermal nodes.

14.4.2 Relay Nodalization Scheme

For the Relay operating independently in its nominal Mars orbit, a 12-node thermal model was adopted.

The external nodes represent the main radiating surfaces of the spacecraft and comprise:

- one node for each external face of the Relay bus;
- one node for each solar panel face;
- one node representing the HGA.

The internal nodes correspond to the most thermally critical subsystems:

- the EPS module, including batteries and PCPU;
- the OBDH module, including all subsystem units;
- the tank.

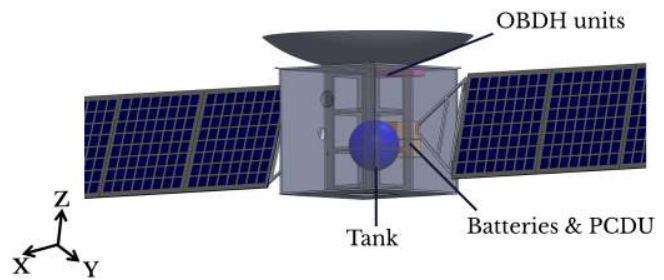


Figure 70: Relay internal thermal nodes.

14.4.3 Radiative Environment and Heat Exchange Modeling

Once the nodal schemes were defined, the radiative exchanges between nodes and external heat sources were computed. The absorbed heat flux on each external node was evaluated by calculating the view factors between spacecraft surfaces and the primary radiative sources, namely the Sun and Mars. These view factors were determined as a function of spacecraft attitude, defined according to the adopted pointing strategy and derived from orbit propagation results. Radiative heat exchange between internal and external nodes was also included in the model. The corresponding view factors were computed using a ray-tracing based approach implemented in *OpenFOAM*. This methodology enables an accurate reconstruction of the relative geometry between spacecraft surfaces and environmental heat sources, ensuring consistent modeling of solar, planetary, and mutual radiative exchanges within the multinode thermal network.

14.5 Design Description

The thermal design of the main structure and instruments is based on a simple, predominantly passive architecture, with active heaters implemented only where necessary to maintain components within critical temperature limits. The system has been sized against worst-case hot and cold operational scenarios, ensuring temperature stability, robustness, and overall operational reliability.

14.5.1 Carrier

- All **external surfaces** of the spacecraft, except for the minimum required radiation area, are covered with **twenty layer of MLI (Multi-Layer Insulation)**, consisting of an outer layer of 2 mil aluminized Kapton over multiple aluminized Kapton layers separated by Dacron nets. The MLI system significantly reduces radiative heat exchange with the external environment, limiting both solar and planetary heat input during hot cases and minimizing heat losses during eclipse and cold operational phases.
- **Internal surfaces** are coated with **high-emissivity black paint** to enhance radiative coupling between internal components. This treatment reduces internal temperature gradients and promotes

a more uniform thermal distribution across the spacecraft structure, improving thermal stability and reducing localized hot spots.

- Waste heat generated by onboard electronics is rejected through **two passive radiators**, each with an effective area of 2.7 m^2 , mounted on the lateral faces of the Carrier. Each radiator is based on **aluminum honeycomb panel** covered with a **second surface mirror (SSM) based on Al/FEP**. Their orientation has been selected to maximize deep-space view factors while minimizing direct solar and planetary infrared flux exposure.
- To optimize heat transfer toward the radiators, **high-conductivity thermal fillers** are implemented between major dissipative units, including the PCDU and the batteries, and the supporting structural panels. These conductive paths reduce thermal resistance, enhance heat spreading, and improve the overall efficiency of the passive thermal rejection system.
- **Propellant tanks** represent one of the most thermally critical elements of the Carrier module due to the risk of propellant freezing during prolonged cold-soak conditions. Radiative isolation is achieved through dedicated **MLI** blankets specifically designed for the tank geometry. In addition, active thermal control is provided through thermostatically controlled **heaters with a total power of 300 W**. This combined passive–active approach ensures that the propellant temperature remains within its strict operational limits throughout all mission phases.
- The thermal radiative properties of front side of the **solar arrays** are dictated by the solar cells themselves, which are characterized by high solar absorptivity and high infrared emissivity while the back is painted with high emittance black paint to minimize operating temperature and enhance electrical efficiency.

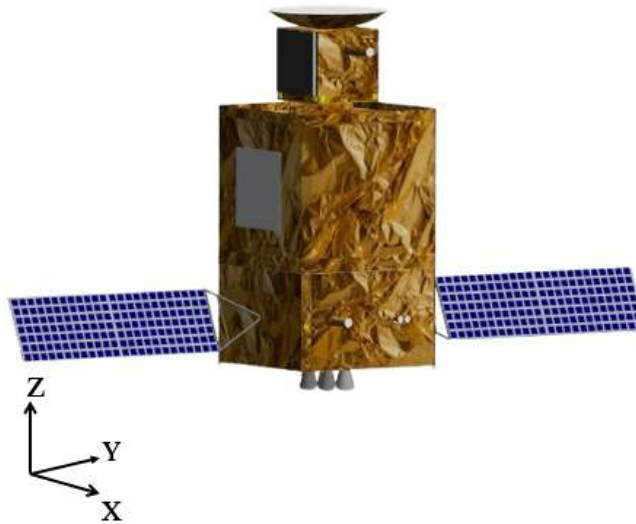


Figure 71: Carrier external thermal design.

14.5.2 Relay

The Relay module adopts the same passive-oriented philosophy:

- The entire spacecraft, except for designated radiative area, is covered with **20 layers of Aluminized Kapton MLI** providing a thermal barrier against the harsh Martian environment.

- **Internal structural panels** are coated with **high-emissivity black paint** to promote radiative coupling, reduce internal gradients, and maintain a uniform temperature distribution.
- On the **-X panel**, less affected by solar flux, the aluminum honeycomb panel is covered with a **SSM based on Al/FEP** and acts as a passive thermal radiator to dissipate the heat generated by the EPS module. The radiator covers an area of 1.7 m^2 , sized to provide the required thermal rejection under worst case conditions.
- The **OBDH elements**, characterized by relatively high power dissipation, are mounted on the +Z face, shielded from the antenna and painted black to maximize radiative heat rejection.
- The **EPS module**, comprising the batteries and the PCDU, is installed on the radiator face. Thermostatically controlled heaters are employed to maintain minimum allowable temperatures under cold case conditions, with a **nominal power of 135 W**. All surfaces of the EPS module that do not face the radiator are covered with **MLI**, preventing unwanted heat loss from the heaters and ensuring efficient thermal management.
- The **hydrazine tank** is fully covered with **MLI** to provide radiative isolation. Its temperature is maintained above $25\text{ }^{\circ}\text{C}$ by on/off controlled heaters with a **nominal power of 40 W**.
- The **parabolic antenna** is coated with **white thermal paint** on the exposed surface, while the back side is covered with **MLI**. The white coating is selected to limit solar heat absorption and to mitigate temperature gradients across the antenna dish, which may arise due to partial illumination and shadowing effects during spacecraft operations. The MLI on the backside reduces heat exchange with the space environment, contributing to a more uniform and controlled thermal behavior of the antenna assembly.
- **Thermal fillers** are applied to improve conductive heat paths between OBDH and EPS components and the structural panels, enhancing overall heat transfer efficiency and ensuring effective thermal management.
- The **solar arrays** rely on the intrinsic thermal properties of the solar cells, characterized by high solar absorptivity and infrared emissivity. The back side is painted with high-emittance black paint to enhance radiative cooling and improve electrical efficiency.

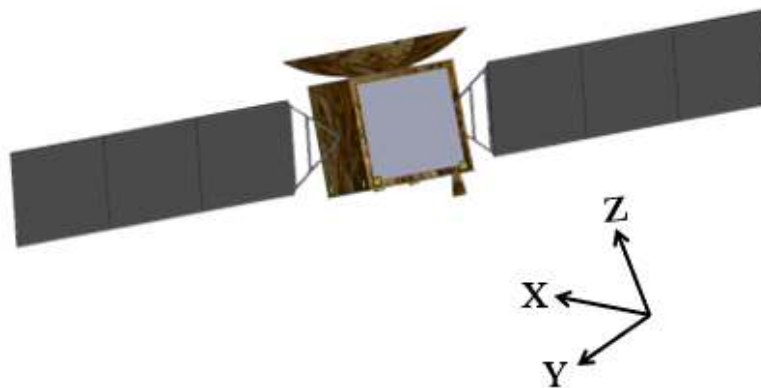


Figure 72: Relay external thermal design.

14.5.3 Thermal Properties

The thermal properties of the selected material and surface finishes are summarized in Table 41. To account for material aging and environmental effects, conservative values are used for thermal properties. Specifically, solar absorptivities (α_S) at beginning of life (BOL) are applied for the cold case while the end of life (EOL) α_S values are used for the hot case. This approach ensures that the thermal performance remains reliable throughout the mission lifetime, even considering potential degradation of the insulation properties.

Material / Surface	α_S BOL ^[112]	α_S EOL ^[112]	ϵ_{IR} ^[112]
Aluminized Kapton MLI	0.30	0.50	0.03–0.05
Z306 Black Paint	0.95	0.95	0.85
A276 White Paint	0.26	0.50	0.85
Aluminum / FEP (SSM)	0.13	0.40	0.78
Solar Cells	0.88	0.95	0.72

Table 41: Optical properties of materials/surfaces.

14.6 Temperature Prediction

The results of the multinodal thermal analysis are summarized in Figure 73, 74 and 75. The table reports, for each thermally critical node, the predicted maximum and minimum temperatures in each phase, with a margin of $\pm 15^\circ\text{C}$, fully compliant with MAR-TCS-010 ^[52]. The analysis demonstrates that, for both Carrier and Relay, all subsystems remain well within their prescribed operational limits throughout all mission phases, confirming the robustness and margin compliant design of the MANTIS thermal architecture.

14.6.1 Interplanetary Cruise

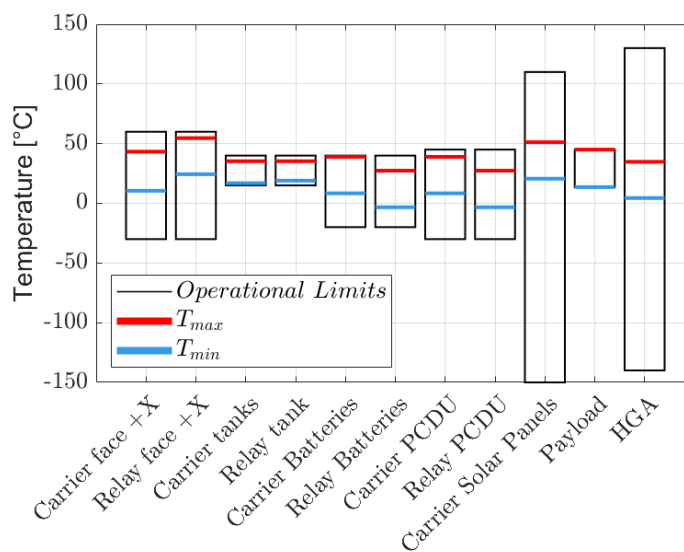


Figure 73: Predicted Temperature for the most critical nodes during Interplanetary.

14.6.2 Release Phase

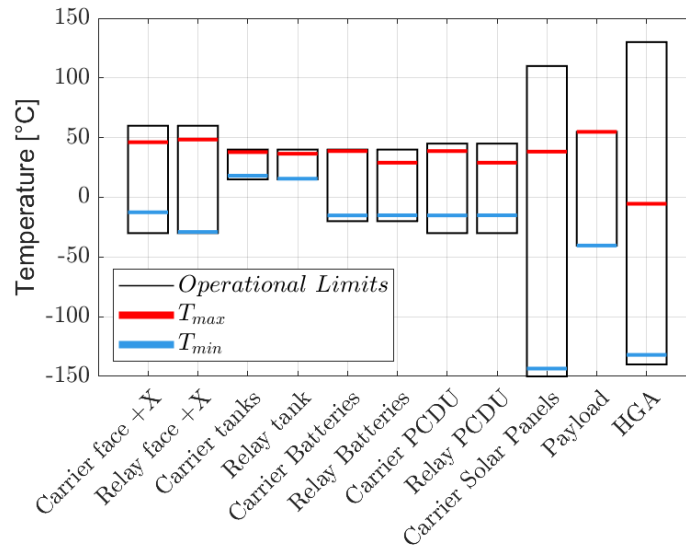


Figure 74: Predicted Temperature for the most critical nodes during Release Phase.

14.6.3 Relay Nominal Orbit

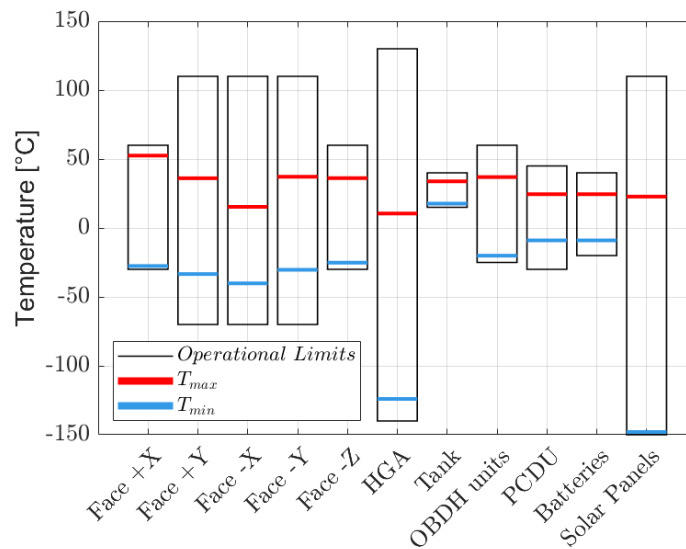


Figure 75: Predicted Temperature for the most critical nodes during Relay Nominal Orbit.

15 Mechanisms

The architectural solutions adopted for the mechanisms and the interfaces of the MANTIS mission are directly driven by a set of system-level constraints. First, both door deployment and relay separation are **one-shot release events**. No re-closure or re-capture capability is foreseen. This imposes a conservative deterministic behaviour and fault containment rather than reconfiguration flexibility. Second, the mechanisms are subject to **long dormant time** prior to activation, indeed they shall be operative after all the previous mission phases. During launch, early orbit, and potentially extended commissioning phases, latches and deployment systems remain inactive while exposed to vibration, thermal cycling and structural loads. In particular these drives the avoidance of continuous power-hold solutions and strong separation between load-bearing and actuation functions. Third, the spacecraft adopts a **stacked configuration**, with the relay mounted on the upper XY plane corresponding to the payload bay door. This introduces tight geometric coupling between structural support and deployment kinematics for a safe payload separation trajectory. The architecture is conceived therefore to minimize over-constraint, tolerance sensitivity and deformation-induced misalignment, particularly across the dual-leaf hinge line. Fourth, **structural and deployment volume constraints** are significant. The bay envelope is limited, and external protrusion during deployment shall remain compatible with spacecraft configuration and payload release shall not influence other payload's safety. Finally, **loads transmitted to the primary structure** during launch shall remain predictable and well-defined. Discrete load introduction points and statically determined interfaces are preferred in order to avoid parasitic stresses due to thermal distortion or cumulative tolerances. These drivers collectively define the guiding principles adopted in the following sections: functional segregation between load path and actuation, symmetric deployment kinematics, deterministic time-bounded logic, and the use of flight-proven building blocks assembled in a mission-tailored configuration.

15.1 Relay Release and Support Frame

15.1.1 System Role and Architectural Philosophy

The **Relay Release and Support Frame (RRSF)** is designed to fulfil two tightly coupled but functionally distinct objectives: to sustain the relay during launch and early orbit phases, and to enable its controlled and robust separation once in operational conditions. From the beginning of the architectural definition, structural load transfer and release actuation have been treated as independent functional layers. In the stowed configuration, launch and in-orbit loads are transferred through discrete structural hardpoints integrated within a **C-shaped support frame**. The use of discrete hardpoints follows a statically determined interface philosophy, avoiding over-constraint and limiting parasitic stresses induced by thermo-elastic deformation during the ground-to-orbit transition. Rather than distributing loads along a continuous circumferential interface, the architecture introduces them locally into reinforced regions of the supporting door panels. Moreover, this choice is coupled with the door concept in order to allow the release payload operations. The adoption of a C-shaped frame mounted on a dual-leaf door results from an architectural trade-off. A closed ring structure would provide geometric continuity but would create a rigid structural bridge across the hinge line, increasing sensitivity to differential deformation between the two leaves under launch loads. Any relative bending, torsion or hinge compliance would be transmitted through the closed loop, potentially inducing secondary stresses and misalignment at the relay interface. Similarly, during door deployment, a continuous ring would impose strict geometric compatibility between the leaves, amplifying the effect of hinge tolerances or asynchronous motion. The C-shaped configuration avoids the formation of a closed structural loop. By preserving an open side, it tolerates limited relative motion between the door leaves without generating constraint-induced stresses. The geometric implementation of the C-shaped support frame on the dual-leaf door is illustrated in Fig. 76.

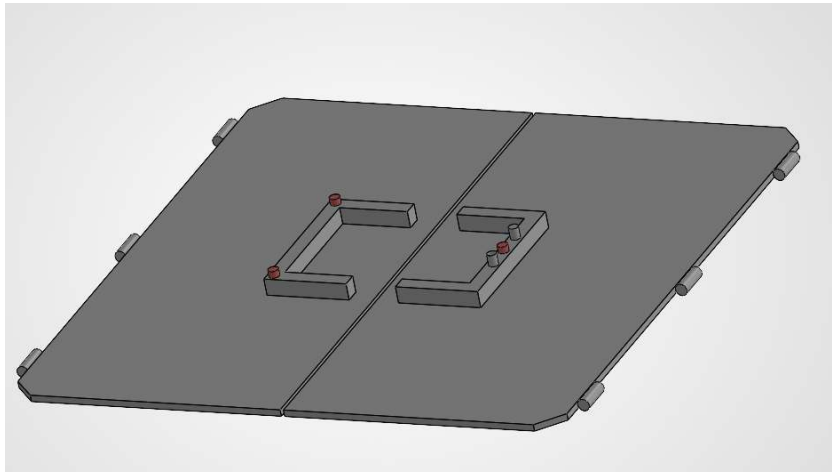


Figure 76: C-shaped Relay Release and Support Frame mounted on the dual-leaf bay door. The open-side geometry avoids the formation of a closed structural loop across the hinge line, reducing sensitivity to differential leaf deformation while preserving primary load-path stiffness.

15.1.2 Retention, Actuation and Controlled Separation

Mechanical retention of the relay during launch is ensured by a positive-locking latch providing the primary load-holding function. The latch is inherently stable without continuous power, ensuring robustness during long dormant phases and eliminating dependency on sustained actuator torque or preload. Nominal actuation is performed through a **non-explosive actuator (NEA) pin-puller**, whose sole function is to remove the mechanical constraint. The selection of a non-explosive solution is driven by shock minimization and compatibility with nearby spacecraft mechanisms. Separation energy is provided independently by **preloaded push-off springs** supported by dedicated seats. The springs are excluded from the primary launch load path, ensuring that their preload remains well-defined and unaffected by structural redistribution. By decoupling constraint removal from energy injection, the initial separation impulse is controlled and repeatable, rather than being dependent on residual elastic relaxation within the latch or surrounding structure. Guide pins constrain the relative motion between relay and frame during the first phase of separation, when clearances are minimal and sensitivity to asymmetries is highest. Their function is purely kinematic: they limit lateral displacement, prevent tilting and reduce the effect of small force imbalances or centre-of-mass offsets. Once sufficient clearance is achieved, the guides disengage and the relay transitions to torque-free rigid-body motion. The structural configuration and the guided initial motion have direct implications for separation dynamics. By limiting deformation-induced misalignment at the hinge line and minimizing angular bias at latch release, the C-shaped architecture reduces the risk of relay-to-door interference during the critical early phase of extraction. The nominal separation direction is aligned with the spacecraft +Z axis, as shown in Fig. 77.

15.1.3 Operational Logic, Monitoring and Redundancy

The release sequence is monitored through dedicated status sensors confirming latch opening and separation completion. Verification is time-bounded. If nominal actuation does not result in confirmed release within the allocated time window, escalation logic may enable a pyrotechnic backup device. The pyrotechnic device is implemented as cold redundancy and remains electrically inhibited during nominal operations. Its activation is gated at system level through explicit FDIR authorization. This dual-path architecture eliminates single-point failure in separation capability while maintaining

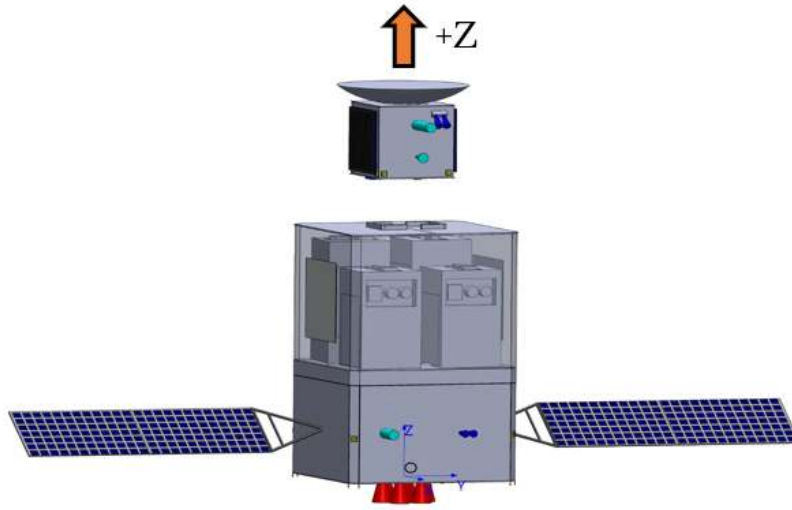


Figure 77: Nominal relay separation along the spacecraft +Z axis. The guided initial motion phase ensures controlled extraction prior to transition to torque-free rigid-body dynamics.

a low-shock nominal release profile. The backup device is not part of routine operations but exists to guarantee separation robustness in the presence of unlikely nominal actuator failure. The release logic and escalation strategy are summarized in Fig. 78.

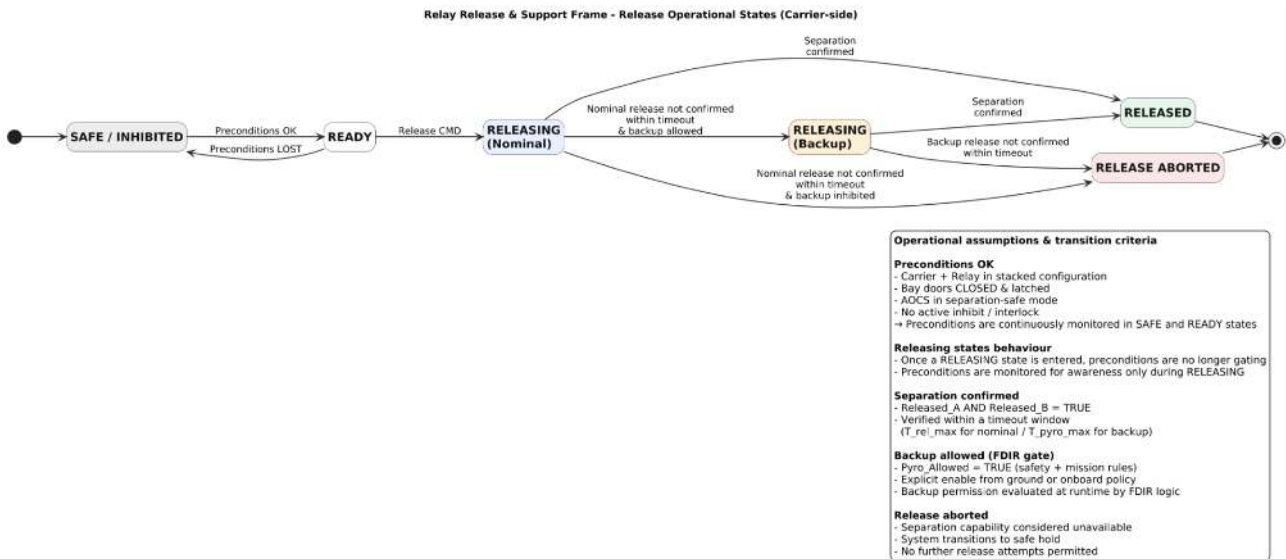


Figure 78: High-level operational state machine of the Relay Release and Support Frame. The architecture separates nominal non-explosive actuation from the cold-redundant pyrotechnic backup path, with time-bounded verification and FDIR-gated escalation logic.

15.1.4 Maturity Assessment

At Phase A level, the activity is limited to architectural definition and functional validation of the concept. Detailed structural sizing, tolerance definition, material selection and qualification aspects are intentionally deferred to subsequent phases. All individual building blocks are based on flight-proven technologies with high component-level TRL. However, the integrated configuration, geometric implementation and interaction with the dual-leaf door system are mission-specific and have not yet been demonstrated at hardware level. The Relay Release and Support Frame is therefore conservatively assessed at TRL 3, with the remaining maturity gap associated with system-level breadboarding, integrated functional verification and detailed design consolidation in later phases.

15.2 Dual-Leaf Payload Bay Door

15.2.1 System Role and Architectural Rationale

The **dual-leaf payload bay door** provides environmental protection during launch and early orbit, while enabling unobstructed relay and payload release once the spacecraft transitions to operational configuration. In the closed state, the door shall ensure structural integrity, maintain the required internal envelope, and remain mechanically compatible with the relay support frame mounted on the door inner surface. In the open state, it shall guarantee deterministic clearance for relay extraction and payload release, and remain dynamically stable without rebound or partial re-closure. As shown in Fig. 79, the symmetric outward rotation of the two leaves provides geometric clearance along the $+Z$ axis while maintaining balanced disturbance about the spacecraft longitudinal plane by rotating symmetrically around $-X$ and $+X$ axis.

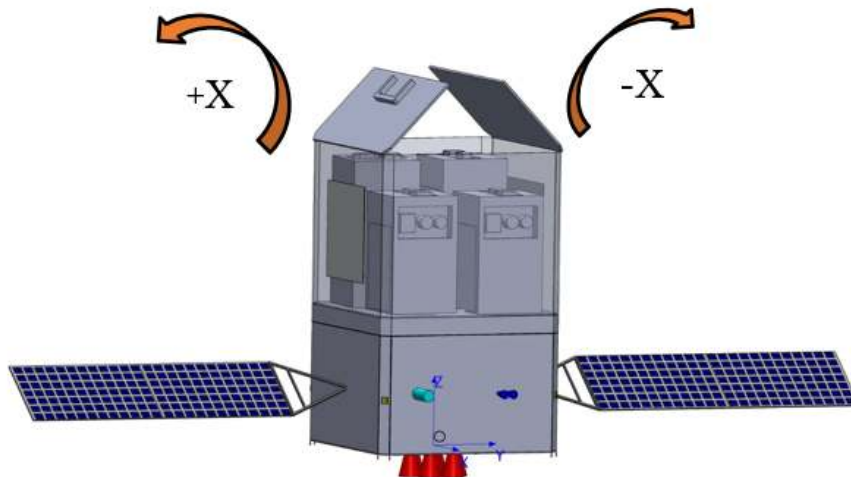


Figure 79: Dual-leaf payload bay door in deployed configuration. Symmetric $\pm X$ rotation ensures balanced disturbance and deterministic clearance for relay extraction.

The adoption of a dual-leaf configuration is primarily driven by envelope efficiency and deployment symmetry. Splitting the door into two independently hinged leaves reduces the required opening envelope compared to a single monolithic panel, limits external protrusion during deployment, but most important reason is the reduction of the hinge-line bending demand. Moreover, symmetric opening

about the spacecraft longitudinal plane minimizes off-axis disturbance torques during deployment and improves attitude stability. From a structural standpoint, the dual-leaf architecture avoids creating a closed structural loop across the hinge line. In a monolithic configuration, differential deformation between the two sides under launch loads would be mechanically coupled, potentially generating parasitic stresses and increasing sensitivity to hinge misalignment. By keeping the two leaves structurally independent, local deformation and hinge compliance do not propagate across the full door width. Although the relay is externally exposed and not laterally confined, maintaining geometric robustness of the support frame improves predictability of the separation axis and limits the introduction of unintended initial angular bias during release.

15.2.2 Operational Logic and State Architecture

The operational logic is structured in three layers: local leaf state machines, a supervisory controller, and system-level gating. Each leaf operates independently through the sequence: *CLOSED & LATCHED* → *UNLATCHING* → *DEPLOYING* → *OPEN & STABLE*, with fault transitions available at each phase. Fault conditions include timeout during unlatching, absence of expected motion, excessive deployment duration, rebound detection, or sensor disagreement. The supervisory state machine coordinates the two leaves. Upon reception of an open command, it broadcasts an open request and starts a global timeout. Transition to *VERIFY OPEN & STABLE* occurs only when both leaves report successful unlatching. The state *OPEN / STABLE* is achieved only if both leaves confirm stable deployment within the allowed time window. If either leaf reports a fault or the timeout expires, the system transitions to *FAULT (HOLD)*. Abort transitions are available at all stages. The supervisory logic coordinating the two independent leaves is formalized in the state machine shown in Figure 80.

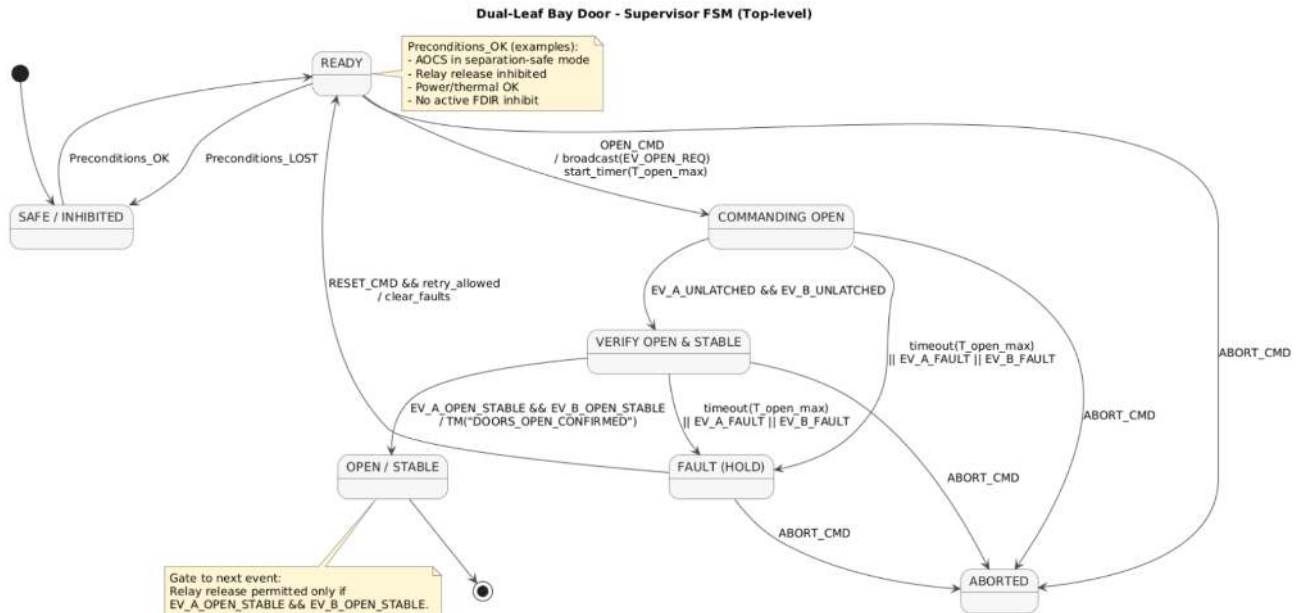


Figure 80: Supervisory state machine coordinating dual-leaf deployment and gating relay release.

Each leaf operates through an independent local state machine, as shown in Figure 81 and Figure 82, enabling segregated fault detection and time-bounded verification prior to global confirmation.

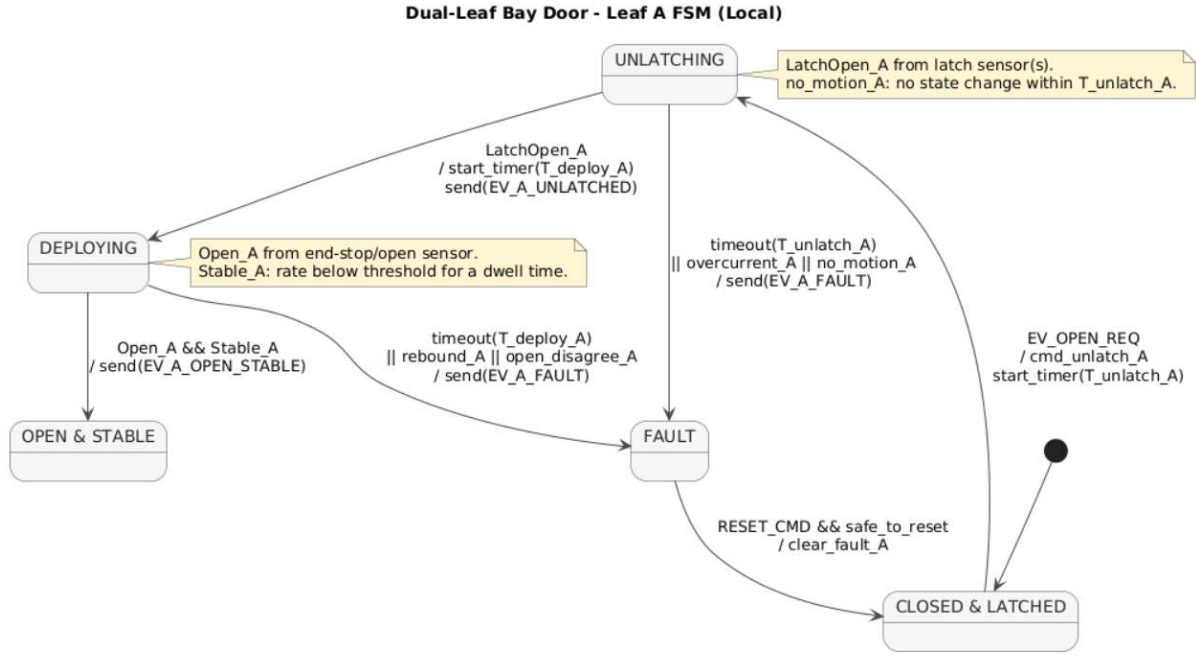


Figure 81: Leaf A local state machine.

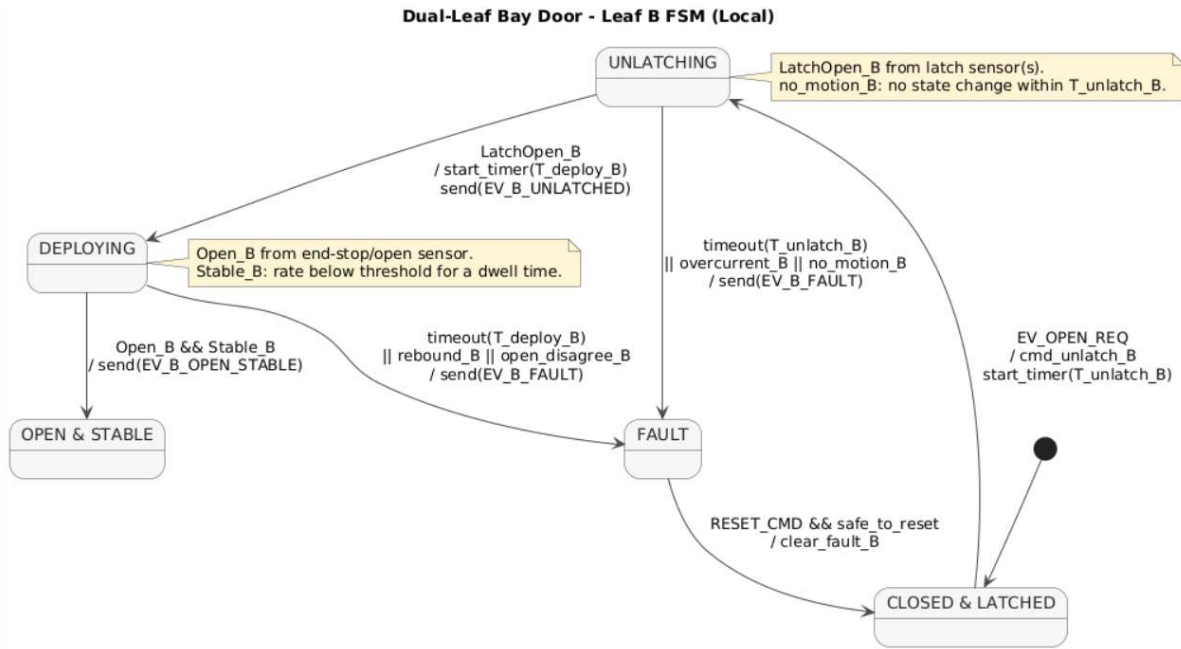


Figure 82: Leaf B local state machine.

15.2.3 Deployment Kinematics and Dynamic Verification

The leaves rotate about hinge axes aligned with the spacecraft longitudinal direction, resulting in outward rotation along opposite $\pm X$ directions. Open-position confirmation is based on two criteria: geometric and dynamic. The variable $Open_A/B$ represents discrete confirmation from end-stop or open-position sensors. However, geometric confirmation alone is insufficient. The variable $Stable_A/B$

represents a dynamic verification criterion, defined as angular rate below a specified threshold for a minimum dwell time after reaching the open position. This prevents rebound or oscillatory motion from being misinterpreted as stable deployment. The state *OPEN & STABLE* is reached only when both geometric and dynamic criteria are satisfied. The conditions *open_disagree_A/B* denote inconsistencies between mechanical position and sensor readings, while *rebound_A/B* captures transient re-closure tendencies detected after initial opening. Timeout parameters ($T_{unlatch}$, T_{deploy} , T_{open_max}) bound the duration of each phase and enable deterministic fault detection. Door opening is a hard prerequisite for relay release. The supervisory logic authorizes the next event only when both leaves report *OPEN & STABLE*.

15.2.4 Actuation and Mechanical Implementation

Each leaf is retained in the closed configuration through dedicated latching mechanisms actuated by pin-puller devices. The latches provide positive mechanical locking under launch loads and do not rely on sustained electrical power. Their function is purely retention: they prevent door opening until an explicit command is issued. Independent actuation per leaf ensures functional segregation and reduces the probability of common-mode failure. Upon reception of an open command, each leaf transitions from the *CLOSED & LATCHED* state to *UNLATCHING*. The variable *LatchOpen_A/B* represents discrete confirmation from latch position sensors indicating successful disengagement. The condition *no_motion_A/B* denotes absence of expected state transition within a predefined unlatch timeout window, triggering a fault condition.

Once unlatched, deployment is driven by preloaded torsion or extension springs integrated within the hinge assemblies. These springs provide controlled opening torque sufficient to overcome hinge friction and minor preload effects. Their function is independent from the latch chain, ensuring that separation of retention and deployment energy remains clearly decoupled.

Hard-stops define the fully open position, while damping elements control the terminal phase of deployment. The damping system limits end-stop impact loads, reduces oscillations, and prevents rebound that could otherwise lead to ambiguous open states.

15.2.5 Architectural Trade-Off

Alternative concepts were evaluated at architectural level:

- **Single monolithic door.** This configuration would simplify the latch chain; however, it would increase panel inertia and hinge torque demand, and would be more sensitive to hinge misalignment. Deployment-induced disturbance torque would consequently increase, and structural reinforcement at the hinge line would become more demanding due to the larger continuous load path.
- **Laterally translating panel with guide rails.** This concept would provide controlled linear kinematics during opening, but at the cost of higher mechanical complexity and increased sensitivity to alignment and tolerance stack-up. In addition, the same concept was evaluated for an alternative side panel of the bay to enable payload release along the *X* or *Y* direction. However, lateral extraction would introduce asymmetric disturbance forces with respect to the spacecraft center of mass and would reduce the effective internal accommodation area available for payload.

The dual-leaf rotating configuration provides a balanced compromise mainly between dynamic symmetry and packaging efficiency, while also remaining fully compatible with the relay support frame previously illustrated.

15.2.6 Functional Responsibility of Main Components

Each component of the door mechanism has a clearly defined role. The latches ensure retention under launch loads. The pin-pullers provide constraint removal without structural load transfer. The hinge assemblies define rotational guidance and maintain alignment throughout deployment. The deployment springs provide controlled opening energy independent of the latch chain. Hard-stops guarantee deterministic end-of-travel geometry. Damping elements control terminal dynamics and mitigate rebound. Position sensors provide geometric confirmation and dynamic stability criteria such that deployment is not only achieved but stabilized before authorizing subsequent events.

15.2.7 Maturity Assessment

The dual-leaf door architecture relies on flight-proven mechanical elements including hinge assemblies, spring-driven deployment, pin-puller latch release, hard-stops, damping devices and discrete sensors. However, the integrated configuration tailored to this mission remains at architectural definition level. In line with Phase A scope, detailed hinge sizing, spring and damping characterization, tolerance analysis and structural verification are deferred to subsequent phases. Thus, also in this case the whole mechanism is conservatively assessed at TRL 3, pending integrated breadboard validation and representative dynamic deployment testing.

15.3 CPDI + PAI (Motorized Lightband) – Mission Service Interface and Payload Release

The CPDI and PAI concepts are introduced to support a *mission service* approach in which the carrier platform is kept as *frozen* as possible, while payload-specific adaptations are confined to a well-bounded accommodation layer.

15.3.1 Architectural role of CPDI and standardization strategy

The CPDI (Carrier–Payload Accommodation Interface) acts as the *single* mechanical/structural and routing interface between the carrier and the payload accommodation hardware. In the baseline configuration, a *single* PAI is installed on the CPDI and drives the sizing mainly in terms of envelope, local stiffness and load introduction. This baseline is shown in Figure 83: the CPDI provides the mounting plane and the structural continuity towards the carrier, while the PAI represents the standardized separation interface towards the payload. The same architecture is scalable to a *multi-payload* configuration, Figure 84, by adopting a CPDI plate customized at mission level (pattern, spacing and local reinforcement), while keeping the PAI as a standardized building block, only scaled in diameter and preloads to match payload class. This split is intentional:

- **Carrier frozen:** the service platform does not change when moving from single to multi-payload mission; only the CPDI accommodation pattern evolves.
- **PAI standardized:** a repeatable interface simplifies integration flow, verification logic, GSE procedures and reduces mission-to-mission redesign.
- **Modularity in late integration:** payloads can be integrated in late phases without re-opening carrier design loops, because the mechanical interface and command logic remain consistent.

15.3.2 PAI mechanism: Non-Explosive Motorized Lightband

The PAI is implemented as a non-explosive motorized lightband release mechanism. It is an *active* separation device: mechanical retention is positive and power-independent in flight, while release is

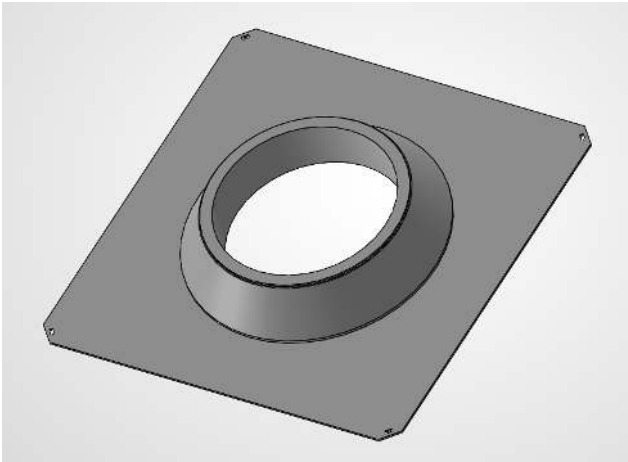


Figure 83: CPDI with single PAI (baseline configuration). The CPDI provides the carrier-side accommodation plane; the PAI defines the standardized separation interface towards the payload.

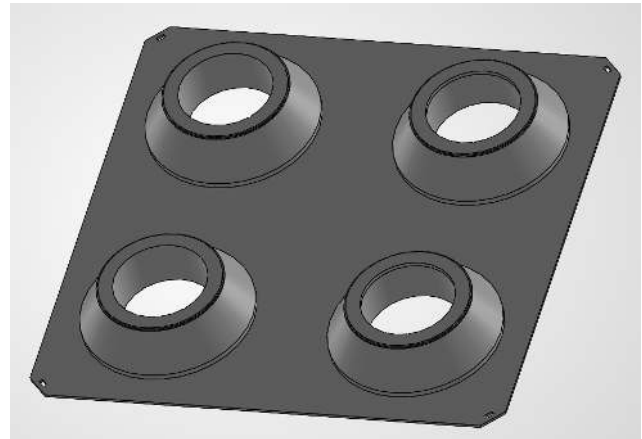


Figure 84: CPDI with multiple PAIs. The CPDI pattern is mission-tailored, while each PAI remains a standardized module, enabling modular payload integration without modifying the carrier platform.

commanded and monitored. Figure 85 captures the functional layout and the key moving parts. The mechanism can be decomposed into eight main elements, each with a single clear responsibility:

1. **Bottom frame:** carrier-side structural base; houses the engagement leaves and provides local stiffness for load introduction.
2. **Separation springs:** store the energy required to generate the separation “kick”; they provide the post-release impulse and are not meant to act as a latch.
3. **Engagement leaves (circular set):** the actual load-carrying retention feature; they latch into the upper ring when held by the retaining ring.
4. **Spring plungers:** provide a deterministic *unlatching* action by pushing the leaves inward once the retaining ring is removed; they ensure leaf disengagement is not left to friction or elastic relaxation.
5. **Motorized release mechanism:** converts electrical command into controlled mechanical motion; its responsibility is constraint removal (not separation energy).
6. **Sliding tube:** kinematic transmission element that drives the internal retraction sequence; it ensures repeatable motion path and end-of-travel definition.
7. **Retaining compression ring:** keeps the engagement leaves expanded and latched during launch; once retracted, it releases the leaves.
8. **Upper ring:** payload-side interface; during integration it compresses the separation springs and is locked by the leaf set; at release it becomes free to translate along the separation axis.

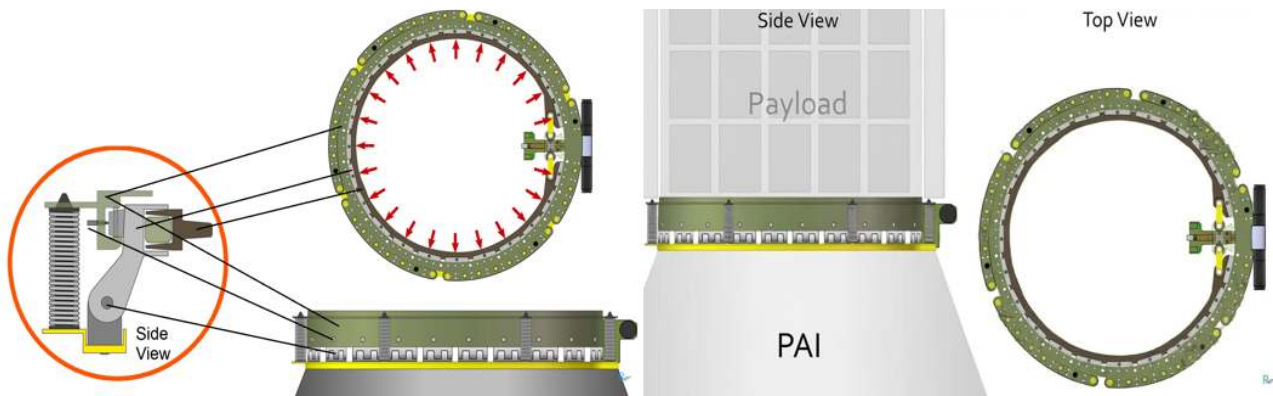


Figure 85: Motorized lightband PAI: functional layout of the upper ring (payload-side), bottom frame (carrier-side), engagement leaves and retaining ring. Release is achieved by retracting the retaining ring, enabling leaf collapse via spring plungers and freeing the upper ring for separation.

Functional sequence (mechanical). During integration, the upper ring is lowered onto the bottom frame, compressing the separation springs until it covers the leaf set. In flight configuration, the retaining ring is positioned so that it forces the leaves to remain engaged into the upper ring, simultaneously preloading the spring plungers. Upon a release command, the motor retracts the sliding tube, which pulls back the retaining ring. Once the retaining ring no longer supports the leaves, the plungers drive the leaves inward, disengaging the latch. At this point the upper ring is mechanically free and the separation springs provide the separation impulse along the nominal axis.

15.3.3 Release command and safety logic: finite-state model

Because the PAI is an active, commanded and safety-critical mechanism, its operation is best represented as a finite-state machine (FSM). The FSM does not model the physics of the separation itself; it makes explicit the *command gating*, the *safety interlocks*, and the *fault management* needed to avoid ambiguous states. The high-level FSM used to formalize these behaviors is reported in Figure 86. The mechanism starts in *SAFE/DISARMED*, in which release is inhibited both logically and at power level (launch, early orbit, and ground phases). A transition to *ARMED* is permitted only after an explicit arm command and verification of system interlocks (spacecraft mode, ADCS stability, absence of FDIR inhibits, power margin and time/epoch constraints). In *ARMED* the mechanism is enabled, but the payload remains fully locked mechanically. Upon *RELEASE_CMD*, the system enters *RELEASE IN PROGRESS*: the motor is energized and the retaining-ring retraction sequence is executed under time and current monitoring. A key safety concept is the *pre-unlatch* region: an abort is only guaranteed before the mechanism crosses the irreversibility threshold (point of no return), because once the disengagement process has started the system cannot guarantee to re-establish a fully locked state. If the end-of-travel is reached and the release signature is consistent, the FSM transitions to *RELEASED/SEPARATED* (latched) and the motor is disabled; the subsequent physical departure is provided by the separation springs. If timeout, overcurrent or no-motion conditions occur during the active phase, the FSM enters *FAULT* (latched) and remains in a fail-safe configuration requiring ground action.

15.3.4 Clearance analysis on separation impulse and spring sizing traceability

The release design is closed by linking the commanded release to the separation dynamics requirement inside the bay. A conservative clearance analysis was adopted: after constraint removal, the payload is treated as a rigid body evolving in torque-free motion, with the translation driven by the

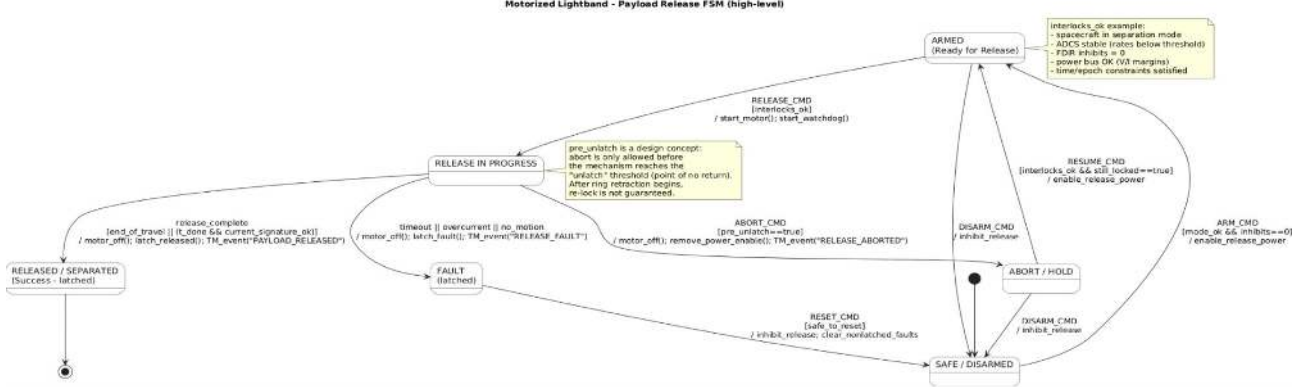


Figure 86: Motorized lightband PAI – high-level release FSM. The model separates SAFE/DISARMED, ARMED and RELEASE IN PROGRESS states, and makes explicit the time-bounded monitoring, fault latching, and the pre-unlatch point of no return (abort permitted only before irreversibility).

initial separation impulse and the attitude evolving freely with constant angular momentum. The bay geometry includes the real lateral walls and the 45° chamfer, and contact is checked only while the payload is within the wall height (i.e., while it is still geometrically confined). Non-idealities are represented through two worst-case drivers: a lateral eccentricity e between the payload centre of mass and the effective line of action of the separation impulse, and (ii) a severity factor α that scales how much of the available impulse is converted into an initial rotational effect (representing spring mismatch, release asynchrony, friction dispersion and local compliance). In this model, e defines the lever arm, while α defines the fraction of “kick” effectively producing angular momentum. Sweeping across $(v_{0z}, v_{0x}, v_{0y}, e, \alpha)$ and selecting the worst direction among $\pm X$ and $\pm Y$ yields a conservative minimum clearance curve. The key outcome is a traceable release requirement: to guarantee a minimum worst-case clearance of 20 mm with no impacts, the separation velocity shall satisfy

$$v_{0z} \geq 1.20 \text{ m/s},$$

while the time-to-exit constraint is already met for $v_{0z} \geq 1.0 \text{ m/s}$. The clearance-driven value is therefore the binding requirement. This requirement is translated into spring sizing by equating the required kinetic energy to the energy stored in the separation springs, including an efficiency factor η accounting for losses (friction, compliance and non-ideal energy transfer). For $N_s = 6$ springs and $\eta = 0.8$, the analysis yields a required total spring energy of approximately 900 J to guarantee $v_{0z} \geq 1.20 \text{ m/s}$. Selecting a release stroke $x_0 = 40 \text{ mm}$ leads to a stiffness requirement of the order of $k \simeq 2.3 \times 10^5 \text{ N/m}$ per spring, corresponding to a release force of about 9.2 kN per spring (total $\sim 55 \text{ kN}$). It is emphasized that the numerical sizing presented above (spring stiffness, preload and resulting release force) refers to the PHAROS single-payload configuration, which drives the current baseline design case in terms of mass and bay geometry. For multi-payload configurations, the required separation velocity and corresponding spring energy scale with payload mass, inertia properties and accommodation geometry. While the PAI architecture remains standardized, the separation spring set (stiffness and preload) is mission-tailored in order to satisfy the same clearance and time-to-exit criteria under the relevant payload class. Depending on the payload class and the corresponding sizing of the separation system, the release velocity along the nominal separation axis is defined within an operational range that can reach up to approximately 2 m/s as a maximum design value.

15.3.5 Separation Dynamics Maturity Roadmap

The clearance-driven requirement derived above corresponds to a Level 1 conservative rigid-body analysis, in which the separation impulse is modeled as an initial condition and the subsequent motion evolves in torque-free dynamics. In subsequent development phases, the separation model will be progressively refined as summarized in Figure 87.

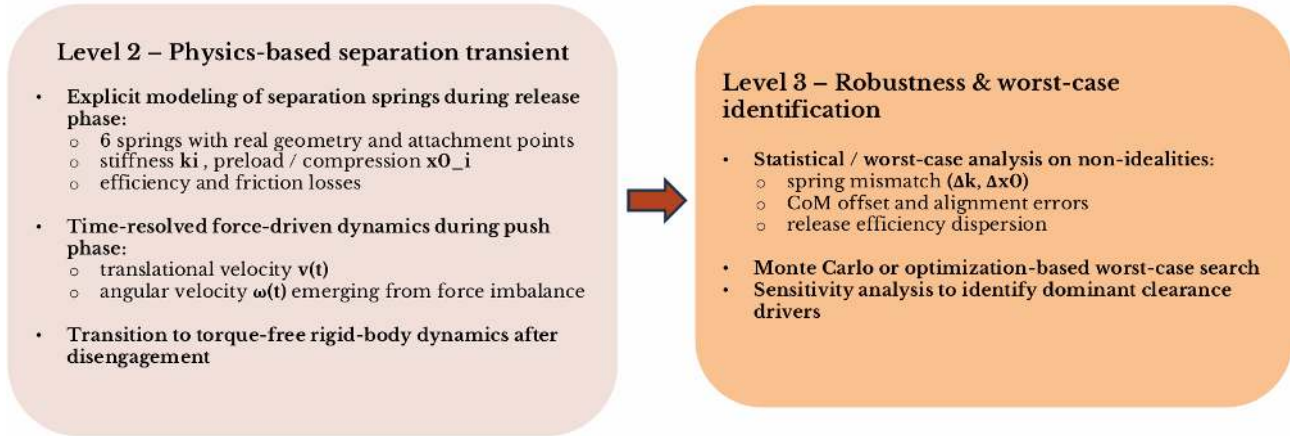


Figure 87: Separation dynamics refinement roadmap. Level 2 introduces explicit spring force modeling during the push phase, while Level 3 addresses statistical robustness and worst-case identification.

Level 2 introduces an explicit physics-based modeling of the release transient, including the six separation springs with their real attachment points, stiffness dispersion, preload $x_{0,i}$ and efficiency losses. The force-driven push phase is resolved in time, allowing translational velocity $v(t)$ and angular velocity $\omega(t)$ to emerge from force imbalance prior to full disengagement, and explicitly capturing the transition from constrained motion to torque-free rigid-body dynamics.

Level 3 addresses robustness and worst-case identification by introducing statistical dispersion and non-idealities, including spring mismatch $(\Delta k, \Delta x_0)$, CoM offset uncertainty, alignment errors and release efficiency variability. Monte Carlo or optimization-based searches are then used to identify dominant clearance drivers and verify margin robustness.

16 Configuration

16.1 MANTIS general configuration and constraints

Configuration constraints The MANTIS shape and configuration are derived from the available volume of the selected launcher and the specific dimensions of the payload. Both the primary and backup launch vehicles feature an extended fairing, characterized by a height of 18 m and a base diameter of 4.6 m. The MANTIS mission is also designed to transport large spacecraft. Consequently, the available payload volume must be maximized to meet this requirement.

Configuration Trade-offs Several internal and external configurations were evaluated to define the final design, with the primary trade-off focused on the accommodation of the spacecraft payloads. The first option considered involved integrating a mounting platform on top of the carrier bus, enabling the payloads to be stacked vertically. Taking the ION Satellite Carrier by D-Orbit ^[113] as a reference, this configuration provides significant flexibility, as it allows for the mounting of either a single large spacecraft or the partitioning of the available base area to accommodate multiple smaller payloads. An alternative option involved mounting the payloads around a common central column, following the design logic of the Rideshare payload adapters used in Falcon launch vehicles ^[114]. The output of this trade-off led to the selection of the first option; the second configuration was discarded as it would not satisfy the requirements for mounting a single large spacecraft, both in terms of available volume and the maintenance of necessary mass symmetry. Starting from the payload accommodation configuration, a second trade-off was conducted to define the geometry of the Carrier Bus. As the core structural element of the mission, the Bus integrates all critical subsystems required to safely deliver the spacecraft to the Martian environment and manage transitions between different orbital bands. The configuration was derived from the available envelope within the launcher fairing. To accommodate large payloads whose base dimensions closely match the fairing's internal diameter, and to ensure an efficient partitioning of the mounting surface for multi-payload scenarios, the Bus module features a square footprint with chamfered corners. This geometry results in a maximum dimension of 4.48 m, optimizing the use of the available launch volume. The final configuration trade-off concerns the integration of the Relay spacecraft. As the Relay is the first satellite scheduled for release upon Mars arrival, its placement is critical. Furthermore, the Relay features a large HGA with a 2.5 m diameter; consequently, it cannot be hosted on the same mounting platform as the other payloads, as it would occupy most of the available volume. This led to the selection of the final configuration, which involves mounting the Relay on a dedicated interface above the payloads. Beyond solving volume constraints, this "stacked" approach allows the Carrier to exploit the Relay's communication subsystem for data transmission during the interplanetary transfer phase.

Payload Integration and Protection As previously noted, payloads are integrated onto the top deck of the Carrier Bus via a CPDI and a customized PAI, which houses all necessary interfaces and separation mechanisms. To ensure environmental protection, the payloads are enclosed within a dedicated bay secured by two bay doors. These doors feature an additional interface and release mechanism specifically for the attachment of the Relay spacecraft.

This architectural layout, characterized by the sequential integration of all elements along the launcher's longitudinal axis, is referred to as **Stacked Configuration**.

16.2 Carrier external configuration

The Carrier configuration changes during the mission lifetime as showed in Figure 92.

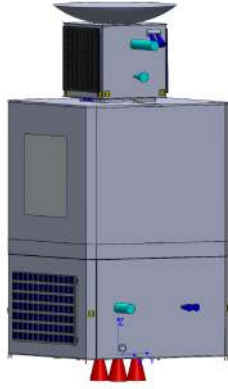


Figure 88: (a) Stowed configuration.

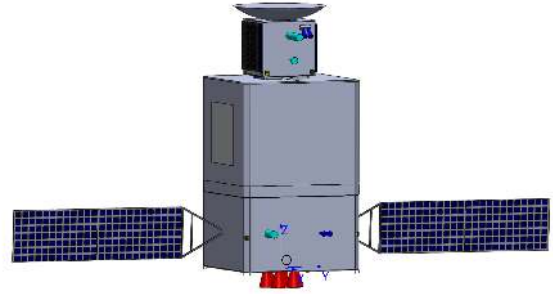


Figure 89: (b) Interplanetary configuration.

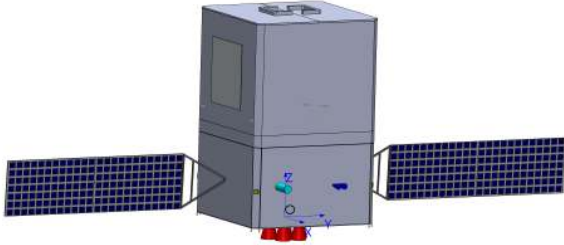


Figure 90: (c) Post-relay release.

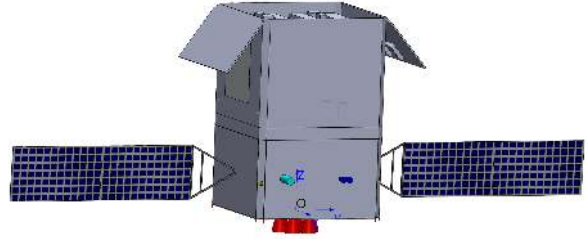


Figure 91: (d) Final release configuration.

Figure 92: MANTIS Spacecraft configurations during different mission phases.

Throughout the mission phases, the Carrier sequentially releases the Relay and the payloads, resulting in significant mass variations. The stacked configuration was specifically designed to maintain maximum mass symmetry at every stage. The only exception occurs during the payload release phase, where symmetry is momentarily reduced; however, payload configuration will be imposed such that the inertia matrix remains well within the operational limits defined by the ADCS subsystem. The complete external configuration of the Carrier is illustrated in Figure 93. Each component is strategically positioned to satisfy specific subsystem requirements. In particular, optical components are oriented to ensure an unobstructed FOV, while the helix antenna is located on the $+X$ face to align with the communication pointing constraints defined by the ADCS subsystem. Furthermore, the primary and secondary thrusters are mounted on the $-Z$ face to align the thrust vector with the spacecraft's CoG, minimizing parasitic torques. The ADCS thrusters are positioned at the corners of the structure to maximize their moment arm, as required by the control system specifications. Finally, the solar panels are mounted symmetrically along the Y .

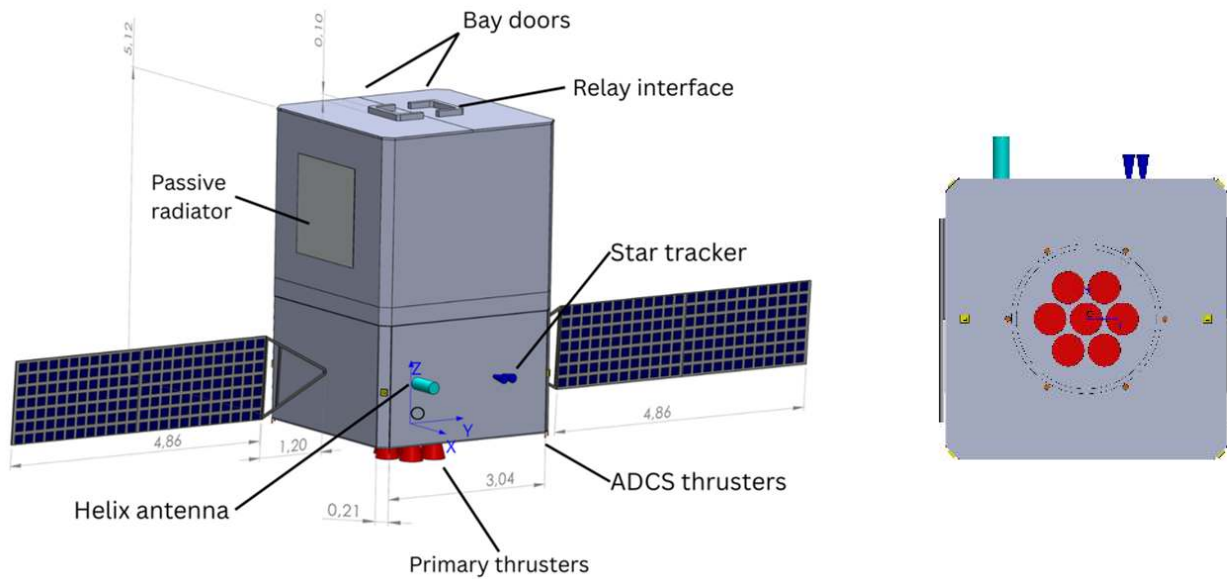


Figure 93: Carrier external configuration and characteristic dimensions in meters.

16.3 Carrier internal configuration

The internal layout of the Carrier is driven by the integration requirements of the various subsystems while maintaining necessary mass symmetry. The configuration is shown in Figure 94. The internal volume of the Bus is mainly occupied by the PS tanks, arranged in a symmetric configuration and polarly mounted by isostatic mounts, which are not represented in the CAD model for clarity.

The RW are integrated onto the upper deck, while the IMU are positioned near the Z-axis to ensure measurement accuracy. To meet thermal requirements, electronic components are mounted beneath the upper deck to remain within their operational temperature ranges. Finally, the payload bay houses all payloads and the cabling required for the relay interface, maintaining a minimum clearance of 20 cm from the walls and a 40 cm separation between individual payloads.

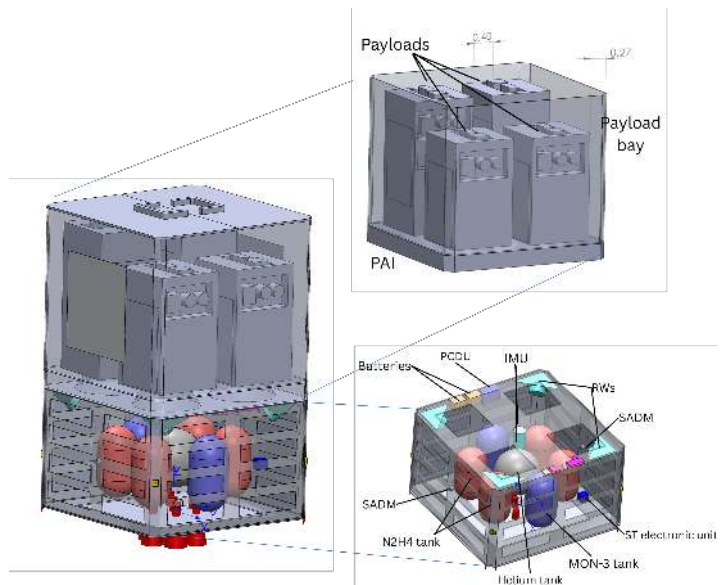


Figure 94: Carrier internal configuration, with a focus on the Carrier Bus and Payload Bay.

16.4 Relay internal and external configuration

The internal and external configuration of the Relay does not represent a critical aspect of the MANTIS mission. As reported in Figure 95 and Figure 96, its components mainly consist of TTMC and OBDH units. The external layout features three antennas: an HGA mounted on the top panel, a choice made to comply with the launcher fairing's maximum volume, and both a helical and a horn antenna, both mounted on the +X face. The OBDH electronics are located on the top deck, directly under the HGA, to minimize cable length and exploit the shadowing of the parabolic element. To comply with the TCS, the batteries and PCDU are mounted on the -X face in correspondence with a radiator. All other internal and external components are mounted following the same logic used for the Carrier satellite.

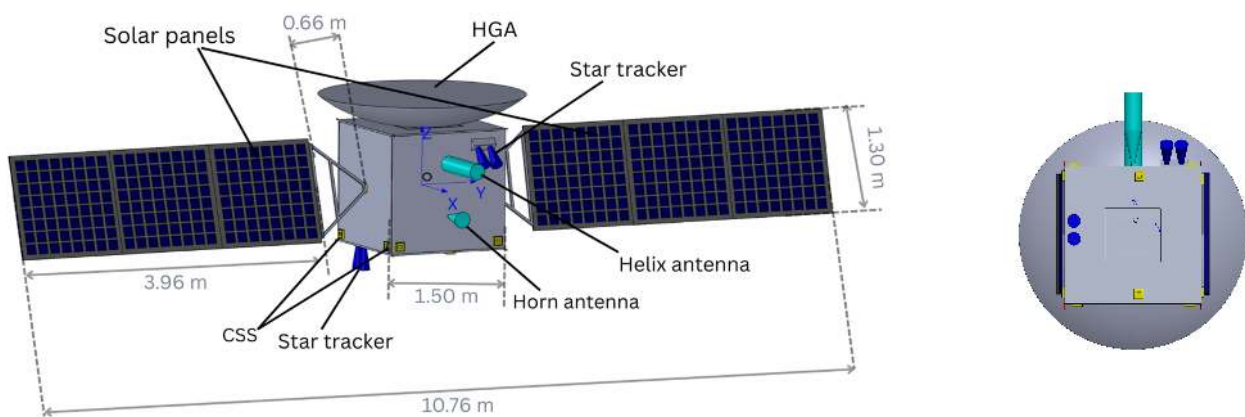


Figure 95: External and bottom configurations of the Relay spacecraft, showing antenna placement and interface details.

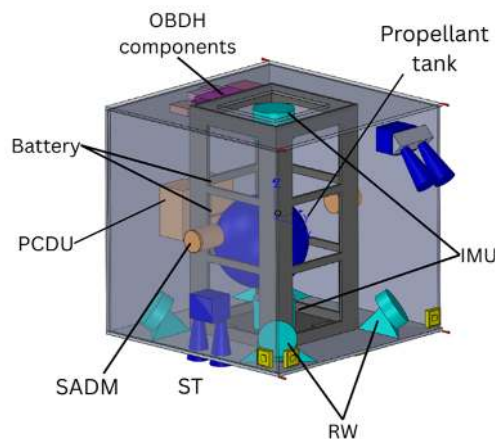


Figure 96: Relay internal configuration.

17 Structures

This section presents the structural design of the MANTIS stacked configuration primary structure. Within this chapter, the structure analyzed is the fully integrated spacecraft, comprising the Carrier, payloads, and the Relay module. The design process begins with the selection of materials, followed by structural verification and testing performed using ANSYS.

17.1 Material selection

The most critical requirement for the MANTIS structure is to remain lightweight while guaranteeing high performance. Given the large scale of the spacecraft, using the wrong materials would lead to an unfeasible increase in structural mass or the failure of key elements, such as the bay doors, the CPDI, or the internal structure. The primary structure of the spacecraft is made by a sandwich panel made by CFRP (Carbon Fiber Reinforced Polymer) face sheets and an Aluminum honeycomb core. As mentioned previously, this technology allows for a significant weight reduction compared to aluminum plates, while providing high stiffness and strong resistance to buckling. On the other hand, sandwich structures are sensitive to local failure modes due to the presence of the adhesive layer, and their manufacturing and production processes are more complex. The selected components of the sandwich structure are:

- **Aluminum honeycomb core with hexagonal cell geometry:** Provides high shear stiffness and shear strength, good compressive strength, and very low effective density, making it ideal for lightweight structural applications. In addition, aluminum honeycomb cores exhibit excellent out-of-plane stiffness, good energy absorption capability, and high thermal resistance, which makes them suitable for space environments and high-temperature operating conditions.
- **CFRP (Carbon Fiber Reinforced Plastic) face sheets:** Primarily responsible for carrying axial loads and bending moments, as well as in-plane shear loads. CFRP face sheets offer a very high stiffness-to-weight and strength-to-weight ratio, good fatigue resistance, and low coefficient of thermal expansion, contributing to the overall structural efficiency and dimensional stability of the sandwich panel.
- **Adhesive layer:** Ensures effective load transfer between the face sheets and the core by transmitting shear stresses across the interfaces. The adhesive is selected to provide high peel strength, adequate fracture toughness, and low creep under sustained loads. Additionally, it must exhibit good environmental resistance (e.g., to temperature variations and moisture) to guarantee long-term structural integrity.

The selection of this material is further justified by its flight heritage in missions operating in the Martian environment. In particular, the primary bus of the Mars Reconnaissance Orbiter (MRO) was manufactured using a sandwich structure composed of an aluminum honeycomb core and composite face sheets. To perform the next analysis, an equivalent material is defined. Even if a global response can be estimated thanks to the orthotropic equivalent material, local failures are neglected, as well as the non-uniform stress distribution inside the panel. The panel equivalent properties are reported in Table 42, where each property is computed as a weighted average over the volume, and tuned to be compatible with the vibration modes of the selected launchers.

Parameter	Description	Value	Unit
<i>Material and Geometric Inputs for the single material</i>			
t_f	Facesheet thickness (CFRP)	0.0005	m
h_c	Core thickness	0.02	m
E_f	Young's modulus (CFRP)	2.9×10^{11}	Pa
G_c	Core shear modulus	3.7×10^7	Pa
ρ_f	CFRP density	1600	kg/m ³
ρ_c	Core density	48	kg/m ³
<i>Equivalent Homogenized Properties for each panel</i>			
h_{tot}	Total panel thickness	0.021	m
ρ_{eq}	Equivalent density	144.14	kg/m ³
E_x	Equivalent Young's moduli	2.9×10^{11}	Pa
E_y	Equivalent Young's moduli	9.45×10^{11}	Pa
E_z	Equivalent Young's moduli	2.55×10^8	Pa
G_{xy}	In-plane shear modulus	5.5×10^9	Pa
G_{xz}, G_{yz}	Out-of-plane shear moduli	3.7×10^9	Pa
ν_{xy}	Equivalent Poisson's ratio	0.27	—
ν_{yz}, ν_{xz}	Equivalent Poisson's ratio	0.001	—

Table 42: Geometric and Mechanical Properties of the Sandwich Panel.

17.2 Carrier and Relay internal structure

The internal structures of the Carrier Bus and Relay, Figure 97 and Figure 98, are designed as a lightweight frame that serves as the spacecraft's primary skeleton. Internal equipment is integrated onto these structures either directly or via isostatic mounts. To minimize structural mass and avoid excessive vibrations or displacements along the longitudinal axis, the design omits horizontal panels or intermediate mounting planes, except for primary elements, such as the CPDI or the Bay doors, relying instead on a streamlined load path.

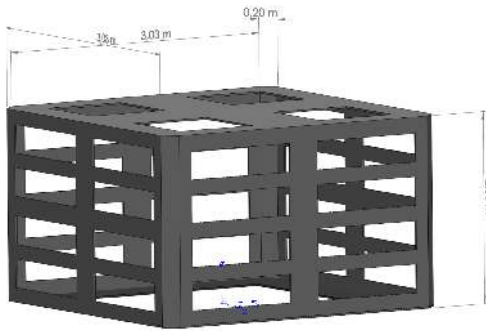


Figure 97: Internal structural skeleton of the Carrier Bus.

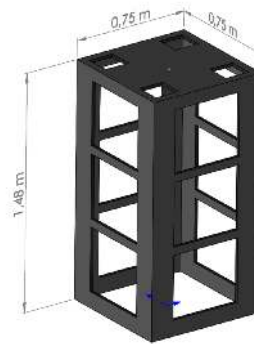


Figure 98: Internal structural skeleton of the Relay spacecraft.

17.3 Structural analysis

The structural analysis is conducted based on the qualification flight loads specified by both the primary and backup launch vehicles. The spacecraft must be qualified to withstand the environment experienced inside the fairing during every mission phase. This qualification is performed by applying the qualification factors provided in the Launcher User Manual (LUM) to the expected environmental loads, ensuring the necessary safety margins are met. The first step in the spacecraft qualification process involves developing the numerical model in Ansys and generating the mesh for Finite Element Method (FEM) analysis. The model focuses exclusively on the primary structure, which consists of the external panels and the internal framework. Once the mesh is finalized, non-structural components are modeled as either point masses or distributed loads on the relevant surfaces. These applied masses include the necessary design margins, ensuring that the total model mass aligns with the launch wet mass specified in the mass budget.

17.3.1 Modal Analysis

A modal analysis is performed on the integrated structure, computing the first 60 modes to ensure that the cumulative effective mass would be $\geq 90\%$. From these results, the primary longitudinal and lateral modes are identified by selecting those with the highest effective mass participation along their respective axes. The output of the analysis and the compliance with the launcher are reported in Table 43.

Modal Parameter	Launcher Requirement	Actual Frequency
First lateral frequency-x	≥ 6 Hz	22.35 Hz
First lateral frequency-y	≥ 6 Hz	29.08 Hz
First longitudinal frequency	≥ 20 Hz	90.98 Hz

Table 43: Output of modal analysis and compliance with launcher requirements [1].

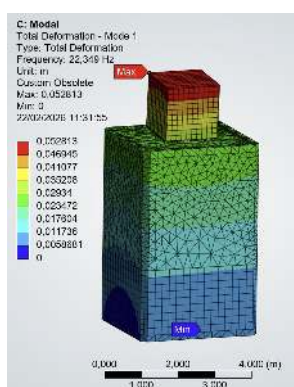


Figure 100: First lateral mode (X-axis).

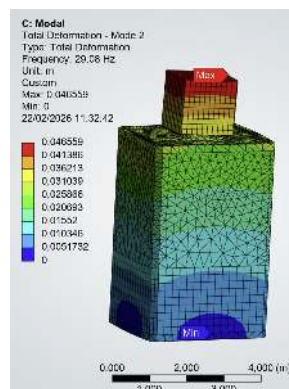


Figure 101: First lateral mode (Y-axis).

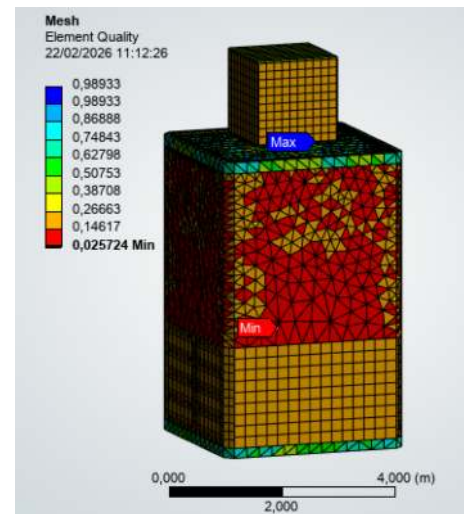


Figure 99: Finite Element Model (FEM) mesh of the primary MANTIS structure.

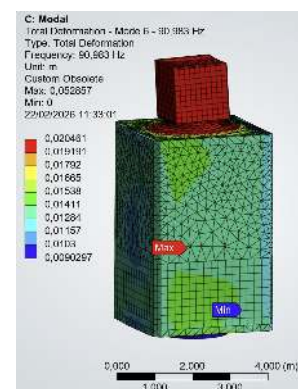


Figure 102: First longitudinal mode (Z-axis).

17.3.2 Quasi Static analysis

The static analysis is computed by applying separately three accelerations which are the maximum qualification loads on each axis comparing the LUM of Ariane 64 and Falcon Heavy. The qualification load is defined as the Quasi Static load multiplied by a 1.25 qualification factor.

Load Direction	Load Magnitude	Maximum Stress	Maximum Displacement
Lateral (X)	± 2.5 g	31.64 MPa	1.36 mm
Lateral (Y)	± 2.5 g	32.06 MPa	2.47 mm
Longitudinal (Z)	± 7.5 g	22.38 MPa	0.49 mm

Table 44: Quasi-Static test results: comprehensive stress and displacement data for all axes.

17.3.3 Sine Analysis

As already explained for the Quasi Static Analysis, the Sine test is carried out by setting an Harmonic Response test case in ANSYS. The load is set as an acceleration on the respective axes coupled with the given frequency band, both extracted by the LUM and then marginated.

Frequency	Load Direction	Load Magnitude	Maximum Stress	Maximum Displacement
2 – 25 Hz	Lateral (X)	1.25 g	11.50 MPa	12.63 mm
2 – 25 Hz	Lateral (Y)	1.25 g	124.68 MPa	11.63 mm
2 – 50 Hz	Longitudinal (Z)	1.25 g	5.50 MPa	190.98 mm

Table 45: Sine vibration test results: comprehensive stress and displacement data for all axes.

While the Y-axis exhibits the highest stress (124.68 MPa) due to greater lateral flexibility, the highest displacement is observed along the longitudinal axis (190.98 mm), indicating a local resonance that should be mitigated by structural damping from mechanical joints, which is not accounted for in this simulation.

17.3.4 Random Analysis

To perform the random analysis, the data from Falcon Heavy^[53] are extracted in order to evaluate the random response given the APSD (Acceleration Power Spectral Density). The random noise is then applied with the same intensity on each axis.

The random profile is applied in three orthogonal directions (X, Y, Z) to simulate the broadband vibration environment experienced during launch, particularly relevant for acoustic and engine-induced excitations. This analysis is crucial for qualifying the spacecraft against fatigue and potential random vibration failures.

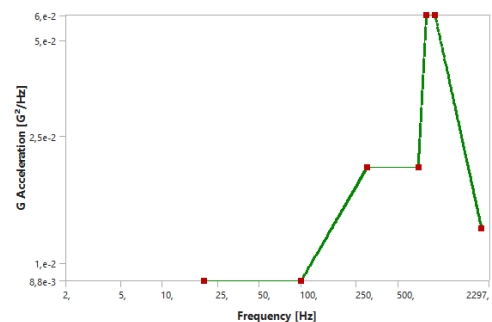


Figure 103: APSD for random analysis.

Load Direction	Statistical Scale	Maximum Stress	Maximum Displacement
Random (X)	3 Sigma (99.73%)	110.34 MPa	8.52 mm
Random (Y)	3 Sigma (99.73%)	220.25 MPa	13.85 mm
Random (Z)	3 Sigma (99.73%)	110.34 MPa	2.46 mm

Table 46: Random vibration analysis results: 3-Sigma peak stresses and displacements.

17.3.5 Analysis Output

To evaluate the results obtained with the structural analysis, the Margin of Safety (MoS) need to be computed. The MoS is a quantitative indicator to determine how much the actual load on the structure is below its maximum allowable limit given by the material properties, after accounting for mandatory safety factors. For composite structures, such as the sandwich panels employed in this design, only the Ultimate MoS is taken into account. This is due to the fact that composite materials do not exhibit a distinct yield point; therefore, a Yield MoS is not applicable. MoS are computed through the following formula:

$$MoS_{ult} = \frac{\sigma_{ultimate}}{\sigma_{actual} \cdot FS} - 1 \quad (2)$$

$\sigma_{ultimate}$ is the ultimate allowable stress material, σ_{actual} is the maximum equivalent stress derived from FEM analysis and FS is the Factor of Safety^[115], which is the products of all margins applied, more in particular is equal to:

$$FS = K_m \cdot K_p \cdot FoS_u = 1.2 \cdot 1.15 \cdot 1.25 = 1.725 \quad (3)$$

Since for this analysis strong assumptions have been made modeling the material as an homogeneous panel the MoS_{ult} is computed both for the Aluminum Honeycomb ($\sigma_{ult}^{Al \text{ Honeycomb}} = 3.59 \text{ MPa}$), and for an UD (Unidirection) CFRP material ($\sigma_{ult}^{UD \text{ CFRP}} = 676 \text{ MPa}$). The stress used to compute the margins is the highest reached during each tests, the results are reported below

Test Case	Max. Stress	MoS _{ult} (Al Honeycomb)	MoS _{ult} (UD CFRP)
Quasi-Static (Y)	32.06 MPa	-0.935	11.223
Sine (Y)	124.68 MPa	-0.983	2.143
Random (Y)	220.25 MPa	-0.991	0.779

Table 47: Summary of Maximum Stresses and Margins of Safety for specific material.

The results are consistent with the initial design assumptions. Since the sandwich panel's properties depend on the interaction between the face sheets and the internal core, the actual structural behavior is expected to fall between the two calculated Margins of Safety. While these discrete results do not provide a singular conclusion on total structural integrity, the potential failure of the honeycomb core is likely mitigated by the superior mechanical performance and load-bearing capacity of the composite skins, which alleviate the stress concentration on the internal core.

17.4 Possible Further Analysis

To finalize the structural qualification and provide a definitive numerical proof of MANTIS's integrity, several critical analyses must be undertaken. A high-fidelity model of the sandwich structure is essential to accurately characterize the interaction between the constituents and determine the optimal composition of the layers. Following the refinement of the material model, the structural verification must be expanded to include acoustic load analysis, buckling analysis, and shock analysis. Furthermore, a Coupled Load Analysis (CLA) is required to investigate the dynamic interactions between MANTIS, the launch vehicle, and the integrated payloads. Upon completion of these simulations, a set of mandatory structural requirements can be established to guide the client spacecraft's structural design and testing protocols, ensuring mission safety and compatibility.

18 PHAROS-M Interface

As previously noted, the first launch is planned to carry the twin-spacecraft PHAROS-M mission. Several interfaces were conducted between the two teams to ensure full compatibility.

Design Constraints and Adopted Solutions: Some aspects of the PHAROS-M design were adopted as constraints for the MANTIS design to ensure interface compatibility and overall mission compliance.

- The final PHAROS-M **dimensions** of $3.10\text{ m} \times 3.10\text{ m} \times 2.79\text{ m}$ were adopted to size the payload bay. The stacked configuration was also driven by the spacecraft's large envelope.
- The nominal PHAROS-M **orbit** of $400\text{ km} \times 400\text{ km}$ altitude did not directly constrain the mission analysis, since aerobraking enables access to a wide range of altitudes. However, the target inclination of 90° drove the pilot launch trajectory design.
- The preliminary maximum **data rate** of 0.9 Mbps defined the antenna sizing and link budget.
- The maximum pre-release **survival power** requirement of 150 W was adopted as the allocated payload power during the transfer phase.

Constraints for PHAROS-M : On the contrary, MANTIS imposed several requirements on PHAROS-M to ensure proper interface compatibility.

- The total allocated payload mass was set to 900 kg, defining the maximum allowable mass for the PHAROS-M spacecraft. This requirement is currently not fully met by the twin mission and remains an open issue to be resolved.
- The UHF band was selected for the proximity link between relay and payload. Consequently, PHAROS-M must adopt a compliant TTMTTC architecture, including a coherent transponder and RHCP.
- Structural requirements were defined to ensure adequate separation of the spacecraft natural frequencies and avoid dynamic coupling.

19 Mission Budgets

19.1 Power Budget

Figure 104 and Figure 105 present the overall power budgets for the Carrier and the Relay spacecraft, respectively. These charts illustrate the specific power consumption allocated to each subsystem across all mission phases, highlighting the variations in energy demand throughout the operational profile. All reported values inherently account for specific component level margin (ranging from 5 % to 20 %) as well as a 30 % system level margin according to the standards^[52]. The comprehensive tables, which include the specific power consumption of the individual components for each subsystem, are provided in Section E.1.

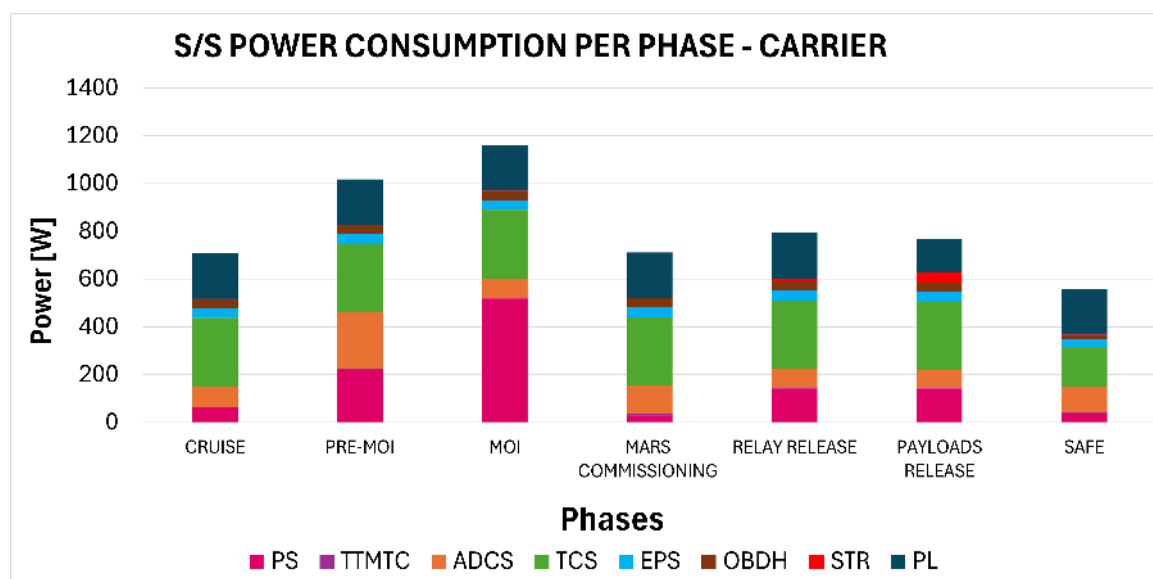


Figure 104: Subsystem Power Consumption per Phase (Carrier).

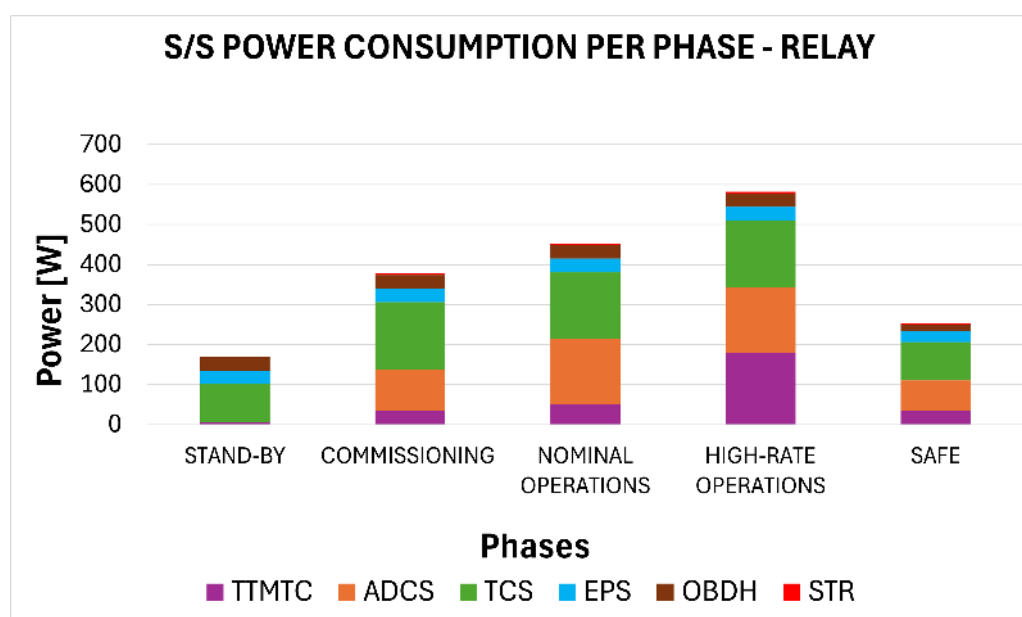


Figure 105: Subsystem Power Consumption per Phase (Relay).

19.2 Mass Budget

The mass budgets of the Relay and the Carrier are summarized in Table 48 and Table 49. Margins have been allocated at subsystem level according to the applicable maturity class, and an additional 30% system-level margin has been applied to the total dry mass in order to ensure compliance with Phase A uncertainty standards.

Subsystem	Mass [kg]	Margin	Margined Mass [kg]
PS	14.02	A	14.71
TTMTC	31.82	A/B, C	35.73
ADCS	43.95	A/B	46.15
TCS	6.13	A/B, C	6.45
EPS	89.37	C	98.31
OBDH	10.25	A	10.76
STR	61.54	D	73.85
MECH	9.16	B, C	10.15
Total Dry	296.11	30%	384.95
Propellant	-	-	69.03
Total Wet	-	-	453.98

Table 48: Relay Mass Budget.

Subsystem	Mass [kg]	Margin	Margined Mass [kg]
PS	263.13	A, D	283.42
TTMTC	12.22	A/B, C	12.88
ADCS	51.85	A/B	54.44
TCS	51.65	A/B	54.23
EPS	129.73	C	142.70
OBDH	1.55	A	1.63
STR	451.99	D	542.39
MECH	46.96	B, C	52.53
Total Dry	1144.22	30%	1487.49
Propellant	-	-	2496.84
Total Wet	-	-	3984.33

Table 49: Carrier Mass Budget.

	Mass [kg]
Relay + Carrier	4438.32
Payload	900
Adapter	80
Total Launch Mass	5418.32
Ariane 64 Launchable Mass	5550
Delta difference	131.68

Table 50: Comparison between system launch mass and Ariane 64 launchable mass.

Table 50 reports the overall launch mass. The total launch mass reaches 5418.24 kg. This value remains below the Ariane 64 launch capability of 5550 kg, leaving a positive margin of 131.76 kg. The

launcher compatibility requirement is therefore satisfied with residual mass allowance.

The complete mass budget breakdown, including all individual components within each subsystem, is reported in Section F.

It is noted that the mass budget presented reflects the latest consolidated iteration across the spacecraft architecture. Due to the concurrent nature of the design process, the detailed sizing of some subsystems was conducted using mass estimates from preceding iterations, resulting in minor numerical discrepancies. Subsequent design phases will close this iterative loop, propagating the updated mass budget to harmonize the sizing of all subsystems.

20 Conclusions

The preliminary analysis conducted within this feasibility study confirms the technical feasibility, programmatic coherence, and economic sustainability of MANTIS as a recurring, service-oriented orbital infrastructure for Mars exploration. The integrated design approach adopted throughout the study demonstrates that a single system can reliably fulfill the dual role of interplanetary space tug and long-term communication node, effectively decoupling payload delivery from relay services and enabling architectural specialization without functional redundancy. The analysis proves the feasibility of a frozen design for repeatable missions.

The Carrier–Relay configuration guarantees the transfer of 900 kg, partitioned into a maximum of four distinct payloads, to the Martian environment, deployment across heterogeneous orbital bands (LMO and MMO), and continuous communication support for all released assets over their operational lifetimes, fully satisfying the defined high-level mission requirements. Moreover, the study proves the capability of constellation deployment. Trajectory design, aerobraking strategy, telecommunications architecture, subsystems sizing and integration, and launcher compatibility have been verified at system level, confirming the internal consistency of the baseline configuration under Martian environmental and operational constraints.

The stacked Carrier–Relay architecture has been validated as the optimal configuration to maximize the usable volume within the Ariane 64 fairing envelope while preserving symmetric mass distribution, a key driver for cruise stability, Mars Orbit Insertion accuracy, and safe aerobraking operations. Subsystem solutions demonstrate architectural robustness across all mission phases, from interplanetary transfer to long-term relay service.

The study identifies specific design constraints, primarily concerning the maximum payload capacity, which is currently capped at 900 kg per mission. Furthermore, a significant limitation is posed by the low TRL 3 of the Payload Bay Door and the Relay Support Frame, both of which remain at the conceptual design stage.

To address the aforementioned constraints, further refinement and higher-fidelity modeling are essential. Future development phases must prioritize detailed structural verification, advanced aerothermodynamic simulations for the aerobraking phase, and progressive cost optimization. Furthermore, critical assessments regarding communication between multiple Relay satellites, and a definitive end-of-life disposal strategy are required. Ultimately, this study establishes a technically robust and strategically credible foundation for advancing MANTIS toward subsequent design and implementation phases.

MANTIS is therefore defined not as a single transfer mission, but as a scalable and reusable orbital infrastructure. By amortizing non-recurring costs over the first six missions, MANTIS significantly reduces marginal access costs to Martian orbit. This strategy, coupled with the provision of persistent communication services, establishes a foundational architectural step toward sustained robotic operations and serves as the logistical backbone for future human exploration of Mars.

A Mission Analysis

A.1 Baseline 1: ΔV Budget

Baseline 1	No. Maneuvers	ΔV [m/s]	Margin	Margined ΔV [m/s]
Aerobraking before relay release	6	0.5	+ 5 %	3.15
Perigee raise for relay release	1	25.5	+ 5 %	26.78
Perigee lower for relay release	1	25.5	+ 5 %	26.78
Aerobraking	1	1	+ 5 %	1.05
MMO perigee raise 1	1	29	+ 5 %	30.45
MMO perigee lower 1	1	29	+ 5 %	30.45
MMO perigee raise 2	1	29	+ 5 %	30.45
MMO perigee lower 2	1	29	+ 5 %	30.45
Perigee rise before plane change	1	30	+ 5 %	31.5
Max Plane change	1	256	+ 5 %	268.8
Perigee lower after plane change	1	30	+ 5 %	31.5
Aerobraking	9	1	+ 5 %	9.45
Aerobraking	2	2	+ 5 %	4.2
Aerobraking	1	4	+ 5 %	4.2
Final perigee rise in LMO	1	47.5	+ 5 %	49.88

A.2 Baseline 2: ΔV Budget

Baseline 2	No. Maneuvers	ΔV [m/s]	Margin	Margined ΔV [m/s]
Aerobraking before relay release	6	0.5	+ 5 %	3.15
Perigee raise for relay release	1	25.5	+ 5 %	26.78
Perigee lower for relay release	1	25.5	+ 5 %	26.78
Aerobraking	1	1	+ 5 %	1.05
MMO perigee raise 1	1	29	+ 5 %	30.45
MMO perigee lower 1	1	29	+ 5 %	30.45
MMO perigee raise 2	1	29	+ 5 %	30.45
MMO perigee lower 2	1	29	+ 5 %	30.45
Aerobraking	9	1	+ 5 %	9.45
Aerobraking	2	2	+ 5 %	4.2
Aerobraking	1	4	+ 5 %	4.2
Final perigee rise in LMO	1	47.5	+ 5 %	49.88
LMO phasing maneuver 1	1	158	+ 5 %	165.9
LMO phasing maneuver 2	1	158	+ 5 %	165.9

A.3 Baseline 3: ΔV Budget

Baseline 3	No. Maneuvers	ΔV [m/s]	Margin	Margined ΔV [m/s]
Aerobraking before relay release	6	0.5	+ 5 %	3.15
Perigee raise for relay release	1	25.5	+ 5 %	26.78
Perigee lower for relay release	1	25.5	+ 5 %	26.78
Aerobraking	1	1	+ 5 %	1.05
MMO perigee raise 1	1	29	+ 5 %	30.45
MMO phasing maneuver 1	1	158	+ 5 %	165.9
MMO phasing maneuver 2	1	158	+ 5 %	165.9
MMO perigee lower 1	1	29	+ 5 %	30.45
MMO perigee raise 2	1	29	+ 5 %	30.45
MMO perigee lower 2	1	29	+ 5 %	30.45
Aerobraking	9	1	+ 5 %	9.45
Aerobraking	2	2	+ 5 %	4.2
Aerobraking	1	4	+ 5 %	4.2
Final perigee rise in LMO	1	47.5	+ 5 %	49.88

B.3 Aerobraking Dynamical Simulator

Figure 108 provides an overview of the nonlinear closed-loop simulator developed to assess the Carrier attitude control performance during the aerobraking phase following Relay separation. The model is implemented in Simulink with a modular architecture composed of the following functional blocks:

- **Environment:** propagates the spacecraft orbit and computes the external disturbance torques. SRP, GG, and Aerodynamic Drag are modeled, the latter being dominant during atmospheric passages. Aerodynamic torque is computed using the time-varying density and relative velocity along the trajectory.
- **Dynamic:** implements the non-linear Euler equations for rigid body dynamics, computing the angular acceleration and velocity based on the sum of control and disturbance torques.
- **Kinematic:** handles the integration of the spacecraft's attitude using quaternion representation.
- **Sensors:** generates realistic measurements of Star Trackers, IMUs, and CSS. True states are corrupted with stochastic noise and constant biases according to representative sensor performance models.
- **Kalman Filter:** a Multiplicative Extended Kalman Filter (MEKF) fuses sensor measurements to estimate attitude and angular velocity while simultaneously identifying and compensating gyro biases.
- **Control & Actuators:** computes the control torque based on the estimated states and models the physical behaviour of the thrusters.

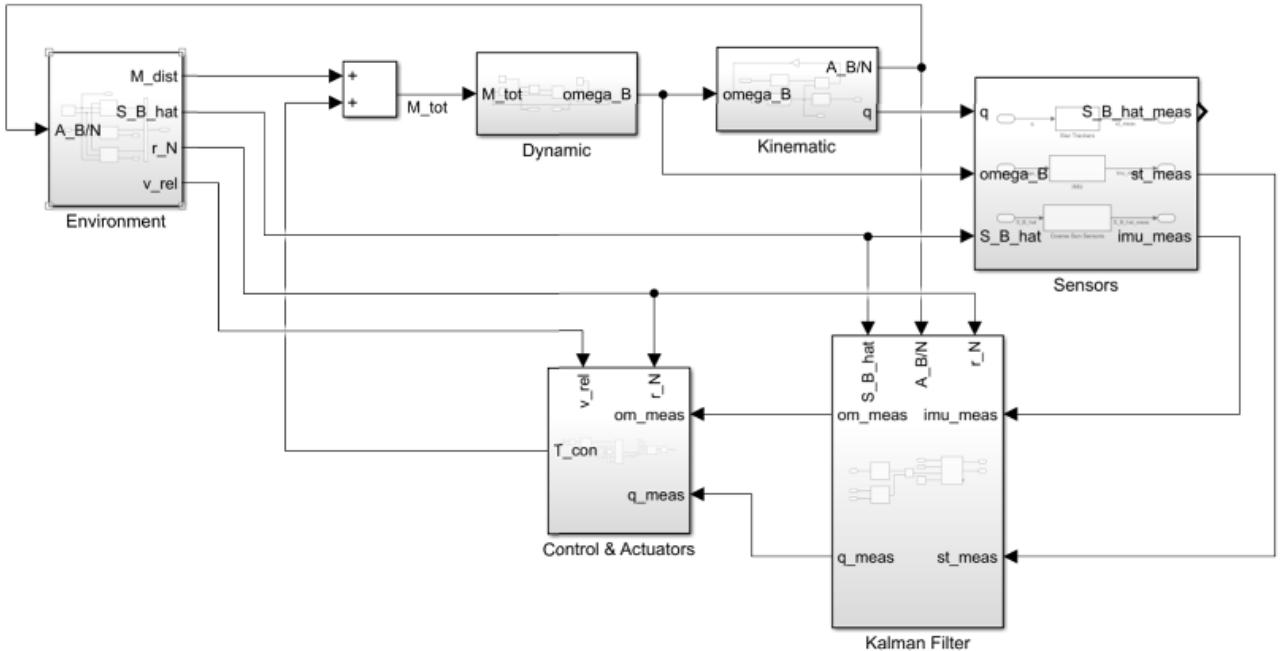


Figure 108: Carrier Dynamical Simulator.

The internal architecture of the **Control & Actuators** block is shown in Figure 109. The attitude control law is based on a Proportional-Integral-Derivative (PID) structure tailored for thruster-based actuation.

The derivative action is not implemented through numerical differentiation of the attitude error. Instead, angular rate measurements from the IMU are directly used as rate feedback, providing improved robustness and noise rejection. The derivative and filter coefficients of the standard PID block are therefore set to zero. The integral term is conditionally enabled: it is disabled when the pointing error exceeds 5° . This anti-windup strategy prevents excessive integral accumulation during large initial errors or acquisition transients, avoiding overshoot once fine pointing is achieved.

The PID gains were tuned iteratively to minimize APE during peak aerodynamic loading while ensuring stability margins under uncertainty. Further gain optimization may be performed in future iterations to balance pointing accuracy and propellant expenditure.

The commanded torque is processed by the **thr_logic** and **Thrusters Control** blocks. Since the actuators operate in on-off mode, a Pulse-Width Pulse-Frequency (PWPF) modulator converts the continuous torque demand into discrete firing pulses. This approach ensures that the average delivered torque matches the commanded value over each control cycle while respecting the minimum impulse bit and actuation constraints.

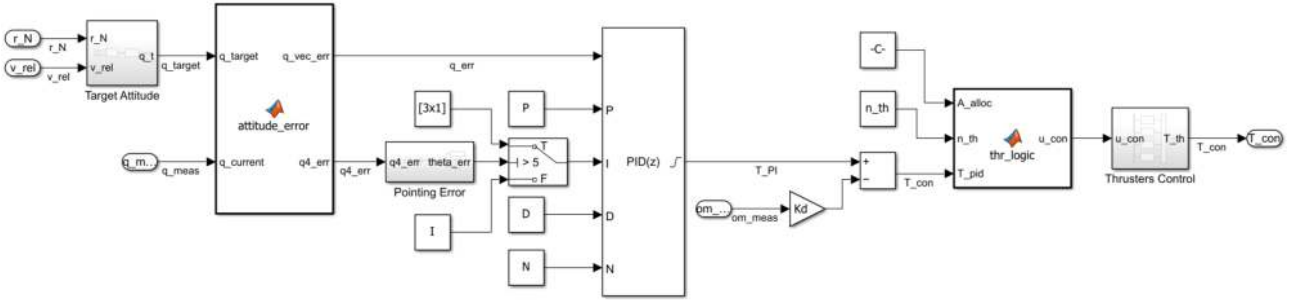


Figure 109: Carrier Dynamical Simulator controller logic.

B.4 Relay Dynamical Simulator

Figure 110 shows the nonlinear closed-loop simulator developed to assess the Relay attitude control performance during the Earth telecommunication phase, representing the most stringent operational mode in terms of pointing accuracy. The simulator is implemented in Simulink and adopts the same modelling philosophy described in Section B.3 for the Carrier. The **Environment**, **Dynamics**, **Kinematics**, **Sensors**, and **MEKF** blocks are conceptually identical; however, only Solar Radiation Pressure and Gravity Gradient torques are modelled, as aerodynamic disturbances are absent in Mars orbit operations. The main architectural difference lies in the actuation chain: attitude control is performed exclusively through reaction wheels.

- **Actuators Control:** computes the required control torque from the estimated states.
- **Actuators:** models reaction wheel dynamics, including torque saturation and representative actuator noise.

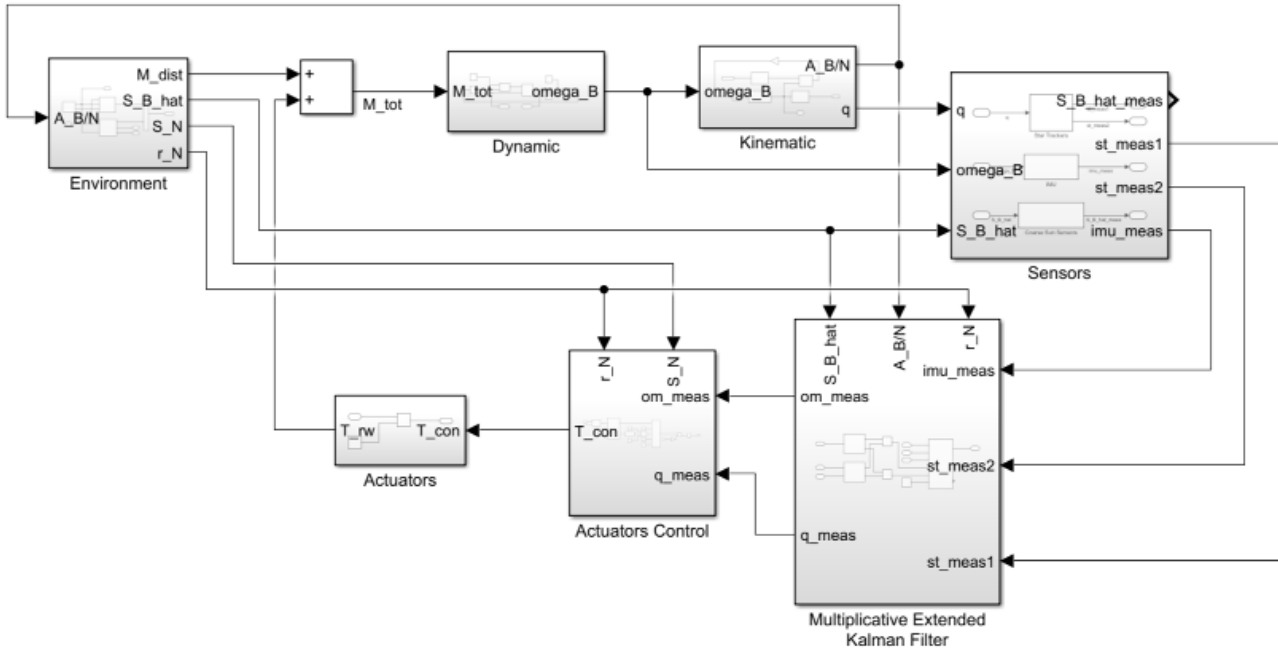


Figure 110: Relay Dynamical Simulator.

The internal structure of the **Actuators Control** block is shown in Figure 111. The control law is based on a PID architecture adapted to reaction-wheel actuation.

As in the Carrier case, angular rate measurements from the IMU are directly used as rate feedback, avoiding numerical differentiation of the attitude error. The derivative and filter coefficients of the standard PID block are therefore set to zero.

The integral action is conditionally enabled and disabled when the pointing error exceeds 5° , preventing windup during large acquisition manoeuvres.

PID gains were tuned to minimize APE while maintaining adequate stability margins with wheel torque saturation limits. Future refinements may include a re-tuning of the gains to reduce reaction wheel momentum build-up, thereby minimizing propellant consumption for desaturation manoeuvres while preserving compliance with the pointing accuracy requirement defined in the budget.

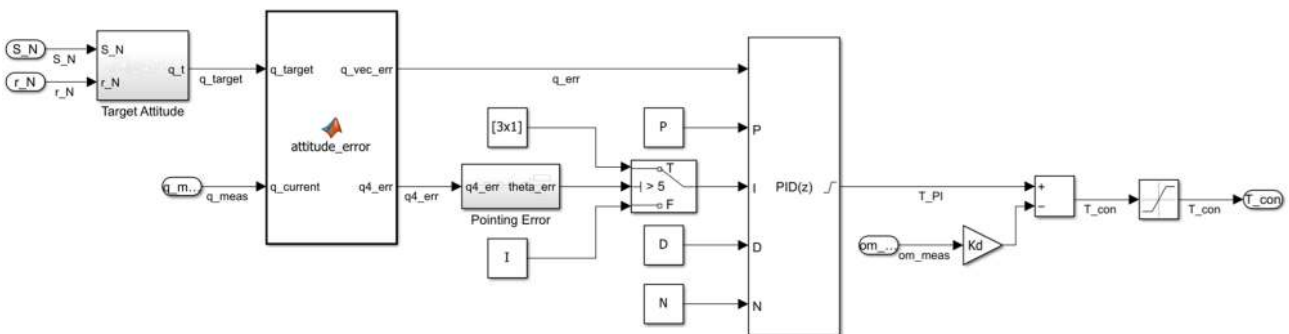


Figure 111: Relay Dynamical Simulator.

C PS results

C.1 Baseline 1

ΔV [m/s]	Prop mass [kg]	Thrusters	Maneuver time [min]	Acceleration [m/s ²]
31.5	49.28	3 x Biprop	2.220	0.237
10.5	16.32	3 x Biprop	0.733	0.239
1218	1574.22	6 x Biprop	42.359	0.479
0.525	0.8166	3 x Monoprop	0.126	0.0696
0.525	0.8164	3 x Monoprop	0.126	0.0697
0.525	0.8162	3 x Monoprop	0.126	0.0697
0.525	0.8160	3 x Monoprop	0.126	0.0697
0.525	0.8158	3 x Monoprop	0.126	0.0697
0.525	0.8156	3 x Monoprop	0.125	0.0697
26.775	41.34	3 x Monoprop	6.322	0.0706
26.775	35.35	3 x Monoprop	5.406	0.0825
1.05	1.378	3 x Monoprop	0.212	0.0826
30.45	39.67	3 x Monoprop	6.062	0.0837
30.45	39.13	3 x Monoprop	5.979	0.0849
30.45	38.59	3 x Monoprop	5.897	0.0861
30.45	38.06	3 x Monoprop	5.816	0.0873
31.5	38.83	3 x Monoprop	5.931	0.0885
268.8	215.85	3 x Biprop	9.279	0.483
31.5	35.22	3 x Monoprop	5.380	0.0976
1.05	1.165	3 x Monoprop	0.179	0.0976
1.05	1.165	3 x Monoprop	0.179	0.0977
1.05	1.164	3 x Monoprop	0.179	0.0977
1.05	1.164	3 x Monoprop	0.179	0.0977
1.05	1.163	3 x Monoprop	0.179	0.0978
1.05	1.162	3 x Monoprop	0.179	0.0979
1.05	1.162	3 x Monoprop	0.179	0.0979
1.05	1.161	3 x Monoprop	0.179	0.0980
1.05	1.161	3 x Monoprop	0.179	0.0980
2.1	2.320	3 x Monoprop	0.357	0.0981
2.1	2.318	3 x Monoprop	0.356	0.0982
4.2	4.629	3 x Monoprop	0.712	0.0984
49.875	54.301	3 x Monoprop	8.261	0.1006

Table 51: Propellant mass, maneuver time and acceleration for Baseline 1.

C.2 Baseline 2

ΔV [m/s]	Prop mass [kg]	Thrusters	Maneuver time [min]	Acceleration [m/s ²]
31.5	48.893	3 x Biprop	2.202	0.238
10.5	16.192	3 x Biprop	0.727	0.241
1218	1561.99	6 x Biprop	42.029	0.483
0.525	0.8103	3 x Monoprop	0.125	0.0702
0.525	0.8101	3 x Monoprop	0.125	0.0702
0.525	0.8099	3 x Monoprop	0.125	0.0702
0.525	0.8097	3 x Monoprop	0.125	0.0702
0.525	0.8095	3 x Monoprop	0.125	0.0703
0.525	0.8093	3 x Monoprop	0.125	0.0703
26.775	41.020	3 x Monoprop	6.273	0.0711
26.775	35.037	3 x Monoprop	5.358	0.0833
1.05	1.365	3 x Monoprop	0.210	0.0833
30.45	39.314	3 x Monoprop	6.007	0.0845
30.45	38.775	3 x Monoprop	5.925	0.0857
30.45	38.244	3 x Monoprop	5.844	0.0865
30.45	37.720	3 x Monoprop	5.764	0.0881
1.05	1.291	3 x Monoprop	0.199	0.0881
1.05	1.291	3 x Monoprop	0.199	0.0881
1.05	1.290	3 x Monoprop	0.198	0.0882
1.05	1.290	3 x Monoprop	0.198	0.0882
1.05	1.289	3 x Monoprop	0.198	0.0883
1.05	1.288	3 x Monoprop	0.198	0.0883
1.05	1.287	3 x Monoprop	0.198	0.0883
1.05	1.287	3 x Monoprop	0.198	0.0884
1.05	1.286	3 x Monoprop	0.198	0.0884
2.1	2.571	3 x Monoprop	0.395	0.0885
2.1	2.569	3 x Monoprop	0.395	0.0886
4.2	5.130	3 x Monoprop	0.789	0.0888
49.875	60.181	3 x Monoprop	9.155	0.0908
165.9	131.940	3 x Biprop	5.764	0.480
165.9	125.330	3 x Biprop	5.475	0.505

Table 52: Propellant mass, maneuver time and acceleration for Baseline 2.

D TTMTTC

D.1 Redundancy, Robustness and Autonomy

ECSS Level	Operational motivation	Telecom availability
Near-Earth Commissioning		
E1	Ground-controlled execution; check-out and verification; no autonomous critical functions	UHF (Carrier) minimal; Relay off
Interplanetary Transfer		
E2	Pre-planned, ground-defined time-tagged sequences; local FDIR for minor contingencies	Relay X-band LGA nominal
Mars Orbit Insertion (MOI)		
E3	Event-based autonomous execution due to incompatibility with real-time ground control during burn	No telecom during burn
Aerobraking Drag Passes		
E3	Autonomous event-based operations during predictable blackout (10–15 min); GNC, thermal and FDIR on-board	No telecom during drag pass; relay contact post-event
Relay–Carrier Separation		
E3	Autonomous execution of critical on-board sequence with local Safe Mode fallback	Limited telecom pre/post separation
Payload Release (Carrier Only)		
E3	Autonomous payload release; not dependent on real-time ground commands	UHF nominal
Relay Operations		
E2-E3	Pre-planned scheduling and event-based autonomy; CFDP-based data handling	X-band HGA nominal; UHF towards payloads

Table 53: Mission execution autonomy levels per mission phase (ECSS-E-ST-70-11C compliant).

D.2 Trade off X and Ka band: comparison link budget

Quantity	Value
Antenna diameter	2.4 m
Antenna efficiency	0.55
Antenna noise temperature	290 K
ESTRACK diameter	34 m
ESTRACK efficiency	0.55
ESTRACK noise temperature	12 K
Zenith distance	$4.01 \cdot 10^8$ km
Elevation	30 deg
Antenna pointing accuracy	0.028 deg
ESTRACK pointing accuracy	0.006 deg
Roll off factor	0.5
Modulation factor	2
Encoding factor	2
X-band uplink	
Frequency	7.2 GHz
BER	10^{-5}
Net data rate	500 kbps
ESTRACK transmitted power	2 kW
X-band downlink	
Frequency	8.4 GHz
BER	10^{-7}
Net data rate	1 Mbps
Transmitted power	102 W
Ka-band downlink	
Frequency	32 GHz
BER	10^{-7}
Net data rate	500 kbps
Transmitted power	35 W

Table 54: Assumptions for link budget.

Quantity	Value
$\frac{Eb}{N_0}_{min}$	8.5 dB
SNR_{min}	3.7391 dB
X-band uplink	
Gross data rate	1 Mbps
Beamwidth	1.5 MHz
TX gain	65.8322 dB
RX gain	42.5551 dB
Losses	-437.0858 dB
$\frac{Eb}{N_0}$	52.2662 dB
SNR	50.5053 dB
X-band downlink	
Gross data rate	500 kbps
Beamwidth	0.75 GHz
TX gain	43.8940 dB
RX gain	67.1711 dB
Losses	-450.2442 dB
$\frac{Eb}{N_0}$	29.5360 dB
SNR	27.7751 dB
Ka-band downlink	
Gross data rate	500 kbps
Beamwidth	0.75 GHz
TX gain	55.5114 dB
RX gain	78.7886 dB
Losses	-486.7782 dB
$\frac{Eb}{N_0}$	11.5916 dB
SNR	9.8307 dB

Table 55: Link budget preliminary design.

D.3 Link Budget Analysis

Link margin for SNR and Eb/N_0 of 3 dB is already included in the results reported. The ground station selected is to guarantee the worst atmosphere losses condition: Kouru for UHF detection in near earth commissioning and New Norcia for X band link.

D.3.1 X-band horn antenna

Table 56: Downlink: Interplanetary.

Parameter	Value
Frequency	8.4 GHz
Data rate	758.3 kbps
Bandwidth	1.1375 Hz
BER	1e-5
Distance	2.4635e8 km
α_{mod}	0.5
α_{enc}	2
D_{ant} transmitter	0.2 m
D_{ant} receiver	35 m
η horn antenna	0.52
η ground antenna	0.55
G_{tx}	22.11 dBi
G_{rx}	67.1711 dBi
HPBW _{tx}	12.79 deg
HPBW _{rx}	0.0714 deg
Atmosphere loss	3.60 dB
Free space loss	284.78 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	1 dBW
Received power P_{rx}	200 dBW
SNR	14.43 dB
SNR_{min}	11.45 dB
E_b/N_0	13.18 dB
$(E_b/N_0)_{min}$	7.2 dB

Table 57: Uplink: Interplanetary.

Parameter	Value
Frequency	7.2 GHz
Data rate	6 kbps
Bandwidth	9 Hz
BER	1e-5
Distance	2.4635e8 km
α_{mod}	0.5
α_{enc}	2
D_{ant} receiver	0.2 m
D_{ant} transmitter	35 m
η horn antenna	0.52
η ground antenna	0.55
G_{tx}	65.8322 dBi
G_{rx}	20.7678 dBi
HPBW _{tx}	0.0833 deg
HPBW _{rx}	14.9208 deg
Atmosphere loss	2.31 dB
Free space loss	283.44 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	16.99 dBW
Received power P_{rx}	120.35 dBW
SNR	21.39 dB
SNR_{min}	11.45 dB
E_b/N_0	20.14 dB
$(E_b/N_0)_{min}$	7.2 dB

Table 58: Downlink: Nominal mode around Mars.

Parameter	Value
Frequency	8.4 GHz
Data rate	0.9167 kbps
Bandwidth	1.375 Hz
BER	1e-5
Distance	4.11e11 km
α_{mod}	0.5
α_{enc}	2
D_{ant} transmitter	0.2 m
D_{ant} receiver	35 m
η horn antenna	0.52
η ground antenna	0.55
G_{tx}	67.17 dBi
G_{rx}	22.11 dBi
HPBW _{tx}	67.17 deg
HPBW _{rx}	0.0714 deg
Atmosphere loss	3.60 dB
Free space loss	289.23 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	10 dBW
Received power P_{rx}	190.81 dBW
SNR	18.16 dB
SNR_{min}	11.449387 dB
E_b/N_0	16.91 dB
$(E_b/N_0)_{min}$	7.2 dB

Table 59: Uplink: Nominal mode around Mars.

Parameter	Value
Frequency	7.2 GHz
Data rate	6 kbps
Bandwidth	9 Hz
BER	1e-5
Distance	4.11e11 km
α_{mod}	0.5
α_{enc}	2
D_{ant} receiver	0.2 m
D_{ant} transmitter	35 m
η horn antenna	0.52
η ground antenna	0.55
G_{tx}	65.83 dBi
G_{rx}	20.77 dBi
HPBW _{tx}	0.0833 deg
HPBW _{rx}	14.92 deg
Atmosphere loss	2.31 dB
Free space loss	287.89 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	16.99 dBW
Received power P_{rx}	189.11 dBW
SNR	16.94 dB
SNR_{min}	11.45 dB
E_b/N_0	15.69 dB
$(E_b/N_0)_{min}$	7.2 dB

In the downlink, the resulting data rate remains moderate since no restrictive visibility constraints are imposed by the ground contact. The available link duration does not drive the need for high throughput. For the uplink, the configuration results in a comfortable link margin.

D.3.2 X-band parabolic antenna

Table 60: Downlink: High data rate.

Parameter	Value
Frequency	8.4 GHz
Data rate	500.39 kbps
Bandwidth	750.5850 Hz
BER	1e-7
Distance	4.11e8 km
α_{mod}	0.5
α_{enc}	2
D_{ant} transmitter	2.5 m
D_{ant} receiver	35 m
η antenna efficiency	0.55
G_{tx}	44.25 dBi
G_{rx}	67.17 dBi
HPBW _{tx}	1 deg
HPBW _{rx}	0.0714 deg
Atmosphere loss	3.6 dB
Free space loss	289.23 dB
Pointing loss	8.53 dB
Cable loss	2 dB
Transmitted power P_{tx}	18 dBW
Received power P_{rx}	173.92 dBW
SNR	12.91 dB
SNR_{min}	12.85 dB
E_b/N_0	11.67 dB
$(E_b/N_0)_{min}$	8.6 dB

The uplink is done using only the X band horn antenna to exploit lower electrical demand.

D.3.3 UHF Relay Link: Carrier

Table 61: Downlink: Near Earth Commissioning.

Parameter	Value
Frequency	440 MHz
Data rate	113.89 kbps
Bandwidth	0.1708 Hz
BER	1e-5
Distance	9.21e3 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} transmitter	0.5 m
D_{ant} transmitter	0.1967 m
D_{ant} receiver	15 m
η antenna	0.52
G_{tx}	8 dBi
G_{rx}	10.08 dBi
HPBW _{tx}	61.80 deg
HPBW _{rx}	0.32 deg
Free space loss	38.95 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	4 dBW
Received power P_{rx}	181 dBW
SNR	18.35 dB
SNR_{min}	11.45 dB
E_b/N_0	17.10 dB
$(E_b/N_0)_{min}$	7.20 dB

Table 62: Uplink: Near Earth Commissioning.

Parameter	Value
Frequency	437 MHz
Data rate	60 kbps
Bandwidth	9 Hz ???
BER	1e-5
Distance	9.21e3 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} receiver	0.5 m
D_{ant} receiver	0.1967 m
D_{ant} transmitted	15 m
η antenna	0.55
G_{tx}	34.14 dBi
G_{rx}	8 dBi
HPBW _{tx}	3.20 deg
HPBW _{rx}	67.7 deg
Free space loss	38.95 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	13.98 dBW
Received power P_{rx}	155.39 dBW
SNR	29.33 dB
SNR_{min}	11.45 dB
E_b/N_0	28.10 dB
$(E_b/N_0)_{min}$	7.20 dB

Table 63: Downlink: Carrier-relay separation and release operations.

Parameter	Value
Frequency	440 MHz
Data rate	167 kbps
Bandwidth	0.125 Hz
BER	1e-5
Distance	14417 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} receiver	0.8 m
D_{ant} receiver	0.2 m
L_{ant} transmitter	0.5 m
D_{ant} transmitter	0.2 m
η antenna	0.55
G_{tx}	3.3 dBi
G_{rx}	10.08 dBi
HPBW _{tx}	61.60 deg
HPBW _{rx}	53.33 deg
Free space loss	189.68 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	6 dBW
Received power P_{rx}	184.49 dBW
SNR	12.16 dB
SNR_{min}	11.45 dB
E_b/N_0	10.91 dB
$(E_b/N_0)_{min}$	7.20 dB

Table 64: Uplink: Carrier-relay separation and release operations.

Parameter	Value
Frequency	437 MHz
Data rate	40 kbps
Bandwidth	6 Hz
BER	1e-5
Distance	14417 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} receiver	0.5 m
D_{ant} receiver	0.2 m
L_{ant} transmitter	0.8 m
D_{ant} transmitter	0.2 m
η antenna	0.55
G_{tx}	10.05 dBi
G_{rx}	8 dBi
HPBW _{tx}	53.52 deg
HPBW _{rx}	61.80 deg
Free space loss	189.62 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	7 dBW
Received power P_{rx}	166.57 dBW
SNR	12.14 dB
SNR_{min}	11.45 dB
E_b/N_0	10.89 dB
$(E_b/N_0)_{min}$	7.20 dB

Concerning the link with the carrier the values are previously shown in the Table 64.

D.3.4 UHF Relay Link: Relay

Table 65: Downlink: telecommand for payloads.

Parameter	Value
Frequency	0.437 Hz
Data rate	4 kbps
Bandwidth	6.5070 Hz
BER	1e-5
Distance	14417 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} transmitter	0.8 m
η antenna	0.55
G_{tx}	10.05 dBi
G_{rx}	8.7 dBi
HPBW _{tx}	53.52 deg
HPBW _{rx}	62 deg
Free space loss	189.62 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	7 dBW
Received power P_{rx}	165.87 dBW
SNR	26.81 dB
SNR_{min}	11.45 dB
E_b/N_0	25.56 dB
$(E_b/N_0)_{min}$	7.2 dB

Table 66: Uplink: High data rate with PHAROS-M.

Parameter	Value
Frequency	440 MHz
Data rate	0.900 Mbps
Bandwidth	1.4644 Hz
BER	1e-7
Distance	14417 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} receiver	0.8 m
η antenna	0.55
G_{tx}	8.7 dBi
G_{rx}	10.08 dBi
HPBW _{tx}	62 deg
HPBW _{rx}	53.34 deg
Free space loss	158.25 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	7 dBW
Received power P_{rx}	135.47 dBW
SNR	33.7 dB
SNR_{min}	12.85 dB
E_b/N_0	32.45 dB
$(E_b/N_0)_{min}$	8.6 dB

Parameter	Value
Frequency	440 MHz
Data rate	1.05 Mbps
Bandwidth	1.4644 Hz
BER	1e-7
Distance	14417 km
α_{mod}	0.5
α_{enc}	2.1690
L_{ant} receiver	0.8 m
η antenna	0.55
G_{tx}	8.7 dBi
G_{rx}	10.08 dBi
HPBW _{tx}	62 deg
HPBW _{rx}	53.34 deg
Free space loss	158.25 dB
Pointing loss	1 dB
Cable loss	2 dB
Transmitted power P_{tx}	7 dBW
Received power P_{rx}	135.47 dBW
SNR	13.03 dB
SNR_{min}	12.85 dB
E_b/N_0	11.78 dB
$(E_b/N_0)_{min}$	8.6 dB

Table 67: Uplink: High data rate maximum.

D.4 Components of the RF chain

D.4.1 X band

Component	Qty	Mass [kg]	Input Power [W]	Notes / Technology Choice
X-band High-Gain Antenna (parabolic)	1	15.0	0	2.5 m class dish. Nominal relay operations.
X-band Medium-Gain Antenna (horn)	1	0.5	0	Wide beamwidth. Used during cruise, MOI and Safe Mode.
SDST - Small Deep Space Transponder	1	3.5	46 (TX peak), 28 nominal	CCSDS-compliant coherent transponder. Integrated LNA, mod/demod, ranging and Doppler. ESTRACK compatible.
X-band TWTA (250 W RF)	1	4.0	400	Required to close the downlink at Mars worst-case geometry.
X-band Diplexer	1	1.0	0.5	Full-duplex TX/RX routing through HGA or LGA. Rated for 250 W RF.

Table 68: Relay X-band Direct-To-Earth communication components.

D.4.2 Relay: UHF

Component	Qty	Mass [kg]	Input Power [W]	Notes
UHF Coherent transponder	1	6.0 each	36 TX peak, 15 RX	CCSDS Prox-1 compliant re-generative radios.
UHF Solid-State Power Amplifier	1	4.2	-	-
UHF Medium-Gain Antenna (helical, RHCP)	1	0.5	0	Relay-to-payload link during nominal relay operations.

Table 69: Relay UHF proximity communication components.

D.4.3 Carrier: UHF

Component	Qty	Notes / Technology Choice
UHF Low-Gain Antenna (helical, RHCP)	1	Quasi - omnidirectional.
UHF Coherent Transponder	1	TRL 9
Solid State Power Amplifier	1	Integrated in UHF transceiver (≈ 5 W RF).
Low Noise Amplifier	1	-
UHF Switch	1	Enables switch of the path during proximity operations.

Table 70: Carrier UHF communication components.

Data Source	Avg Volume [Mb/day]	# Contacts	Contact window	Data rate
Near-Earth Commissioning				
Carrier ADCS HK (high-rate)	0.20	6	10 min (worst)	69 bps
Carrier EPS HK	0.015			
Carrier TCS HK	0.01			
Carrier PS HK	0.01			
Carrier OBDH HK	0.01			
FDIR	0.005			
Interplanetary Transfer				
Carrier ADCS HK (low-rate)	0.05	2	20–30 min (worst)	70.8 bps
Carrier TCS HK	0.03			
Carrier EPS HK	0.02			
Carrier PS HK	0.02			
Carrier GNC	0.02			
Carrier OBDH HK	0.02			
FDIR	0.01			
Mars Orbit Insertion (MOI)				
ADCS HK (high-rate)	0.50	1	10–20 min (worst)	4.17 kbps
PS HK	1.0			
TCS HK (high-rate)	0.80			
FDIR	0.20			
Relay–Carrier Separation				
Carrier Telemetry HK	0.30	2	20–30 min (worst)	375 bps
Separation Logs	0.60			
Payload Release (Carrier Only)				
Carrier ADCS HK	0.40	3	20–30 min (worst)	333 bps
Carrier Telemetry HK	0.20			
Payload Interface HK	0.60			

Table 71: Carrier TMTC data budget with contact assumptions.

E EPS

E.1 Power Budget Tables

Component	N°	Peak power			Phases/Modes												Power required																			
		Unit	Power Margin [Z]	Margined Power [W]	Fraction of Peak power required						CRUISE			PRE-MOI			PLS RELEASE			RELAY RELEASE			PLS RELEASE			SAFE MODE										
					MOI	Peak	Avg	MOI	Peak	Avg	MOI	Peak	Avg	CRUISE	Peak	Avg	PRE-MOI	Peak	Avg	MOI	Peak	Avg	MOI	Peak	Avg	MOI	Peak	Avg								
PS	6	61,00	5	384,50	0,50	0,10	0,40	1,00	1,00	0,00	0,00	0,00	0,00	0,00	0,00	1921,5	38,45	153,72	384,50	384,50	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00							
	6	15,00	5	94,50	0,50	0,10	0,15	0,15	0,15	0,00	0,00	0,00	0,00	0,00	0,00	47,25	9,45	141,8	141,8	141,8	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00							
	3	52,00	5	163,80	0,00	0,00	0,00	0,00	0,00	1,00	0,10	1,00	0,50	1,00	0,50	0,00	0,00	0,00	0,00	0,00	163,80	163,80	81,90	163,80	81,90	0,00	0,00	0,00	0,00							
	3	17,20	5	54,18	0,00	0,00	0,00	0,00	0,00	1,00	0,10	1,00	0,50	1,00	0,50	0,00	0,00	0,00	0,00	0,00	54,18	54,18	27,09	54,18	27,09	0,00	0,00	0,00	0,00							
	SUB-TOTAL						299,40			47,88			167,90			398,48			217,98			217,98			108,99			217,98								
TTMTC	1	6,54	10	6,97	1,00	0,20	1,00	0,80	0,00	0,00	1,00	0,80	1,00	0,50	0,00	0,00	6,97	1,39	6,97	5,58	0,00	0,00	6,97	5,58	6,97	3,49	0,00	0,00	0,00							
	1	1,00	10	1,10	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	1,00	0,50	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	1,10	0,55	0,00	0,00								
	SUB-TOTAL						6,97			1,39			6,97			5,58			6,97			5,58			6,97			3,49			1,10					
ADCS	4	150,00	5	630,00	1,00	0,05	1,00	0,20	1,00	0,05	1,00	0,10	1,00	0,05	1,00	0,05	630,00	61,50	630,00	630,00	31,50	630,00	63,00	630,00	31,50	630,00	31,50	0,00	0,00	0,00						
	4	16,00	5	67,20	1,00	0,10	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	67,20	6,72	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00							
	4	6,50	5	26,46	0,00	0,00	1,00	1,00	0,20	0,20	0,00	0,00	0,00	0,00	0,00	0,00	26,46	26,46	5,29	5,29	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00							
	1	13,50	5	14,18	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18							
	1	13,50	5	14,18	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18	14,18							
Star Tracker	8	0,00	5	0,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	1,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00						
	SUB-TOTAL						725,55			66,57			180,81			663,64			651,4			658,35			59,85			658,35			59,85					
	TCS																																			
HEATER Tank	6	50,00	5	315,00	1,00	0,70	1,00	0,70	1,00	0,70	1,00	0,70	1,00	0,70	1,00	0,70	315,00	220,50	315,00	220,50	315,00	220,50	315,00	220,50	315,00	220,50	315,00	220,50	0,00	0,00						
	SUB-TOTAL						315,00			220,50			315,00			220,50			315,00			220,50			315,00			220,50			315,00					
	60,00	5	63,00	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	63,00	31,50	63,00	31,50	63,00	31,50	63,00	31,50	63,00	31,50	63,00	31,50	63,00	31,50							
	SUB-TOTAL						63,00			31,50			63,00			31,50			63,00			31,50			63,00			31,50			63,00					
	ORBH																																			
DEC	2	10,00	5	21,00	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	21,00	10,50	21,00	10,50	21,00	10,50	21,00	10,50	21,00	10,50	21,00	10,50	0,00	0,00						
	1	18,00	5	18,90	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	18,90	9,45	18,90	9,45	18,90	9,45	18,90	9,45	18,90	9,45	18,90	9,45	18,90	9,45						
	1	10,00	5	10,50	1,00	0,70	1,00	0,70	1,00	0,70	1,00	0,70	1,00	0,70	1,00	0,70	10,50	7,35	10,50	7,35	10,50	7,35	10,50	7,35	10,50	7,35	10,50	7,35	10,50	7,35						
	1	0,60	5	0,65	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	0,65	0,50	0,65	0,50	0,65	0,50	0,65	0,50	0,65	0,50	0,65	0,50	0,65	0,50						
	1	0,50	5	0,53	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	0,53	0,42	0,53	0,42	0,53	0,42	0,53	0,42	0,53	0,42	0,53	0,42	0,53	0,42						
SpaceWire Router	1	1,00	5	1,05	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,00	0,80	1,05	0,84	1,05	0,84	1,05	0,84	1,05	0,84	1,05	0,84	1,05	0,84	1,05	0,84						
	SUB-TOTAL						52,61			29,06			52,61			29,06			52,61			29,06			52,61			29,06			52,61					
	STR																																			
	1	170,00	10	187,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00							
	2	100,00	10	220,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00						
Solar Arrays Drive Mech	2	3,50	5	7,55	1,00	0,10	1,00	0,10	1,00	0,10	1,00	0,10	1,00	0,10	1,00	0,10	7,55	0,74	7,55	0,74	7,55	0,74	7,55	0,74	7,55	0,74	7,55	0,74	7,55	0,74						
	SUB-TOTAL						7,55			0,74			7,55			0,74			7,55			0,74			7,55			0,74			7,55					
	PAYLOAD						210,00			147,00			210,00			147,00			210,00			147,00			210,00			147,00			210,00					
	SUB-TOTAL						210,00			147,00			210,00			147,00			210,00			147,00			210,00			147,00			210,00					
	SYSTEM						1619,88			544,64			1507,63			783,08			1710,07			892,42			1531,26			547,53			1718,26					
SYSTEM LEVEL	30						2105,84			708,04			1959,92			1018,01			2223,09			1160,14			1990,64			711,78			2233,74					

Figure 112: Carrier Power Budget.

Component	Peak power			Phases/Modes																
	Unit Power [W]	Margin [%]	Margined Power [W]	Fraction of Peak power required						Power required										
				STAND-BY		COMMISSIONING		NOMINAL OPERATIONS		HIGH-RATE OPERATIONS		STAND-BY		COMMISSIONING		NOMINAL OPERATIONS		HIGH-RATE OPERATIONS		SAFE MODE
N°	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg	Peak	Avg
ITMTC																				
	1	252,38	20	302,86	0,00	0,00	0,00	0,00	1,00	0,40	0,00	0,00	0,00	0,00	0,00	0,00	302,86	121,14		
	1	40,00	10	44,00	1,00	0,10	1,00	0,40	1,00	0,50	0,00	0,00	44,00	4,40	44,00	17,60	44,00	22,00	0,00	0,00
	1	20,05	10	22,06	0,00	0,00	1,00	0,40	1,00	0,80	0,00	0,00	22,06	8,82	22,06	17,64	22,06	17,64		
SUB-TOTAL																				
											44,00	4,40	66,06	26,42	66,06	39,64	324,91	138,79	26,40	
ADCS																				
	4	90,00	5	378,00	0,00	0,00	1,00	0,10	1,00	0,20	0,00	0,00	378,00	37,80	378,00	75,60	378,00	75,60		
	4	16,00	5	67,20	0,00	0,00	0,00	0,00	1,00	0,10	0,00	0,00	0,00	0,00	0,00	67,20	67,20	6,72		
	1	13,50	5	14,18	0,00	0,00	1,00	1,00	1,00	1,00	0,00	0,00	14,18	14,18	14,18	14,18	14,18	14,18		
	2	13,50	5	28,35	0,00	0,00	1,00	1,00	1,00	1,00	0,00	0,00	28,35	28,35	28,35	28,35	28,35	28,35		
	8	0,00	5	0,00	0,00	0,00	1,00	1,00	1,00	1,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00		
SUB-TOTAL																				
											0,00	0,00	420,53	80,33	487,73	124,85	487,73	124,85	58,70	
ITCS																				
	3	45,00	5	141,75	1,00	0,40	1,00	0,70	1,00	0,70	0,70	141,75	56,70	141,75	99,23	141,75	99,23	99,23		
	1	40,00	5	42,00	1,00	0,40	1,00	0,70	1,00	0,70	0,70	42,00	16,80	42,00	29,40	42,00	29,40	29,40		
SUB-TOTAL																				
											183,75	73,50	183,75	128,63	183,75	128,63	183,75	128,63	73,50	
	50,00	5	52,50	1,00	0,50	1,00	0,50	1,00	0,50	1,00	0,50	52,50	26,25	52,50	26,25	52,50	26,25	26,25		
SUB-TOTAL																				
											52,50	26,25	52,50	26,25	52,50	26,25	52,50	26,25	21,00	
OBDH																				
	2	10,00	5	21,00	1,00	0,50	1,00	0,50	1,00	0,50	0,50	21,00	10,50	21,00	10,50	21,00	10,50			
	1	15,00	5	15,75	1,00	0,50	1,00	0,50	1,00	0,50	0,50	15,75	7,88	15,75	7,88	15,75	7,88			
	1	10,00	5	10,50	1,00	0,70	1,00	0,70	1,00	0,70	0,70	10,50	7,35	10,50	7,35	10,50	7,35			
	1	0,60	5	0,63	1,00	0,80	1,00	0,80	1,00	0,80	0,80	0,63	0,50	0,63	0,50	0,63	0,50			
SUB-TOTAL																				
											47,88	26,23	47,88	26,23	47,88	26,23	47,88	26,23	13,11	
ISTR																				
	2	7,00	5	14,70	0,00	0,00	1,00	0,20	1,00	0,20	0,00	0,00	14,70	2,94	14,70	2,94	14,70	2,94		
SUB-TOTAL																				
											0,00	0,00	14,70	2,94	14,70	2,94	14,70	2,94	2,94	
SYSTEM																				
											328,13	130,38	785,41	290,79	852,61	348,53	1111,47	447,68	195,65	
SYSTEM LEVEL																				
	30										496,57	160,40	1091,03	373,03	1108,39	453,09	1444,91	581,98	954,34	

Figure 113: Relay Power Budget.

E.2 Batteries Dynamical Simulator

Figure 114 shows the Simulink model utilized to perform dynamical power simulations of the batteries.

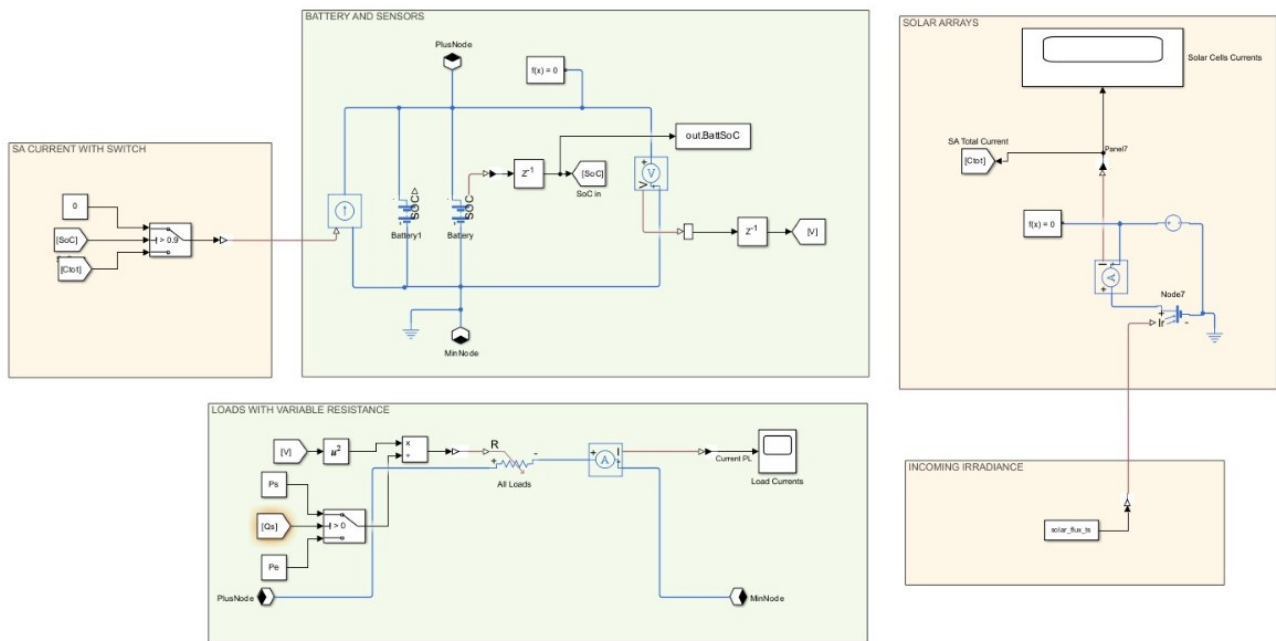


Figure 114: Batteries Dynamical Power Simulator.

F Mass Budget

Spacecraft	Component	Number of units	Mass per unit [kg]	Margin	Mass with Margin [kg]
Relay				DMM	
	ADCS				
	IMU (Astrix 1090)	2	4,5000	5%	9,4500
	Star Tracker (Hydra TC)	2	6,7000	5%	14,0700
	Coarse Sun Sensor	10	0,2150	5%	2,2575
	Reaction Wheel (RSI 12-75/60)	4	4,8500	5%	20,3700
	OBDH				
	OBC (Single-core rad-hard	2	1,0000	5%	2,1000
	Mass Memory (SSR)	1	7,0000	5%	7,3500
	FPGA (RTG4)	1	0,3000	5%	0,3150
	MRAM memory	1	0,1500	5%	0,1575
	MIL-STD-1553 Coupler	1	0,4000	5%	0,4200
	SpaceWire Router	1	0,4000	5%	0,4200
	TMTC				
	HGA	1	15,0000	20%	18,0000
	LGA horn antenna	1	0,5000	10%	0,5500
	UHF antenna	1	1,0000	10%	1,1000
	SDST	1	3,2000	5%	3,3600
	TWTA	1	0,8950	5%	0,9398
	Diplexer	1	1,0000	5%	1,0500
	Transponder	1	6,0000	5%	6,3000
	SSA	1	4,2000	5%	4,4100
	Converter	2	0,0050	5%	0,0105
	Low Noise	1	0,0100	5%	0,0105
	PS				
	Attitude thrusters (MR-103J)	8	0,3700	5%	3,1080
	Filter	2	0,2500	5%	0,5250
	Latch Valves	4	0,4000	5%	1,6800
	Pyrotechnick Normally Closed	2	0,3000	5%	0,6300
	Service Valves	2	0,0600	5%	0,1260
	Propellant and Pressurant Tank	1	6,8500	5%	7,1925
	Helium mass	-	0,0932	20%	0,1118
	Cables	-			1,3373
	EPS				
	Solar Arrays	2	18,1520	10%	39,9344
	Batteries (Yardney NCP 25-1)	2	14,0488	10%	30,9074
	PCDU	1	11,8780	10%	13,0658
	Harness	-	13,0920	10%	14,4012
	TCS				
	MLI	-	5,9100	5%	6,2055
	SSM	1	0,0116	10%	0,0122
	Heaters (KHLVA 104)	1	0,0009	5%	0,0010
	Heaters (KHLVA 303)	3	0,0021	5%	0,0066
	Thermal filler	1	0,2030	10%	0,2233
	STRUCT				
	Primary structure	1	47,5430	20%	57,0516
	Interface with Carrier	1	1,0000	20%	1,2000
	Guide bushingliners	2	0,0500	20%	0,1200
	YOKE	2	6,4500	20%	15,4800
	MECH				
	Gravity)	2	4,2000	10%	9,2400
	HDRM NEReA 8 (Sener)	4	0,1900	20%	0,9120
	Total dry mass		296,1124		
	Dry mass + system margin			30%	384,9461
	Propellant mass		65,3670	5,5%	68,9622
	Wet mass				453,9083

Figure 115: Relay Mass Budget.

Carrier				DMM	
	ADCS				
	IMU (Astrix 1090)	2	4,5000	5%	9,4500
	Star Tracker (Hydra TC)	1	6,7000	5%	7,0350
	Coarse Sun Sensor	10	0,2150	5%	2,2575
	Reaction Wheel (RSI 30-280/30)	4	8,5000	5%	35,7000
	OBDH				
	OBC (Single-core rad-hard PowerPC)	1	1,0000	5%	1,0500
	MRAM memory	1	0,1500	5%	0,1575
	MIL-STD-1553 Coupler	1	0,4000	5%	0,4200
	TMTC				
	Diplexer	1	1,0000	5%	1,0500
	Transponder	1	6,0000	5%	6,3000
	SSA	1	4,2000	5%	4,4100
	Converter	2	0,0050	5%	0,0105
	Low Noise	1	0,0100	5%	0,0105
	UHF antenna	1	1,0000	10%	1,1000
	PS				
	Attitude thrusters (MR-111G)	8	0,3700	5%	3,1080
	Main Bipropellant thrusters	7	5,4400	5%	39,9840
	Medium Monopropellant thrusters (MR-107T)	6	1,0100	5%	6,3630
	Filter	5	0,2500	5%	1,3125
	Latch Valve	16	0,4000	5%	6,7200
	Pyrotechnick Normally Closed Valve	15	0,3000	5%	4,7250
	Service Valve	14	0,0600	5%	0,8820
	Quad check valve	2	0,8000	5%	1,6800
	Calibrated Orifice	14	0,0200	5%	0,2940
	Pyrotechnick Normally Open Valve	8	0,3000	5%	2,5200
	Pressure regulator	2	1,5000	5%	3,1500
	Hydrazine tank	4	18,6100	5%	78,1620
	MON-3 tank	2	19,6800	5%	41,3280
	Helium tank	1	40,1400	20%	48,1680
	Helium mass	-	16,0500	20%	19,2600
	Cables	-			25,7657
	EPS				
	Solar Arrays	2	25,6240	10%	56,3728
	Batteries (Yardney NCP 25-1)	2	21,0730	10%	46,3606
	PCDU	1	17,9790	10%	19,7769
	Harness	-	18,3560	10%	20,1916
	TCS				
	SSM	1	0,0037	5%	0,0038
	Heater (KHLVA-105)	6	0,0012	5%	0,0013
	MLI	-	51,6420	5%	54,2241

Figure 116: Carrier Mass Budget.

	STRUCT				
	Primary structure	1	341,3600	20%	409,6320
	CPDI	1	16,1700	20%	19,4040
	PAI	1	65,0000	20%	78,0000
	YOKE	2	14,7300	20%	35,3520
	MECH				
	Relay Release & Support Frame:				
	NEA pin-puller (Sener)	3	0,0400	10%	0,1320
	Release latch HW	3	0,0250	20%	0,0900
	Pyro backup device (Airbus DS)	1	0,1500	10%	0,1650
	Hardpoint brackets	3	0,4000	20%	1,4400
	Door load spreaders	2	0,5000	20%	1,2000
	Push-off springs	3	0,1000	20%	0,3600
	Spring seats	3	0,1500	20%	0,5400
	Guide pins	2	0,1000	20%	0,2400
	Status sensors (Sensata technologies)	4	0,0200	5%	0,0840
	Local harness & brackets	1	0,3000	20%	0,3600
	SADM-SEPTA36G-2 (Beyond Gravity)	2	5,2000	10%	11,4400
	HDRM NEReA 10 (Sener)	2	0,1900	10%	0,4180
	PAI release mech:				
	Clamp-band assembly (band+tensioning HW)	1	18,0000	10%	19,8000
	Non-pyrotechnic release actuator	2	2,5000	10%	5,5000
	Actuator-to-clamp interface HW (pins, levers, cams)	2	1,5000	20%	3,6000
	Clamp-band local guides / retainers (1 set)	1	1,0000	20%	1,2000
	Release status sensors	2	0,1500	10%	0,3300
	Sensor local brackets (relase only)	2	0,2000	20%	0,4800
	Local release harness and connectors	1	1,0000	20%	1,2000
	Mechanism fasteners	1	2,0000	10%	2,2000
	Dual-leaf payload bay door:				
	Latch catches	4	0.08	20%	0,3800
	Pre load springs	2	0.05	5%	0,1100
	Damping device	2	0.25	10%	0,5500
	Hinge assembly	2	0.32	5%	0,6700
	Local sensors	4	0.01	10%	0,0440
	Total dry mass		1144,2253		
	Dry mass + system margin			30%	1487,4928
	Propellant mass		2369,6872	5,5%	2500,0200
	Wet mass				3987,5128
	Payload		900,0000		
	Adapter		80,0000		
	Total launchable mass				5421,4211

Figure 117: Carrier Mass Budget.

References

- [1] Arianespace, Evry-Courcouronnes Cedex-France. *Ariane 6 User's Manual*, February 2021. Issue 2 Revision 0.
- [2] D. Heynderickx and M. Kruglanski. *Space Environment Information System (SPENVIS) User Guide*. European Space Agency, 2018.
- [3] Mars transportation environment definition document. NASA Technical Memorandum NASA/TM-2001-210935, National Aeronautics and Space Administration (NASA), Marshall Space Flight Center, Marshall Space Flight Center, Alabama, USA, March 2001. Prepared by the Engineering Systems Department, Engineering Directorate. Report ID on NTRS: 20010046858. 114 pp.
- [4] G. De Angelis and F. A. Cucinotta. Planetary and interplanetary environmental models for radiation analysis. Technical Report NASA/TP-2005-213715, National Aeronautics and Space Administration (NASA), Johnson Space Center, Houston, TX, USA, July 2005. NASA Technical Publication. Presented at the 36th COSPAR Scientific Assembly, Beijing, 2006.
- [5] E. Grün, K. Matthews, and N. M. Lanznaster. Comparison of meteoroid flux models for near earth space. *Advances in Space Research*, 40(11):1780–1789, 2007. Includes discussion of the Divine–Staubach model and comparisons with interplanetary meteoroid engineering models.
- [6] Jet Propulsion Laboratory. Report from mars: Mariner iv, 1964–1965. Technical Report NASA SP-139, National Aeronautics and Space Administration (NASA), Washington, D.C., USA, 1967. Comprehensive mission report including engineering data, scientific results, and micrometeoroid impact observations during the Earth–Mars transfer.
- [7] E. Grün, H. Fechtig, M. Svedhem, J. Kissel, D. Linkert, H. G. Eichhorn, J. Giese, H. K. Schmidt, and R. A. Zook. Collisional balance of the meteoritic complex. *Icarus*, 62:244–272, 1985.
- [8] David R. Williams. Mars fact sheet. <https://web.archive.org/web/20250820142225/https://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html>, 2025. Last updated: 19 May 2025.
- [9] European Space Agency. Spacecraft fall silent as mars disappears behind the sun. https://www.esa.int/Enabling_Support/Operations/Spacecraft_fall_silent_as_Mars_disappears_behind_the_Sun, YYYY. Accessed: YYYY-MM-DD.
- [10] Richard W Zurek and Suzanne E Smrekar. An overview of the mars reconnaissance orbiter (mro) science mission. *Journal of Geophysical Research: Planets*, 112(E5), 2007.
- [11] Stacia M Long, Tung-Han You, C Allen Halsell, Ramachand S Bhat, Stuart W Demcak, Eric J Graat, Earl S Higa, Dolan E Highsmith, Neil A Mottinger, and Moriba K Jah. Mars reconnaissance orbiter aerobraking daily operations and collision avoidance. In *Proceedings of the 20th International Symposium on Space Flight Dynamics*, 2007.
- [12] Goddard Space Flight Center NASA. Mgs aerobraking, 2017. Last access: 2025-10-13.
- [13] S. E. Laubach et al. 2001 mars odyssey aerobraking commentary. Technical Report 20030014800, NASA Technical Reports Server (NTRS), 2003. Last access: 2025-10-13.
- [14] AF Nagy, Daniel Winterhalter, Konrad Sauer, TE Cravens, S Brecht, C Mazelle, D Crider, E Kallio, Alexander Zakharov, Eduard Dubinin, et al. The plasma environment of mars. *Space Science Reviews*, 111(1):33–114, 2004.
- [15] Stas Barabash, Mats Holmström, Robin Ramstad, Yoshifumi Futaana, and Andrii Voshchepynets. The induced magnetosphere of mars and the near-mars environment as revealed by mars express. *Space Science Reviews*, 221(6):79, 2025.
- [16] SW Bougher, D Pawlowski, JM Bell, S Nelli, T McDunn, JR Murphy, M Chizek, and A Ridley. Mars global ionosphere-thermosphere model: Solar cycle, seasonal, and diurnal variations of the mars upper atmosphere. *Journal of Geophysical Research: Planets*, 120(2):311–342, 2015.

-
- [17] Hermann J Opgenoorth, David J Andrews, M Fränz, Mark Lester, NJT Edberg, D Morgan, F Duru, O Witasse, and AO Williams. Mars ionospheric response to solar wind variability. *Journal of Geophysical Research: Space Physics*, 118(10):6558–6587, 2013.
- [18] European Space Agency. Mars’ ionosphere shaped by crustal magnetic fields, November 2016. Last access: 2025-10-13.
- [19] Olga Melnik and Michel Parrot. Propagation of electromagnetic waves through the martian ionosphere. *Journal of Geophysical Research: Space Physics*, 104(A6):12705–12714, 1999.
- [20] JW Gao, ZJ Rong, Lucy Klinger, XZ Li, D Liu, and Y Wei. A spherical harmonic martian crustal magnetic field model combining data sets of maven and mgs. *Earth and Space Science*, 8(10):e2021EA001860, 2021.
- [21] NASA. Mariner 9, 2025. Accessed: October 11, 2025.
- [22] NASA. Mars odyssey, 2025. Accessed: October 11, 2025.
- [23] European Space Agency. Mars express, 2025. Last access: 2025-10-16.
- [24] NASA. Mars reconnaissance orbiter, 2025. Accessed: October 11, 2025.
- [25] NASA. Mars science laboratory: Curiosity rover, 2025. Accessed: October 11, 2025.
- [26] NASA. Mars orbiter mission (mangalyaan), 2025. Accessed: October 11, 2025.
- [27] NASA. Mars atmosphere and volatile evolution (maven), 2025. Accessed: October 11, 2025.
- [28] European Space Agency (ESA). Exomars factsheet, 2025. Accessed: October 11, 2025.
- [29] NASA. Mars 2020: Perseverance rover, 2025. Accessed: October 11, 2025.
- [30] The Planetary Society. Tianwen-1 and zhurong, china’s mars orbiter and rover, 2025. Accessed: October 11, 2025.
- [31] eoPortal. Sherpa rideshare mission, 2025. Last access: 2025-10-14.
- [32] Stephen Clark. Cluster of international satellites ready for ride into orbit on vega rocket, June27 2020. Accessed: October 14, 2025.
- [33] D-Orbit S.p.A. D-orbit launches ion’s first orbital transportation mission into a mid-inclination orbit. <https://www.dorbit.space/media/3/42.pdf>, January31 2023. Accessed: 2025-10-14.
- [34] Momentum Space. Our missions, 2025. Last access: 2025-10-14.
- [35] Northrop Grumman. Espastar, 2025. Accessed: 2025-10-14.
- [36] ESA. Towards low-cost missions to mars, 21/01/2025. Last access: 2025-10-11.
- [37] Technical report 32-1550, mariner mars 1971 project final report. Technical report, NASA, 1973.
- [38] Maggie Aderin-Pocock. What happened to beagle 2?, 2022. Last access: 2025-10-16.
- [39] Toni Tolker-Nielsen. Exomars 2016 - schiaparelli anomaly inquiry. Technical report, ESA, 2017.
- [40] NASA. Perseverance rover’s entry, descent and landing profile, 05/01/2021. Last access: 2025-10-12.
- [41] Momentum Inc. Investor presentation, 2020. Accessed: 2025-10-14.
- [42] Jos Heyman. The continued rise of cubesats. <http://satmagazine.com/story.php?number=192493859#:~:text=Although%20not%20a%20launch%20vehicle,they%20are%20into%20different%20orbits>, July 2015. Accessed: 2025-10-19.
- [43] Richard Zurek, Leslie Tamppari, M Dan Johnston, Scott Murchie, Alfred McEwen, Shane Byrne, Roberto Seu, Nathaniel Putzig, David Kass, Michael Malin, et al. Mro overview: Sixteen years in mars orbit. *Icarus*, 419:116102, 2024.

-
- [44] The planetary society. Cost of the mars reconnaissance orbiter. Last access: 2025-10-11.
- [45] R. W. Saaty. The analytic hierarchy process—what it is and how it is used. *Mathematical Modelling*, 9(3-5):161–176, 1987.
- [46] Matthew W. Gerberich and Steven R. Oleson. Estimation model of spacecraft parameters and cost based on a statistical analysis of COMPASS designs. *American Institute of Aeronautics and Astronautics*, 2013. AIAA Student Member; COMPASS Lead Engineer.
- [47] European Space Agency. Mars express: The scientific payload. Technical Report ESA SP–1291, European Space Agency (ESA), 2004. Describes Mars Express and the single release of the Beagle-2 lander from a fixed elliptical orbit.
- [48] Jet Propulsion Laboratory. Mars reconnaissance orbiter mission description. Technical Report JPL D–31688, National Aeronautics and Space Administration (NASA), 2006. Describes the single-orbit architecture and aerobraking operations of the Mars Reconnaissance Orbiter.
- [49] European Space Agency. Bepicolombo mission to mercury. Technical Report ESA SP–1320, European Space Agency (ESA), 2020. Describes the composite spacecraft architecture (MTM, MPO, MMO) and the unique shared protection and transfer strategy during deep-space cruise.
- [50] Council of European Aerospace Societies (CEAS) / Deutsche Gesellschaft für Luft- und Raumfahrt (DGLR). Ceas newsletter issue 3 – 2024. https://www.dglr.de/fileadmin/daten-dglr/informieren/meldungen/ceasnewsletter/ceas_newsletter_2024-3.pdf, September 2024.
- [51] Frederic Masson. Ariane 6 and space tugs: An enabler for european exploration missions. In *Proceedings of the 68th IAC (International Astronautical Congress)*, September 2017. IAC-17.D2.8.A5.4.2.
- [52] HRE-XE. Margin philosophy for mars exploration studies. Technical Report ESA-E3P-MSR-RS-001, European Space Agency (ESA), Noordwijk, The Netherlands, December 2017. Issue 1, Revision 0.
- [53] Space Exploration Technologies Corp. (SpaceX). Falcon user’s guide. <https://www.spacex.com/media/falcon-users-guide-2025-05-09.pdf>, May 2025. Applicable to Falcon 9 and Falcon Heavy configurations.
- [54] Athul Pradeepkumar Girija. Launch vehicle high-energy performance dataset. Dataset, School of Aeronautics and Astronautics, Purdue University, 2025. West Lafayette, IN, USA.
- [55] Institute of Space and Astronautical Science (ISAS), Japan Aerospace Exploration Agency (JAXA). Nozomi (planet-b) — japan’s first mars explorer. <https://www.isas.jaxa.jp/en/missions/spacecraft/past/nozomi.html>, 2025. Accessed 12 Nov 2025.
- [56] D. T. Lyons, J. G. Beerer, P. Esposito, M. D. Johnston, and W. H. Willcockson. Mars global surveyor: Aerobraking mission overview. *Journal of Spacecraft and Rockets*, 36(3):307–313, 1999.
- [57] Maximilien Berthet. Aerobraking: Review of the state of the art and required developments. *Acta Astronautica*, 2025.
- [58] M. Denis, P. Schmitz, S. Sangiorgi, R. Guilanya, P. Mitschdoerfer, M. Montagna, H. Renault, and N. Kutrowski. Thousand times through the atmosphere of mars: Aerobraking the exomars trace gas orbiter. In *2018 SpaceOps Conference*, Marseille, France, 2018. AIAA. Paper AIAA 2018-2713.
- [59] S. Demcak, B. Young, E. Graat, R. Beswick, K. Criddle, R. Ionasescu, R. S. Hughes, et al. Navigation design and operations of maven aerobraking. In *AIAA Scitech 2020 Forum*, Orlando, Florida, USA, 2020. AIAA. Paper AIAA 2020-0471.
- [60] Daniele Barberi Spirito, Jacopo Prinetto, Andrea Capannolo, and Michèle Lavagna. Data relay constellation for high-performance links supply to future martian missions. *Advances in Space Research*, 73(11):5696–5713, 2024. Open Access under CC BY-NC-ND license. Politecnico di Milano, Department of Aerospace Science and Technology.

-
- [61] C. D. Edwards, P. R. Barela, R. E. Gladden, et al. Replenishing the mars relay network. In *Proceedings of the 2014 IEEE Aerospace Conference*, pages 1–13, 2014. Describes the architecture and operational performance of the Mars Relay Network (MRO, Odyssey, TGO) supporting multiple surface assets.
- [62] ECSS Secretariat / ESA. ECSS-E-ST-70-41C Rev.1 DIR1: Telemetry and telecommand packet utilization, 2024. Draft standard – 11 December 2024.
- [63] Scott R Starin and John Eterno. Attitude determination and control systems. 2011.
- [64] Paul A Mason and Scott R Starin. Sdo delta h mode design and analysis. In *Proceedings of the 20th International Symposium on Space Flight Dynamics*, 2007.
- [65] Mathias Lauer, Sabine Kielbassa, and Ulrich Herfort. Assessment of aocs in-orbit performance for mars express and rosetta. In *18th International Symposium on Space Flight Dynamics*, volume 548, page 21, 2004.
- [66] Sodern. *WISEUR HP: High Performance Star Tracker – Technical Specification*. Sodern, 2025. Accessed: 2025-12-22.
- [67] Bradford Space. *Coarse Sun Sensor Datasheet*. Bradford Space, 2017. Accessed: 2025-12-22.
- [68] Airbus Defence and Space. *Astrix 1090 Datasheet*. Airbus Defence and Space, 2020. Accessed: 2025-12-22.
- [69] Collins Aerospace. *RSI 12-75-60 Reaction Wheel Datasheet*. Collins Aerospace, 2021. Accessed: 2025-12-22.
- [70] Aerojet Rocketdyne. *MR-103J 1N Monopropellant Thruster Datasheet*. Aerojet Rocketdyne, 2020. Accessed: 2025-12-22.
- [71] Collins Aerospace. *RSI 30-280-30 Reaction Wheel Assembly Datasheet*. Collins Aerospace, 2021. Accessed: August 31, 2026.
- [72] Rozann W. Saaty. The analytic hierarchy process—what it is and how it is used. *Mathematical Modelling*, 9(3-5):161–176, 1987.
- [73] L3Harris Technologies. Ar bipropellant rocket engines datasheet. [Datasheet_thrusters/Chemical/l3harris-ar-bipropellant-rocket-engines.pdf](#), 2023.
- [74] L3Harris Technologies. Monopropellant rocket engines data sheet. <https://www.l3harris.com/sites/default/files/2025-10/l3harris-ar-monopropellant-rocket-engines.pdf>, 2025.
- [75] Kyun Ho Lee. Numerical comparison of exhaust plume flow behaviors of small monopropellant and bipropellant thrusters. *PLOS ONE*, 2017.
- [76] Pedro J. Herráiz, José M^a Fernández, and José R. Villa. Development of a MATLAB[®] plume impingement tool for fast system analysis. In *8th European Conference for Aeronautics and Aerospace Sciences (EUCASS)*, Madrid, Spain, 2019. EUCASS.
- [77] Titanium material ti-6al-4v. <https://www.saglammetal.com/en/titanium-alloys/other-alloys/ti-gr5-grade-5>.
- [78] MT Aerospace AG. E3000 family – 1241 / 5901 bi-propellant pmd tank datasheet. https://www.mt-aerospace.de/files/mta/tankkatalog/Datenblatt_E3000-Family-124-590.pdf.
- [79] ArianeGroup - Orbital Propulsion Centre. Surface tension tank ost 31/0 hydrazine propellant tank brochure. <https://www.space-propulsion.com/brochures/propellant-tanks/104-1771t-n2h4-tank-ost-31-0.pdf>.
- [80] National Aeronautics and Space Administration. Mars relay network. <https://science.nasa.gov/mars/mars-relay-network>, 2025. Accessed: 2025-11-13.
- [81] Jim Taylor, Dennis K Lee, and Shervin Shambayati. Mars reconnaissance orbiter. *Deep Space Communications*, pages 193–250, 2016.

-
- [82] Jet Propulsion Laboratory. “prolific nasa mars orbiter passes big data milestone”. <https://www.jpl.nasa.gov/news/prolific-nasa-mars-orbiter-passes-big-data-milestone/>, Nov 2013. Accessed: 2025-11-13.
- [83] National Aeronautics and Space Administration. Mars odyssey. <https://science.nasa.gov/mission/odyssey/>, 2025. Accessed: 2025-11-13.
- [84] European Space Agency (ESA). Exomars/tgo operations, 2016. Accessed: November 13, 2025.
- [85] CD Edwards, JT Adams, DJ Bell, R Cesarone, R DePaula, JF Durning, TA Ely, RY Leung, CA McGraw, and SN Rosell. Strategies for telecommunications and navigation in support of mars exploration. *Acta Astronautica*, 48(5-12):661–668, 2001.
- [86] Charles D Edwards, Philip R Barela, Roy E Gladden, Charles H Lee, and Ramon De Paula. Replenishing the mars relay network. In *2014 IEEE Aerospace Conference*, pages 1–13. IEEE, 2014.
- [87] Charles D. Edwards. Relay communications for mars exploration. *International Journal of Satellite Communications and Networking*, 25(2):111–145, 2007.
- [88] Ramon P De Paula. Telecommunications systems evolution for mars exploration. In *54th International Astronautical Congress of the International Astronautical Federation, the International Academy of Astronautics, and the International Institute of Space Law*, pages M–4, 2003.
- [89] Salvatore Alfano, Daniel Negron, and James L. Moore. Rapid determination of satellite visibility periods. *Journal of the Astronautical Sciences*, 40(3):281–296, 1992.
- [90] Valentina Marchese, Karthik V Mani, Luca Vigna, Alessandro Novero, Francesca Ingiosi, Federico Miglioretti, Simone Simonetti, Stefania Cornara, Pablo Hermosin Herrera, Stefania Tonetti, et al. Microsatellite constellation for mars communication and navigation. 2021.
- [91] James R. Wertz and Wiley J. Larson. *Space Mission Analysis and Design*. Space Technology Library. Microcosm Press and Kluwer Academic Publishers, 3rd edition, 1999.
- [92] F. Castellini, G. Bellei, and B. Godard. Flight dynamics operational experience from ExoMars TGO aerobraking campaign at mars. In *2018 SpaceOps Conference*, Marseille, France, May–June 2018. American Institute of Aeronautics and Astronautics.
- [93] P. G. Antreasian, D. T. Baird, J. S. Border, P. D. Burkhart, E. J. Graat, M. K. Jah, R. A. Mase, T. P. McElrath, and B. M. Portock. 2001 Mars Odyssey orbit determination during interplanetary cruise. In *AAS/AIAA Astrodynamics Specialist Conference*, AAS 01-356, Quebec City, Canada, July–August 2001. Univelt, Inc.
- [94] Tung-Han You, Eric Graat, Allen Halsell, Dolan Highsmith, Stacia Long, Ram Bhat, Stuart Demcak, Earl Higa, Neil Mottinger, and Moriba Jah. Mars reconnaissance orbiter interplanetary cruise navigation. In *20th International Symposium on Space Flight Dynamics*, Annapolis, Maryland, September 2007. Also presented at Space 2007 Conference & Exposition, AIAA-2007-6093, Long Beach, California.
- [95] CESI S.p.A. Triple-junction solar cell for space applications (ctj30), 2020.
- [96] European Space Agency (ESA) and Royal Belgian Institute for Space Aeronomy (BIRA-IASB). SPENVIS - Space Environment Information System. <https://www.spenvis.oma.be/>, 2026.
- [97] Marshall Smart, Bugga Ratnakumar, Larry Whitcanack, Elizabeth Dewell, Loren Jones, Christopher Salvo, Frank Puglia, Robert Gitzendanner, and S Cohen. Performance testing of yardney li-ion cells and batteries in support of jpl’s 2009 mars science laboratory mission. In *6th International Energy Conversion Engineering Conference (IECEC)*, page 5781, 2008.
- [98] Magellan Aerospace. *Power Control Unit*. Magellan Aerospace, Mississauga, ON, Canada, 2026. Scheda tecnica - PCU Digital Version.
- [99] Maja Marminge. Design of a resettable and scalable hold-down release mechanism for space systems, 2025.

-
- [100] NANOSTAR Project. Thermal design - preliminary design. <https://nanostarproject.gitlab.io/main/source/preliminary-design/thermal.html>, 2020.
- [101] SatNow. Spacecraft & satellite propellant tanks - MON propellant filter. <https://www.satnow.com/search/propellant-tanks/filters?page=1&country=global&spropellant=;MON;>, 2026. Accessed: 2026-02-19.
- [102] SatNow. Spacecraft & satellite propellant tanks - Helium filter. <https://www.satnow.com/search/propellant-tanks/filters?page=1&country=global&spropellant=;Helium;>, 2026. Accessed: 2026-02-19.
- [103] BAE Systems. Rad750[®] radiation-hardened powerpc microprocessor. Technical Report CS-16-F78, BAE Systems, Space Electronics, Manassas, VA, USA, 2025. Product datasheet, cleared for open publication on 07/08.
- [104] AAC CLYDE SPACE. *TRX-U UHF transceiver datasheet*. AAC CLYDE SPACE, 2022. Accessed: August 31, 2026.
- [105] Teledyne. *TDLNA0840SEP Low Noise Amplifier Datasheet*. Teledyne, 2025. Accessed: August 31, 2026.
- [106] Moog Inc. Radiation-tolerant advanced scalable solid state recorder, 2022. Accessed 2025.
- [107] Microchip Technology Inc. Radiation-tolerant fpgas. Technical Report DS00003023D, Microchip Technology Inc., Chandler, AZ, USA, 2022. Product brochure, Rev. DS00003023D, November 2022.
- [108] General Dynamics Mission System. Spacecraft terminal for x-band and ka-band deep space transmission, 2026. Accessed 2026.
- [109] Anteral. *DPLX-SATSX-01 datasheet*. Anteral, 2023. Accessed: August 31, 2026.
- [110] Stellant. *86160Hz Traveling Wave Tubes Datasheet*. Stellant, 2024. Accessed: August 31, 2026.
- [111] Celestia TTI. *GaN UHF SSPA*. Celestia TTI, 2024. Accessed: August 31, 2026.
- [112] David G Gilmore. Spacecraft thermal control handbook, fundamental technologies, vol. i the aerospace press/american institute of aeronautics and astronautics, 2002.
- [113] D-Orbit. Launch & deployment, 2026. Accessed: 2026-02-21.
- [114] Space Exploration Technologies Corp. (SpaceX). Rideshare payload user’s guide, September 2024. Version 10. Accessed: 2026-02-21.
- [115] Space engineering – structural factors of safety for spaceflight hardware, 2019.